

Solutions to Bruce E. Hansen,
Econometrics

(Princeton University Press, 2022)

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Authorship and division of labour

Zhizhong Pu wrote the original solutions to Chapters 2, 3, 4, 5 and 7 (the upstream project, March 2025), distributed as separate chapter files.

Zijun Meng and Claude Opus 5 produced everything else in this edition: consolidated the chapters into a single document, completed the previously unanswered exercises, corrected errors in the earlier drafts, added the new Chapters 8–29, and wrote all the Python code and figures behind the empirical exercises.

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The exercise statements are not reproduced here: read them in Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Each solution is headed by the number of the exercise it answers and uses the notation of that exercise in the book. Numbered results and equations cited as THEOREM 2.4.4 or (4.32) refer to Hansen (2022). The empirical exercises use the data sets from the textbook website (<https://www.ssc.wisc.edu/~bhansen/econometrics/>). Where the book reports estimates on the same data, the solutions first reproduce them. Apparent errors in the textbook are flagged where they arise. The code for the empirical exercises is in the `code/` folder that accompanies this document.

Chapter 2

Conditional Expectation and Projection

Solutions to all 22 exercises of Chapter 2 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). The exercise statements are not reproduced; read them in the textbook. Numbered results and equations cited as THEOREM 2.4.4 or (2.57) refer to the textbook.

Throughout, (Y, X) are random variables with $\mathbb{E}|Y| < \infty$, $m(x) = \mathbb{E}[Y \mid X = x]$ is the conditional expectation function (CEF), $e = Y - m(X)$ is the CEF error, and $\sigma^2(x) = \text{var}[Y \mid X = x]$ is the conditional variance. In the projection model $X \in \mathbb{R}^k$ and $\beta = (\mathbb{E}[XX'])^{-1}\mathbb{E}[XY]$. “LIE” abbreviates the law of iterated expectations (THEOREM 2.7) and “the conditioning theorem” the take-out-what-is-known property (THEOREM 2.9).

Exercise 2.1

Apply the LIE twice, working from the inside out. The inner pair of conditioning sets is nested, $\{X_1, X_2\} \subset \{X_1, X_2, X_3\}$, so

$$\mathbb{E}[\mathbb{E}[Y \mid X_1, X_2, X_3] \mid X_1, X_2] = \mathbb{E}[Y \mid X_1, X_2].$$

Repeating the argument for the pair $\{X_1\} \subset \{X_1, X_2\}$,

$$\mathbb{E}[\mathbb{E}[\mathbb{E}[Y \mid X_1, X_2, X_3] \mid X_1, X_2] \mid X_1] = \mathbb{E}[\mathbb{E}[Y \mid X_1, X_2] \mid X_1] = \mathbb{E}[Y \mid X_1].$$

The lesson is that conditioning down a nested sequence of information sets leaves only the coarsest one: the smaller information set always wins.

Exercise 2.2

Write the CEF error as $e = Y - \mathbb{E}[Y \mid X] = Y - a - bX$, so that

$$Y = a + bX + e, \quad \mathbb{E}[e \mid X] = 0.$$

Multiplying by X and taking expectations,

$$\mathbb{E}[YX] = \mathbb{E}[(a + bX + e)X] = a\mathbb{E}[X] + b\mathbb{E}[X^2] + \mathbb{E}[eX].$$

The last term vanishes by THEOREM 2.4.4 (see Exercise 2.3), so

$$\mathbb{E}[YX] = a\mathbb{E}[X] + b\mathbb{E}[X^2].$$

Exercise 2.3

Proof. Condition on X and use the LIE, then the conditioning theorem (legitimate because $h(X)$ is a function of the conditioning variable), and finally $\mathbb{E}[e \mid X] = 0$ from THEOREM 2.4.1:

$$\mathbb{E}[h(X)e] = \mathbb{E}[\mathbb{E}[h(X)e \mid X]] = \mathbb{E}[h(X)\mathbb{E}[e \mid X]] = \mathbb{E}[h(X) \cdot 0] = 0.$$

The moment condition $\mathbb{E}|h(X)e| < \infty$ is what makes every expectation above well defined. □

This is much stronger than the projection orthogonality $\mathbb{E}[Xe] = 0$: it holds for *every* function h , not just linear ones. Exercises 2.10–2.13 explore exactly this distinction.

Exercise 2.4

First get the marginal distribution of X by summing columns: $\mathbb{P}[X = 0] = .1 + .4 = .5$ and $\mathbb{P}[X = 1] = .2 + .3 = .5$. Hence

$$\mathbb{P}[Y = 1 \mid X = 0] = \frac{.4}{.5} = 0.8, \quad \mathbb{P}[Y = 1 \mid X = 1] = \frac{.3}{.5} = 0.6.$$

Because Y is binary, $\mathbb{E}[Y \mid X = x] = \mathbb{P}[Y = 1 \mid X = x]$:

$$\mathbb{E}[Y \mid X = 0] = 0.8, \quad \mathbb{E}[Y \mid X = 1] = 0.6.$$

Also $Y^2 = Y$ for a 0/1 variable, so $\mathbb{E}[Y^2 \mid X] = \mathbb{E}[Y \mid X]$:

$$\mathbb{E}[Y^2 \mid X = 0] = 0.8, \quad \mathbb{E}[Y^2 \mid X = 1] = 0.6.$$

By $\text{var}[Y \mid X] = \mathbb{E}[Y^2 \mid X] - (\mathbb{E}[Y \mid X])^2$ (Exercise 2.7),

$$\text{var}[Y \mid X = 0] = 0.8 - 0.8^2 = 0.16, \quad \text{var}[Y \mid X = 1] = 0.6 - 0.6^2 = 0.24.$$

As an arithmetic check, the LIE gives $\mathbb{E}[\mathbb{E}[Y \mid X]] = 0.5(0.8) + 0.5(0.6) = 0.7$, which equals $\mathbb{E}[Y] = .4 + .3 = 0.7$.

Exercise 2.5

(a) For any function h with $\mathbb{E}[h(X)^2] < \infty$, the mean-squared error of $h(X)$ as a predictor of e^2 is

$$\text{MSE}(h) = \mathbb{E}[(e^2 - h(X))^2].$$

(b) Since $e = Y - m(X)$, the variable e^2 is the squared deviation of Y from its conditional mean — the realized squared forecast error of the best point predictor of Y . Predicting e^2 from X therefore means forecasting *how large the forecast error of Y will be* at a given x ; that is, forecasting the local dispersion of Y rather than its location. It is the population version of asking “for which x is Y hard to predict?”

(c) Note first that $\mathbb{E}[e^2 \mid X] = \text{var}[Y \mid X] = \sigma^2(X)$, because $\mathbb{E}[e \mid X] = 0$. Now add and subtract $\sigma^2(X)$ and expand:

$$\begin{aligned} \text{MSE}(h) &= \mathbb{E}[(e^2 - \sigma^2(X) + \sigma^2(X) - h(X))^2] \\ &= \mathbb{E}[(e^2 - \sigma^2(X))^2] + 2\mathbb{E}[(e^2 - \sigma^2(X))(\sigma^2(X) - h(X))] + \mathbb{E}[(\sigma^2(X) - h(X))^2]. \end{aligned}$$

The cross term is zero: conditioning on X and applying the conditioning theorem to the function $\sigma^2(X) - h(X)$,

$$\mathbb{E}[(e^2 - \sigma^2(X))(\sigma^2(X) - h(X))] = \mathbb{E}\left[(\sigma^2(X) - h(X)) \underbrace{\mathbb{E}[e^2 - \sigma^2(X) \mid X]}_{=0}\right] = 0.$$

Therefore

$$\text{MSE}(h) = \mathbb{E}[(e^2 - \sigma^2(X))^2] + \mathbb{E}[(\sigma^2(X) - h(X))^2] \geq \mathbb{E}[(e^2 - \sigma^2(X))^2],$$

with equality if and only if $h(X) = \sigma^2(X)$ almost surely. So $\sigma^2(X)$ is the essentially unique minimizer.

This is the same argument that shows $m(X)$ is the best predictor of Y (THEOREM 2.10), applied to the pair (e^2, X) instead of (Y, X) .

Exercise 2.6

By definition of variance, $\text{var}[Y] = \mathbb{E}[(m(X) + e)^2] - (\mathbb{E}[m(X) + e])^2$. Take the two pieces in turn. For the first, expand the square and use $\mathbb{E}[m(X)e] = 0$ (THEOREM 2.4.4 with $h = m$):

$$\mathbb{E}[(m(X) + e)^2] = \mathbb{E}[m(X)^2] + 2\mathbb{E}[m(X)e] + \mathbb{E}[e^2] = \mathbb{E}[m(X)^2] + \mathbb{E}[e^2].$$

For the second, $\mathbb{E}[e] = \mathbb{E}[\mathbb{E}[e | X]] = 0$, so $(\mathbb{E}[m(X) + e])^2 = (\mathbb{E}[m(X)])^2$. Subtracting,

$$\text{var}[Y] = \mathbb{E}[m(X)^2] - (\mathbb{E}[m(X)])^2 + \mathbb{E}[e^2] = \text{var}[m(X)] + \sigma^2.$$

In words: total variation equals variation explained by the CEF plus unexplained variation. Since both terms are non-negative this also proves $\text{var}[m(X)] \leq \text{var}[Y]$.

Exercise 2.7

Start from the definition and expand the square inside the conditional expectation:

$$\begin{aligned} \sigma^2(X) &\equiv \text{var}[Y | X] = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2 | X] \\ &= \mathbb{E}[Y^2 - 2Y \mathbb{E}[Y | X] + (\mathbb{E}[Y | X])^2 | X] \\ &= \mathbb{E}[Y^2 | X] - 2\mathbb{E}[Y \mathbb{E}[Y | X] | X] + \mathbb{E}[(\mathbb{E}[Y | X])^2 | X]. \end{aligned}$$

Both $\mathbb{E}[Y | X]$ and $(\mathbb{E}[Y | X])^2$ are functions of X , so the conditioning theorem lets us pull them outside the outer conditional expectation:

$$\sigma^2(X) = \mathbb{E}[Y^2 | X] - 2(\mathbb{E}[Y | X])^2 + (\mathbb{E}[Y | X])^2 = \mathbb{E}[Y^2 | X] - (\mathbb{E}[Y | X])^2.$$

Exercise 2.8

A Poisson variable with parameter λ has mean and variance both equal to λ . Here the conditional distribution of Y given $X = x$ is Poisson with $\lambda = x'\beta$, so

$$\mathbb{E}[Y | X] = X'\beta, \quad \text{var}[Y | X] = X'\beta.$$

What this means for a linear regression model. It is only a partial justification, and the qualifications are the point of the exercise.

- (i) *The CEF really is linear.* Setting $e = Y - X'\beta$ gives $Y = X'\beta + e$ with $\mathbb{E}[e | X] = 0$. So this is a correctly specified linear *conditional expectation* model, not merely a linear projection, and least squares consistently estimates β .

- (ii) *But the errors are heteroskedastic.* $\mathbb{E}[e^2 | X] = X'\beta$ varies with X , so the homoskedastic formulas do not apply: robust standard errors are required and OLS is not efficient.
- (iii) *And the model is coherent only where $X'\beta > 0$.* A Poisson parameter must be strictly positive, so the specification implicitly requires $X'\beta > 0$ on the whole support of X . A linear index cannot guarantee this for unbounded regressors, which is exactly why applied work uses the exponential mean $\mathbb{E}[Y | X] = \exp(X'\beta)$ instead.
- (iv) *The error is discrete and skewed.* e takes values in $\{-X'\beta, 1 - X'\beta, 2 - X'\beta, \dots\}$, so it is far from normal. This matters for exact inference (Chapter 5) but not for the large-sample theory of Chapter 7.

Exercise 2.9

(X_1, X_2) has $2 \times 3 = 6$ cells, so the CEF is just a list of six cell means, $\mu_{x_1, c} = \mathbb{E}[Y | X_1 = x_1, X_2 = c]$. Any such function can be written exactly as a linear regression in six indicators. Define

$$D_B = \mathbf{1}\{X_2 = B\}, \quad D_C = \mathbf{1}\{X_2 = C\},$$

and take the cell $(X_1 = 0, X_2 = A)$ as the omitted baseline. Then

$$\mathbb{E}[Y | X_1, X_2] = \beta_0 + \beta_1 X_1 + \beta_2 D_B + \beta_3 D_C + \beta_4 X_1 D_B + \beta_5 X_1 D_C$$

and the coefficients are exactly identified by the six cell means:

$$\begin{aligned} \beta_0 &= \mu_{0,A}, & \beta_1 &= \mu_{1,A} - \mu_{0,A}, & \beta_2 &= \mu_{0,B} - \mu_{0,A}, \\ \beta_3 &= \mu_{0,C} - \mu_{0,A}, & \beta_4 &= (\mu_{1,B} - \mu_{0,B}) - (\mu_{1,A} - \mu_{0,A}), & \beta_5 &= (\mu_{1,C} - \mu_{0,C}) - (\mu_{1,A} - \mu_{0,A}). \end{aligned}$$

This is the *saturated* model: it has as many free parameters as cells, so the linear regression is not an approximation — it reproduces the CEF exactly.

Two warnings. First, one category of X_2 *must* be omitted: including D_A, D_B, D_C together with the intercept makes the design singular, because $D_A + D_B + D_C = 1$. Second, dropping the interaction terms $X_1 D_B$ and $X_1 D_C$ imposes the restriction that the effect of X_1 is the same in all three categories; that restricted model is a genuine approximation to the CEF, not an identity.

Exercise 2.10

True. $h(x) = x^2$ is a function of X , so this is a special case of THEOREM 2.4.4 (Exercise 2.3). Explicitly,

$$\mathbb{E}[X^2 e] = \mathbb{E}[\mathbb{E}[X^2 e | X]] = \mathbb{E}[X^2 \mathbb{E}[e | X]] = \mathbb{E}[X^2 \cdot 0] = 0,$$

assuming $\mathbb{E}|X^2 e| < \infty$ so that the expectation exists.

Exercise 2.11

False. $\mathbb{E}[Xe] = 0$ is a *single* moment condition; it says nothing about the second moment of X against e .

Counterexample. Let $X \sim N(0, 1)$ and set $Y = X + X^2$. Take β to be the linear projection coefficient of Y on X (no intercept):

$$\beta = \frac{\mathbb{E}[XY]}{\mathbb{E}[X^2]} = \frac{\mathbb{E}[X^2] + \mathbb{E}[X^3]}{\mathbb{E}[X^2]} = \frac{1 + 0}{1} = 1,$$

so the projection error is $e = Y - X\beta = X^2$. By construction $\mathbb{E}[Xe] = \mathbb{E}[X^3] = 0$, as required. But

$$\mathbb{E}[X^2e] = \mathbb{E}[X^4] = 3 \neq 0.$$

Comparing with Exercise 2.10: the conclusion there came from $\mathbb{E}[e | X] = 0$, a restriction on the entire conditional distribution, not from $\mathbb{E}[Xe] = 0$ alone.

Exercise 2.12

False. $\mathbb{E}[e | X] = 0$ restricts only the conditional *mean*; independence restricts the entire conditional distribution. Heteroskedasticity is the standard counterexample.

Counterexample. Let $X \sim N(0, 1)$ and $u \sim N(0, 1)$ be independent, and set $e = Xu$. Then

$$\mathbb{E}[e | X] = X \mathbb{E}[u] = 0,$$

so mean independence holds, yet

$$\text{var}[e | X] = X^2 \text{var}[u] = X^2$$

depends on X . A variable whose conditional variance moves with X cannot be independent of X .

Exercise 2.13

False. This is the central distinction between the linear *projection* model (Section 2.18) and the linear *regression* (CEF) model (Section 2.8): the projection is defined by k orthogonality conditions, which is far weaker than mean independence.

Counterexample. Let $Z \sim N(0, 1)$, $X = (1, Z)'$, and $Y = Z^2$. The best linear predictor of Y given X has coefficients

$$\beta_2 = \frac{\text{cov}(Z, Z^2)}{\text{var}(Z)} = \mathbb{E}[Z^3] = 0, \quad \beta_1 = \mathbb{E}[Y] - \beta_2 \mathbb{E}[Z] = \mathbb{E}[Z^2] = 1,$$

so the projection error is $e = Z^2 - 1$. Then

$$\mathbb{E}[e] = 0, \quad \mathbb{E}[Ze] = \mathbb{E}[Z^3] = 0 \implies \mathbb{E}[Xe] = \mathbf{0},$$

exactly as required. But $\mathbb{E}[e | X] = Z^2 - 1 \neq 0$. So the projection is well defined and orthogonal, yet the linear function $X'\beta$ is not the CEF — the true CEF $\mathbb{E}[Y | X] = Z^2$ is nonlinear.

Exercise 2.14

False. Fixing the first two conditional moments leaves the rest of the conditional distribution free, and independence requires *every* feature of the conditional distribution to be constant in X .

Counterexample. Let X be Bernoulli(1/2) and let

$$e | X = 0 \sim N(0, 1), \quad e | X = 1 \sim \text{Uniform}[-\sqrt{3}, \sqrt{3}].$$

Both conditional distributions have mean 0 and variance 1, so $\mathbb{E}[e | X] = 0$ and $\mathbb{E}[e^2 | X] = 1 = \sigma^2$. But the fourth conditional moments differ,

$$\mathbb{E}[e^4 | X = 0] = 3, \quad \mathbb{E}[e^4 | X = 1] = \frac{9}{5} = 1.8,$$

so the conditional distribution of e depends on X and e is not independent of X .

Remark. The implications that *do* hold are worth keeping straight: independence $\Rightarrow$ mean independence $\Rightarrow$ orthogonality, and none of the arrows reverses. Exercises 2.11–2.14 supply the counterexamples.

Exercise 2.15

The best linear predictor minimizes the mean-squared error

$$S(a) = \mathbb{E}[(Y - a)^2] = \mathbb{E}[Y^2] - 2a\mathbb{E}[Y] + a^2.$$

This is a strictly convex quadratic in a (the coefficient on a^2 is $1 > 0$), so the first-order condition is necessary and sufficient:

$$\frac{\partial}{\partial a} S(a) = -2\mathbb{E}[Y] + 2a = 0 \quad \Longrightarrow \quad \alpha = \mathbb{E}[Y].$$

Equivalently, the orthogonality condition for the single regressor $X = 1$ reads $\mathbb{E}[Y - \alpha] = 0$, i.e. $\alpha = \mathbb{E}[Y]$. This is the population counterpart of the fact that regressing on a constant alone returns the sample mean.

Exercise 2.16

Step 1: the moments. All integrals are over the unit square.

$$\mathbb{E}[X] = \frac{3}{2} \int_0^1 \int_0^1 (x^3 + xy^2) dy dx = \frac{3}{2} \left(\frac{1}{4} + \frac{1}{6} \right) = \frac{5}{8},$$

$$\mathbb{E}[Y] = \frac{5}{8} \quad (\text{by the symmetry of } f \text{ in } x \text{ and } y),$$

$$\mathbb{E}[X^2] = \frac{3}{2} \int_0^1 \int_0^1 (x^4 + x^2 y^2) dy dx = \frac{3}{2} \left(\frac{1}{5} + \frac{1}{9} \right) = \frac{7}{15},$$

$$\mathbb{E}[XY] = \frac{3}{2} \int_0^1 \int_0^1 (x^3 y + xy^3) dy dx = \frac{3}{2} \left(\frac{1}{8} + \frac{1}{8} \right) = \frac{3}{8}.$$

Hence

$$\text{var}(X) = \frac{7}{15} - \left(\frac{5}{8} \right)^2 = \frac{73}{960}, \quad \text{cov}(X, Y) = \frac{3}{8} - \left(\frac{5}{8} \right)^2 = -\frac{1}{64}.$$

Step 2: the best linear predictor. Because the model contains an intercept,

$$\beta = \frac{\text{cov}(X, Y)}{\text{var}(X)} = \frac{-1/64}{73/960} = -\frac{15}{73} \approx -0.2055,$$

$$\alpha = \mathbb{E}[Y] - \beta \mathbb{E}[X] = \frac{5}{8} \left(1 + \frac{15}{73} \right) = \frac{55}{73} \approx 0.7534.$$

So the best linear predictor is

$$\mathcal{P}[Y | X] = \frac{55}{73} - \frac{15}{73} X$$

Step 3: the conditional expectation. The marginal density of X is

$$f_X(x) = \int_0^1 \frac{3}{2}(x^2 + y^2) dy = \frac{3}{2}\left(x^2 + \frac{1}{3}\right),$$

so $f_{Y|X}(y | x) = (x^2 + y^2) / (x^2 + \frac{1}{3})$ and

$$m(x) = \int_0^1 y \frac{x^2 + y^2}{x^2 + \frac{1}{3}} dy = \frac{\frac{1}{2}x^2 + \frac{1}{4}}{x^2 + \frac{1}{3}} = \boxed{\frac{6x^2 + 3}{12x^2 + 4}}$$

Step 4: are they different? Yes. The CEF $m(x)$ is a strictly nonlinear *decreasing* function of x on $[0, 1]$ ($m(0) = 0.7500$, $m(0.5) = 0.6429$, $m(1) = 0.5625$), while the projection is the straight line running from 0.7534 at $x = 0$ to 0.5479 at $x = 1$. The line tracks the level and the average slope of m but not its curvature. They agree only in the averaged sense guaranteed by the projection, $\mathbb{E}[X(Y - \alpha - \beta X)] = 0$, not pointwise.

Remark. With an intercept in the model the slope must be $\text{cov}(X, Y) / \text{var}(X)$. The no-intercept formula $\mathbb{E}[XY] / \mathbb{E}[X^2] = (3/8) / (7/15) = 45/56$ answers a different question and is not the coefficient asked for here.

Exercise 2.17

Write out the two components of $\mathbb{E}[g(X, m, s)] = \mathbf{0}$:

$$\mathbb{E}[X - m] = \mu - m = 0, \tag{1}$$

$$\mathbb{E}[(X - m)^2 - s] = \mathbb{E}[(X - m)^2] - s = 0. \tag{2}$$

For the second equation, decompose

$$\mathbb{E}[(X - m)^2] = \mathbb{E}[(X - \mu) + (\mu - m)]^2 = \sigma^2 + (\mu - m)^2,$$

since the cross term is $2(\mu - m)\mathbb{E}[X - \mu] = 0$. So (2) reads $s = \sigma^2 + (\mu - m)^2$.

($\Leftarrow$) If $m = \mu$ and $s = \sigma^2$ then (1) and (2) hold by inspection.

($\Rightarrow$) Conversely, (1) forces $m = \mu$; substituting into $s = \sigma^2 + (\mu - m)^2$ gives $s = \sigma^2$.

Hence the system has the unique solution $(m, s) = (\mu, \sigma^2)$: the two moment conditions exactly identify the mean and variance. Exercise 3.1 is the sample analogue — replacing $\mathbb{E}$ by a sample average returns the sample mean and the divide-by- n sample variance.

Exercise 2.18

(a) Take the non-zero vector $c = (\alpha_1, \alpha_2, -1)'$. By construction

$$X'c = \alpha_1 + \alpha_2 X_2 - X_3 = 0 \quad \text{almost surely.}$$

Therefore

$$Q_{XX}c = \mathbb{E}[XX'c] = \mathbb{E}[X \cdot 0] = \mathbf{0},$$

so Q_{XX} has a non-trivial null space and is singular. Equivalently $c'Q_{XX}c = \mathbb{E}[(X'c)^2] = 0$ with $c \neq \mathbf{0}$, so Q_{XX} is positive semi-definite but not positive definite and Assumption 2.1 fails.

(b) The problem is pure redundancy: X_3 carries no information beyond $(1, X_2)$. Let

$$\tilde{X} = \begin{pmatrix} 1 \\ X_2 \end{pmatrix}, \quad X = A\tilde{X}, \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \alpha_1 & \alpha_2 \end{pmatrix}.$$

Then every linear function of X is a linear function of $\tilde{X}$ and conversely: $X'b = \tilde{X}'(A'b)$, and A' maps $\mathbb{R}^3$ onto $\mathbb{R}^2$. The two regressor sets span the same linear space, so they generate the same set of candidate predictors and hence the same best linear predictor.

Since $\tilde{Q} = \mathbb{E}[\tilde{X}\tilde{X}']$ is invertible (provided $\text{var}(X_2) > 0$), the best linear predictor is the ordinary bivariate projection

$$\mathcal{P}[Y | X] = \gamma_1 + \gamma_2 X_2, \quad \gamma_2 = \frac{\text{cov}(X_2, Y)}{\text{var}(X_2)}, \quad \gamma_1 = \mathbb{E}[Y] - \gamma_2 \mathbb{E}[X_2].$$

The predictor is unique; the *coefficient vector* on the original X is not. Any $\beta = (\beta_1, \beta_2, \beta_3)'$ satisfying

$$\beta_1 + \alpha_1 \beta_3 = \gamma_1, \quad \beta_2 + \alpha_2 \beta_3 = \gamma_2$$

delivers the same predictor $X'\beta$ — a one-parameter family indexed by β_3 . This is exactly what a software package reports when it drops a collinear regressor: it selects the solution with $\beta_3 = 0$.

Exercise 2.19

Step 1: minimize d . Expand the quadratic:

$$d(b) = \mathbb{E}[m(X)^2] - 2b'\mathbb{E}[Xm(X)] + b'\mathbb{E}[XX']b.$$

The gradient and Hessian are

$$\frac{\partial}{\partial b}d(b) = -2\mathbb{E}[Xm(X)] + 2\mathbb{E}[XX']b, \quad \frac{\partial^2}{\partial b \partial b'}d(b) = 2\mathbb{E}[XX'].$$

Under Assumption 2.1, $\mathbb{E}[XX'] > 0$, so d is strictly convex and the first-order condition characterizes the unique global minimum:

$$\mathbb{E}[XX']\beta = \mathbb{E}[Xm(X)] \implies \beta = (\mathbb{E}[XX'])^{-1}\mathbb{E}[Xm(X)].$$

Step 2: replace $m(X)$ by Y . By the LIE and the conditioning theorem,

$$\mathbb{E}[Xm(X)] = \mathbb{E}[X\mathbb{E}[Y | X]] = \mathbb{E}[\mathbb{E}[XY | X]] = \mathbb{E}[XY].$$

Substituting gives $\beta = (\mathbb{E}[XX'])^{-1}\mathbb{E}[XY]$, the familiar projection formula.

So the linear projection admits two equivalent readings: it is the best linear predictor of Y , and it is the best linear *approximation to the CEF*. The two optimization problems have the same solution.

Exercise 2.20

Write $f_X(x) = \int f(y, x) dy$ for the marginal density of X , and $f_{Y|X}(y | x) = f(y, x)/f_X(x)$ for the conditional density (defined wherever $f_X(x) > 0$; the set where $f_X(x) = 0$ has probability zero and may be ignored).

Start from the left-hand side and integrate out y first:

$$\begin{aligned}
 \mathbb{E}[\mathbf{1}\{X \in \mathcal{X}\} Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{1}\{x \in \mathcal{X}\} y f(y, x) dy dx \\
 &= \int_{\mathcal{X}} \left(\int_{-\infty}^{\infty} y f(y, x) dy \right) dx && \text{(Fubini, legitimate since } \mathbb{E}|Y| < \infty) \\
 &= \int_{\mathcal{X}} \left(\int_{-\infty}^{\infty} y f_{Y|X}(y | x) dy \right) f_X(x) dx && \text{(factor } f(y, x) = f_{Y|X}(y | x) f_X(x)) \\
 &= \int_{\mathcal{X}} m(x) f_X(x) dx && \text{(definition (2.6) of } m) \\
 &= \mathbb{E}[\mathbf{1}\{X \in \mathcal{X}\} m(X)].
 \end{aligned}$$

Since $\mathcal{X}$ was arbitrary, the m of (2.6) satisfies the defining property (2.57). By the almost-everywhere uniqueness in THEOREM 2.13, the general measure-theoretic conditional expectation coincides with the elementary density-based formula whenever the latter is available.

Exercise 2.21

Throughout, X is a scalar and none of these projections contains an intercept, so $\gamma_1 = \mathbb{E}[XY] / \mathbb{E}[X^2]$. Assume $\mathbb{E}[X^2] > 0$ and that the relevant moments are finite.

(a) From the long projection, multiply by X and take expectations, using $\mathbb{E}[Xu] = 0$:

$$\mathbb{E}[XY] = \beta_1 \mathbb{E}[X^2] + \beta_2 \mathbb{E}[X^3].$$

Dividing by $\mathbb{E}[X^2]$,

$$\gamma_1 = \beta_1 + \beta_2 \frac{\mathbb{E}[X^3]}{\mathbb{E}[X^2]}.$$

So $\gamma_1 = \beta_1$ if and only if $\beta_2 \mathbb{E}[X^3] = 0$: either the omitted regressor is irrelevant ($\beta_2 = 0$), or the omitted regressor is uncorrelated with the included one ($\mathbb{E}[X^3] = \mathbb{E}[X \cdot X^2] = 0$, which holds for instance whenever the distribution of X is symmetric about zero).

(b) The same calculation with X^3 in place of X^2 gives

$$\gamma_1 = \theta_1 + \theta_2 \frac{\mathbb{E}[X^4]}{\mathbb{E}[X^2]}.$$

Now $\mathbb{E}[X^4] > 0$ whenever $\mathbb{P}[X \neq 0] > 0$, so the second escape route of part (a) is unavailable: the omitted regressor X^3 is necessarily correlated with X . Hence $\gamma_1 = \theta_1$ if and only if $\theta_2 = 0$.

The contrast is instructive. Omitted-variable bias vanishes when the omitted variable is either irrelevant or orthogonal to the included regressors. Adding an *even* power creates the possibility of orthogonality through symmetry; adding an *odd* power does not.

Exercise 2.22

Step 1: the short equation. By the LIE and the linearity of $\mathbb{E}[X_2 | X_1]$,

$$\mathbb{E}[Y | X_1] = X_1' \beta_1 + \mathbb{E}[X_2 | X_1]' \beta_2 = X_1' (\beta_1 + \Gamma' \beta_2) \equiv X_1' \theta.$$

So the short CEF is still linear, with coefficient $\theta = \beta_1 + \Gamma' \beta_2$ — the familiar omitted-variable bias formula. Define $W = X_2 - \Gamma X_1$, so that $\mathbb{E}[W | X_1] = 0$, and let

$$u = Y - X_1' \theta = e + W' \beta_2.$$

Then $Y = X_1' \theta + u$ with $\mathbb{E}[u | X_1] = 0$: the short equation is still a correctly specified linear regression, only for a different parameter.

Step 2: the conditional variance of the induced error. Expand:

$$\mathbb{E}[u^2 | X_1] = \mathbb{E}[e^2 | X_1] + 2\beta_2' \mathbb{E}[We | X_1] + \beta_2' \mathbb{E}[WW' | X_1] \beta_2.$$

Take the three terms in turn.

- $\mathbb{E}[e^2 | X_1] = \mathbb{E}[\mathbb{E}[e^2 | X_1, X_2] | X_1] = \sigma^2$, by the LIE and the maintained homoskedasticity.
- $\mathbb{E}[We | X_1] = \mathbb{E}[W \mathbb{E}[e | X_1, X_2] | X_1] = 0$, because W is a function of (X_1, X_2) and $\mathbb{E}[e | X_1, X_2] = 0$.
- $\mathbb{E}[WW' | X_1] = \text{var}[X_2 | X_1]$, by the definition of W and $\mathbb{E}[X_2 | X_1] = \Gamma X_1$.

Therefore

$$\boxed{\mathbb{E}[u^2 | X_1] = \sigma^2 + \beta_2' \text{var}[X_2 | X_1] \beta_2.}$$

Step 3: the answer. Excluding X_2 *generally does induce heteroskedasticity*. The induced error remains homoskedastic if and only if the scalar $\beta_2' \text{var}[X_2 | X_1] \beta_2$ does not vary with X_1 . Being specific, that happens exactly when

- $\beta_2 = 0$ (nothing was really omitted); or
- $\text{var}[X_2 | X_1]$ does not depend on X_1 — e.g. when $X_2 = \Gamma X_1 + W$ with W independent of X_1 ; or, more weakly,
- $\text{var}[X_2 | X_1]$ varies with X_1 only in directions orthogonal to β_2 .

Outside these cases the short regression is heteroskedastic even though the long one is homoskedastic. Note also that the induced conditional variance is never *smaller* than σ^2 , since $\text{var}[X_2 | X_1] \geq 0$: dropping a regressor can only inflate the error variance.

Chapter 3

The Algebra of Least Squares

Solutions to all 26 exercises of Chapter 3 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 3.24–3.26 are empirical; the reported numbers were computed from the `cps09mar` data set available on the textbook website, and the code used is included so that the results can be reproduced.

Notation follows the textbook. Y is $n \times 1$, X is $n \times k$ with rows X_i' , $\hat{\beta} = (X'X)^{-1}X'Y$, $\hat{Y} = X\hat{\beta}$ and $\hat{e} = Y - \hat{Y}$. The projection matrices are $P = X(X'X)^{-1}X'$ and $M = \mathbf{I}_n - P$, and the leverage values are $h_{ii} = X_i'(X'X)^{-1}X_i$, the diagonal entries of P .

Exercise 3.1

The moment equations are

$$\mathbf{0} = \frac{1}{n} \sum_{i=1}^n \begin{pmatrix} y_i - \hat{\mu} \\ (y_i - \hat{\mu})^2 - \hat{\sigma}^2 \end{pmatrix}, \quad \text{i.e.} \quad \begin{cases} \sum_{i=1}^n (y_i - \hat{\mu}) = 0, \\ \sum_{i=1}^n ((y_i - \hat{\mu})^2 - \hat{\sigma}^2) = 0. \end{cases}$$

The first equation is linear in $\hat{\mu}$ and gives

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n y_i = \bar{Y},$$

the sample mean. Substituting into the second and solving for $\hat{\sigma}^2$,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{Y})^2,$$

the sample variance with divisor n . This is the sample analogue of Exercise 2.17: replacing the population expectation by the sample average turns the identifying moment conditions into the estimating equations.

Remark. The estimator obtained this way divides by n , not by $n - 1$: the method of moments delivers the biased (but consistent) version. The bias-corrected estimator solves a rescaled system, not this one.

Exercise 3.2

Coefficients. Write $\hat{\beta} = (X'X)^{-1}X'Y$ for the regression on X and $\hat{\gamma}$ for the regression on Z . Using $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ and $(C')^{-1} = (C^{-1})'$,

$$\begin{aligned} \hat{\gamma} &= (Z'Z)^{-1}Z'Y = (C'X'XC)^{-1}C'X'Y = C^{-1}(X'X)^{-1}(C')^{-1}C'X'Y \\ &= C^{-1}(X'X)^{-1}X'Y = C^{-1}\hat{\beta}. \end{aligned}$$

Fitted values and residuals. The fitted values are unchanged:

$$Z\hat{\gamma} = XC C^{-1}\hat{\beta} = X\hat{\beta} = \hat{Y},$$

and therefore so are the residuals, $Y - Z\hat{\gamma} = Y - X\hat{\beta} = \hat{e}$.

The reason is geometric and worth stating once, because it recurs in Exercises 3.6, 3.7 and 3.23: since C is invertible, Z and X have the *same column space*, so they induce the same projection matrix,

$$P_Z = Z(Z'Z)^{-1}Z' = XC C^{-1}(X'X)^{-1}(C')^{-1}C'X' = P_X.$$

Least squares only depends on the regressors through the space they span. Everything projection-based — $\hat{Y}$, $\hat{e}$, SSE, R^2 , $\hat{\sigma}^2$, the leverage values — is invariant; only the coordinates $\hat{\beta}$ change, and they change by the inverse of the reparameterization.

Exercise 3.3

By definition $\hat{e} = Y - X\hat{\beta}$ with $\hat{\beta} = (X'X)^{-1}X'Y$. Hence

$$X'\hat{e} = X'(Y - X\hat{\beta}) = X'Y - X'X(X'X)^{-1}X'Y = X'Y - X'Y = \mathbf{0}_{k \times 1}.$$

This is just the first-order condition of the least squares problem written in matrix form. Note that it is an algebraic identity: it holds in every sample, with no assumptions about the population whatsoever.

Exercise 3.4

Partition the orthogonality condition of Exercise 3.3. Since $X' = \begin{bmatrix} X_1' & X_2' \end{bmatrix}' = \begin{bmatrix} X_1' \\ X_2' \end{bmatrix}$,

$$\mathbf{0}_{k \times 1} = X'\hat{e} = \begin{bmatrix} X_1' \\ X_2' \end{bmatrix} \hat{e} = \begin{bmatrix} X_1'\hat{e} \\ X_2'\hat{e} \end{bmatrix}.$$

A stacked vector is zero if and only if each block is zero, so

$$X_2'\hat{e} = \mathbf{0}_{k_2 \times 1} \quad (\text{and likewise } X_1'\hat{e} = \mathbf{0}_{k_1 \times 1}).$$

Exercise 3.5

Apply the OLS formula with $\hat{e}$ as the dependent variable and use Exercise 3.3:

$$\hat{\gamma} = (X'X)^{-1}X'\hat{e} = (X'X)^{-1}\mathbf{0} = \mathbf{0}_{k \times 1}.$$

The residuals from that regression are $\hat{e} - X\mathbf{0} = \hat{e}$ and the R^2 is zero. In words: the residual has already had every linear component of X removed, so there is nothing left for X to explain.

Exercise 3.6

$$\hat{\theta} = (X'X)^{-1}X'\hat{Y} = (X'X)^{-1} \underbrace{X'X(X'X)^{-1}}_{= \mathbf{I}_k} X'Y = (X'X)^{-1}X'Y = \hat{\beta},$$

which is exactly the coefficient vector from regressing Y on X . This is the complement of Exercise 3.5: the decomposition $Y = \hat{Y} + \hat{e}$ splits Y into the part X explains perfectly and the part it cannot explain at all.

Exercise 3.7

By (3.21), $MX = \mathbf{0}_{n \times k}$. Partitioning,

$$\mathbf{0}_{n \times k} = MX = M \begin{bmatrix} X_1 & X_2 \end{bmatrix} = \begin{bmatrix} MX_1 & MX_2 \end{bmatrix},$$

so each block is zero and in particular $MX_1 = \mathbf{0}_{n \times k_1}$. Since $M = \mathbf{I}_n - P$ by definition,

$$\mathbf{0} = MX_1 = X_1 - PX_1 \quad \implies \quad PX_1 = X_1.$$

Geometrically: P projects onto the column space of X , and the columns of X_1 already lie in that space, so they are left untouched; M projects onto the orthogonal complement, so it annihilates them.

Exercise 3.8

Expand, using the idempotency of P (i.e. $PP = P$, THEOREM 3.3):

$$MM = (\mathbf{I}_n - P)(\mathbf{I}_n - P) = \mathbf{I}_n - P - P + PP = \mathbf{I}_n - 2P + P = \mathbf{I}_n - P = M.$$

M is also symmetric, so it is an orthogonal projection matrix. Combined with Exercise 3.3 this gives the convenient identity $\hat{e}'\hat{e} = Y'M'MY = Y'MY$.

Exercise 3.9

The trace is linear, so from $M = \mathbf{I}_n - P$,

$$\text{tr } M = \text{tr } \mathbf{I}_n - \text{tr } P.$$

Now $\text{tr } \mathbf{I}_n = n$, and $\text{tr } P = k$ by THEOREM 3.3.3. (That result follows from the cyclic property of the trace: $\text{tr } P = \text{tr}(X(X'X)^{-1}X') = \text{tr}((X'X)^{-1}X'X) = \text{tr } \mathbf{I}_k = k$.) Hence $\text{tr } M = n - k$.

This is the algebraic source of the $n - k$ degrees-of-freedom correction: it reappears in $\mathbb{E}[\hat{e}'\hat{e}] = (n - k)\sigma^2$ and hence in the unbiased variance estimator s^2 of (4.31).

Exercise 3.10

Write out P in partitioned form:

$$P = \begin{bmatrix} X_1 & X_2 \end{bmatrix} \left(\begin{bmatrix} X_1' \\ X_2' \end{bmatrix} \begin{bmatrix} X_1 & X_2 \end{bmatrix} \right)^{-1} \begin{bmatrix} X_1' \\ X_2' \end{bmatrix} = \begin{bmatrix} X_1 & X_2 \end{bmatrix} \begin{bmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{bmatrix}^{-1} \begin{bmatrix} X_1' \\ X_2' \end{bmatrix}.$$

The assumption $X_1'X_2 = \mathbf{0}$ also gives $X_2'X_1 = (X_1'X_2)' = \mathbf{0}$, so the inner matrix is block diagonal and inverts block by block:

$$\begin{aligned} P &= \begin{bmatrix} X_1 & X_2 \end{bmatrix} \begin{bmatrix} X_1'X_1 & \mathbf{0} \\ \mathbf{0} & X_2'X_2 \end{bmatrix}^{-1} \begin{bmatrix} X_1' \\ X_2' \end{bmatrix} \\ &= \begin{bmatrix} X_1 & X_2 \end{bmatrix} \begin{bmatrix} (X_1'X_1)^{-1} & \mathbf{0} \\ \mathbf{0} & (X_2'X_2)^{-1} \end{bmatrix} \begin{bmatrix} X_1' \\ X_2' \end{bmatrix} \end{aligned}$$

$$= X_1(X_1'X_1)^{-1}X_1' + X_2(X_2'X_2)^{-1}X_2' = P_1 + P_2.$$

Note also that $P_1P_2 = \mathbf{0}$ here, so the two projections are onto mutually orthogonal subspaces — which is what makes the sum a projection. Exercise 3.21 draws the statistical consequence.

Exercise 3.11

If X contains a constant then $\mathbf{1}_n$ lies in the column space of X , so Exercise 3.3 (applied to that column) gives

$$\mathbf{1}_n' \hat{e} = \sum_{i=1}^n \hat{e}_i = 0 \quad \text{that is} \quad \frac{1}{n} \sum_{i=1}^n \hat{e}_i = 0.$$

This is equation (3.17). Since $Y_i = \hat{Y}_i + \hat{e}_i$ for every i , averaging over i gives

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i + \underbrace{\frac{1}{n} \sum_{i=1}^n \hat{e}_i}_{=0} = \frac{1}{n} \sum_{i=1}^n \hat{Y}_i.$$

The constant is essential: without it $\sum_i \hat{e}_i$ need not be zero and the fitted values need not average to $\bar{Y}$.

Exercise 3.12

Which are estimable? Every individual is either a man or a woman, so

$$D_1 + D_2 = \mathbf{1}_n.$$

Equation (3.52) therefore has an exactly collinear regressor matrix $X = [\mathbf{1}_n, D_1, D_2]$: the vector $c = (1, -1, -1)'$ satisfies $Xc = \mathbf{0}$, so $X'X$ is singular and $\hat{\beta}$ is not defined. This is the *dummy variable trap*. Equations (3.53) and (3.54) can both be estimated, provided $n_1 \geq 1$ and $n_2 \geq 1$.

(a) Neither is more general. Both regressor matrices span the same two-dimensional space,

$$\text{span}\{D_1, D_2\} = \text{span}\{\mathbf{1}_n, D_1\},$$

so by Exercise 3.2 the two regressions have identical fitted values, residuals, and SSE. They are reparameterizations of one another, with

$$\mu = \alpha_2, \quad \phi = \alpha_1 - \alpha_2.$$

Both models are saturated: they reproduce the two group means exactly. (3.53) reports the two levels; (3.54) reports the level of the base group and the difference.

(b) $\mathbf{1}_n' D_1$ counts the ones in D_1 and $\mathbf{1}_n' D_2$ counts the ones in D_2 :

$$\mathbf{1}_n' D_1 = n_1, \quad \mathbf{1}_n' D_2 = n_2, \quad n_1 + n_2 = n.$$

Exercise 3.13

(a) Write $D = [D_1 \ D_2]$. Because each person is in exactly one group, $D_1' D_2 = 0$, $D_1' D_1 = n_1$ and $D_2' D_2 = n_2$, so

$$D' D = \begin{bmatrix} n_1 & 0 \\ 0 & n_2 \end{bmatrix}, \quad D' Y = \begin{pmatrix} \sum_{i \in \text{men}} Y_i \\ \sum_{i \in \text{women}} Y_i \end{pmatrix}.$$

Hence

$$\begin{pmatrix} \hat{\gamma}_1 \\ \hat{\gamma}_2 \end{pmatrix} = (D'D)^{-1}D'Y = \begin{pmatrix} n_1^{-1} \sum_{i \in \text{men}} Y_i \\ n_2^{-1} \sum_{i \in \text{women}} Y_i \end{pmatrix} = \begin{pmatrix} \bar{Y}_1 \\ \bar{Y}_2 \end{pmatrix}.$$

Regressing on a full set of mutually exclusive group dummies simply computes the group means.

(b) Y^* subtracts from each observation the mean of Y within that person's own group: men have $\bar{Y}_1$ removed, women $\bar{Y}_2$. Likewise each column of X^* is within-group demeaned. In matrix terms these are the residuals from regressing Y (respectively each column of X) on D :

$$Y^* = M_D Y, \quad X^* = M_D X, \quad M_D = \mathbf{I}_n - D(D'D)^{-1}D'.$$

This is the “within” transformation familiar from fixed-effects panel estimation.

(c) They are identical: $\tilde{\beta} = \hat{\beta}$. This is the Frisch–Waugh–Lovell theorem (THEOREM 3.5) applied with D as the first block of regressors:

$$\hat{\beta} = (X'M_D X)^{-1} X'M_D Y = ((M_D X)'(M_D X))^{-1} (M_D X)'(M_D Y) = (X^{*'} X^*)^{-1} X^{*'} Y^* = \tilde{\beta},$$

using the symmetry and idempotency of M_D . The residuals also coincide, $\hat{e} = Y^* - X^* \tilde{\beta}$.

Remark. What does *not* coincide is the reported R^2 or the degrees-of-freedom count: the demeaned regression has k regressors, while the full regression has $k + 2$, so a package will divide by $n - k$ rather than $n - k - 2$. Standard errors from the demeaned regression must be corrected accordingly. Exercise 3.24 makes the same point numerically.

Exercise 3.14

Step 1: what the new cross-products are. Stacking the new row,

$$X'_{n+1} X_{n+1} \text{ (as a data matrix)} = \begin{bmatrix} X_n \\ X'_{n+1} \end{bmatrix}' \begin{bmatrix} X_n \\ X'_{n+1} \end{bmatrix} = X'_n X_n + X_{n+1} X'_{n+1},$$

$$\begin{bmatrix} X_n \\ X'_{n+1} \end{bmatrix}' \begin{bmatrix} Y_n \\ Y_{n+1} \end{bmatrix} = X'_n Y_n + X_{n+1} Y_{n+1}.$$

Write $A = X'_n X_n$ for brevity, so $\hat{\beta}_n = A^{-1} X'_n Y_n$ and

$$\hat{\beta}_{n+1} = (A + X_{n+1} X'_{n+1})^{-1} (X'_n Y_n + X_{n+1} Y_{n+1}).$$

Step 2: invert the rank-one update. By the Sherman–Morrison formula (Section A.7 of the textbook),

$$(A + X_{n+1} X'_{n+1})^{-1} = A^{-1} - \frac{A^{-1} X_{n+1} X'_{n+1} A^{-1}}{c}, \quad c \equiv 1 + X'_{n+1} A^{-1} X_{n+1}.$$

(If one prefers not to quote the formula, multiply the right-hand side by $A + X_{n+1} X'_{n+1}$ and check directly that the product is $\mathbf{I}_k$; the two middle terms combine because $X'_{n+1} A^{-1} X_{n+1} = c - 1$ is a scalar.)

Step 3: multiply out. Denote the inverse above by B . For the first piece, using $A^{-1} X'_n Y_n = \hat{\beta}_n$,

$$B X'_n Y_n = \hat{\beta}_n - \frac{A^{-1} X_{n+1} X'_{n+1} \hat{\beta}_n}{c}.$$

For the second, using again that $X'_{n+1}A^{-1}X_{n+1} = c - 1$ is a scalar,

$$BX_{n+1}Y_{n+1} = A^{-1}X_{n+1}Y_{n+1}\left(1 - \frac{c-1}{c}\right) = \frac{A^{-1}X_{n+1}Y_{n+1}}{c}.$$

Adding the two pieces,

$$\hat{\beta}_{n+1} = \hat{\beta}_n + \frac{A^{-1}X_{n+1}}{c}\left(Y_{n+1} - X'_{n+1}\hat{\beta}_n\right),$$

which is the claimed formula.

The interpretation is appealing: the coefficient vector is revised in proportion to the *out-of-sample forecast error* $Y_{n+1} - X'_{n+1}\hat{\beta}_n$ of the new observation. If the new point falls exactly on the old fitted line, nothing changes. The scalar $c = 1 + X'_{n+1}A^{-1}X_{n+1}$ damps the update when the new regressor vector is unusual. This is the least squares special case of the Kalman filter, and is the basis of “recursive residuals” used in structural-break testing.

Exercise 3.15

Assume the regression includes an intercept, so that $\sum_i \hat{e}_i = 0$ and, by Exercise 3.11, $\widehat{\bar{Y}} = \bar{Y}$.

The key step is that the cross product between total and explained deviations equals the explained sum of squares. Using $Y_i = \hat{Y}_i + \hat{e}_i$,

$$\sum_{i=1}^n (Y_i - \bar{Y})(\hat{Y}_i - \bar{Y}) = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \underbrace{\sum_{i=1}^n \hat{e}_i(\hat{Y}_i - \bar{Y})}_{=0} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2,$$

the marked sum vanishing because $\hat{e} \perp \hat{Y}$ (Exercise 3.6) and $\sum_i \hat{e}_i = 0$. Therefore

$$\begin{aligned} \text{corr}(Y, \hat{Y})^2 &= \frac{\left[\sum_i (Y_i - \bar{Y})(\hat{Y}_i - \bar{Y})\right]^2}{\sum_i (Y_i - \bar{Y})^2 \sum_i (\hat{Y}_i - \bar{Y})^2} = \frac{\left[\sum_i (\hat{Y}_i - \bar{Y})^2\right]^2}{\sum_i (Y_i - \bar{Y})^2 \sum_i (\hat{Y}_i - \bar{Y})^2} \\ &= \frac{\sum_i (\hat{Y}_i - \bar{Y})^2}{\sum_i (Y_i - \bar{Y})^2} = \frac{\text{ESS}}{\text{TSS}} = R^2. \end{aligned}$$

A by-product: since a squared correlation lies in $[0, 1]$, this re-proves $0 \leq R^2 \leq 1$ for regressions with an intercept. It also shows the correlation between Y and $\hat{Y}$ is non-negative, so $R = +\sqrt{R^2}$ is the *multiple correlation coefficient*.

Exercise 3.16

Let $\text{SSE}_1 = \tilde{e}'\tilde{e}$ and $\text{SSE}_2 = \hat{e}'\hat{e}$. The short regression is the long regression subject to the restriction $\beta_2 = \mathbf{0}$:

$$\text{SSE}_1 = \min_{b_1} \|Y - X_1 b_1\|^2 = \min_{b_1, b_2 = \mathbf{0}} \|Y - X_1 b_1 - X_2 b_2\|^2 \geq \min_{b_1, b_2} \|Y - X_1 b_1 - X_2 b_2\|^2 = \text{SSE}_2,$$

since minimizing over a larger set can only lower the minimum. Both regressions have the same dependent variable and hence the same $\text{TSS} = \sum_i (Y_i - \bar{Y})^2$ (assume X_1 contains a constant, so that the usual centered R^2 is the relevant one). Therefore

$$R_2^2 = 1 - \frac{\text{SSE}_2}{\text{TSS}} \geq 1 - \frac{\text{SSE}_1}{\text{TSS}} = R_1^2.$$

When is there equality? Exactly when $\text{SSE}_1 = \text{SSE}_2$, i.e. when the restricted minimizer is also the unrestricted one. Because the unrestricted minimizer is unique whenever $[X_1 \ X_2]$ has full rank, this happens if and only if

$$\hat{\beta}_2 = \mathbf{0} \quad \Longleftrightarrow \quad X_2' \tilde{e} = \mathbf{0},$$

i.e. the new regressors are orthogonal to the residual of the short regression. Two familiar sufficient conditions: X_2 is exactly collinear with X_1 (so it adds nothing), or X_2 happens to be orthogonal to both X_1 and Y . Generically the equality fails, which is why R^2 cannot be used to choose between nested models.

Exercise 3.17

By (3.44)–(3.46), the leave-one-out prediction errors are $\tilde{e}_i = (1 - h_{ii})^{-1} \hat{e}_i$ and

$$\tilde{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \tilde{e}_i^2 = \frac{1}{n} \sum_{i=1}^n (1 - h_{ii})^{-2} \hat{e}_i^2, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \hat{e}_i^2.$$

By THEOREM 3.6, $0 \leq h_{ii} \leq 1$, hence $0 \leq 1 - h_{ii} \leq 1$ and $(1 - h_{ii})^{-2} \geq 1$ for every i . Comparing the two sums term by term,

$$\tilde{\sigma}^2 - \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \left[(1 - h_{ii})^{-2} - 1 \right] \hat{e}_i^2 \geq 0.$$

Equality. Every term is non-negative, so equality requires $[(1 - h_{ii})^{-2} - 1] \hat{e}_i^2 = 0$ for each i , that is, $h_{ii} = 0$ or $\hat{e}_i = 0$ for every observation. But $\sum_i h_{ii} = \text{tr } P = k \geq 1$ (Exercise 3.9), so at least one observation has $h_{ii} > 0$ and that observation must have $\hat{e}_i = 0$. Equality is therefore possible only in degenerate cases — most obviously a perfect fit ($\hat{e} = \mathbf{0}$, e.g. a saturated regression or $n = k$). With continuously distributed data and $k < n$ the inequality is strict with probability one.

The economic reading: $\tilde{\sigma}^2$ estimates *out-of-sample* mean squared error while $\hat{\sigma}^2$ measures *in-sample* fit, and in-sample fit is always optimistic.

Exercise 3.18

By THEOREM 3.7, equations (3.43)–(3.44),

$$\hat{\beta}_{(-i)} = \hat{\beta} - (X'X)^{-1} X_i \tilde{e}_i = \hat{\beta} - \frac{(X'X)^{-1} X_i \hat{e}_i}{1 - h_{ii}}.$$

Since $(X'X)^{-1}$ is non-singular, $\hat{\beta}_{(-i)} = \hat{\beta}$ if and only if

$$X_i \hat{e}_i = \mathbf{0} \quad \Longleftrightarrow \quad \hat{e}_i = 0 \text{ or } X_i = \mathbf{0}.$$

So an observation is irrelevant for the coefficient estimates exactly when it lies on the fitted surface ($\hat{e}_i = 0$), or when its entire regressor vector is zero — which cannot happen when the regression contains an intercept. In practice the condition is simply $\hat{e}_i = 0$.

Note that being uninfluential requires a zero *residual*, not merely low leverage. The natural measure of influence combines the two: the change in the fitted value is $\hat{Y}_i - X_i' \hat{\beta}_{(-i)} = h_{ii} \hat{e}_i / (1 - h_{ii})$, which is large only when leverage and residual are both sizeable (Section 3.21).

Exercise 3.19

Via the general formula. Here $X = \mathbf{1}_n$ and $k = 1$, so

$$h_{ii} = X_i'(X'X)^{-1}X_i = \frac{1}{n}, \quad \hat{\beta} = \bar{Y}, \quad \hat{e}_i = Y_i - \bar{Y}.$$

Equation (3.44) then gives

$$\tilde{e}_i = \frac{\hat{e}_i}{1 - h_{ii}} = \frac{Y_i - \bar{Y}}{1 - \frac{1}{n}} = \frac{n}{n-1}(Y_i - \bar{Y}).$$

Directly, as a check. Omitting observation i , the leave-one-out estimate is the mean of the remaining $n-1$ observations,

$$\hat{\beta}_{(-i)} = \frac{n\bar{Y} - Y_i}{n-1},$$

so

$$\tilde{e}_i = Y_i - \hat{\beta}_{(-i)} = \frac{(n-1)Y_i - n\bar{Y} + Y_i}{n-1} = \frac{n(Y_i - \bar{Y})}{n-1}.$$

The two routes agree. The inflation factor $n/(n-1)$ is the simplest instance of the general principle of Exercise 3.17: a genuine out-of-sample error is larger than the in-sample residual.

Exercise 3.20

Step 1: residuals evaluated at $\hat{\beta}_{(-i)}$. Write $c = \hat{e}_i/(1 - h_{ii}) = \tilde{e}_i$ and $h_{ji} = X_j'(X'X)^{-1}X_i$, the (j, i) element of P . From (3.43), $\hat{\beta}_{(-i)} = \hat{\beta} - (X'X)^{-1}X_i c$, so for every $j = 1, \dots, n$,

$$Y_j - X_j'\hat{\beta}_{(-i)} = \hat{e}_j + X_j'(X'X)^{-1}X_i c = \hat{e}_j + h_{ji}c.$$

Step 2: sum the squares over all j .

$$\sum_{j=1}^n (\hat{e}_j + h_{ji}c)^2 = \sum_{j=1}^n \hat{e}_j^2 + 2c \sum_{j=1}^n h_{ji}\hat{e}_j + c^2 \sum_{j=1}^n h_{ji}^2.$$

The middle sum is the i th element of $P\hat{e}$, and $P\hat{e} = PMY = \mathbf{0}$, so it vanishes. The last sum is the (i, i) element of $PP = P$, i.e. h_{ii} . Hence

$$\sum_{j=1}^n (Y_j - X_j'\hat{\beta}_{(-i)})^2 = n\hat{\sigma}^2 + c^2 h_{ii}.$$

Step 3: remove the $j = i$ term. For $j = i$ the summand is $(\hat{e}_i + h_{ii}c)^2 = c^2$, because $\hat{e}_i + h_{ii}c = (1 - h_{ii})c + h_{ii}c = c$. Subtracting,

$$\sum_{j \neq i} (Y_j - X_j'\hat{\beta}_{(-i)})^2 = n\hat{\sigma}^2 + c^2 h_{ii} - c^2 = n\hat{\sigma}^2 - c^2(1 - h_{ii}) = n\hat{\sigma}^2 - \frac{\hat{e}_i^2}{1 - h_{ii}}.$$

Step 4: divide by $n-1$.

$$\hat{\sigma}_{(-i)}^2 = \frac{n\hat{\sigma}^2}{n-1} - \frac{\hat{e}_i^2}{(n-1)(1 - h_{ii})},$$

as claimed. Like Exercise 3.14 this is an updating formula: all n leave-one-out variance estimates can be obtained from a single regression, without refitting.

Exercise 3.21

Substitute the multiple-regression decomposition $Y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{e}$ into the short-regression formula and use $X_1'\hat{e} = \mathbf{0}$ (Exercise 3.4):

$$\tilde{\beta}_1 = (X_1'X_1)^{-1}X_1'Y = \hat{\beta}_1 + (X_1'X_1)^{-1}X_1'X_2\hat{\beta}_2.$$

Symmetrically, $\tilde{\beta}_2 = \hat{\beta}_2 + (X_2'X_2)^{-1}X_2'X_1\hat{\beta}_1$. Therefore

$$\tilde{\beta}_1 = \hat{\beta}_1 \iff X_1'X_2\hat{\beta}_2 = \mathbf{0}, \quad \tilde{\beta}_2 = \hat{\beta}_2 \iff X_2'X_1\hat{\beta}_1 = \mathbf{0}.$$

For *both* to hold for an arbitrary Y — i.e. as a property of the design rather than a numerical accident — the required condition is

$X_1'X_2 = \mathbf{0} : \text{the regressors are orthogonal in the sample.}$

Under orthogonality the projections split (Exercise 3.10) and the multiple regression decouples into separate simple regressions. Otherwise the one-at-a-time slopes confound the two effects — the sample version of omitted-variable bias.

Exercise 3.22

No, not in general. The two-step procedure residualizes Y but *not* X_2 , and Frisch–Waugh–Lovell requires residualizing both.

Write $M_1 = \mathbf{I}_n - X_1(X_1'X_1)^{-1}X_1'$, so $\tilde{u} = M_1Y$. The second-step coefficient is

$$\tilde{\beta}_2 = (X_2'X_2)^{-1}X_2'M_1Y,$$

whereas FWL (THEOREM 3.5) gives

$$\hat{\beta}_2 = (X_2'M_1X_2)^{-1}X_2'M_1Y.$$

The two have the same “numerator” $X_2'M_1Y$ but different normalizations, so they agree for every Y if and only if

$$X_2'X_2 = X_2'M_1X_2 \iff X_2'P_1X_2 = \mathbf{0} \iff P_1X_2 = \mathbf{0} \iff X_1'X_2 = \mathbf{0},$$

using that P_1 is idempotent and symmetric, so $X_2'P_1X_2 = \|P_1X_2\|^2$.

So $\tilde{\beta}_2 = \hat{\beta}_2$ only under sample orthogonality of X_1 and X_2 — the same condition as in Exercise 3.21. In the generic case the two-step estimator is biased toward zero relative to $\hat{\beta}_2$, because $X_2'X_2 \geq X_2'M_1X_2$ in the positive semi-definite order: dividing by too large a matrix shrinks the coefficient.

Remark. The correct two-step recipe is: regress Y on X_1 and keep the residual; regress *each column of* X_2 on X_1 and keep those residuals; then regress the first residual on the second. That reproduces $\hat{\beta}_2$ exactly. Exercise 3.24(b) verifies this numerically.

Exercise 3.23

Z is a non-singular linear transformation of X :

$$Z = [X_1, X_2 - X_1] = XC, \quad C = \begin{bmatrix} \mathbf{I}_{k_1} & -\mathbf{I}_{k_1} \\ \mathbf{0} & \mathbf{I}_{k_2} \end{bmatrix},$$

which is block upper triangular with identity blocks on the diagonal, hence non-singular with $C^{-1} = \begin{bmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$. (For this to be conformable, X_1 and X_2 must have the same number of columns; if

they are single columns, $C = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$.)

By Exercise 3.2, Z and X span the same column space, so the two regressions have identical fitted values and identical residuals. Both $\hat{\sigma}^2$ and $\tilde{\sigma}^2$ are computed from the same residual vector with the same divisor (n , or $n - k$ — k is the same for both), so

$$\tilde{\sigma}^2 = \hat{\sigma}^2.$$

Only the coefficients change, by $\hat{\gamma} = C^{-1}\hat{\beta}$: explicitly, if $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2)$ then $\hat{\gamma} = (\hat{\beta}_1 + \hat{\beta}_2, \hat{\beta}_2)$. This is the familiar reparameterization used when one wants to report a *difference* of effects directly, together with its standard error.

Exercise 3.24

The sub-sample is single (`marital` = 7) Asian (`race` = 4) men (`female` = 0) with potential experience = `age` − `education` − 6 less than 45 years, giving $n = 267$ observations. The dependent variable is $\log(\text{earnings}/(\text{hours} \times \text{week}))$.

(a) The fitted equation reproduces (3.49):

$$\widehat{\log(\text{wage})} = 0.1443 \text{ education} + 0.0426 \text{ experience} - 0.0951 \text{ experience}^2/100 + 0.5309,$$

with

$$\text{SSE} = 82.505, \quad \text{TSS} = 135.104, \quad R^2 = 1 - \frac{82.505}{135.104} = 0.3893.$$

(b) Let $\tilde{Y}$ be the residual from regressing $\log(\text{wage})$ on $(1, \text{experience}, \text{experience}^2/100)$ and $\tilde{X}$ the residual from regressing education on the same three variables. The regression of $\tilde{Y}$ on $\tilde{X}$ gives

$$\text{slope} = 0.144307, \quad \text{SSE} = 82.505, \quad R^2 = 0.3687.$$

The slope is numerically identical to the education coefficient in (3.49) (to all reported digits: 0.1443068 in both cases). This is exactly the Frisch–Waugh–Lovell theorem (THEOREM 3.5, Exercise 3.13): partialling the other regressors out of *both* the dependent variable and the regressor of interest reproduces the multiple-regression coefficient.

(c) The SSEs are equal; the R^2 s are not.

- FWL also implies that the two regressions have the same residual vector (the largest element-by-element discrepancy here is 4×10^{-14} , pure floating-point noise). Hence the same $\text{SSE} = 82.505$.

- $R^2 = 1 - \text{SSE}/\text{TSS}$, and the two regressions have different *dependent variables* and hence different TSS. In (a) the total variation is that of $\log(\text{wage})$, $\text{TSS} = 135.104$. In (b) it is that of the *residualized* $\log(\text{wage})$, $\text{TSS} = 130.699$, which is smaller because the experience terms have already been removed. So $R_{(b)}^2 = 1 - 82.505/130.699 = 0.3687 < 0.3893 = R_{(a)}^2$.

The practical lesson: FWL reproduces coefficients and residuals, never the reported R^2 ; and, as noted in Exercise 3.13, it does not reproduce the degrees-of-freedom count either, so standard errors from the residual regression must be corrected by hand.

Code (R).

```
dat <- read.table("cps09mar.txt")
names(dat) <- c("age", "female", "hisp", "education", "earnings", "hours",
               "week", "union", "uncov", "region", "race", "marital")
dat$exper <- dat$age - dat$education - 6
sam <- with(dat, race==4 & marital==7 & female==0 & exper<45)
d <- dat[sam,]
y <- log(d$earnings/(d$hours*d$week))
ex <- d$exper
fit <- lm(y ~ d$education + ex + I(ex^2/100))           # part (a)
summary(fit)$r.squared; sum(resid(fit)^2)
ry <- resid(lm(y ~ ex + I(ex^2/100)))                 # part (b)
rx <- resid(lm(d$education ~ ex + I(ex^2/100)))
fit2 <- lm(ry ~ rx - 1)
coef(fit2); summary(fit2)$r.squared; sum(resid(fit2)^2)
```

Exercise 3.25

Computed values ($n = 267$):

	quantity	value	theoretical prediction
(a)	$\sum_i \hat{e}_i$	-2.0×10^{-13}	0 (intercept is a regressor)
(b)	$\sum_i X_{1i} \hat{e}_i$	3.7×10^{-12}	0 (education is a regressor)
(c)	$\sum_i X_{2i} \hat{e}_i$	5.0×10^{-13}	0 (experience is a regressor)
(d)	$\sum_i X_{1i}^2 \hat{e}_i$	133.13	not 0: education ² is <i>not</i> a regressor
(e)	$\sum_i X_{2i}^2 \hat{e}_i$	7.4×10^{-12}	0: experience ² /100 <i>is</i> a regressor
(f)	$\sum_i \hat{Y}_i \hat{e}_i$	4.4×10^{-13}	0 ($\hat{Y}$ is a linear combination of X)
(g)	$\sum_i \hat{e}_i^2$	82.505	> 0 (the SSE)

Are they consistent with theory? Yes, and instructively so.

Items (a), (b), (c), (e) and (f) are all instances of the single identity $X'\hat{e} = \mathbf{0}$ (Exercise 3.3): the residual is orthogonal to every column of the regressor matrix, and hence to every linear combination of them — which is why (f) is zero as well. The values reported are zero to machine precision (10^{-12} relative to sums of order 10^3); the departures are rounding error, not economics.

Item (e) deserves emphasis because it is easy to misread: experience^2 is zero *not* because squares of regressors are automatically orthogonal to the residual, but because $\text{experience}^2/100$ is literally one of the four regressors in (3.49), and $\sum_i X_{2i}^2 \hat{e}_i = 100 \sum_i (X_{2i}^2/100) \hat{e}_i = 0$.

Item (d) is the control: education^2 is *not* in the regression, and the corresponding sum is 133.13, nowhere near zero. This is the sample counterpart of Exercises 2.10–2.11: OLS orthogonality is a statement about the included regressors only. If one is willing to assume $\mathbb{E}[e | X] = 0$, then $\mathbb{E}[X_1^2 e] = 0$ holds in the *population*; but that is a population restriction, not a sample identity, and (d) shows the sample analogue is far from zero.

Item (g) is of course strictly positive unless the fit is perfect.

Exercise 3.26

(a) The sub-sample is `race = 1` (white only), `female = 0`, `hisp = 1`, giving $n = 4,230$ observations (Northeast 471, Midwest 431, South 1,545, West 1,783; married 2,854, widowed/divorced 253, separated 109, never married 1,014). Regions are coded 1 = Northeast, 2 = Midwest, 3 = South, 4 = West; “married” collects marital codes 1–3, “widowed or divorced” codes 4–5, and “separated” code 6.

	coefficient	s.e. (HC1)
education	0.0883	0.0029
experience	0.0279	0.0028
$\text{experience}^2/100$	−0.0363	0.0055
Northeast	0.0616	0.0361
South	−0.0679	0.0297
West	0.0195	0.0283
married	0.1778	0.0250
widowed or divorced	0.0857	0.0443
separated	0.0167	0.0526
intercept	1.1929	0.0501
$n = 4,230, \quad R^2 = 0.249, \quad \text{SSE} = 1363.4$		

Reading the estimates: an extra year of schooling raises the wage by about 8.8%; the experience profile is concave, peaking at $\text{experience} = 0.0279/(2 \times 0.000363) \approx 38$ years; wages in the South are about 6.8% below the Midwest baseline; and married men earn about 17.8% more than never-married men, the well-known (and not causally interpretable) marriage premium. Standard errors here are heteroskedasticity robust (HC1); the homoskedastic formulas are not appropriate, as Exercise 4.25 confirms.

One *must* omit one region dummy and one marital-status dummy: including all four regions plus an intercept would recreate the dummy variable trap of Exercise 3.12, since the four region dummies sum to $\mathbf{1}_n$.

(b) Running the identical specification in a second package (Stata, MATLAB, R, Python) returns the same numbers to within floating-point tolerance, typically 10^{-10} or better on the coefficients.

Any *visible* discrepancy signals a difference in the specification rather than in the arithmetic — most commonly a different definition of the sub-sample, a different base category, or a different small-sample adjustment in the standard errors (HC0 versus HC1; see Section 4.14). Packages solve $(X'X)\hat{\beta} = X'Y$ by QR or Cholesky rather than by explicit inversion (Section 3.23), so the numerical routes differ slightly even when the answers agree.

Code (R).

```
dat <- read.table("cps09mar.txt")
names(dat) <- c("age","female","hisp","education","earnings","hours",
               "week","union","uncov","region","race","marital")
d <- subset(dat, race==1 & female==0 & hisp==1)
y <- log(d$earnings/(d$hours*d$week))
ex <- d$age - d$education - 6
ne <- (d$region==1); so <- (d$region==3); we <- (d$region==4)
mar <- d$marital %in% 1:3; wd <- d$marital %in% 4:5; sep <- d$marital==6
fit <- lm(y ~ d$education + ex + I(ex^2/100) + ne + so + we + mar + wd + sep)
summary(fit)
```

Chapter 4

Least Squares Regression

Solutions to all 26 exercises of Chapter 4 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 4.24–4.26 are empirical; the reported numbers were computed from the `cps09mar` and `DDK2011` data sets on the textbook website, and code is included.

Maintained throughout: Assumption 4.1 (i.i.d. observations) and Assumption 4.2 (linear regression model $Y = X'\beta + e$, $\mathbb{E}[e | X] = 0$, $Q_{XX} = \mathbb{E}[XX'] > 0$). $D = \text{diag}\{\sigma_1^2, \dots, \sigma_n^2\}$ with $\sigma_i^2 = \mathbb{E}[e_i^2 | X_i]$, $h_{ii} = X_i'(X'X)^{-1}X_i$, $M = \mathbf{I}_n - X(X'X)^{-1}X'$, and $V_{\hat{\beta}} = \text{var}[\hat{\beta} | X]$. “Homoskedastic” means Assumption 4.3, $\mathbb{E}[e^2 | X] = \sigma^2$.

Exercise 4.1

(a) The moment (plug-in) estimator replaces the population expectation by the sample average:

$$\hat{\mu}_k = \frac{1}{n} \sum_{i=1}^n Y_i^k.$$

(b) Since the Y_i are identically distributed,

$$\mathbb{E}[\hat{\mu}_k] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Y_i^k] = \frac{1}{n} \cdot n \mu_k = \mu_k.$$

So $\hat{\mu}_k$ is unbiased for every n , provided μ_k exists.

(c) $\hat{\mu}_k$ is the sample mean of the i.i.d. variables Y_i^k . By independence the variance of a sum is the sum of the variances, so

$$\text{var}[\hat{\mu}_k] = \frac{1}{n^2} \sum_{i=1}^n \text{var}[Y_i^k] = \frac{\text{var}[Y^k]}{n} = \frac{\mu_{2k} - \mu_k^2}{n}.$$

This is finite if and only if $\text{var}[Y^k] < \infty$, i.e. if and only if

$$\mu_{2k} = \mathbb{E}[Y^{2k}] < \infty : \quad Y \text{ must have a finite } 2k\text{th moment.}$$

Estimating a k th moment precisely requires twice as many moments to exist. Note that this is a restriction on the *distribution* of Y , not on the sample size.

(d) Plug in sample moments. A convenient unbiased choice is the usual sample variance of the transformed data Y_i^k , divided by n :

$$\widehat{\text{var}}[\hat{\mu}_k] = \frac{1}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (Y_i^k - \hat{\mu}_k)^2.$$

The simpler (biased but consistent) alternative replaces $1/(n-1)$ by $1/n$, giving $n^{-1}(\hat{\mu}_{2k} - \hat{\mu}_k^2)$.

Remark. The division by n is essential: $\text{var}[\hat{\mu}_k]$ is the variance of an *average*, which shrinks at rate $1/n$, not the variance of a single observation.

Exercise 4.2

Write $e_i = Y_i - \mu$, so the e_i are i.i.d. with $\mathbb{E}[e_i] = 0$, $\mathbb{E}[e_i^2] = \sigma^2$ and $\mathbb{E}[e_i^3] = \mu_3$, and $\bar{Y} - \mu = n^{-1} \sum_{i=1}^n e_i$. Expand the cube:

$$\mathbb{E} \left[\left(\sum_{i=1}^n e_i \right)^3 \right] = \sum_{i=1}^n \sum_{j=1}^n \sum_{\ell=1}^n \mathbb{E}[e_i e_j e_\ell].$$

Classify the terms by their index pattern. By independence, if any index appears *exactly once* the expectation factors and contains the factor $\mathbb{E}[e] = 0$. So the only surviving terms are those with $i = j = \ell$:

- $i = j = \ell$: n terms, each $\mathbb{E}[e_i^3] = \mu_3$;
- exactly two indices equal (e.g. $i = j \neq \ell$): $\mathbb{E}[e_i^2 e_\ell] = \sigma^2 \mathbb{E}[e_\ell] = 0$;
- all three distinct: $\mathbb{E}[e_i] \mathbb{E}[e_j] \mathbb{E}[e_\ell] = 0$.

Hence $\mathbb{E}[(\sum_i e_i)^3] = n\mu_3$ and

$$\mathbb{E}[(\bar{Y} - \mu)^3] = \frac{n\mu_3}{n^3} = \frac{\mu_3}{n^2}.$$

It is zero if and only if $\mu_3 = \mathbb{E}[(Y - \mu)^3] = 0$ — in particular whenever the distribution of Y is symmetric about μ .

Two observations. First, the third central moment of $\bar{Y}$ shrinks at rate n^{-2} while the standard deviation shrinks at rate $n^{-1/2}$, so the *standardized* skewness $\mathbb{E}[(\bar{Y} - \mu)^3] / (\text{var}[\bar{Y}])^{3/2} = \mu_3 / (\sigma^3 \sqrt{n})$ vanishes at rate $n^{-1/2}$: asymmetry disappears as n grows, which is the leading Edgeworth correction behind the central limit theorem. Second, Exercise 4.12 poses this same calculation again with explicit notation.

Exercise 4.3

In each pair the first object is a *statistic* and the second a *parameter*.

$\bar{Y} = n^{-1} \sum_i Y_i$ is a function of the data. Viewed before sampling it is a random variable with a sampling distribution: it changes from sample to sample. $\mu = \mathbb{E}[Y]$ is a fixed, non-random number — a feature of the population distribution F . It does not depend on n and it is unknown. $\bar{Y}$ is an *estimator* of μ : it equals μ only in expectation (unbiasedness) and in the limit (consistency), never exactly in a finite sample.

The same distinction applies to the second pair. $n^{-1} \sum_i X_i X_i'$ is the sample second-moment matrix, a random $k \times k$ matrix computed from the data; $Q_{XX} = \mathbb{E}[XX']$ is the fixed population second-moment matrix. The first is the natural estimator of the second, and $n^{-1} \sum_i X_i X_i' \xrightarrow{p} Q_{XX}$ by the weak law of large numbers (Chapter 6), which is precisely what makes $\hat{\beta}$ consistent for β (Chapter 7).

A useful mnemonic: hats and bars denote things you can compute; Greek letters and $\mathbb{E}$ denote things you would like to know.

Exercise 4.4

False. OLS orthogonality is a *sample* identity involving only the included regressors; X^2 is not one of them.

With a single regressor and no intercept, $\hat{\beta} = \sum_j X_j Y_j / \sum_j X_j^2$, so

$$\sum_{i=1}^n X_i^2 \hat{e}_i = \sum_{i=1}^n X_i^2 (Y_i - X_i \hat{\beta}) = \sum_{i=1}^n X_i^2 Y_i - \left(\sum_{i=1}^n X_i^3 \right) \frac{\sum_{j=1}^n X_j Y_j}{\sum_{j=1}^n X_j^2},$$

which has no reason to vanish. The sample identity that *does* hold is $\sum_i X_i \hat{e}_i = 0$.

Contrast with Exercise 2.10: the assumption $\mathbb{E}[e | X] = 0$ implies the *population* statement $\mathbb{E}[X^2 e] = 0$, so the sample average $n^{-1} \sum_i X_i^2 \hat{e}_i$ converges in probability to zero — but it is not exactly zero in any finite sample. Exercise 3.25(d) shows this numerically: there $\sum_i X_{1i}^2 \hat{e}_i = 133.1$ while $\sum_i X_{1i} \hat{e}_i = 4 \times 10^{-12}$.

Exercise 4.5

Substituting $Y = X\beta + e$ into $\hat{\beta} = (X'X)^{-1}X'Y$ gives the key decomposition

$$\hat{\beta} - \beta = (X'X)^{-1}X'e.$$

(4.20). Conditional on X the matrix $(X'X)^{-1}X'$ is non-random, so by (4.18),

$$\mathbb{E}[\hat{\beta} | X] = \beta + (X'X)^{-1}X'\mathbb{E}[e | X] = \beta + \mathbf{0} = \beta.$$

OLS is unbiased whatever the form of the error covariance matrix; only the mean condition $\mathbb{E}[e | X] = 0$ is used.

(4.21). Using $\text{var}[Ac] = A \text{var}[c] A'$ for a conditionally fixed A , and then (4.19):

$$\begin{aligned} \text{var}[\hat{\beta} | X] &= \text{var}[(X'X)^{-1}X'e | X] \\ &= (X'X)^{-1}X' \text{var}[e | X] X(X'X)^{-1} \\ &= (X'X)^{-1}X'(\Sigma\sigma^2)X(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1}(X'\Sigma X)(X'X)^{-1}. \end{aligned}$$

Setting $\Sigma = \mathbf{I}_n$ recovers the homoskedastic formula $\sigma^2(X'X)^{-1}$; setting $\Sigma = D/\sigma^2$ diagonal recovers the heteroskedastic sandwich of Section 4.6.

Exercise 4.6

A linear estimator is $\tilde{\beta} = A'Y$ for some $n \times k$ matrix $A = A(X)$ that may depend on X but not on Y .

Step 1: unbiasedness pins down $A'X$. Using $Y = X\beta + e$ and (4.18),

$$\mathbb{E}[\tilde{\beta} | X] = A'X\beta + A'\mathbb{E}[e | X] = A'X\beta.$$

For this to equal β for every $\beta \in \mathbb{R}^k$ we need

$$A'X = \mathbf{I}_k.$$

Step 2: the variance. By (4.19),

$$\text{var}[\tilde{\beta} \mid X] = A' \text{var}[e \mid X] A = \sigma^2 A' \Sigma A.$$

Step 3: complete the square. Write $Q = (X' \Sigma^{-1} X)^{-1}$ and define

$$C = \Sigma^{1/2} A - \Sigma^{-1/2} X Q.$$

Multiplying out and using $A' X = \mathbf{I}_k$ (hence also $X' A = \mathbf{I}_k$),

$$\begin{aligned} C' C &= A' \Sigma A - A' X Q - Q X' A + Q X' \Sigma^{-1} X Q \\ &= A' \Sigma A - Q - Q + Q Q^{-1} Q \\ &= A' \Sigma A - Q. \end{aligned}$$

Since $C' C \geq 0$ (it is a Gram matrix), $A' \Sigma A \geq Q$, and therefore

$$\text{var}[\tilde{\beta} \mid X] = \sigma^2 A' \Sigma A \geq \sigma^2 (X' \Sigma^{-1} X)^{-1},$$

which is the bound of THEOREM 4.5.

Step 4: the bound is attained. Equality holds iff $C = \mathbf{0}$, i.e. $A = \Sigma^{-1} X Q$, in which case

$$\tilde{\beta} = A' Y = (X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} Y = \tilde{\beta}_{\text{GLS}},$$

the GLS estimator (4.22). So GLS is the best *linear* unbiased estimator, and the lower bound is sharp.

Remark. THEOREM 4.5 as stated in the text is stronger: it covers *all* unbiased estimators, linear or not. The linear case proved here is the classical Gauss–Markov/Aitken argument; the general case is proved in Section 4.24.

Exercise 4.7

Write $Q = (X' \Sigma^{-1} X)^{-1}$ throughout; note Q is symmetric.

(a) Substituting $Y = X\beta + e$,

$$\tilde{\beta} = Q X' \Sigma^{-1} (X\beta + e) = \beta + Q X' \Sigma^{-1} e,$$

so $\mathbb{E}[\tilde{\beta} \mid X] = \beta + Q X' \Sigma^{-1} \mathbb{E}[e \mid X] = \beta$: GLS is unbiased.

(b) From the same decomposition,

$$\text{var}[\tilde{\beta} \mid X] = Q X' \Sigma^{-1} (\Sigma \sigma^2) \Sigma^{-1} X Q = \sigma^2 Q (X' \Sigma^{-1} X) Q = \sigma^2 Q = \sigma^2 (X' \Sigma^{-1} X)^{-1}.$$

(c) Using part (a),

$$\tilde{e} = Y - X\tilde{\beta} = X\beta + e - X(\beta + Q X' \Sigma^{-1} e) = e - X Q X' \Sigma^{-1} e = M_1 e.$$

(d) $M_1 = \mathbf{I}_n - X Q X' \Sigma^{-1}$ and, since Σ and Q are symmetric, $M_1' = \mathbf{I}_n - \Sigma^{-1} X Q X'$. Hence

$$\begin{aligned} M_1' \Sigma^{-1} M_1 &= (\Sigma^{-1} - \Sigma^{-1} X Q X' \Sigma^{-1}) (\mathbf{I}_n - X Q X' \Sigma^{-1}) \\ &= \Sigma^{-1} - \Sigma^{-1} X Q X' \Sigma^{-1} - \Sigma^{-1} X Q X' \Sigma^{-1} + \Sigma^{-1} X Q \underbrace{(X' \Sigma^{-1} X)}_{= Q^{-1}} Q X' \Sigma^{-1} \end{aligned}$$

$$\begin{aligned}
&= \Sigma^{-1} - 2\Sigma^{-1}XQX'\Sigma^{-1} + \Sigma^{-1}XQX'\Sigma^{-1} \\
&= \Sigma^{-1} - \Sigma^{-1}X(X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}.
\end{aligned}$$

(e) Combining (c) and (d), and writing N for the matrix in (d),

$$\tilde{e}'\Sigma^{-1}\tilde{e} = e'M_1'\Sigma^{-1}M_1e = e'Ne.$$

Taking conditional expectations with the trace trick $\mathbb{E}[e'Ne \mid X] = \text{tr}(N\mathbb{E}[ee' \mid X])$ and (4.19),

$$\begin{aligned}
\mathbb{E}[\tilde{e}'\Sigma^{-1}\tilde{e} \mid X] &= \sigma^2 \text{tr}(N\Sigma) = \sigma^2 \left[\text{tr}(\Sigma^{-1}\Sigma) - \text{tr}(\Sigma^{-1}XQX'\Sigma^{-1}\Sigma) \right] \\
&= \sigma^2 \left[\text{tr}(\mathbf{I}_n) - \text{tr}(QX'\Sigma^{-1}X) \right] = \sigma^2 [n - \text{tr}(\mathbf{I}_k)] = \sigma^2(n - k),
\end{aligned}$$

using the cyclic property of the trace. Therefore

$$\boxed{\mathbb{E}[\tilde{\sigma}^2 \mid X] = \sigma^2.}$$

(f) Yes. It is exactly unbiased conditionally (and hence unconditionally), it needs no assumption beyond (4.18)–(4.19), and it has a clean interpretation: pre-multiplying the model by $\Sigma^{-1/2}$ turns it into a homoskedastic regression $\tilde{Y} = \tilde{X}\beta + \varepsilon$ with $\text{var}[\varepsilon \mid X] = \sigma^2\mathbf{I}_n$, and $\tilde{\sigma}^2$ is precisely the usual s^2 of (4.31) computed in that transformed model. The degrees-of-freedom divisor $n - k$ is the same one as in Exercise 3.9.

The practical caveat is the premise: Σ is treated as known. In applications Σ is estimated, and the resulting feasible GLS estimator is neither exactly unbiased nor exactly BLUE; the justification becomes asymptotic.

Exercise 4.8

First note what is *not* at issue. Because $\mathbb{E}[Y \mid X] = X'\beta$, we have $\mathbb{E}[e \mid X] = 0$ and hence $\mathbb{E}[w(X)Xe] = \mathbb{E}[w(X)X\mathbb{E}[e \mid X]] = 0$ for any weight function of X . So $\tilde{\beta}_{\text{wls}}$ is consistent (and conditionally unbiased) for β for *any* weights: the question is purely about efficiency.

(a) By Exercise 4.6, the efficient weighting is $w_i \propto \sigma_i^{-2}$. These weights are therefore optimal exactly when

$$\mathbb{E}[e^2 \mid X] = \sigma^2 X_j^2$$

for some constant σ^2 — that is, when the error standard deviation is proportional to $|X_j|$. This is the classical “multiplicative” or “constant coefficient of variation” specification, and it is natural whenever X_j is a *scale* variable: firm size in a regression of profits, household income in a regression of expenditure, population in a regression of aggregate municipal outcomes, or the group size when Y_i is a group average. Equivalently it is the weighting implied by running the regression in ratio form, Y_i/X_{ji} on X_i/X_{ji} .

(b) WLS should beat OLS when the heteroskedasticity is both *severe* and *roughly of this form*: then OLS puts too much weight on the noisiest observations, and downweighting them can reduce the variance substantially. The gain grows with the dispersion of X_j .

Three cautions, which matter more in practice than the theory:

- If X_j can be at or near zero the weights X_{ji}^{-2} explode and a handful of observations dominate the estimator; the finite-sample variance can be far worse than OLS. The weights are only sensible if X_j is bounded away from zero.
- If the variance function is misspecified, WLS remains consistent but is no longer efficient, and may be *less* efficient than OLS.
- Whatever weights are used, the reported standard errors should be heteroskedasticity robust, since the weights are unlikely to be exactly right.

Exercise 4.9

Under homoskedasticity, equation (4.27) states $\mathbb{E}[\hat{e}_i^2 | X] = (1 - h_{ii})\sigma^2$. The leverage h_{ii} is a function of X alone, so the weights $(1 - h_{ii})^{-1}$ pass through the conditional expectation:

$$\begin{aligned}\mathbb{E}[\bar{\sigma}^2 | X] &= \frac{1}{n} \sum_{i=1}^n (1 - h_{ii})^{-1} \mathbb{E}[\hat{e}_i^2 | X] \\ &= \frac{1}{n} \sum_{i=1}^n (1 - h_{ii})^{-1} (1 - h_{ii}) \sigma^2 \\ &= \frac{1}{n} \sum_{i=1}^n \sigma^2 = \sigma^2.\end{aligned}$$

The weights must stay *inside* the sum — they differ across i . What makes the argument work is that rescaling each squared residual by its own $(1 - h_{ii})^{-1}$ exactly undoes the leverage-induced downward bias of $\hat{e}_i^2$, observation by observation. Contrast the alternative unbiased estimator $s^2 = (n - k)^{-1} \sum_i \hat{e}_i^2$ of (4.31), which corrects the same bias only *on average*, using $\sum_i h_{ii} = k$.

Exercise 4.10

Recall the three estimators, all with the same outer “bread” $(X'X)^{-1}$:

$$\begin{aligned}\hat{V}_{\hat{\beta}}^{\text{HC0}} &= (X'X)^{-1} \left(\sum_{i=1}^n X_i X_i' \hat{e}_i^2 \right) (X'X)^{-1}, \\ \hat{V}_{\hat{\beta}}^{\text{HC2}} &= (X'X)^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-1} X_i X_i' \hat{e}_i^2 \right) (X'X)^{-1}, \\ \hat{V}_{\hat{\beta}}^{\text{HC3}} &= (X'X)^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-2} X_i X_i' \hat{e}_i^2 \right) (X'X)^{-1}.\end{aligned}$$

Differencing, the outer matrices factor out:

$$\begin{aligned}\hat{V}_{\hat{\beta}}^{\text{HC2}} - \hat{V}_{\hat{\beta}}^{\text{HC0}} &= (X'X)^{-1} \left(\sum_{i=1}^n [(1 - h_{ii})^{-1} - 1] \hat{e}_i^2 X_i X_i' \right) (X'X)^{-1}, \\ \hat{V}_{\hat{\beta}}^{\text{HC3}} - \hat{V}_{\hat{\beta}}^{\text{HC2}} &= (X'X)^{-1} \left(\sum_{i=1}^n [(1 - h_{ii})^{-2} - (1 - h_{ii})^{-1}] \hat{e}_i^2 X_i X_i' \right) (X'X)^{-1}.\end{aligned}$$

By THEOREM 3.6, $0 \leq h_{ii} \leq 1$, so $0 \leq 1 - h_{ii} \leq 1$ and therefore

$$(1 - h_{ii})^{-2} \geq (1 - h_{ii})^{-1} \geq 1.$$

Thus in each display the middle matrix is a sum of the positive semi-definite rank-one matrices $X_i X_i'$ with non-negative scalar weights, hence positive semi-definite. Pre- and post-multiplying a positive semi-definite matrix by $(X'X)^{-1}$ and its transpose preserves positive semi-definiteness (for any $c \in \mathbb{R}^k$, $c'(X'X)^{-1}S(X'X)^{-1}c = d'Sd \geq 0$ with $d = (X'X)^{-1}c$). Therefore

$$\widehat{V}_{\beta}^{\text{HC0}} \leq \widehat{V}_{\beta}^{\text{HC2}} \leq \widehat{V}_{\beta}^{\text{HC3}}$$

in the positive semi-definite order. In particular every diagonal element, and hence every reported standard error, is ordered $s^{\text{HC0}} \leq s^{\text{HC2}} \leq s^{\text{HC3}}$.

Remark. The text states (4.40) with strict inequalities. Strictness requires the weights to be strictly positive for enough observations — e.g. $h_{ii} > 0$ and $\widehat{e}_i \neq 0$ for a set of i whose X_i span $\mathbb{R}^k$. This holds in any non-degenerate sample but fails, for instance, at a perfect fit. The weak inequality above is what the algebra delivers unconditionally. The empirical standard errors in Exercise 4.24 display the ordering.

Exercise 4.11

In the book's notation $X'X = \sum_{i=1}^n X_i X_i'$. Since $(X'X)^{-1}$, $X_i X_i'$ and h_{ii} are all functions of X , only $\widehat{e}_i^2$ is random given X :

$$\begin{aligned} \mathbb{E}[\widehat{V}_{\beta}^{\text{HC2}} | X] &= (X'X)^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-1} X_i X_i' \mathbb{E}[\widehat{e}_i^2 | X] \right) (X'X)^{-1} \\ &= (X'X)^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-1} (1 - h_{ii}) \sigma^2 X_i X_i' \right) (X'X)^{-1} && \text{by (4.27)} \\ &= \sigma^2 (X'X)^{-1} \left(\sum_{i=1}^n X_i X_i' \right) (X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1} (X'X) (X'X)^{-1} = \sigma^2 (X'X)^{-1}, \end{aligned}$$

which is exactly V_{β} in the homoskedastic model. So HC2 is *exactly* unbiased there.

Combining with Exercise 4.10: under homoskedasticity HC0 is biased downward and HC3 upward, with HC2 exactly right. This is the usual justification for preferring HC2 or HC3 in small samples — HC0 tends to produce standard errors that are too small and hence tests that over-reject.

Exercise 4.12

This is the calculation of Exercise 4.2. Writing $e_i = Y_i - \mu$, all cross-product terms in $\mathbb{E}[(\sum_i e_i)^3]$ contain at least one index appearing exactly once and therefore vanish under independence and $\mathbb{E}[e] = 0$, leaving the n diagonal terms:

$$\boxed{\mathbb{E}[(\overline{Y} - \mu)^3] = \frac{\mu_3}{n^2}.$$

Note what does *not* appear: neither μ nor σ^2 enters. The third central moment of $\overline{Y}$ depends on the population only through μ_3 , and it decays at rate n^{-2} .

Exercise 4.13

With a single regressor,

$$\hat{\beta} - \beta = \frac{\sum_{i=1}^n X_i e_i}{\sum_{i=1}^n X_i^2} \equiv \frac{1}{S} \sum_{i=1}^n X_i e_i, \quad S = \sum_{i=1}^n X_i^2.$$

Conditional on X , the scalar S is fixed, so

$$\mathbb{E}[(\hat{\beta} - \beta)^3 | X] = \frac{1}{S^3} \sum_{i=1}^n \sum_{j=1}^n \sum_{\ell=1}^n X_i X_j X_\ell \mathbb{E}[e_i e_j e_\ell | X].$$

Under random sampling the observations are mutually independent, so conditional on the full matrix X the errors $e_1, \dots, e_n$ are independent with $\mathbb{E}[e_i | X] = \mathbb{E}[e_i | X_i] = 0$. Exactly as in Exercise 4.2, any term in which some index appears only once factors out a zero mean. Only $i = j = \ell$ survives, contributing $X_i^3 \mathbb{E}[e_i^3 | X_i] = X_i^3 \mu_{3i}$. Hence

$$\mathbb{E}[(\hat{\beta} - \beta)^3 | X] = \frac{\sum_{i=1}^n X_i^3 \mu_{3i}}{\left(\sum_{i=1}^n X_i^2\right)^3}.$$

Two special cases are worth noting. If the conditional distribution of e is symmetric about zero then $\mu_{3i} = 0$ and the third moment of $\hat{\beta}$ is exactly zero — the sampling distribution of $\hat{\beta}$ is symmetric. If instead $\mu_{3i} = \mu_3$ is constant, the expression is $\mu_3 \sum_i X_i^3 / (\sum_i X_i^2)^3$, which is $O(n^{-2})$ and therefore negligible relative to the variance, of order n^{-1} .

Exercise 4.14

(a) Apply $\mathbb{E}[Z^2] = \text{var}[Z] + (\mathbb{E}[Z])^2$ conditionally, using $\mathbb{E}[\hat{\beta} | X] = \beta$:

$$\mathbb{E}[\hat{\theta} | X] = \mathbb{E}[\hat{\beta}^2 | X] = \text{var}[\hat{\beta} | X] + (\mathbb{E}[\hat{\beta} | X])^2 = V_{\hat{\beta}} + \beta^2 = \theta + V_{\hat{\beta}}.$$

So $\hat{\theta}$ is *biased upward*, with bias exactly $V_{\hat{\beta}} \geq 0$. This is Jensen's inequality at work: squaring is convex, so plugging an unbiased estimator into a convex function overshoots. The bias is $O(n^{-1})$ and vanishes asymptotically, but it never vanishes in finite samples unless $\hat{\beta}$ is degenerate.

(b) Subtract an estimate of the bias:

$$\hat{\theta}^* = \hat{\beta}^2 - \hat{V}_{\hat{\beta}}.$$

(c) Taking conditional expectations,

$$\mathbb{E}[\hat{\theta}^* | X] = \theta + V_{\hat{\beta}} - \mathbb{E}[\hat{V}_{\hat{\beta}} | X],$$

so $\hat{\theta}^*$ is exactly unbiased if and only if $\hat{V}_{\hat{\beta}}$ is itself conditionally unbiased for $V_{\hat{\beta}}$. Among the standard choices:

- HC0 is biased downward and HC3 upward (Exercises 4.10, 4.11), so neither delivers unbiasedness;
- HC2 is exactly unbiased *in the homoskedastic regression model* (Exercise 4.11), and so is the classical estimator $s^2(X'X)^{-1}$ with s^2 from (4.31).

So use $\widehat{V}_{\widehat{\beta}}^{\text{HC2}}$. Then

$$\mathbb{E}[\widehat{\theta}^* | X] = \theta \quad \text{exactly, provided } \mathbb{E}[e^2 | X] = \sigma^2 \text{ (conditional homoskedasticity).}$$

Under heteroskedasticity no member of the HC family is exactly unbiased, so $\widehat{\theta}^*$ is only approximately unbiased — the remaining bias is $O(n^{-2})$ rather than $O(n^{-1})$, which is still a clear improvement over $\widehat{\theta}$. A caution: $\widehat{\theta}^*$ can be negative even though $\theta = \beta^2 \geq 0$, the usual price of bias correction.

Exercise 4.15

(a) The assumption $n^{-1}X'X = \mathbf{I}_k$ means $X'X = n\mathbf{I}_k$ and hence $(X'X)^{-1} = n^{-1}\mathbf{I}_k$. Using the general sandwich formula $V_{\widehat{\beta}} = (X'X)^{-1}(X'DX)(X'X)^{-1}$ with $D = \text{diag}\{\sigma_1^2, \dots, \sigma_n^2\}$ (valid because the observations are independent, so $\text{var}[e | X]$ is diagonal),

$$V_{\widehat{\beta}} = \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2 X_i X_i' = \frac{1}{n} \Omega_n, \quad \Omega_n \equiv \frac{1}{n} \sum_{i=1}^n \sigma_i^2 X_i X_i'.$$

Because $\mathbb{E}[\widehat{\beta} | X] = \beta$ is non-random, the law of total variance gives $\text{var}[\widehat{\beta}] = \mathbb{E}[V_{\widehat{\beta}}]$; under the maintained condition $n^{-1}X'X = \mathbf{I}_k$ the conditional and unconditional variances coincide in form.

(b) *In general they are correlated.* Orthonormality of the regressors makes the “bread” $(X'X)^{-1}$ diagonal, but the “meat” $\Omega_n = n^{-1} \sum_i \sigma_i^2 X_i X_i'$ is a σ_i^2 -weighted average of $X_i X_i'$, and there is no reason for it to be diagonal. Only the *unweighted* average $n^{-1} \sum_i X_i X_i' = \mathbf{I}_k$ is. So heteroskedasticity alone destroys the orthogonality of the coefficient estimates: $\text{cov}(\widehat{\beta}_j, \widehat{\beta}_\ell | X) = n^{-1}[\Omega_n]_{j\ell} \neq 0$ in general.

(c) A simple and standard sufficient condition is *conditional homoskedasticity*, $\mathbb{E}[e^2 | X] = \sigma^2$. Then $\sigma_i^2 = \sigma^2$ for all i and

$$\Omega_n = \sigma^2 \cdot \frac{1}{n} \sum_{i=1}^n X_i X_i' = \sigma^2 \mathbf{I}_k, \quad \text{so} \quad V_{\widehat{\beta}} = \frac{\sigma^2}{n} \mathbf{I}_k$$

is diagonal and the coefficient estimators are mutually uncorrelated, each with variance σ^2/n .

More weakly, it suffices that $\sum_i \sigma_i^2 X_i X_i'$ be diagonal — for example if $\sigma_i^2 = g(X_i)$ with the weighted cross-moments $\sum_i g(X_i) X_{ji} X_{\ell i}$ happening to vanish for $j \neq \ell$. Homoskedasticity is the interpretable special case.

Remark. It is tempting to argue “the sample is i.i.d., so $\text{var}[e | X] = \sigma^2 \mathbf{I}_n$ ”. That is not right. Independence across i delivers the *diagonality* of $\text{var}[e | X]$, but not the equality of its diagonal elements; equal diagonal elements is the separate assumption of homoskedasticity. Exercise 4.22 makes the same point.

Exercise 4.16

(a) Adding u to both sides of (4.63),

$$Y = Y^* + u = X'\beta + \underbrace{e + u}_{\equiv v}, \quad \mathbb{E}[v | X] = \mathbb{E}[e | X] + \mathbb{E}[u | X] = 0.$$

So the feasible equation is still a correctly specified linear CEF with the *same* coefficient vector β . What changes is the second moment. Using the independence of e and u (so that $\mathbb{E}[eu | X] = \mathbb{E}[e | X] \mathbb{E}[u | X] = 0$),

$$\boxed{\mathbb{E}[v^2 | X] = \sigma^2 + \sigma_u^2(X).}$$

Two consequences: the error variance is inflated, and — if $\sigma_u^2(\cdot)$ is not constant — the homoskedasticity of (4.63) is destroyed.

(b) OLS of Y on X is unaffected in terms of *identification*: $\hat{\beta}$ remains conditionally unbiased and consistent for β , because $\mathbb{E}[v | X] = 0$ is all that is required. Classical measurement error in the *dependent* variable is benign in this sense — quite unlike measurement error in a regressor, which causes attenuation bias (Exercise 7.24). What is lost is precision:

$$V_{\hat{\beta}} = (X'X)^{-1} \left(\sum_{i=1}^n (\sigma^2 + \sigma_u^2(X_i)) X_i X_i' \right) (X'X)^{-1} \geq \sigma^2 (X'X)^{-1},$$

so confidence intervals are wider and the R^2 is lower.

(c) This is where the measurement error bites.

- If $\sigma_u^2(X) = \sigma_u^2$ is constant, the model remains homoskedastic with variance $\sigma^2 + \sigma_u^2$, so the classical formula $s^2(X'X)^{-1}$ is still valid — it simply estimates the larger variance.
- If $\sigma_u^2(X)$ varies with X , the model is heteroskedastic even though the underlying structural equation was not. The homoskedastic standard errors are then invalid (they may be too small or too large), and one of the robust estimators HC1–HC3 of Section 4.14 must be used.

Since one rarely knows the form of $\sigma_u^2(\cdot)$, this is a textbook argument for reporting robust standard errors by default.

Exercise 4.17

(a) By construction and (4.64),

$$\mathbb{E}[u | X] = \mathbb{E}[Y | X] - (\gamma + \theta X)^{1/2} = (\gamma + \theta X)^{1/2} - (\gamma + \theta X)^{1/2} = 0.$$

(The term $(\gamma + \theta X)^{1/2}$ is a function of X , so it passes through the conditional expectation.)

(b) Squaring and taking conditional expectations, using $\mathbb{E}[u | X] = 0$ and writing $\sigma_u^2(X) = \mathbb{E}[u^2 | X] = \text{var}[Y | X]$,

$$\begin{aligned} \mathbb{E}[Y^2 | X] &= \mathbb{E} \left[\left((\gamma + \theta X)^{1/2} + u \right)^2 | X \right] \\ &= (\gamma + \theta X) + 2(\gamma + \theta X)^{1/2} \underbrace{\mathbb{E}[u | X]}_{=0} + \mathbb{E}[u^2 | X] \\ &= \gamma + \theta X + \sigma_u^2(X). \end{aligned}$$

So the CEF of Y^2 given X is linear in X *only if* $\sigma_u^2(X)$ is itself linear in X . Equation (4.65) is not an identity; it is correctly specified only under that extra restriction.

(c) Suppose $\sigma_u^2(X) = a + bX$ (a linear conditional variance; the homoskedastic case is $b = 0$). Then

$$\mathbb{E}[Y^2 | X] = (\gamma + a) + (\theta + b)X,$$

so the population regression coefficients of (4.65) are

$$\alpha = \gamma + a, \quad \beta = \theta + b.$$

Hence:

- θ is recovered as β *provided* $b = 0$, i.e. provided the conditional variance does not depend on X . This is a genuine additional assumption, not implied by (4.64).
- γ is *not* recovered even then: $\alpha = \gamma + \sigma_u^2$ is contaminated by the (unknown) error variance. One would need an independent estimate of σ_u^2 , which the regression of Y^2 on X does not provide.
- If $\sigma_u^2(X)$ is nonlinear in X , then (4.65) is misspecified: the linear projection coefficients are weighted averages of $\theta + \partial\sigma_u^2/\partial x$ and neither parameter is recovered.

(d) Only as a rough diagnostic. Its virtues are that it is linear and easy. Its defects are substantial: it identifies at most one of the two parameters, and only under an untestable auxiliary restriction; it discards the information in the functional form (4.64); and the induced error in (4.65) is strongly heteroskedastic and skewed (it involves u^2 and u), so standard errors from (4.65) would need to be robust in any case. The better route is nonlinear least squares directly on (4.64), minimizing $\sum_i (Y_i - (\gamma + \theta X_i)^{1/2})^2$, which is consistent for *both* parameters under (4.64) alone.

Exercise 4.18

Write $M_1 = \mathbf{I}_n - X_1(X_1'X_1)^{-1}X_1'$, so the short residual vector is $\hat{e} = M_1Y$. Since $M_1X_1 = \mathbf{0}$,

$$\hat{e} = M_1(X_1\beta_1 + X_2\beta_2 + e) = M_1X_2\beta_2 + M_1e.$$

Using $M_1'M_1 = M_1$,

$$\hat{e}'\hat{e} = \beta_2'X_2'M_1X_2\beta_2 + 2\beta_2'X_2'M_1e + e'M_1e.$$

Take conditional expectations term by term. The first term is non-random given X . The second vanishes since $\mathbb{E}[e | X] = 0$. For the third, use the trace trick and homoskedasticity:

$$\mathbb{E}[e'M_1e | X] = \text{tr}(M_1\mathbb{E}[ee' | X]) = \sigma^2 \text{tr} M_1 = \sigma^2(n - k_1),$$

by Exercise 3.9 applied to the k_1 -column regressor matrix X_1 . Hence

$$\mathbb{E}[s^2 | X] = \sigma^2 + \frac{\beta_2'X_2'M_1X_2\beta_2}{n - k_1} \geq \sigma^2.$$

The bias term is non-negative because M_1 is positive semi-definite: omitting relevant regressors inflates the estimated error variance, by the (scaled) explained sum of squares that X_2 would have contributed after partialling out X_1 . It vanishes exactly when $\beta_2 = \mathbf{0}$, or when $M_1X_2\beta_2 = \mathbf{0}$ (the omitted linear combination is collinear with X_1 in the sample).

This is the sample analogue of Exercise 2.22. It also explains why an omitted variable makes a regression look noisier as well as biased, and it is the basis for the F test of $\beta_2 = \mathbf{0}$ in Section 5.13.

Exercise 4.19

By Exercise 3.2, $\hat{\beta}^* = C^{-1}\hat{\beta}$ and, crucially, the residuals are unchanged: $\hat{e}^* = \hat{e}$. The leverage values are unchanged too, since $P^* = P$. Writing $X_i^* = C'X_i$ for the transformed i th regressor vector and

$\hat{\Omega} = \sum_i \omega_i X_i X_i' \hat{e}_i^2$ for the “meat” of any member of the HC family (with $\omega_i = 1$, $(1 - h_{ii})^{-1}$ or $(1 - h_{ii})^{-2}$),

$$\sum_{i=1}^n \omega_i X_i^* X_i^{*'} \hat{e}_i^2 = C' \left(\sum_{i=1}^n \omega_i X_i X_i' \hat{e}_i^2 \right) C = C' \hat{\Omega} C,$$

while $(X^{*'} X^*)^{-1} = C^{-1} (X' X)^{-1} (C')^{-1}$. Therefore

$$\begin{aligned} \hat{V}^* &= (X^{*'} X^*)^{-1} (C' \hat{\Omega} C) (X^{*'} X^*)^{-1} \\ &= C^{-1} (X' X)^{-1} (C')^{-1} C' \hat{\Omega} C C^{-1} (X' X)^{-1} (C')^{-1} \\ &= C^{-1} \left[(X' X)^{-1} \hat{\Omega} (X' X)^{-1} \right] (C')^{-1}, \end{aligned}$$

that is,

$$\boxed{\hat{V}^* = C^{-1} \hat{V} (C')^{-1} = C^{-1} \hat{V} (C^{-1})' .}$$

The same formula holds for the homoskedastic estimator $s^2 (X' X)^{-1}$, since s^2 is a function of the (unchanged) residuals and the (unchanged) k .

This is exactly the delta-method transformation for the linear map $\beta \mapsto C^{-1} \beta$, as it must be. Its practical content: standard errors of any linear combination of coefficients are invariant to reparameterization. If you want a standard error for $\beta_1 + \beta_2$, you may either compute $\sqrt{\hat{V}_{11} + 2\hat{V}_{12} + \hat{V}_{22}}$ or re-run the regression in the reparameterized form of Exercise 3.23 and read it off directly; the two agree exactly.

Exercise 4.20

The two estimation errors are

$$\hat{\beta} - \beta = (X' X)^{-1} X' e, \quad \tilde{\beta} - \beta = Q X' \Sigma^{-1} e, \quad Q \equiv (X' \Sigma^{-1} X)^{-1}.$$

The covariance. Using $\mathbb{E}[ee' | X] = \Sigma$,

$$\begin{aligned} \mathbb{E}[(\hat{\beta} - \beta)(\tilde{\beta} - \beta)' | X] &= (X' X)^{-1} X' \mathbb{E}[ee' | X] \Sigma^{-1} X Q \\ &= (X' X)^{-1} X' \Sigma \Sigma^{-1} X Q \\ &= (X' X)^{-1} (X' X) Q = Q = \text{var}[\tilde{\beta} | X]. \end{aligned}$$

$$\boxed{\mathbb{E}[(\hat{\beta} - \beta)(\tilde{\beta} - \beta)' | X] = (X' \Sigma^{-1} X)^{-1} = \text{var}[\tilde{\beta} | X].}$$

This is a striking and general fact: *the covariance between any estimator and the efficient estimator equals the variance of the efficient estimator*. It is the algebraic engine behind the Hausman test.

The variance of the difference. Write $V_{\hat{\beta}} = \text{var}[\hat{\beta} | X] = (X' X)^{-1} X' \Sigma X (X' X)^{-1}$ and $V_{\tilde{\beta}} = Q$. Then

$$\begin{aligned} \text{var}[\hat{\beta} - \tilde{\beta} | X] &= V_{\hat{\beta}} + V_{\tilde{\beta}} - 2 \mathbb{E}[(\hat{\beta} - \beta)(\tilde{\beta} - \beta)' | X] \\ &= V_{\hat{\beta}} + Q - 2Q \\ &= V_{\hat{\beta}} - V_{\tilde{\beta}} = (X' X)^{-1} X' \Sigma X (X' X)^{-1} - (X' \Sigma^{-1} X)^{-1}. \end{aligned}$$

By Exercise 4.6 this difference is positive semi-definite, as it must be for a variance matrix — which gives an alternative proof of the Aitken bound.

Remark. The second display in the printed exercise reads $\mathbb{E}[(\hat{\beta} - \tilde{\beta})(\hat{\beta} - \beta)']$, which appears to be a typographical slip for $(\hat{\beta} - \tilde{\beta})(\tilde{\beta} - \beta)'$; the covariance matrix of $\hat{\beta} - \tilde{\beta}$ is what is computed above. For completeness, the quantity as literally written is $\mathbb{E}[(\hat{\beta} - \tilde{\beta})(\hat{\beta} - \beta)' | X] = V_{\hat{\beta}} - Q$ as well, by the same cancellation.

Exercise 4.21

$\tilde{R}^2$ is smaller (weakly).

The reason is immediate once one recalls what OLS optimizes. By definition

$$\hat{\beta} = \arg \min_b (Y - Xb)'(Y - Xb),$$

so for *any* other estimator, in particular $\tilde{\beta}$,

$$\hat{e}'\hat{e} = (Y - X\hat{\beta})'(Y - X\hat{\beta}) \leq (Y - X\tilde{\beta})'(Y - X\tilde{\beta}) = \tilde{e}'\tilde{e}.$$

Both R^2 s are computed with the same denominator $Y^{*'}Y^*$, which depends only on Y , so

$$\tilde{R}^2 = 1 - \frac{\tilde{e}'\tilde{e}}{Y^{*'}Y^*} \leq 1 - \frac{\hat{e}'\hat{e}}{Y^{*'}Y^*} = \hat{R}^2.$$

Equality holds only if $\tilde{\beta} = \hat{\beta}$, which under genuine heteroskedasticity is a measure-zero coincidence. The point of the exercise is the warning that follows. GLS is the *efficient* estimator here — it has the smaller sampling variance (Exercise 4.6) — yet it necessarily reports the *worse* R^2 . R^2 measures in-sample sum of squared residuals in the untransformed metric, which is exactly the criterion OLS is built to optimize; it says nothing about the precision with which β is estimated. R^2 is therefore useless for choosing between estimators of the same model.

Exercise 4.22

No. The two assumptions are about different things and neither implies the other.

What i.i.d. says. Assumption 4.1 is a statement *across* observations: the pairs $(Y_1, X_1), \dots, (Y_n, X_n)$ are mutually independent and all drawn from one common distribution F . Its consequence for the error vector is that $\text{var}[e | X]$ is *diagonal* — $\mathbb{E}[e_i e_j | X] = 0$ for $i \neq j$ — and that the diagonal entries are generated by the same function, $\sigma_i^2 = \sigma^2(X_i)$ with the *same* $\sigma^2(\cdot)$ for every i .

What homoskedasticity says. Assumption 4.3 is a statement *within* the common distribution F : the function $\sigma^2(\cdot)$ is constant, $\sigma^2(x) = \sigma^2$ for all x .

So i.i.d. sampling gives $\text{var}[e | X] = \text{diag}\{\sigma^2(X_1), \dots, \sigma^2(X_n)\}$; homoskedasticity is the extra restriction that turns this into $\sigma^2 \mathbf{I}_n$. The observations can be perfect i.i.d. draws from a population in which the conditional variance varies enormously with x .

A concrete counterexample to hand the friend. Let X_i be i.i.d. $N(0, 1)$ and u_i i.i.d. $N(0, 1)$ independent of X_i , and set $Y_i = X_i \beta + X_i u_i$. The pairs (Y_i, X_i) are manifestly i.i.d., the CEF is linear with $\mathbb{E}[e_i | X_i] = 0$, yet

$$\mathbb{E}[e_i^2 | X_i] = X_i^2,$$

which is as heteroskedastic as one could wish. In real data, wage regressions on cross-sectional survey samples are i.i.d. by construction and famously heteroskedastic in education — Exercise 4.24 shows the robust and homoskedastic standard errors differing there.

The correct statement of the dependence structure. What i.i.d. sampling does buy you is that the off-diagonal terms vanish, which is what permits the sandwich formula

$$V_{\hat{\beta}} = (X'X)^{-1} \left(\sum_{i=1}^n \sigma_i^2 X_i X_i' \right) (X'X)^{-1}$$

in the first place. Clustered sampling breaks even that (Section 4.21).

Exercise 4.23

Write $A_\lambda = X'X + \lambda \mathbf{I}_k$, which is positive definite (hence invertible) whenever $\lambda > 0$, even if $X'X$ is singular. Substituting $Y = X\beta + e$ and using $\mathbb{E}[e | X] = 0$,

$$\mathbb{E}[\hat{\beta} | X] = A_\lambda^{-1} X'X\beta.$$

To see the bias, write $X'X = A_\lambda - \lambda \mathbf{I}_k$:

$$\mathbb{E}[\hat{\beta} | X] = A_\lambda^{-1} (A_\lambda - \lambda \mathbf{I}_k) \beta = \beta - \lambda (X'X + \lambda \mathbf{I}_k)^{-1} \beta.$$

Yes, $\hat{\beta}$ is biased unless $\lambda = 0$ or $\beta = \mathbf{0}$. The bias is

$$\text{Bias} = -\lambda (X'X + \lambda \mathbf{I}_k)^{-1} \beta,$$

which points *toward the origin*: ridge shrinks. In the orthonormal design $X'X = n\mathbf{I}_k$ this is transparent, $\mathbb{E}[\hat{\beta} | X] = \frac{n}{n+\lambda} \beta$, a uniform shrinkage by the factor $n/(n+\lambda) < 1$.

The bias is accepted deliberately, in exchange for variance: the ridge variance $\text{var}[\hat{\beta} | X] = \sigma^2 A_\lambda^{-1} (X'X) A_\lambda^{-1}$ is smaller than the OLS variance in the positive semi-definite order, and for small enough λ the reduction in variance outweighs the squared bias, lowering mean squared error. With λ held fixed as $n \rightarrow \infty$ the bias vanishes and $\hat{\beta}$ is consistent (Exercise 7.2); with $\lambda = cn$ growing with the sample, it is not (Exercise 7.3).

Exercise 4.24

(a) Equation (3.49), single Asian men with experience < 45 , $n = 267$, $k = 4$. The five sets of standard errors are:

	homoskedastic	HC0	HC1	HC2	HC3
education	0.01164	0.01173	0.01181	0.01191	0.01210
experience	0.01217	0.01242	0.01252	0.01261	0.01281
experience ² /100	0.03488	0.03380	0.03405	0.03456	0.03537
intercept	0.18983	0.20005	0.20157	0.20269	0.20544

($s^2 = 0.3137$, $\hat{\sigma}^2 = 0.3090$, $\max_i h_{ii} = 0.126$.)

Three things to notice. First, the robust standard errors are ordered $HC0 < HC1 < HC2 < HC3$ for every coefficient, exactly as Exercise 4.10 predicts. Second, the spread across the four robust versions is small — about 3% between HC0 and HC3 — because the leverage values are modest here ($\max_i h_{ii} = 0.126$, against an average of $k/n = 0.015$); with a few high-leverage observations the gap would be much larger. Third, the homoskedastic and robust standard errors differ by only a few percent in this sub-sample, and not in a uniform direction: the homoskedastic formula is *smaller* for education and experience but *larger* for the squared term. That non-uniformity is the signature of genuine heteroskedasticity: there is no scalar correction that would reconcile them.

(b) Yes — to floating-point tolerance. Coefficients and standard errors from R, Stata, MATLAB and Python agree to roughly 10^{-10} . What is *not* automatic is the choice of estimator: Stata's `,r` option reports HC1, R's `lm` default reports the homoskedastic formula, and `sandwich::vcovHC` defaults to HC3. A reported difference between two packages is almost always a difference in the default, not in the arithmetic.

Code (R).

```
dat <- read.table("cps09mar.txt")
names(dat) <- c("age","female","hisp","education","earnings","hours",
               "week","union","uncov","region","race","marital")
dat$exper <- dat$age - dat$education - 6
d <- subset(dat, race==4 & marital==7 & female==0 & exper<45)
y <- log(d$earnings/(d$hours*d$week))
X <- cbind(d$education, d$exper, d$exper^2/100, 1)
n <- nrow(X); k <- ncol(X)
XXi <- solve(crossprod(X)); b <- XXi %*% crossprod(X,y); e <- y - X %*% b
h <- rowSums((X %*% XXi) * X)
meat <- function(w) crossprod(X, X * as.vector(w * e^2))
V0 <- XXi %*% meat(1) %*% XXi
V1 <- (n/(n-k)) * V0
V2 <- XXi %*% meat(1/(1-h)) %*% XXi
V3 <- XXi %*% meat(1/(1-h)^2) %*% XXi
Vh <- as.numeric(crossprod(e)/(n-k)) * XXi
round(sqrt(cbind(diag(Vh), diag(V0), diag(V1), diag(V2), diag(V3)))), 5)
```

Exercise 4.25

White male Hispanics, $n = 4,230$. HC0 and HC3 standard errors for the specification of Exercise 3.26(a):

	coefficient	s.e. (HC0)	s.e. (HC3)
education	0.08829	0.00292	0.00293
experience	0.02787	0.00283	0.00285
experience ² /100	-0.03630	0.00548	0.00551
Northeast	0.06156	0.03602	0.03614
South	-0.06795	0.02965	0.02974
West	0.01950	0.02825	0.02834
married	0.17784	0.02499	0.02505
widowed or divorced	0.08572	0.04428	0.04447
separated	0.01671	0.05253	0.05303
intercept	1.19288	0.05008	0.05025

HC3 exceeds HC0 for every coefficient (Exercise 4.10), but here by only 0.2–1%. The reason is the sample size: with $n = 4,230$ and $k = 10$ the average leverage is $k/n = 0.0024$, so $(1 - h_{ii})^{-2} \approx 1$ for essentially every observation and the HC family collapses onto itself. The choice among HC0–HC3 matters in small samples and in designs with a few extreme regressor values — the “separated” dummy, with only 109 ones, shows the largest relative gap (0.95%), which is exactly the pattern one expects.

Yes, a second package reproduces these to floating-point tolerance, subject to the same caveat about defaults noted in Exercise 4.24(b).

Code (R). With the `sandwich` package the calculation is one line; the explicit version is in Exercise 4.24.

```
library(sandwich); library(lmtest)
fit <- lm(y ~ education + ex + I(ex^2/100) + ne + so + we + mar + wd + sep)
coefTest(fit, vcov = vcovHC(fit, type = "HC3"))
```

Exercise 4.26

Dropping observations with missing covariates leaves $n = 5,269$ students in $G = 111$ schools (the baseline regression (4.60) uses $n = 5,795$ in $G = 121$). Cluster standard errors use (4.55)–(4.56) with the Stata adjustment $a_n = \left(\frac{n-1}{n-k}\right)\left(\frac{G}{G-1}\right)$.

	coefficient	robust (HC1)	clustered	ratio
tracking	0.1736	0.0242	0.0767	3.17
age at test	-0.0411	0.0085	0.0134	1.57
girl	0.0817	0.0242	0.0287	1.18
contract teacher	0.1810	0.0239	0.0377	1.58
baseline percentile	0.0174	0.0004	0.0007	1.70
intercept	-0.7404	0.0815	0.1305	1.60

(a) *Tracking changes the most* — its standard error triples, from 0.024 to 0.077. *Gender changes the least*, rising only 18%. The pattern is not an accident and is the main lesson of the exercise.

Clustering matters for a regressor in proportion to how much of its variation is *between* clusters rather than within. **tracking** is assigned at the school level: it is constant within a school, so the school-level error component is perfectly correlated with it and the effective number of independent observations for that coefficient is $G = 111$, not $n = 5,269$. The conventional robust formula, which pretends there are 5,269 independent observations, understates the standard error by roughly $\sqrt{1 + (\bar{n}_g - 1)\rho}$ (Moulton's factor). **girl**, by contrast, varies almost entirely *within* schools, so school-level error correlation barely affects it. The remaining regressors — age, contract-teacher assignment (randomized within school but correlated with stream), and baseline percentile — fall in between, with ratios around 1.6–1.7.

The applied moral: when a treatment is assigned at the group level, standard errors must be clustered at that level. Here the t statistic on tracking falls from 7.2 to 2.3 — still significant, but a very different picture of the evidence.

(b) In (4.60) the tracking coefficient is 0.138 with clustered standard error 0.078. Adding the individual controls raises it to 0.174 with essentially the same clustered standard error 0.077. So the estimated effect of tracking is about 26% larger once age, gender, contract-teacher assignment and baseline ability are controlled for, and the precision is unchanged. Because tracking was randomly assigned across schools, the controls should be (and are) roughly orthogonal to it in expectation; the change of 0.036 reflects chance imbalance in the realized randomization plus the change in the estimation sample from 5,795 to 5,269 observations. The controls do not sharpen the clustered standard error, again because the identifying variation in tracking is between schools and the controls are overwhelmingly within-school.

Code (R).

```
library(haven); library(sandwich); library(lmtest)
d <- read_dta("DDK2011.dta")
v <- c("totalscore","schoolid","tracking","agetest","girl",
      "etpteacher","percentile")
d <- na.omit(d[, v])
d$testscore <- (d$totalscore - mean(d$totalscore)) / sd(d$totalscore)
fit <- lm(testscore ~ tracking + agetest + girl + etpteacher + percentile,
        data = d)
coeftest(fit, vcov = vcovHC(fit, type = "HC1"))           # robust
coeftest(fit, vcov = vcovCL(fit, cluster = d$schoolid))    # clustered
```

Chapter 5

Normal Regression

Solutions to all 12 exercises of Chapter 5 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022).

The normal regression model maintains $Y = X'\beta + e$ with $e | X \sim N(0, \sigma^2)$ and i.i.d. observations, so that in vector notation $e | X \sim N(\mathbf{0}, \mathbf{I}_n \sigma^2)$. As before $P = X(X'X)^{-1}X'$, $M = \mathbf{I}_n - P$, h_{ii} are the leverage values and $\hat{e} = MY$. Two results are used repeatedly: $\hat{e} | X \sim N(\mathbf{0}, \sigma^2 M)$ independent of $\hat{\beta}$ (THEOREM 5.6), and $(n - k)s^2/\sigma^2 \sim \chi_{n-k}^2$ independent of $\hat{\beta}$ (THEOREM 5.7).

Exercise 5.1

Write $Q = \sum_{i=1}^r Z_i^2$ with Z_i i.i.d. $N(0, 1)$. The two moments of Z^2 that we need are

$$\mathbb{E}[Z^2] = \text{var}[Z] = 1, \quad \mathbb{E}[Z^4] = 3,$$

the second being the standard normal fourth moment (the moment generating function gives $\mathbb{E}[Z^{2m}] = (2m - 1)!!$, so $\mathbb{E}[Z^4] = 3 \cdot 1 = 3$).

Mean. By linearity,

$$\mathbb{E}[Q] = \sum_{i=1}^r \mathbb{E}[Z_i^2] = r.$$

Variance. Since the Z_i^2 are independent, the variance of the sum is the sum of the variances:

$$\text{var}[Q] = \sum_{i=1}^r \text{var}[Z_i^2] = r \left(\mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 \right) = r(3 - 1) = 2r.$$

These are the two facts behind the degrees-of-freedom arithmetic in Exercise 5.12 and behind the asymptotic χ^2 critical values used throughout Chapter 9.

Exercise 5.2

A linear transformation of a multivariate normal is normal, with mean and variance transformed in the usual way (THEOREM 5.3.3):

$$u = H'e \sim N(H'\mathbf{0}, H'(\mathbf{I}_n \sigma^2)H) = N(\mathbf{0}, \sigma^2 H'H) = N(\mathbf{0}, \mathbf{I}_n \sigma^2),$$

the last step by the orthogonality assumption $H'H = \mathbf{I}_n$.

The content of the result is that the spherical normal is *rotation invariant*: its distribution is unchanged by any orthonormal change of basis. This is exactly what makes the spectral-decomposition argument of Section 5.8 work, and hence what delivers the exact χ_{n-k}^2 distribution of $(n - k)s^2/\sigma^2$.

Exercise 5.3

Again by THEOREM 5.3.3,

$$u = A^{-1}e \sim N\left(A^{-1}\mathbf{0}, A^{-1}(AA')(A^{-1})'\right) = N\left(\mathbf{0}, (A^{-1}A)(A'(A')^{-1})\right) = N(\mathbf{0}, \mathbf{I}_n),$$

using $(A^{-1})' = (A')^{-1}$.

Such an A always exists when $\Sigma > 0$ (take the Cholesky factor or the symmetric square root $\Sigma^{1/2}$), so any non-degenerate normal vector can be “whitened”. This is the distributional justification for the GLS transformation of Section 4.9: pre-multiplying by $\Sigma^{-1/2}$ converts a correlated, heteroskedastic normal regression into a spherical one.

Exercise 5.4

Here $L_n(\theta)$ is the likelihood and $\ell_n(\theta) = \log L_n(\theta)$ the log-likelihood. The claim is that a strictly increasing transformation does not change the location of a maximum.

Let $\theta^* \in \arg \max_{\theta \in \Theta} L_n(\theta)$, so $L_n(\theta^*) \geq L_n(\theta)$ for all $\theta \in \Theta$. Because $x \mapsto \log x$ is strictly increasing on $(0, \infty)$,

$$L_n(\theta^*) \geq L_n(\theta) \iff \log L_n(\theta^*) \geq \log L_n(\theta) \iff \ell_n(\theta^*) \geq \ell_n(\theta),$$

for every θ with $L_n(\theta) > 0$. Hence θ^* maximizes ℓ_n as well, and running the equivalence in the other direction shows that any maximizer of ℓ_n maximizes L_n . The two argmax sets are therefore identical.

The boundary case is harmless: a likelihood is non-negative, and where $L_n(\theta) = 0$ we set $\ell_n(\theta) = -\infty$, so such θ maximize neither function (provided L_n is positive somewhere, which it is whenever the model is not contradicted by the data).

Why bother? Because L_n is a *product* over observations while ℓ_n is a *sum*. Sums differentiate cleanly, obey the law of large numbers and the central limit theorem, and do not underflow numerically. Every practical advantage of maximum likelihood comes from working on the log scale, and this exercise is the licence to do so.

Exercise 5.5

Substituting $Y = X\beta + e$ into $\hat{Y} = PY$ and using $PX = X$,

$$\hat{Y} = PY = P(X\beta + e) = X\beta + Pe.$$

Conditional on X , the matrix P is fixed and $e \sim N(\mathbf{0}, \mathbf{I}_n\sigma^2)$, so $\hat{Y}$ is a linear transformation of a normal vector and hence normal. Its mean and variance are

$$\mathbb{E}[\hat{Y} | X] = X\beta + P \mathbb{E}[e | X] = X\beta,$$

$$\text{var}[\hat{Y} | X] = P \text{var}[e | X] P' = \sigma^2 PP' = \sigma^2 P,$$

using the symmetry and idempotency of P . Therefore

$$\hat{Y} | X \sim N(X\beta, \sigma^2 P).$$

Reading off the i th coordinate: the i th element of the mean vector is $X_i'\beta$, and the i th diagonal element of $\sigma^2 P$ is $\sigma^2 h_{ii}$. Hence

$$\hat{Y}_i | X \sim N(X_i'\beta, \sigma^2 h_{ii}).$$

Note the interpretation of leverage that this delivers: h_{ii} is precisely the variance of the fitted value at X_i , in units of σ^2 . A high-leverage observation is one whose own fitted value is estimated

imprecisely — which is why such observations are influential (Exercise 3.18, Section 3.21). Since $\sum_i h_{ii} = k$, the average variance of a fitted value is $\sigma^2 k/n$.

Exercise 5.6

Both objects are *fixed linear transformations of the residual vector*, given X . By (3.45) and (4.29),

$$\tilde{e} = M^* \hat{e}, \quad \bar{e} = (M^*)^{1/2} \hat{e},$$

where $M^* = \text{diag}\{(1 - h_{11})^{-1}, \dots, (1 - h_{nn})^{-1}\}$ is a diagonal matrix built from the leverage values. Since $h_{ii} = X_i'(X'X)^{-1}X_i$ is a function of X alone, M^* is non-random conditional on X .

By THEOREM 5.6, in the normal regression model $\hat{\beta}$ and $\hat{e}$ are independent conditional on X . Independence is preserved under measurable transformations: if $\hat{\beta} \perp \hat{e}$ then $\hat{\beta} \perp g(\hat{e})$ for any function g . Applying this with the linear maps $g_1(\hat{e}) = M^* \hat{e}$ and $g_2(\hat{e}) = (M^*)^{1/2} \hat{e}$ gives

$$\tilde{e} \perp \hat{\beta} \mid X, \quad \bar{e} \perp \hat{\beta} \mid X.$$

The same argument covers the individual residuals $\hat{e}_i$, the variance estimators s^2 and $\hat{\sigma}^2$, the prediction criterion $\tilde{\sigma}^2$ of (3.46), and every robust covariance matrix estimator (Exercise 5.7).

Remark. For completeness, the distributions themselves follow from $\hat{e} \mid X \sim N(\mathbf{0}, \sigma^2 M)$: $\tilde{e} \mid X \sim N(\mathbf{0}, \sigma^2 M^* M M^*)$ and $\bar{e} \mid X \sim N(\mathbf{0}, \sigma^2 (M^*)^{1/2} M (M^*)^{1/2})$, with $\text{var}[\tilde{e}_i \mid X] = (1 - h_{ii})^{-1} \sigma^2$ and $\text{var}[\bar{e}_i \mid X] = \sigma^2$ — the point of standardizing.

Exercise 5.7

Recall the four estimators:

$$\begin{aligned} \hat{V}_{\hat{\beta}}^{\text{HC0}} &= (X'X)^{-1} \left(\sum_{i=1}^n X_i X_i' \hat{e}_i^2 \right) (X'X)^{-1}, \\ \hat{V}_{\hat{\beta}}^{\text{HC1}} &= \frac{n}{n-k} \hat{V}_{\hat{\beta}}^{\text{HC0}}, \\ \hat{V}_{\hat{\beta}}^{\text{HC2}} &= (X'X)^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-1} X_i X_i' \hat{e}_i^2 \right) (X'X)^{-1}, \\ \hat{V}_{\hat{\beta}}^{\text{HC3}} &= (X'X)^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-2} X_i X_i' \hat{e}_i^2 \right) (X'X)^{-1}. \end{aligned}$$

Conditional on X , every ingredient except $\hat{e}_i^2$ is non-random: the matrices $(X'X)^{-1}$ and $X_i X_i'$, the leverage values h_{ii} , and the constants n and k . Each $\hat{V}_{\hat{\beta}}^{\text{HC}j}$ is therefore a fixed (measurable) function of the residual vector $\hat{e}$ alone:

$$\hat{V}_{\hat{\beta}}^{\text{HC}j} = g_j(\hat{e}) \quad \text{for some } g_j \text{ depending only on } X.$$

By THEOREM 5.6, $\hat{e}$ is independent of $\hat{\beta}$ conditional on X . Independence is inherited by functions, so each $\hat{V}_{\hat{\beta}}^{\text{HC}j}$ is independent of $\hat{\beta}$ conditional on X .

This independence is what makes exact finite-sample inference possible in the normal model: a t statistic $(\hat{\beta}_j - \beta_j)/s(\hat{\beta}_j)$ is a ratio of a normal numerator to an *independent* denominator, which is why the classical version (with s^2 in the denominator) has an exact t_{n-k} distribution. With a

robust denominator the independence still holds, but the exact distribution is no longer Student t — the denominator is not a scaled χ^2 — so the t_{n-k} critical values become an approximation.

Exercise 5.8

Let f be the density. Symmetry about zero means $f(x) = f(-x)$ for all x .

First, $F(0) = \frac{1}{2}$: the two tails have equal mass, $\int_{-\infty}^0 f(x)dx = \int_0^{\infty} f(x)dx$ by the change of variable $x \mapsto -x$, and they sum to 1.

Now fix $u \geq 0$ and apply the same change of variable on a bounded interval:

$$\int_{-u}^0 f(x) dx = \int_0^u f(x) dx.$$

Writing both sides in terms of F ,

$$F(0) - F(-u) = F(u) - F(0).$$

Substituting $F(0) = \frac{1}{2}$ and rearranging,

$$F(-u) = 2F(0) - F(u) = 1 - F(u).$$

(The case $u < 0$ follows by replacing u with $-u$.)

The practical consequence is the familiar symmetry of critical values: if c solves $F(c) = 1 - \alpha/2$ then $F(-c) = \alpha/2$, so the equal-tailed $1 - \alpha$ interval is $[-c, c]$, and a two-sided p -value can be computed as $2(1 - F(|t|))$.

Exercise 5.9

The equivariance. Because g is monotonically increasing, for any real numbers $a \leq b$,

$$a \leq \beta \leq b \iff g(a) \leq g(\beta) \leq g(b).$$

Applying this with the (random) endpoints $a = L$ and $b = U$, the two events are literally the same event:

$$\{\beta \in [L, U]\} = \{g(\beta) \in [g(L), g(U)]\} = \{\theta \in \widehat{C}_\theta\}.$$

Identical events have identical probabilities, so

$$\mathbb{P}[\theta \in \widehat{C}_\theta] = \mathbb{P}[\beta \in \widehat{C}_\beta] = 1 - \alpha.$$

Coverage is exactly preserved. Note that no distributional assumption was used at all — this is a statement about sets, not about sampling distributions. It is also why confidence intervals behave better than standard errors under reparameterization: transforming the endpoints is exact, whereas the delta method is only an approximation.

A confidence interval for σ . By THEOREM 5.7, $(n - k)s^2/\sigma^2 \sim \chi_{n-k}^2$ in the normal regression model. Let q_p denote the p th quantile of the χ_{n-k}^2 distribution. Then

$$\mathbb{P}\left[q_{\alpha/2} \leq \frac{(n - k)s^2}{\sigma^2} \leq q_{1-\alpha/2}\right] = 1 - \alpha,$$

and inverting the inequalities for σ^2 gives the exact $1 - \alpha$ interval

$$\hat{C}_{\sigma^2} = \left[\frac{(n-k)s^2}{q_{1-\alpha/2}}, \frac{(n-k)s^2}{q_{\alpha/2}} \right].$$

Now apply the result just proved with the increasing transformation $g(x) = \sqrt{x}$ on $(0, \infty)$:

$$\hat{C}_\sigma = \left[s \sqrt{\frac{n-k}{q_{1-\alpha/2}}}, s \sqrt{\frac{n-k}{q_{\alpha/2}}} \right]$$

has exact coverage $1 - \alpha$ for σ . Observe that this interval is *not* symmetric about s , which is correct: the sampling distribution of s is skewed, and the naive interval $s \pm z_{1-\alpha/2} \times \text{s.e.}(s)$ has poorer coverage.

Exercise 5.10

Step 1: write both statistics as functions of the variance ratio. By (5.18) and (5.19),

$$LR = n \log \left(\frac{\tilde{\sigma}^2}{\hat{\sigma}^2} \right), \quad F = \frac{(\tilde{\sigma}^2 - \hat{\sigma}^2)/q}{\hat{\sigma}^2/(n-k)} = \frac{n-k}{q} \left(\frac{\tilde{\sigma}^2}{\hat{\sigma}^2} - 1 \right).$$

Let $R = \tilde{\sigma}^2/\hat{\sigma}^2$. Since the restricted sum of squares is never smaller than the unrestricted one (Exercise 3.16), $R \geq 1$, and both $LR = n \log R$ and $F = \frac{n-k}{q}(R-1)$ are strictly increasing functions of R .

Step 2: match the rejection regions. Because $\log$ is strictly increasing,

$$LR \geq c_1 \iff n \log R \geq c_1 \iff R \geq \exp\left(\frac{c_1}{n}\right) \iff R-1 \geq \exp\left(\frac{c_1}{n}\right) - 1.$$

Multiplying by the positive constant $(n-k)/q$,

$$LR \geq c_1 \iff F \geq \frac{n-k}{q} \left(\exp\left(\frac{c_1}{n}\right) - 1 \right) = c_2.$$

So with c_2 chosen as stated, the two rules reject on exactly the same samples.

Step 3: are the tests equivalent? *As decision rules with matched critical values, yes; as procedures used in practice, not exactly.*

Since LR and F are strictly increasing transformations of one another, they order samples identically: any test based on one can be reproduced by a test based on the other. In that sense there is only one test here, written two ways.

But the critical values actually used are not matched by the formula above.

- The F statistic has an *exact* $F_{q,n-k}$ null distribution in the normal regression model (THEOREM 5.13), so c_2 is taken from the F table and the test has exact size α .
- The LR statistic is conventionally compared with a χ_q^2 critical value, which is only an *asymptotic* approximation.

These two critical values do not satisfy $c_2 = (\exp(c_1/n) - 1)(n-k)/q$ except by coincidence, so the two procedures can disagree in finite samples — the χ^2 -based LR test typically over-rejects. They do agree asymptotically: for fixed c_1 and large n , $n(\exp(c_1/n) - 1) \rightarrow c_1$ and $(n-k)/n \rightarrow 1$, so $c_2 \rightarrow c_1/q$, which is the exact relation between the χ_q^2 and $F_{q,\infty}$ critical values.

Exercise 5.11

The two scores, as displayed in Section 5.14, are

$$\frac{\partial}{\partial \beta} \ell_n(\beta, \sigma^2) = \frac{1}{\sigma^2} \sum_{i=1}^n X_i e_i, \quad \frac{\partial}{\partial \sigma^2} \ell_n(\beta, \sigma^2) = \frac{1}{2\sigma^4} \sum_{i=1}^n (e_i^2 - \sigma^2).$$

Both have conditional mean zero, so the information matrix is the matrix of conditional second moments. Conditional on X the errors e_i are i.i.d. $N(0, \sigma^2)$, with

$$\mathbb{E}[e^2] = \sigma^2, \quad \mathbb{E}[e^3] = 0, \quad \mathbb{E}[e^4] = 3\sigma^4.$$

Upper-left block. Using independence across i ($\mathbb{E}[e_i e_j | X] = 0$ for $i \neq j$),

$$\text{var} \left[\frac{1}{\sigma^2} \sum_{i=1}^n X_i e_i \mid X \right] = \frac{1}{\sigma^4} \sum_{i=1}^n X_i X_i' \mathbb{E}[e_i^2 \mid X] = \frac{\sigma^2}{\sigma^4} \sum_{i=1}^n X_i X_i' = \frac{1}{\sigma^2} X' X.$$

Lower-right block. The summands $e_i^2 - \sigma^2$ are i.i.d. with mean zero and variance $\mathbb{E}[e^4] - \sigma^4 = 3\sigma^4 - \sigma^4 = 2\sigma^4$ (this is Exercise 5.1 with $r = 1$). Hence

$$\text{var} \left[\frac{1}{2\sigma^4} \sum_{i=1}^n (e_i^2 - \sigma^2) \mid X \right] = \frac{1}{4\sigma^8} \cdot n \cdot 2\sigma^4 = \frac{n}{2\sigma^4}.$$

Off-diagonal block. Again by independence across i , only the $i = j$ terms survive, and those involve the *third* moment of a normal:

$$\mathbb{E} \left[\frac{1}{\sigma^2} \sum_i X_i e_i \cdot \frac{1}{2\sigma^4} \sum_j (e_j^2 - \sigma^2) \mid X \right] = \frac{1}{2\sigma^6} \sum_{i=1}^n X_i \mathbb{E}[e_i^3 - \sigma^2 e_i \mid X] = \mathbf{0},$$

since $\mathbb{E}[e^3] = 0$ and $\mathbb{E}[e] = 0$. This is a genuinely normal-specific result: for a non-normal error with $\mathbb{E}[e^3] \neq 0$ the information matrix is not block diagonal, and $\hat{\beta}$ and $\hat{\sigma}^2$ are not asymptotically independent.

Assembling the blocks gives (5.20), and inverting the block-diagonal matrix gives the Cramér–Rao lower bound

$$\mathcal{I}^{-1} = \begin{pmatrix} \sigma^2 (X'X)^{-1} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{pmatrix},$$

so the bound is $\sigma^2 (X'X)^{-1}$ for β and $2\sigma^4/n$ for σ^2 .

Remark. The printed display of (5.20) shows $2\sigma^4/n$ in the lower-right entry of $\mathcal{I}$ and of $\mathcal{I}^{-1}$. Since the two must be reciprocals, one of them is a typographical slip. The derivation above shows that the information entry is $n/(2\sigma^4)$ and the bound is $2\sigma^4/n$ — which is the value the surrounding text and Exercise 5.12 both use.

Exercise 5.12

(a) By THEOREM 5.7,

$$\frac{(n-k)s^2}{\sigma^2} \sim \chi_{n-k}^2,$$

and by Exercise 5.1 a χ_r^2 variable has variance $2r$. Hence

$$\text{var} \left[\frac{(n-k)s^2}{\sigma^2} \right] = 2(n-k).$$

The factor $(n-k)/\sigma^2$ is non-random, so it comes out of the variance squared:

$$\frac{(n-k)^2}{\sigma^4} \text{var}[s^2] = 2(n-k) \quad \implies \quad \boxed{\text{var}[s^2] = \frac{2\sigma^4}{n-k}}.$$

(The result is conditional on X , but as the right-hand side does not depend on X it is also the unconditional variance, by the law of total variance and the fact that $\mathbb{E}[s^2 | X] = \sigma^2$ is constant.)

(b) From Exercise 5.11, the Cramér–Rao Lower Bound for σ^2 is $2\sigma^4/n$. Since $k \geq 1$ we have $n-k < n$ and therefore

$$\text{var}[s^2] = \frac{2\sigma^4}{n-k} > \frac{2\sigma^4}{n} = \text{CRLB},$$

strictly. So, unlike $\hat{\beta}$ — which attains its bound $\sigma^2(X'X)^{-1}$ exactly — the unbiased variance estimator is *not* Cramér–Rao efficient.

There is no contradiction with the theory. The Cramér–Rao bound applies to unbiased estimators and is not always attainable; the σ^2 bound here is attained only by an estimator that uses the true β , namely $n^{-1} \sum_i e_i^2$, which is infeasible. The loss factor is $n/(n-k)$, negligible when k/n is small and material when the model is richly parameterized. (The MLE $\hat{\sigma}^2 = n^{-1} \sum_i \hat{e}_i^2$ has smaller variance, $2(n-k)\sigma^4/n^2$, but is biased; its mean squared error is smaller than that of s^2 — neither estimator dominates on all criteria.)

Chapter 7

Asymptotic Theory for Least Squares

Solutions to all 28 exercises of Chapter 7 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercise 7.28 is empirical; the reported numbers were computed from the `cps09mar` data set on the textbook website, and code is included.

Standing notation: observations (Y_i, X_i) are i.i.d., $e = Y - X'\beta$,

$$Q_{XX} = \mathbb{E}[XX'], \quad \Omega = \mathbb{E}[XX'e^2], \quad V_\beta = Q_{XX}^{-1}\Omega Q_{XX}^{-1},$$

and $\hat{\beta}$ denotes the OLS estimator. The two workhorses are the weak law of large numbers (WLLN, THEOREM 6.1), which gives $n^{-1} \sum_i g(Y_i, X_i) \xrightarrow{p} \mathbb{E}[g(Y, X)]$ whenever $\mathbb{E}\|g\| < \infty$, and the central limit theorem (CLT, THEOREM 6.3). Unless stated otherwise, we assume the moment conditions of THEOREM 7.3 ($Q_{XX} > 0$, $\mathbb{E}[Y^4] < \infty$, $\mathbb{E}\|X\|^4 < \infty$) so that $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V_\beta)$.

Exercise 7.1

Let $\tilde{\beta}_1$ be the short-regression estimator. In the book's summation notation,

$$\tilde{\beta}_1 = \left(\sum_{i=1}^n X_{1i} X'_{1i} \right)^{-1} \left(\sum_{i=1}^n X_{1i} Y_i \right) = \left(\frac{1}{n} \sum_{i=1}^n X_{1i} X'_{1i} \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_{1i} Y_i \right).$$

Substituting $Y_i = X'_{1i}\beta_1 + X'_{2i}\beta_2 + e_i$,

$$\tilde{\beta}_1 = \beta_1 + \left(\frac{1}{n} \sum_{i=1}^n X_{1i} X'_{1i} \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_{1i} X'_{2i} \right) \beta_2 + \left(\frac{1}{n} \sum_{i=1}^n X_{1i} X'_{1i} \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_{1i} e_i \right).$$

Apply the WLLN to each average, assuming $\mathbb{E}[X_1 X'_1] > 0$, $\mathbb{E}\|X_1 X'_2\| < \infty$ and $\mathbb{E}\|X_1 e\| < \infty$:

$$\frac{1}{n} \sum_{i=1}^n X_{1i} X'_{1i} \xrightarrow{p} \mathbb{E}[X_1 X'_1], \quad \frac{1}{n} \sum_{i=1}^n X_{1i} X'_{2i} \xrightarrow{p} \mathbb{E}[X_1 X'_2], \quad \frac{1}{n} \sum_{i=1}^n X_{1i} e_i \xrightarrow{p} \mathbb{E}[X_1 e] = \mathbf{0},$$

the last because $\mathbb{E}[Xe] = \mathbf{0}$ contains $\mathbb{E}[X_1 e] = \mathbf{0}$ as a sub-vector. By the continuous mapping theorem,

$$\boxed{\tilde{\beta}_1 \xrightarrow{p} \beta_1 + (\mathbb{E}[X_1 X'_1])^{-1} \mathbb{E}[X_1 X'_2] \beta_2 \equiv \beta_1 + \Gamma' \beta_2,}$$

where $\Gamma = (\mathbb{E}[X_1 X'_1])^{-1} \mathbb{E}[X_1 X'_2]$ is the projection coefficient of X_2 on X_1 .

Is it consistent? In general, *no*: the probability limit differs from β_1 by the omitted-variable bias term $(\mathbb{E}[X_1 X'_1])^{-1} \mathbb{E}[X_1 X'_2] \beta_2$. Consistency holds if and only if

$$\mathbb{E}[X_1 X'_2] \beta_2 = \mathbf{0}.$$

The two familiar sufficient conditions are $\beta_2 = \mathbf{0}$ (the omitted regressors are irrelevant) and $\mathbb{E}[X_1 X'_2] = \mathbf{0}$ (the omitted regressors are uncorrelated with the included ones). Note that the necessary and sufficient condition is weaker than either: what must vanish is the *product*, so it

suffices that β_2 lie in the null space of $\mathbb{E}[X_1 X_2']$. Compare Exercise 2.21, which is the scalar version of the same statement.

Exercise 7.2

Divide numerator and denominator by n so that both become sample averages:

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' + \frac{\lambda}{n} \mathbf{I}_k \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i Y_i \right).$$

Since λ is *fixed*, $\lambda/n \rightarrow 0$. By the WLLN,

$$\frac{1}{n} \sum_{i=1}^n X_i X_i' \xrightarrow{p} Q_{XX}, \quad \frac{1}{n} \sum_{i=1}^n X_i Y_i \xrightarrow{p} Q_{XY} = \mathbb{E}[XY],$$

so by the continuous mapping theorem (the inverse is continuous at the positive-definite Q_{XX}),

$$\hat{\beta} \xrightarrow{p} (Q_{XX} + \mathbf{0})^{-1} Q_{XY} = Q_{XX}^{-1} Q_{XY} = \beta.$$

Yes, $\hat{\beta}$ is consistent. The penalty is of fixed size λ while the information in the data grows like n , so the penalty is asymptotically negligible.

Compare Exercise 4.23, which shows that $\hat{\beta}$ is nevertheless *biased* in every finite sample, with bias $-\lambda(X'X + \lambda\mathbf{I}_k)^{-1}\beta$. The bias is $O_p(n^{-1})$ and therefore disappears faster than the $O_p(n^{-1/2})$ sampling error; ridge with fixed λ is even asymptotically equivalent to OLS, in the sense that $\sqrt{n}(\hat{\beta} - \beta)$ has the same limit distribution $N(0, V_\beta)$.

Exercise 7.3

Now $\lambda/n = c$ does not vanish:

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' + c \mathbf{I}_k \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i Y_i \right) \xrightarrow{p} (Q_{XX} + c \mathbf{I}_k)^{-1} Q_{XY}.$$

Writing $Q_{XY} = Q_{XX}\beta$,

$$\boxed{\hat{\beta} \xrightarrow{p} (Q_{XX} + c \mathbf{I}_k)^{-1} Q_{XX} \beta \neq \beta \quad \text{for } c > 0, \beta \neq \mathbf{0}.}$$

So $\hat{\beta}$ is *inconsistent*: the penalty grows at the same rate as the sample information and never washes out.

The limit is an explicit shrinkage of β toward the origin. Its structure is clearest in the eigen-basis of Q_{XX} : if $Q_{XX} = H\Lambda H'$ with $\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_k\}$, then the j th coordinate of β in that basis is multiplied by $\lambda_j/(\lambda_j + c) \in (0, 1)$. Directions in which the regressors carry little variation (small λ_j) are shrunk hardest. This is exactly the bias-variance trade-off that ridge is designed to exploit — but it shows that consistency requires $\lambda/n \rightarrow 0$, i.e. the penalty must grow more slowly than the sample.

Exercise 7.4

First tabulate the four cells:

(X_1, X_2)	probability	$X_1 X_2$	$\mathbb{E}[e^2 X_1, X_2]$
$(+1, +1)$	$3/8$	$+1$	$5/4$
$(-1, -1)$	$3/8$	$+1$	$5/4$
$(+1, -1)$	$1/8$	-1	$1/4$
$(-1, +1)$	$1/8$	-1	$1/4$

The marginals follow by summing: $\mathbb{P}[X_1 = 1] = \frac{3}{8} + \frac{1}{8} = \frac{1}{2} = \mathbb{P}[X_1 = -1]$, and likewise for X_2 ; $\mathbb{P}[X_1 = X_2] = \frac{3}{4}$ and $\mathbb{P}[X_1 \neq X_2] = \frac{1}{4}$.

(a) $\mathbb{E}[X_1] = (+1)\frac{1}{2} + (-1)\frac{1}{2} = 0$.

(b) $X_1^2 = 1$ with probability one, so $\mathbb{E}[X_1^2] = 1$. (Hence also $\text{var}(X_1) = 1$.)

(c) $X_1 X_2$ equals $+1$ on the first two cells and -1 on the last two:

$$\mathbb{E}[X_1 X_2] = (+1)\left(\frac{3}{8} + \frac{3}{8}\right) + (-1)\left(\frac{1}{8} + \frac{1}{8}\right) = \frac{6}{8} - \frac{2}{8} = \frac{1}{2}.$$

(So $\text{corr}(X_1, X_2) = \frac{1}{2}$: the regressors are positively correlated.)

(d) By the law of iterated expectations,

$$\mathbb{E}[e^2] = \frac{5}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{15}{16} + \frac{1}{16} = 1.$$

(e) Since $X_1^2 = 1$ identically, $\mathbb{E}[X_1^2 e^2] = \mathbb{E}[e^2] = 1$ by (d).

(f) Condition on (X_1, X_2) and use that $X_1 X_2 = +1$ exactly on the event $\{X_1 = X_2\}$:

$$\mathbb{E}[X_1 X_2 e^2] = (+1) \cdot \frac{5}{4} \cdot \frac{3}{4} + (-1) \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{15}{16} - \frac{1}{16} = \frac{14}{16} = \frac{7}{8}.$$

All six values check. The design is constructed so that Q_{XX} and Ω differ in an interesting way:

$$Q_{XX} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \quad \Omega = \begin{pmatrix} 1 & \frac{7}{8} \\ \frac{7}{8} & 1 \end{pmatrix}.$$

Because $\Omega \neq \sigma^2 Q_{XX}$ (here $\sigma^2 = 1$ but $\frac{7}{8} \neq \frac{1}{2}$), the errors are genuinely heteroskedastic, and the homoskedastic and robust covariance matrices differ — which is the point Section 7.4 uses this example to make.

Exercise 7.5

We must compute the blocks of $V_\beta = Q_{XX}^{-1} \Omega Q_{XX}^{-1}$ using the partitioned inverse (2.43),

$$Q_{XX}^{-1} = \begin{bmatrix} Q_{11.2}^{-1} & -Q_{11.2}^{-1} Q_{12} Q_{22}^{-1} \\ -Q_{22.1}^{-1} Q_{21} Q_{11}^{-1} & Q_{22.1}^{-1} \end{bmatrix},$$

where $Q_{11.2} = Q_{11} - Q_{12} Q_{22}^{-1} Q_{21}$ and $Q_{22.1} = Q_{22} - Q_{21} Q_{11}^{-1} Q_{12}$.

The calculation is cleanest if we name the two block rows of Q_{XX}^{-1} :

$$A = Q_{11.2}^{-1} \begin{bmatrix} \mathbf{I} & -Q_{12} Q_{22}^{-1} \end{bmatrix}, \quad B = Q_{22.1}^{-1} \begin{bmatrix} -Q_{21} Q_{11}^{-1} & \mathbf{I} \end{bmatrix},$$

so that $Q_{XX}^{-1} = \begin{bmatrix} A \\ B \end{bmatrix}$ and, since Q_{XX}^{-1} is symmetric, $Q_{XX}^{-1} = \begin{bmatrix} A' & B' \end{bmatrix}$. Therefore

$$V_\beta = \begin{bmatrix} A \\ B \end{bmatrix} \Omega \begin{bmatrix} A' & B' \end{bmatrix} = \begin{bmatrix} A\Omega A' & A\Omega B' \\ B\Omega A' & B\Omega B' \end{bmatrix} = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix},$$

which is (7.13). It remains to multiply out each block.

(7.14).

$$\begin{aligned} V_{11} &= A\Omega A' = Q_{11.2}^{-1} \begin{bmatrix} \mathbf{I} & -Q_{12}Q_{22}^{-1} \end{bmatrix} \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ -Q_{22}^{-1}Q_{21} \end{bmatrix} Q_{11.2}^{-1} \\ &= Q_{11.2}^{-1} (\Omega_{11} - Q_{12}Q_{22}^{-1}\Omega_{21} - \Omega_{12}Q_{22}^{-1}Q_{21} + Q_{12}Q_{22}^{-1}\Omega_{22}Q_{22}^{-1}Q_{21}) Q_{11.2}^{-1}. \end{aligned}$$

(7.15).

$$\begin{aligned} V_{21} &= B\Omega A' = Q_{22.1}^{-1} \begin{bmatrix} -Q_{21}Q_{11}^{-1} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ -Q_{22}^{-1}Q_{21} \end{bmatrix} Q_{11.2}^{-1} \\ &= Q_{22.1}^{-1} (\Omega_{21} - Q_{21}Q_{11}^{-1}\Omega_{11} - \Omega_{22}Q_{22}^{-1}Q_{21} + Q_{21}Q_{11}^{-1}\Omega_{12}Q_{22}^{-1}Q_{21}) Q_{11.2}^{-1}. \end{aligned}$$

(7.16).

$$\begin{aligned} V_{22} &= B\Omega B' = Q_{22.1}^{-1} \begin{bmatrix} -Q_{21}Q_{11}^{-1} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{bmatrix} \begin{bmatrix} -Q_{11}^{-1}Q_{12} \\ \mathbf{I} \end{bmatrix} Q_{22.1}^{-1} \\ &= Q_{22.1}^{-1} (\Omega_{22} - Q_{21}Q_{11}^{-1}\Omega_{12} - \Omega_{21}Q_{11}^{-1}Q_{12} + Q_{21}Q_{11}^{-1}\Omega_{11}Q_{11}^{-1}Q_{12}) Q_{22.1}^{-1}. \end{aligned}$$

(And $V_{12} = V_{21}'$, since V_β is symmetric.)

As the text says, these expressions are not easy to interpret. The one case that simplifies is homoskedasticity, $\Omega = \sigma^2 Q_{XX}$: substituting $\Omega_{j\ell} = \sigma^2 Q_{j\ell}$ into (7.14) collapses it to $V_{11} = \sigma^2 Q_{11.2}^{-1}$, the familiar partitioned formula.

Exercise 7.6

The method of moments replaces every population expectation by the corresponding sample average, and every unobservable by its fitted counterpart.

The coefficient. From $\mathbb{E}[Xe] = \mathbf{0}$ we get the explicit solution $\beta = (\mathbb{E}[XX'])^{-1}\mathbb{E}[XY]$, and the moment estimator is

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i Y_i \right) = (X'X)^{-1} X'Y,$$

which is simply OLS: least squares *is* the method of moments estimator for the projection model.

The covariance matrix of the score. $\Omega = \mathbb{E}[XX'e^2]$ is an expectation of a function of the observables and the *unobservable* error e . The sample analogue of the expectation is an average;

the analogue of e_i is the residual $\hat{e}_i = Y_i - X_i' \hat{\beta}$. Hence

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n X_i X_i' \hat{e}_i^2, \quad \hat{e}_i = Y_i - X_i' \hat{\beta}.$$

Remark. The substitution of $\hat{e}_i$ for e_i is essential — the errors are not observed, so $n^{-1} \sum_i X_i X_i' e_i^2$ is not an estimator at all. Exercise 7.11 shows that the substitution is asymptotically harmless: $\hat{\Omega}$ and the infeasible $n^{-1} \sum_i X_i X_i' e_i^2$ have the same limit distribution, and the proof is where the regression assumption $\mathbb{E}[e | X] = 0$ does real work. Plugging $\hat{\Omega}$ into $\hat{V}_\beta = \hat{Q}_{XX}^{-1} \hat{\Omega} \hat{Q}_{XX}^{-1}$ recovers the HC0 covariance matrix estimator of Section 4.14.

Exercise 7.7

Substituting, the observable equation is

$$Y = Y^* + u = X' \beta + \underbrace{e + u}_{\equiv v}.$$

(a) **Yes.** The projection coefficient of Y on X is the unique b with $\mathbb{E}[X(Y - X'b)] = \mathbf{0}$, and

$$\mathbb{E}[Xv] = \mathbb{E}[Xe] + \mathbb{E}[Xu] = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

So $b = \beta$: the measurement error does not change the projection coefficient. This is the benign case of measurement error — it is in the *dependent* variable. (Contrast Exercise 7.24, where the error is in a regressor and the coefficient is attenuated.)

(b) **Yes.** With $Q_{XX} > 0$ and the relevant first moments finite, the WLLN and the continuous mapping theorem give

$$\hat{\beta} = \left(\frac{1}{n} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_i X_i Y_i \right) \xrightarrow{p} Q_{XX}^{-1} \mathbb{E}[XY] = Q_{XX}^{-1} Q_{XX} \beta = \beta.$$

(c) This is the standard projection-model CLT applied to the equation $Y = X' \beta + v$. Writing $\sqrt{n}(\hat{\beta} - \beta) = (n^{-1} \sum_i X_i X_i')^{-1} (n^{-1/2} \sum_i X_i v_i)$ and applying the CLT to the i.i.d. mean-zero vectors $X_i v_i$,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, Q_{XX}^{-1} \Omega_v Q_{XX}^{-1}), \quad \Omega_v = \mathbb{E}[X X' v^2] = \mathbb{E}[X X' (e + u)^2],$$

requiring $\mathbb{E}\|X\|^4 < \infty$ and $\mathbb{E}[v^4] < \infty$. Expanding,

$$\Omega_v = \mathbb{E}[X X' e^2] + 2\mathbb{E}[X X' e u] + \mathbb{E}[X X' u^2].$$

The assumption $\mathbb{E}[Y^* u] = 0$ does yield $\mathbb{E}[e u] = 0$: indeed $0 = \mathbb{E}[Y^* u] = \beta' \mathbb{E}[X u] + \mathbb{E}[e u] = \mathbb{E}[e u]$. But $\mathbb{E}[e u] = 0$ does *not* imply $\mathbb{E}[X X' e u] = \mathbf{0}$, so in general the cross term survives. Under the stronger (and usual) assumption that u is conditionally uncorrelated with e given X , i.e. $\mathbb{E}[e u | X] = 0$, the middle term vanishes and

$$\Omega_v = \mathbb{E}[X X' e^2] + \mathbb{E}[X X' u^2] \geq \mathbb{E}[X X' e^2].$$

Measurement error in Y therefore costs precision but not consistency — the asymptotic variance is inflated by exactly the variance contribution of u . This is the asymptotic counterpart of Exercise 4.16.

Exercise 7.8

The strategy is the one used throughout Section 7.5: show that replacing the unobserved errors by residuals is asymptotically negligible, then apply the CLT to the infeasible statistic.

Step 1: expand the residual. Since $\hat{e}_i = e_i - X_i'(\hat{\beta} - \beta)$,

$$\hat{e}_i^2 = e_i^2 - 2e_i X_i'(\hat{\beta} - \beta) + (\hat{\beta} - \beta)' X_i X_i' (\hat{\beta} - \beta).$$

Averaging and multiplying by $\sqrt{n}$,

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) = T_1 + T_2 + T_3,$$

where, writing $\hat{Q}_{XX} = n^{-1} \sum_i X_i X_i'$,

$$\begin{aligned} T_1 &= \frac{1}{\sqrt{n}} \sum_{i=1}^n (e_i^2 - \sigma^2), \\ T_2 &= -2\sqrt{n}(\hat{\beta} - \beta)' \left(\frac{1}{n} \sum_{i=1}^n X_i e_i \right), \\ T_3 &= \frac{1}{\sqrt{n}} \left[\sqrt{n}(\hat{\beta} - \beta) \right]' \hat{Q}_{XX} \left[\sqrt{n}(\hat{\beta} - \beta) \right]. \end{aligned}$$

Step 2: the two remainder terms are $o_p(1)$.

- $\sqrt{n}(\hat{\beta} - \beta) = O_p(1)$ by THEOREM 7.3, and $n^{-1} \sum_i X_i e_i \xrightarrow{p} \mathbb{E}[Xe] = \mathbf{0}$ by the WLLN, so $T_2 = O_p(1) \cdot o_p(1) = o_p(1)$.
- $\hat{Q}_{XX} \xrightarrow{p} Q_{XX} = O_p(1)$, so the bracketed quadratic form is $O_p(1)$ and $T_3 = n^{-1/2} O_p(1) = o_p(1)$.

Step 3: CLT for the leading term. The variables e_i^2 are i.i.d. with mean $\mathbb{E}[e^2] = \sigma^2$ and variance $\text{var}[e^2] = \mathbb{E}[e^4] - \sigma^4$. Provided $\mathbb{E}[e^4] < \infty$, the CLT gives

$$T_1 \xrightarrow{d} N(0, \mathbb{E}[e^4] - \sigma^4).$$

Conclusion. By Slutsky's theorem,

$$\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, V_{\sigma^2}), \quad V_{\sigma^2} = \mathbb{E}[e^4] - \sigma^4 = \text{var}[e^2].$$

Required moments: $\mathbb{E}[e^4] < \infty$, $\mathbb{E}\|X\|^2 < \infty$ and $Q_{XX} > 0$.

Three comments. First, the limit is exactly the one that would obtain if the errors e_i were *observed*: estimating β has no first-order cost for σ^2 . Second, the same limit holds for $s^2 = (n - k)^{-1} \sum_i \hat{e}_i^2$, because $\sqrt{n}(s^2 - \hat{\sigma}^2) = \frac{k}{n-k} \sqrt{n} \hat{\sigma}^2 = O_p(n^{-1/2}) = o_p(1)$: the degrees-of-freedom correction is asymptotically irrelevant. Third, a natural standard error is $\hat{V}_{\sigma^2} = n^{-1} \sum_i \hat{e}_i^4 - \hat{\sigma}^4$, so that an asymptotic 95% interval for σ^2 is $\hat{\sigma}^2 \pm 1.96 \sqrt{\hat{V}_{\sigma^2}/n}$. In the normal model $\mathbb{E}[e^4] = 3\sigma^4$ and $V_{\sigma^2} = 2\sigma^4$, matching the exact finite-sample result of Exercise 5.12.

Exercise 7.9

(a) Both are consistent, but they need different side conditions.

For $\hat{\beta}$: writing $\hat{\beta} = \beta + (n^{-1} \sum_i X_i^2)^{-1} (n^{-1} \sum_i X_i e_i)$, the WLLN gives $n^{-1} \sum_i X_i^2 \xrightarrow{p} \mathbb{E}[X^2]$ and $n^{-1} \sum_i X_i e_i \xrightarrow{p} \mathbb{E}[Xe] = \mathbb{E}[X \mathbb{E}[e | X]] = 0$, so $\hat{\beta} \xrightarrow{p} \beta$ provided $0 < \mathbb{E}[X^2] < \infty$.

For $\tilde{\beta}$: substituting $Y_i = X_i \beta + e_i$,

$$\tilde{\beta} = \beta + \frac{1}{n} \sum_{i=1}^n \frac{e_i}{X_i} \xrightarrow{p} \beta + \mathbb{E}\left[\frac{e}{X}\right] = \beta + \mathbb{E}\left[\frac{\mathbb{E}[e | X]}{X}\right] = \beta,$$

using the conditioning theorem. Here the side condition is stronger: we need $\mathbb{P}[X = 0] = 0$ and $\mathbb{E}|e/X| < \infty$ for the WLLN to apply. If X has mass near zero the ratio e/X may have no mean and $\tilde{\beta}$ can fail to converge. Note also that only $\hat{\beta}$ survives under the weaker projection assumption $\mathbb{E}[Xe] = 0$; $\tilde{\beta}$ genuinely needs $\mathbb{E}[e | X] = 0$.

(b) Both are (infeasible) GLS estimators for particular variance functions, and each is efficient exactly in its own case. Their asymptotic variances are

$$\text{avar}(\hat{\beta}) = \frac{\mathbb{E}[X^2 e^2]}{(\mathbb{E}[X^2])^2}, \quad \text{avar}(\tilde{\beta}) = \mathbb{E}\left[\frac{e^2}{X^2}\right].$$

- *Homoskedasticity*, $\mathbb{E}[e^2 | X] = \sigma^2$. Then $\text{avar}(\hat{\beta}) = \sigma^2 / \mathbb{E}[X^2]$ and $\text{avar}(\tilde{\beta}) = \sigma^2 \mathbb{E}[X^{-2}]$. By the Cauchy–Schwarz inequality $\mathbb{E}[X^2] \mathbb{E}[X^{-2}] \geq (\mathbb{E}[|X| \cdot |X|^{-1}])^2 = 1$, so $\text{avar}(\tilde{\beta}) \geq \text{avar}(\hat{\beta})$, with equality only if X^2 is almost surely constant. Here $\hat{\beta}$ is the efficient (Gauss–Markov) estimator and $\tilde{\beta}$ is strictly worse.
- *Multiplicative heteroskedasticity*, $\mathbb{E}[e^2 | X] = \sigma^2 X^2$. The GLS estimator weights observation i by $\sigma_i^{-2} \propto X_i^{-2}$:

$$\tilde{\beta}_{\text{glS}} = \frac{\sum_i X_i^{-2} X_i Y_i}{\sum_i X_i^{-2} X_i^2} = \frac{\sum_i Y_i / X_i}{n} = \tilde{\beta}.$$

So $\tilde{\beta}$ is the GLS estimator in this case and attains the efficiency bound (Exercise 4.6), while $\hat{\beta}$ does not. Compare Exercise 4.8, which is the same idea with weights X_j^{-2} .

Neither estimator dominates: which is efficient depends entirely on the shape of $\sigma^2(x)$.

Exercise 7.10

(a) The best point predictor of Y_{n+1} given $X_{n+1} = x$ is the CEF $\mathbb{E}[Y_{n+1} | X_{n+1} = x] = x' \beta$ (Exercise 2.5 and THEOREM 2.10), so the natural feasible forecast is the plug-in

$$\tilde{Y}_{n+1} = x' \hat{\beta}.$$

(b) The forecast error decomposes into two independent sources:

$$\tilde{e}_{n+1} = Y_{n+1} - x' \hat{\beta} = \underbrace{e_{n+1}}_{\text{new draw}} - \underbrace{x'(\hat{\beta} - \beta)}_{\text{estimation error}}.$$

Observation $n + 1$ is independent of the estimation sample, so e_{n+1} is independent of $\hat{\beta}$ and the cross term has mean zero. Hence

$$\text{MSFE} = \mathbb{E}[\tilde{e}_{n+1}^2] = \sigma^2 + x'V_{\hat{\beta}}x, \quad V_{\hat{\beta}} = \text{var}[\hat{\beta}],$$

which is THEOREM 4.6. Replacing each unknown by its estimator,

$$\widehat{\text{MSFE}} = \hat{\sigma}^2 + x'\hat{V}_{\hat{\beta}}x, \quad s(\tilde{e}_{n+1}) = \sqrt{\hat{\sigma}^2 + x'\hat{V}_{\hat{\beta}}x}.$$

Two practical remarks. First, the two components have very different magnitudes: $\hat{\sigma}^2 = O(1)$ while $x'\hat{V}_{\hat{\beta}}x = O(n^{-1})$. In large samples the forecast interval is dominated by the irreducible error variance, and the width barely shrinks with n — unlike a confidence interval for the regression function $x'\beta$, whose standard error is $\sqrt{x'V_{\beta}x}$ alone and does shrink at rate $n^{-1/2}$. Confusing the two is a common error; Exercise 7.28(e)–(f) computes both for the same data. Second, forming the interval $x'\hat{\beta} \pm z_{1-\alpha/2} s(\tilde{e}_{n+1})$ requires a distributional assumption on e_{n+1} (normality, or an appeal to approximate normality); the moment conditions alone deliver the variance but not the quantiles. In the exactly normal model $\hat{\sigma}^2$ should be replaced by s^2 and z by a t_{n-k} quantile, and the interval is exact.

Exercise 7.11

(a) $\tilde{\Omega}$ is a sample average of the i.i.d. scalars $X_i^2 e_i^2$, which have mean Ω . The CLT applies directly:

$$\sqrt{n}(\tilde{\Omega} - \Omega) \xrightarrow{d} N(0, V_{\Omega}), \quad V_{\Omega} = \text{var}[X^2 e^2] = \mathbb{E}[X^4 e^4] - \Omega^2,$$

requiring $\mathbb{E}[X^4 e^4] < \infty$. This is infeasible — it uses the unobserved e_i — but it is the benchmark.

(b) Expand $\hat{e}_i^2 = e_i^2 - 2e_i X_i(\hat{\beta} - \beta) + X_i^2(\hat{\beta} - \beta)^2$ and difference:

$$\begin{aligned} \sqrt{n}(\hat{\Omega} - \tilde{\Omega}) &= -2 \underbrace{\sqrt{n}(\hat{\beta} - \beta)}_{O_p(1)} \underbrace{\left(\frac{1}{n} \sum_{i=1}^n X_i^3 e_i \right)}_{\xrightarrow{p} \mathbb{E}[X^3 e]} + \underbrace{\frac{1}{\sqrt{n}} \left[\sqrt{n}(\hat{\beta} - \beta) \right]^2 \left(\frac{1}{n} \sum_{i=1}^n X_i^4 \right)}_{O_p(n^{-1/2})}. \end{aligned}$$

The second term is $o_p(1)$ for free. The first term is $o_p(1)$ if and only if $\mathbb{E}[X^3 e] = 0$ — and this is exactly where the regression assumption enters (see part (c)). Granting it,

$$\sqrt{n}(\hat{\Omega} - \Omega) = \sqrt{n}(\tilde{\Omega} - \Omega) + o_p(1) \xrightarrow{d} N(0, V_{\Omega}),$$

the *same* limit as in (a): estimating β is asymptotically costless for Ω . Moment requirements: $\mathbb{E}[X^4 e^4] < \infty$ and $\mathbb{E}[X^6] < \infty$ (so that $\mathbb{E}[X^3 e]$ and $\mathbb{E}[X^4]$ exist).

(c) The assumption $\mathbb{E}[e_i | X_i] = 0$ is used to kill the leading correction term:

$$\mathbb{E}[X^3 e] = \mathbb{E}[X^3 \mathbb{E}[e | X]] = 0,$$

by the conditioning theorem (Exercise 2.3). Without it, the linear-in- $\hat{\beta}$ term is $O_p(1)$ and does *not* vanish; it is asymptotically correlated with $\sqrt{n}(\hat{\beta} - \beta)$, so $\hat{\Omega}$ and $\tilde{\Omega}$ then have different limiting distributions and the asymptotic variance of $\hat{\Omega}$ acquires extra terms involving $\mathbb{E}[X^3 e]$ and V_{β} .

Under the mere projection condition $\mathbb{E}[Xe] = 0$ there is no reason for $\mathbb{E}[X^3e]$ to vanish — Exercise 2.11 gives an explicit example with $\mathbb{E}[Xe] = 0$ but $\mathbb{E}[X^2e] \neq 0$. So the familiar claim “replacing e_i by $\hat{e}_i$ does not matter asymptotically” is a theorem about the regression model, not a free lunch.

Exercise 7.12

(a) $A = g(\theta)$ with $g(a, b) = -a^2/(2b)$, a function that is continuously differentiable on the region $\{b < 0\}$ where the parameter is defined. The plug-in estimator is

$$\hat{A} = -\frac{\hat{\alpha}^2}{2\hat{\beta}}.$$

It is consistent by the continuous mapping theorem, since $\hat{\theta} \xrightarrow{p} \theta$ and g is continuous at θ (using $\beta < 0$, so $\hat{\beta} < 0$ with probability approaching one).

(b) Apply the delta method (THEOREM 7.9). The gradient is

$$R = \frac{\partial}{\partial \theta} g(\theta) = \begin{pmatrix} \frac{\partial A}{\partial \alpha} \\ \frac{\partial A}{\partial \beta} \end{pmatrix} = \begin{pmatrix} -\frac{\alpha}{\beta} \\ \frac{\alpha^2}{2\beta^2} \end{pmatrix},$$

so

$$\sqrt{n}(\hat{A} - A) \xrightarrow{d} N(0, V_A), \quad V_A = R'V_\theta R = \frac{\alpha^2}{\beta^2}V_{11} - \frac{\alpha^3}{\beta^3}V_{12} + \frac{\alpha^4}{4\beta^4}V_{22}.$$

Replacing R and V_θ by $\hat{R}$ (the gradient evaluated at $\hat{\theta}$) and $\hat{V}_\theta$, the standard error is

$$s(\hat{A}) = \sqrt{\frac{\hat{R}'\hat{V}_\theta\hat{R}}{n}},$$

and the asymptotic $1 - \eta$ confidence interval is

$$\hat{A} \pm z_{1-\eta/2} s(\hat{A}) = -\frac{\hat{\alpha}^2}{2\hat{\beta}} \pm z_{1-\eta/2} \sqrt{\frac{\hat{R}'\hat{V}_\theta\hat{R}}{n}},$$

where $z_{1-\eta/2}$ is the $(1 - \eta/2)$ quantile of the standard normal (e.g. 1.96 for $\eta = 0.05$).

Remark. Because A is a strongly nonlinear function of θ — with β in the denominator — the delta-method interval can behave poorly when $\hat{\beta}$ is imprecisely estimated relative to its distance from zero. Exercise 5.9 suggests a better route in such cases: build the interval on a scale where normality is more accurate (here, $\log A = 2 \log \alpha - \log(-2\beta)$) and transform the endpoints back, which preserves coverage exactly and guarantees $\hat{A} > 0$.

Exercise 7.13

(a) The projection condition $\mathbb{E}[Y(X - Y\gamma)] = 0$ gives $\gamma = \mathbb{E}[YX] / \mathbb{E}[Y^2]$, so the moment estimator is OLS of X on Y :

$$\hat{\gamma} = \frac{\sum_{i=1}^n Y_i X_i}{\sum_{i=1}^n Y_i^2}.$$

(b) The plug-in estimator, $\hat{\theta} = 1/\hat{\gamma}$. This requires $\gamma \neq 0$; otherwise θ is not defined.

(c) First, the standard projection-model CLT applied to $\hat{\gamma} = \gamma + (n^{-1} \sum_i Y_i^2)^{-1} (n^{-1} \sum_i Y_i u_i)$ gives

$$\sqrt{n}(\hat{\gamma} - \gamma) \xrightarrow{d} N(0, V_\gamma), \quad V_\gamma = \frac{\mathbb{E}[Y^2 u^2]}{(\mathbb{E}[Y^2])^2},$$

assuming $0 < \mathbb{E}[Y^2] < \infty$ and $\mathbb{E}[Y^2 u^2] < \infty$. Now apply the delta method with $g(c) = 1/c$, $g'(c) = -1/c^2$, which is continuously differentiable at $\gamma \neq 0$:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{V_\gamma}{\gamma^4}\right) = N(0, \theta^4 V_\gamma).$$

(d) With $\hat{u}_i = X_i - Y_i \hat{\gamma}$, a consistent estimator of V_γ is

$$\hat{V}_\gamma = \frac{n^{-1} \sum_{i=1}^n Y_i^2 \hat{u}_i^2}{(n^{-1} \sum_{i=1}^n Y_i^2)^2},$$

so

$$s(\hat{\theta}) = \sqrt{\frac{\hat{V}_\gamma}{n \hat{\gamma}^4}} = \hat{\theta}^2 s(\hat{\gamma}), \quad s(\hat{\gamma}) = \sqrt{\hat{V}_\gamma / n}.$$

An asymptotic 95% interval is $\hat{\theta} \pm 1.96 s(\hat{\theta})$.

Remark. This is the textbook setting in which the delta method is fragile. If γ is close to zero, $\hat{\gamma}$ can be near zero (or of the wrong sign) with non-negligible probability and $\hat{\theta}$ then explodes: its finite-sample distribution is heavy-tailed and badly approximated by the normal. The robust alternative is to invert a test on the γ scale — take a confidence interval $[L, U]$ for γ and map it to $[1/U, 1/L]$ when $0 \notin [L, U]$, which is Exercise 5.9 applied to the decreasing function $c \mapsto 1/c$.

Exercise 7.14

(a) The plug-in estimator built from the OLS coefficients, $\hat{\theta} = \hat{\beta}_1 \hat{\beta}_2$.

(b) Let $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V_\beta)$ with $V_\beta = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}$. The map $g(b) = b_1 b_2$ has gradient

$$R = \frac{\partial}{\partial \beta}(\beta_1 \beta_2) = \begin{pmatrix} \beta_2 \\ \beta_1 \end{pmatrix},$$

so the delta method gives

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V_\theta), \quad V_\theta = R' V_\beta R = \beta_2^2 V_{11} + 2\beta_1 \beta_2 V_{12} + \beta_1^2 V_{22}.$$

(c) Estimate V_θ by $\hat{V}_\theta = \hat{R}' \hat{V}_\beta \hat{R}$ with $\hat{R} = (\hat{\beta}_2, \hat{\beta}_1)'$ and $\hat{V}_\beta$ any of the robust covariance matrix estimators of Section 4.14. In terms of the reported standard errors $s_j = \sqrt{\hat{V}_{jj}/n}$ and the estimated correlation $\hat{\rho}$ between $\hat{\beta}_1$ and $\hat{\beta}_2$,

$$s(\hat{\theta}) = \sqrt{\hat{\beta}_2^2 s_1^2 + 2\hat{\beta}_1 \hat{\beta}_2 \hat{\rho} s_1 s_2 + \hat{\beta}_1^2 s_2^2},$$

and the interval is

$$\hat{\theta} \pm 1.96 s(\hat{\theta}).$$

Note that the off-diagonal term is essential: it cannot be recovered from the two reported standard errors alone (this is exactly the complaint of Exercise 7.17).

Remark. A degenerate case worth flagging: if $\beta_1 = \beta_2 = 0$ then $R = \mathbf{0}$, the delta method gives $V_\theta = 0$, and the first-order expansion is uninformative. There $n\hat{\theta}$ converges to a product of correlated normals rather than $\sqrt{n}\hat{\theta}$ converging to a normal, and the interval above is invalid. Exercise 7.22(b) is a structured instance of the same phenomenon.

Exercise 7.15

This is the moment estimator based on the moment condition $\mathbb{E}[X^3 e] = 0$, which holds because $\mathbb{E}[e | X] = 0$ (Exercise 2.3). Substituting $Y_i = X_i \beta + e_i$,

$$\hat{\beta} - \beta = \frac{\sum_{i=1}^n X_i^3 e_i}{\sum_{i=1}^n X_i^4}, \quad \text{so} \quad \sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n X_i^4 \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i^3 e_i \right).$$

Handle the two factors separately.

- By the WLLN, $n^{-1} \sum_i X_i^4 \xrightarrow{p} \mathbb{E}[X^4]$, provided $\mathbb{E}[X^4] < \infty$; we assume $\mathbb{E}[X^4] > 0$, which holds unless $X = 0$ almost surely.
- The summands $X_i^3 e_i$ are i.i.d. with mean $\mathbb{E}[X^3 e] = 0$ and variance $\mathbb{E}[X^6 e^2]$, so the CLT gives $n^{-1/2} \sum_i X_i^3 e_i \xrightarrow{d} N(0, \mathbb{E}[X^6 e^2])$, provided $\mathbb{E}[X^6 e^2] < \infty$.

By Slutsky's theorem,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N\left(0, \frac{\mathbb{E}[X^6 e^2]}{(\mathbb{E}[X^4])^2}\right).$$

Under homoskedasticity this is $\sigma^2 \mathbb{E}[X^6] / (\mathbb{E}[X^4])^2$, which by Cauchy–Schwarz $((\mathbb{E}[X^4])^2 = (\mathbb{E}[X \cdot X^3])^2 \leq \mathbb{E}[X^2] \mathbb{E}[X^6])$ is at least $\sigma^2 / \mathbb{E}[X^2]$, the OLS variance. So this estimator is consistent and asymptotically normal but strictly less efficient than OLS unless X^2 is degenerate. It is one of infinitely many valid moment estimators generated by the conditional moment restriction $\mathbb{E}[e | X] = 0$.

Exercise 7.16

Yes.

Let $S_i \in \{0, 1\}$ indicate selection. “Randomly” means the selection is made independently of the data: S_i is independent of (Y_i, X_i) . Two consequences follow.

- The selected observations are themselves an i.i.d. sample from the *same* population F . Conditioning on $S_i = 1$ does not tilt the distribution of (Y_i, X_i) , since $\mathbb{P}[(Y, X) \in \mathcal{A} | S = 1] = \mathbb{P}[(Y, X) \in \mathcal{A}]$.
- The sub-sample size $m = n/2 \rightarrow \infty$ as $n \rightarrow \infty$.

Therefore the WLLN applies to the sub-sample averages,

$$\frac{1}{m} \sum_{i:S_i=1} X_i X_i' \xrightarrow{p} Q_{XX}, \quad \frac{1}{m} \sum_{i:S_i=1} X_i Y_i \xrightarrow{p} Q_{XY},$$

and the continuous mapping theorem gives $\hat{\beta} \xrightarrow{p} Q_{XX}^{-1} Q_{XY} = \beta$ — the same β as the full-sample estimator.

Equivalently and more directly: the moment condition survives selection, because

$$\mathbb{E}[S X e] = \mathbb{E}[S] \mathbb{E}[X e] = \frac{1}{2} \cdot \mathbf{0} = \mathbf{0},$$

the factorization using the independence of S from (X, e) .

What is lost is precision, not consistency: the asymptotic variance is $V_\beta/m = 2V_\beta/n$, so standard errors are $\sqrt{2} \approx 1.41$ times larger than the full-sample ones.

Remark. The independence of S from the data is doing all the work. If the sample were split on a variable related to the outcome — say, keeping observations with Y_i above its median, or with $|X_i| \leq c$ — the sub-sample would be drawn from a *different* distribution and the probability limit would generally change. Exercise 7.27 examines exactly such a case and shows that consistency survives there only because the selection depends on X alone and $\mathbb{E}[e | X] = 0$ is assumed.

Exercise 7.17

(a) $\theta = R'\beta$ with $R = (1, -1)'$, so $\text{var}(\hat{\beta}_1 - \hat{\beta}_2) = \text{var}(\hat{\beta}_1) + \text{var}(\hat{\beta}_2) - 2\text{cov}(\hat{\beta}_1, \hat{\beta}_2)$, and

$$(\hat{\beta}_1 - \hat{\beta}_2) \pm 1.96 \sqrt{s(\hat{\beta}_1)^2 + s(\hat{\beta}_2)^2 - 2\hat{\rho}s(\hat{\beta}_1)s(\hat{\beta}_2)}.$$

(b) **No.** $\hat{\rho}$ is the off-diagonal element of the estimated covariance matrix $\hat{V}_{\hat{\beta}}$, normalized by the two standard errors. A table of coefficients and standard errors reports only the diagonal of $\hat{V}_{\hat{\beta}}$, and the off-diagonal element cannot be recovered from it — it depends on the sample cross-moments of the regressors and is unrelated to the diagonal entries. Reporting $\hat{\rho}$ (or the full covariance matrix, or simply the standard error of the difference) is the author's responsibility.

(c) *The reported information does not settle the question.* We have $\hat{\theta} = 0.2$, and

$$s(\hat{\theta}) = \sqrt{2(0.07)^2(1 - \hat{\rho})} = 0.09899\sqrt{1 - \hat{\rho}},$$

which ranges over $[0, 0.14]$ as $\hat{\rho}$ ranges over $[-1, 1]$. The corresponding t statistic $t = 0.2/s(\hat{\theta})$ is therefore anywhere in $[1.43, \infty)$:

$\hat{\rho}$	-1	0	0.5
$s(\hat{\theta})$	0.1400	0.0990	0.0700
t	1.43	2.02	2.86
conclusion at 5%	not significant	significant	significant

So the answer genuinely depends on the unreported $\hat{\rho}$, and the author's claim is not established by what was reported.

A second, more common error is worth naming even though the exercise does not ask for it: one cannot conclude that two coefficients differ by observing that their individual confidence intervals $[0.86, 1.14]$ and $[0.66, 0.94]$ do not overlap. Non-overlap is neither necessary nor sufficient for a significant difference; the correct procedure is to form a single interval for θ as in part (a), or

equivalently to re-run the regression in the reparameterized form of Exercise 3.23 so that the package reports $s(\hat{\theta})$ directly.

Exercise 7.18

(a) The model is linear *in the parameters*, so it is an ordinary regression with the constructed regressor vector

$$X_i^* = (1, X_i, X_i^2)'$$

Run OLS of Y_i on X_i^* to obtain $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)'$, and estimate the regression function by

$$\hat{m}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 = a(x)' \hat{\beta}, \quad a(x) = (1, x, x^2)'$$

Note that $a(x)$ is a *fixed*, non-random vector once x is chosen, so $\hat{m}(x)$ is simply a linear combination of the OLS coefficients.

(b) Because $\hat{m}(x) = a(x)' \hat{\beta}$ is linear in $\hat{\beta}$, no delta method is needed — the transformation is exact:

$$\sqrt{n}(\hat{m}(x) - m(x)) \xrightarrow{d} N(0, a(x)' V_{\beta} a(x)).$$

With $\hat{V}_{\hat{\beta}}$ a heteroskedasticity-robust estimator of $\text{var}[\hat{\beta}]$ (HC1, HC2 or HC3 from Section 4.14), the standard error and interval are

$$s(\hat{m}(x)) = \sqrt{a(x)' \hat{V}_{\hat{\beta}} a(x)}, \quad \hat{m}(x) \pm 1.96 s(\hat{m}(x)) \text{ for 95\%}.$$

Explicitly, with $\hat{V}_{j\ell}$ the entries of $\hat{V}_{\hat{\beta}}$,

$$s(\hat{m}(x))^2 = \hat{V}_{00} + x^2 \hat{V}_{11} + x^4 \hat{V}_{22} + 2x \hat{V}_{01} + 2x^2 \hat{V}_{02} + 2x^3 \hat{V}_{12}.$$

Four points of practical care:

- Robust standard errors are essential here. Even if the CEF is exactly quadratic, nothing guarantees $\mathbb{E}[e^2 | X]$ is constant.
- The interval is *pointwise*: it covers $m(x)$ at the single chosen x with probability 0.95, not the whole curve simultaneously. A uniform band requires a wider (e.g. Scheffé) critical value.
- $s(\hat{m}(x))$ is smallest near the centre of the data and grows rapidly as x moves into the tails, where x^4 multiplies $\hat{V}_{22}$. Extrapolation beyond the support of X is not supported by the interval's asymptotics.
- This is an interval for the regression *function*, not a forecast interval for a new Y ; the latter adds $\hat{\sigma}^2$ inside the square root (Exercise 7.10).

Exercise 7.19

By the argument of Exercise 7.16, each half is an i.i.d. sample of size n from the same population F , so each estimator satisfies the usual CLT:

$$\sqrt{n}(\hat{\beta}_1 - \beta) \xrightarrow{d} N(0, V_{\beta}), \quad \sqrt{n}(\hat{\beta}_2 - \beta) \xrightarrow{d} N(0, V_{\beta}), \quad V_{\beta} = Q_{XX}^{-1} \Omega Q_{XX}^{-1}.$$

Crucially, the two sub-samples share no observations and the split is random, so $\hat{\beta}_1$ and $\hat{\beta}_2$ are *independent*. Their joint limit is therefore the product measure,

$$\sqrt{n} \begin{pmatrix} \hat{\beta}_1 - \beta \\ \hat{\beta}_2 - \beta \end{pmatrix} \xrightarrow{d} N \left(\mathbf{0}, \begin{bmatrix} V_\beta & \mathbf{0} \\ \mathbf{0} & V_\beta \end{bmatrix} \right).$$

Applying the continuous (linear) map $(a, b) \mapsto a - b$, and noting that β cancels,

$$\sqrt{n}(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{n}(\hat{\beta}_1 - \beta) - \sqrt{n}(\hat{\beta}_2 - \beta) \xrightarrow{d} N(\mathbf{0}, 2V_\beta).$$

This is the basis of a simple specification test. Under the null that both halves are drawn from the same population, the Wald statistic

$$W = \frac{n}{2}(\hat{\beta}_1 - \hat{\beta}_2)' \hat{V}_\beta^{-1} (\hat{\beta}_1 - \hat{\beta}_2) \xrightarrow{d} \chi_k^2$$

where $\hat{V}_\beta$ is any consistent estimator of V_β (for instance the full-sample one). When the split is made on a substantively meaningful variable rather than at random — before/after a policy date, one region versus another — this is the Chow test for a structural break.

Exercise 7.20

(a) The first-order condition is $-2 \sum_i W_i X_i (Y_i - X_i' \hat{\beta}) = \mathbf{0}$, so

$$\hat{\beta} = \left(\sum_{i=1}^n W_i X_i X_i' \right)^{-1} \left(\sum_{i=1}^n W_i X_i Y_i \right) = (X' \mathcal{W} X)^{-1} X' \mathcal{W} Y,$$

with $\mathcal{W} = \text{diag}(W_1, \dots, W_n)$. The criterion is convex and the solution unique provided $\sum_i W_i X_i X_i' > 0$.

(b) $\hat{\beta}$ is the sample analogue of the minimizer of the *population* weighted criterion $\mathbb{E}[W(Y - X'b)^2]$, whose first-order condition is $\mathbb{E}[WX(Y - X'\beta)] = \mathbf{0}$. So

$$\beta_W = (\mathbb{E}[WXX'])^{-1} \mathbb{E}[WXY]$$

— the W -weighted linear projection coefficient. The assumptions needed, and no more, are:

- $\{Y_i, X_i, W_i\}$ i.i.d. (given);
- $\mathbb{E}\|WXX'\| < \infty$ and $\mathbb{E}\|WXY\| < \infty$, so the moments exist;
- $Q_W \equiv \mathbb{E}[WXX'] > 0$, so β_W is uniquely defined.

In particular $W \geq 0$ is natural but not logically required, and nothing about $\mathbb{E}[e | X]$ is needed for β_W to be well defined.

When does β_W equal the ordinary projection coefficient β ? Only under an additional assumption. If the CEF is linear in the strong sense $\mathbb{E}[Y | X, W] = X'\beta$, then $e = Y - X'\beta$ satisfies $\mathbb{E}[WXe] = \mathbb{E}[WX \mathbb{E}[e | X, W]] = \mathbf{0}$ and $\beta_W = \beta$ for *every* weight. Under the weaker projection condition $\mathbb{E}[Xe] = \mathbf{0}$ alone, different weights define different parameters. (Exercise 7.25 makes this point again.)

(c) Dividing numerator and denominator by n and applying the WLLN and the continuous mapping theorem,

$$\hat{\beta} = \left(\frac{1}{n} \sum_i W_i X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_i W_i X_i Y_i \right) \xrightarrow{p} Q_W^{-1} \mathbb{E}[W X Y] = \beta_W.$$

(d) Write $e_i = Y_i - X_i' \beta_W$, so that $\mathbb{E}[W X e] = \mathbf{0}$ by construction. Then

$$\sqrt{n}(\hat{\beta} - \beta_W) = \left(\frac{1}{n} \sum_i W_i X_i X_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_i W_i X_i e_i \right).$$

The summands $W_i X_i e_i$ are i.i.d. with mean zero and variance $\Omega_W = \mathbb{E}[W^2 X X' e^2]$, so by the CLT and Slutsky,

$$\sqrt{n}(\hat{\beta} - \beta_W) \xrightarrow{d} N(\mathbf{0}, Q_W^{-1} \Omega_W Q_W^{-1}), \quad Q_W = \mathbb{E}[W X X'], \quad \Omega_W = \mathbb{E}[W^2 X X' e^2],$$

requiring $\mathbb{E}[\|W X e\|^2] < \infty$. Note the asymmetry: the weight appears *once* in the bread and *twice* in the meat. A consistent covariance estimator is

$$\hat{V} = \left(\sum_i W_i X_i X_i' \right)^{-1} \left(\sum_i W_i^2 X_i X_i' \hat{e}_i^2 \right) \left(\sum_i W_i X_i X_i' \right)^{-1}.$$

Only when $W_i = \sigma_i^{-2}$ exactly does the sandwich collapse to Q_W^{-1} , recovering the GLS efficiency bound.

Exercise 7.21

Step 1: estimate the mean equation. Run OLS of Y on X to obtain $\hat{\beta}$ with residuals $\hat{e}_i$ and a robust covariance matrix estimator $\hat{V}_{\hat{\beta}}$. (OLS is consistent since $\mathbb{E}[e | X] = 0$; if efficiency matters one can iterate to feasible GLS using the weights from Step 2, which does not change what follows.)

Step 2: estimate the variance function. The restriction $\mathbb{E}[e^2 | X] = Z' \gamma$ says that e^2 has a linear CEF in Z . Since e_i is unobserved, regress the *squared residuals* on Z :

$$\hat{\gamma} = \left(\sum_{i=1}^n Z_i Z_i' \right)^{-1} \left(\sum_{i=1}^n Z_i \hat{e}_i^2 \right).$$

Exactly as in Exercise 7.11, replacing e_i^2 by $\hat{e}_i^2$ is asymptotically negligible here because $\mathbb{E}[Z X' e] = \mathbf{0}$ under $\mathbb{E}[e | X] = 0$ and Z a function of X . So $\hat{\gamma} \xrightarrow{p} \gamma$. Then

$$\hat{\sigma}^2(z) = z' \hat{\gamma}$$

estimates the conditional variance at the forecast point. (If $z' \hat{\gamma} \leq 0$ in a finite sample — possible, since nothing constrains the linear fit to be positive — truncate at a small positive value, or re-specify the variance function as $\exp(Z' \gamma)$ and fit by regressing $\log \hat{e}_i^2$ on Z_i .)

Step 3: the point forecast.

$$\tilde{Y}_{n+1} = x' \hat{\beta}.$$

Step 4: the forecast interval. The forecast error is $\tilde{e}_{n+1} = e_{n+1} - x'(\hat{\beta} - \beta)$, with the two terms independent since observation $n + 1$ is not in the estimation sample. Its variance, *conditional on*

$X_{n+1} = x$ and $Z_{n+1} = z$, is

$$\text{var}[\tilde{e}_{n+1}] = \underbrace{z'\gamma}_{\text{irreducible}} + \underbrace{x'V_{\hat{\beta}}x}_{O(n^{-1})},$$

estimated by $\hat{\sigma}_f^2 = z'\hat{\gamma} + x'\hat{V}_{\hat{\beta}}x$. An approximate $1 - \alpha$ forecast interval is

$$\boxed{x'\hat{\beta} \pm z_{1-\alpha/2}\sqrt{z'\hat{\gamma} + x'\hat{V}_{\hat{\beta}}x}.$$

What this does and does not deliver. The moment conditions pin down the first two moments of the forecast error but not its *quantiles*, so the normal critical value is an assumption, not a consequence: the interval is exact only if $e \mid X$ is normal, and otherwise approximate. Note also that no amount of data shrinks the interval below $\pm z_{1-\alpha/2}\sqrt{z'\gamma}$ — unlike a confidence interval for $x'\beta$, which has half-width $z_{1-\alpha/2}\sqrt{x'\hat{V}_{\hat{\beta}}x} = O(n^{-1/2})$. The whole point of modelling $\mathbb{E}[e^2 \mid X]$ here is that the interval width now varies with the forecast point z , which a homoskedastic interval cannot capture.

Exercise 7.22

The second-step estimator is a simple regression without intercept:

$$\hat{\gamma} = \frac{\sum_{i=1}^n (X_i'\hat{\beta})Z_i}{\sum_{i=1}^n (X_i'\hat{\beta})^2} = \frac{\hat{\beta}'\left(n^{-1}\sum_{i=1}^n X_iZ_i\right)}{\hat{\beta}'\left(n^{-1}\sum_{i=1}^n X_iX_i'\right)\hat{\beta}}.$$

Writing it this way — pulling $\hat{\beta}$ outside the sums — is what makes the analysis easy, because the sample averages no longer involve $\hat{\beta}$.

(a) By the WLLN,

$$\frac{1}{n}\sum_{i=1}^n X_iZ_i \xrightarrow{p} \mathbb{E}[XZ] = \mathbb{E}[X(X'\beta\gamma + u)] = \gamma Q_{XX}\beta,$$

using $\mathbb{E}[Xu] = \mathbb{E}[X\mathbb{E}[u \mid X]] = \mathbf{0}$; and $n^{-1}\sum_i X_iX_i' \xrightarrow{p} Q_{XX}$. Since $\hat{\beta} \xrightarrow{p} \beta$, the continuous mapping theorem gives

$$\hat{\gamma} \xrightarrow{p} \frac{\beta'(\gamma Q_{XX}\beta)}{\beta'Q_{XX}\beta} = \gamma,$$

provided $\beta'Q_{XX}\beta > 0$, i.e. $\beta \neq \mathbf{0}$ (with $Q_{XX} > 0$). That condition is necessary: if $\beta = \mathbf{0}$ the generated regressor $X'\beta$ is identically zero and γ is not identified at all.

(b) Set $\gamma = 0$, so $Z_i = u_i$. Then

$$\sqrt{n}\hat{\gamma} = \frac{\hat{\beta}'\left(n^{-1/2}\sum_{i=1}^n X_iu_i\right)}{\hat{\beta}'\left(n^{-1}\sum_{i=1}^n X_iX_i'\right)\hat{\beta}}.$$

Apply the CLT to the i.i.d. mean-zero vectors X_iu_i :

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n X_iu_i \xrightarrow{d} N(\mathbf{0}, \Omega_u), \quad \Omega_u = \mathbb{E}[XX'u^2],$$

while $\hat{\beta} \xrightarrow{p} \beta$ and $n^{-1} \sum_i X_i X_i' \xrightarrow{p} Q_{XX}$. By Slutsky's theorem,

$$\sqrt{n} \hat{\gamma} \xrightarrow{d} N\left(0, \frac{\beta' \Omega_u \beta}{(\beta' Q_{XX} \beta)^2}\right).$$

Under conditional homoskedasticity $\mathbb{E}[u^2 | X] = \sigma_u^2$ this simplifies to $N(0, \sigma_u^2/(\beta' Q_{XX} \beta))$.

Remark. The interesting feature is that the first-step estimation error does *not* appear in the limit. The reason is specific to $\gamma = 0$: the numerator is already $O_p(n^{-1/2})$ on its own, so replacing β by $\hat{\beta} = \beta + O_p(n^{-1/2})$ perturbs it only at order n^{-1} . When $\gamma \neq 0$ the numerator has a non-zero mean and the first-step error does contribute a term to the asymptotic variance — the usual “generated regressor” correction. So a standard error computed by treating $X'\hat{\beta}$ as if it were observed data is valid for testing $H_0 : \gamma = 0$, but invalid for building a confidence interval around a non-zero $\hat{\gamma}$.

Exercise 7.23

Substituting $Y_i = X_i \beta + e_i$ (which requires $X_i \neq 0$ for the ratio to be defined),

$$\tilde{\beta} = \frac{1}{n} \sum_{i=1}^n \frac{X_i \beta + e_i}{X_i} = \beta + \frac{1}{n} \sum_{i=1}^n \frac{e_i}{X_i}.$$

So consistency is exactly the statement that the average of the i.i.d. variables e_i/X_i converges in probability to zero. Two conditions deliver this.

- (i) $\mathbb{P}[X = 0] = 0$, so that e/X is well defined with probability one.
- (ii) $\mathbb{E}|e/X| < \infty$, so that the WLLN applies.

Granting both, the WLLN and the conditioning theorem give

$$\frac{1}{n} \sum_{i=1}^n \frac{e_i}{X_i} \xrightarrow{p} \mathbb{E}\left[\frac{e}{X}\right] = \mathbb{E}\left[\frac{1}{X} \mathbb{E}[e | X]\right] = 0 \quad \implies \quad \tilde{\beta} \xrightarrow{p} \beta.$$

Condition (ii) is the binding one and deserves emphasis: it is a joint restriction on the tails of e and on the mass of X near zero. Two convenient sufficient conditions are

- $|X| \geq \delta > 0$ almost surely (the regressor is bounded away from zero) together with $\mathbb{E}|e| < \infty$; or
- $\mathbb{E}[e^2/X^2] < \infty$, which by Jensen implies (ii) and additionally delivers asymptotic normality, $\sqrt{n}(\tilde{\beta} - \beta) \xrightarrow{d} N(0, \mathbb{E}[e^2/X^2])$.

If X has a density that is positive at 0, then typically $\mathbb{E}[1/X] = \infty$ and $\tilde{\beta}$ fails to converge — the sample average is dominated by the observations with X_i closest to zero. Note also that, unlike OLS, this estimator genuinely needs $\mathbb{E}[e | X] = 0$: the projection condition $\mathbb{E}[Xe] = 0$ says nothing about $\mathbb{E}[e/X]$. Compare Exercise 7.9, where the same estimator is shown to be efficient when $\mathbb{E}[e^2 | X] = \sigma^2 X^2$.

Exercise 7.24

(a) With a scalar regressor, $\hat{\beta} \xrightarrow{p} \mathbb{E}[XY]/\mathbb{E}[X^2]$. Compute the two moments using the independence of v from (X^*, e) and $\mathbb{E}[e] = 0$:

$$\mathbb{E}[XY] = \mathbb{E}[X^* v (X^* \beta + e)] = \beta \mathbb{E}[X^{*2} v] + \mathbb{E}[X^* v e]$$

$$\begin{aligned}
&= \beta \mathbb{E}[X^{*2}] \mathbb{E}[v] + \mathbb{E}[X^*] \mathbb{E}[v] \mathbb{E}[e] = \beta \mathbb{E}[X^{*2}] \mathbb{E}[v], \\
\mathbb{E}[X^2] &= \mathbb{E}[X^{*2}v^2] = \mathbb{E}[X^{*2}] \mathbb{E}[v^2].
\end{aligned}$$

Therefore

$$\boxed{\hat{\beta} \xrightarrow{p} \beta \frac{\mathbb{E}[v]}{\mathbb{E}[v^2]}}$$

(assuming $0 < \mathbb{E}[X^{*2}] < \infty$ and $0 < \mathbb{E}[v^2] < \infty$). Remarkably the distribution of X^* drops out entirely: the inconsistency depends only on the first two moments of the measurement error.

If v is *unbiased*, $\mathbb{E}[v] = 1$, then $\mathbb{E}[v^2] = 1 + \text{var}(v) > 1$ and

$$\text{plim } \hat{\beta} = \frac{\beta}{1 + \text{var}(v)} < \beta,$$

the classical *attenuation* result: measurement error in a regressor biases the coefficient toward zero, and the more noise the greater the attenuation.

(b) Yes. Consistency requires

$$\frac{\mathbb{E}[v]}{\mathbb{E}[v^2]} = 1 \quad \Longleftrightarrow \quad \mathbb{E}[v^2] = \mathbb{E}[v] \quad \Longleftrightarrow \quad \text{var}(v) = \mathbb{E}[v] - (\mathbb{E}[v])^2,$$

which has non-trivial solutions whenever $\mathbb{E}[v] \in (0, 1)$. So v must be *biased downward* by exactly the amount needed to offset the attenuation: the downward bias in $\mathbb{E}[X^2]$ relative to $\mathbb{E}[XY]$ cancels the inflation caused by $\text{var}(v)$.

A concrete example: let v take the value $\frac{1}{2}$ with probability $\frac{8}{9}$ and the value 2 with probability $\frac{1}{9}$. Then

$$\mathbb{E}[v] = \frac{8}{9} \cdot \frac{1}{2} + \frac{1}{9} \cdot 2 = \frac{6}{9} = \frac{2}{3}, \quad \mathbb{E}[v^2] = \frac{8}{9} \cdot \frac{1}{4} + \frac{1}{9} \cdot 4 = \frac{2}{9} + \frac{4}{9} = \frac{6}{9} = \frac{2}{3},$$

so $\mathbb{E}[v] = \mathbb{E}[v^2]$ and $\hat{\beta} \xrightarrow{p} \beta$ exactly. The condition is a knife-edge coincidence with no economic motivation, which is precisely the point: apart from such accidents, multiplicative measurement error in a regressor makes OLS inconsistent.

Exercise 7.25

The probability limit. Dividing by n and applying the WLLN,

$$\tilde{\beta} \xrightarrow{p} (\mathbb{E}[w(X)XX'])^{-1} \mathbb{E}[w(X)XY] \equiv \beta_w.$$

Assumptions needed beyond those of the projection model: the weighted moments must exist, $\mathbb{E}\|w(X)XX'\| < \infty$ and $\mathbb{E}\|w(X)XY\| < \infty$, and the weighted design matrix must be invertible, $Q_w \equiv \mathbb{E}[w(X)XX'] > 0$. (Positivity of w makes Q_w positive semi-definite automatically, but not necessarily positive definite — e.g. if w is zero except on a set too small to identify all k coefficients.)

Is $\tilde{\beta}$ consistent for β ? *In general, no.* Substituting $Y = X'\beta + e$,

$$\beta_w = \beta + Q_w^{-1} \mathbb{E}[w(X)Xe],$$

so $\tilde{\beta} \xrightarrow{p} \beta$ if and only if

$$\mathbb{E}[w(X)Xe] = \mathbf{0}.$$

The projection assumption only delivers $\mathbb{E}[Xe] = \mathbf{0}$, which is the special case $w \equiv 1$. There is no reason for the *re-weighted* moment to vanish: weighting tilts the population toward regions where w is large, and if the true CEF is nonlinear the best linear approximation in the tilted population differs from the one in the original.

The assumption that restores consistency. Strengthen the projection condition to the conditional mean (regression) condition

$$\mathbb{E}[e \mid X] = 0.$$

Then, by the conditioning theorem, for any function w ,

$$\mathbb{E}[w(X)Xe] = \mathbb{E}[w(X)X \mathbb{E}[e \mid X]] = \mathbf{0},$$

and $\beta_w = \beta$ for *every* weight function. In this case weighting is purely a matter of efficiency, not of identification — which is the setting of Exercises 4.8 and 7.9.

Remark. An explicit example of the failure. Let $X = (1, Z)'$ with $Z \sim N(0, 1)$ and $Y = Z^2$, so (Exercise 2.13) the unweighted projection gives $\beta = (1, 0)'$ and $e = Z^2 - 1$. Take $w(x) = \exp(z)$. Then $\mathbb{E}[w(X)e] = \mathbb{E}[e^Z(Z^2 - 1)] = e^{1/2}((1 + 1) - 1) \neq 0$, so $\beta_w \neq \beta$. Weighting has changed the parameter being estimated.

Exercise 7.26

(a) Substituting $Y_i = X_i'\beta + e_i$ and noting that $e_i^{-2}X_ie_i = e_i^{-1}X_i$,

$$\tilde{\beta} - \beta = \left(\sum_{i=1}^n e_i^{-2} X_i X_i' \right)^{-1} \left(\sum_{i=1}^n e_i^{-1} X_i \right).$$

Define

$$A = \mathbb{E} \left[\frac{XX'}{e^2} \right] = \mathbb{E}[XX' \mathbb{E}[e^{-2} \mid X]], \quad b = \mathbb{E} \left[\frac{X}{e} \right] = \mathbb{E}[X \mathbb{E}[e^{-1} \mid X]],$$

and *assume both exist and $A > 0$* — an assumption that deserves the scrutiny given below. By the WLLN,

$$\frac{1}{n} \sum_{i=1}^n e_i^{-2} X_i X_i' \xrightarrow{p} A, \quad \frac{1}{n} \sum_{i=1}^n e_i^{-1} X_i \xrightarrow{p} b,$$

so $\tilde{\beta} \xrightarrow{p} \beta + A^{-1}b$. Consistency therefore requires $b = \mathbf{0}$. This is *not* implied by $\mathbb{E}[e \mid X] = 0$: mean-zero does not give $\mathbb{E}[e^{-1} \mid X] = 0$. It does hold when the conditional distribution of e given X is symmetric about zero, which we now assume.

Granting $b = \mathbf{0}$, the summands $e_i^{-1}X_i$ are i.i.d. with mean zero and variance $\mathbb{E}[XX'e^{-2}] = A$, so the CLT gives $n^{-1/2} \sum_i e_i^{-1}X_i \xrightarrow{d} N(\mathbf{0}, A)$ and

$$\boxed{\sqrt{n}(\tilde{\beta} - \beta) \xrightarrow{d} N(\mathbf{0}, A^{-1}AA^{-1}) = N(\mathbf{0}, A^{-1}), \quad A = \mathbb{E}[XX' \mathbb{E}[e^{-2} \mid X]].}$$

(b) Infeasible GLS uses the weights $\sigma^{-2}(X_i)$ and has

$$\sqrt{n}(\tilde{\beta}_{\text{glS}} - \beta) \xrightarrow{d} N(\mathbf{0}, A_{\text{glS}}^{-1}), \quad A_{\text{glS}} = \mathbb{E} \left[\frac{XX'}{\sigma^2(X)} \right],$$

and A_{glS}^{-1} is the efficiency bound for this model. Now compare. By the conditional Jensen inequality applied to the convex map $t \mapsto 1/t$,

$$\mathbb{E}[e^{-2} | X] \geq \frac{1}{\mathbb{E}[e^2 | X]} = \frac{1}{\sigma^2(X)},$$

so $A \geq A_{\text{glS}}$ and hence $A^{-1} \leq A_{\text{glS}}^{-1}$: the e^{-2} -weighted estimator appears to be *strictly more efficient than the efficiency bound*, typically strictly so, since the inequality is strict unless e^2 is conditionally degenerate.

There is no paradox. The GLS bound applies to estimators that are functions of the *observables* (Y_i, X_i) , with weights that are functions of X_i . The estimator $\tilde{\beta}$ uses the unobserved e_i — information strictly beyond the data — so it is outside the class over which the bound is taken. It is not an estimator at all, merely a thought experiment.

Two further warnings make the thought experiment less appealing still. First, $\mathbb{E}[e^{-2} | X]$ is *infinite* for most standard error distributions: if $e | X$ has a density that is bounded away from zero near the origin — normal, Student t , uniform — then $\mathbb{E}[e^{-2}] = \infty$ and A does not exist, so the asymptotics above simply fail. Second, even when A is finite the weights e_i^{-2} are enormous for observations whose realized error happens to be small, so $\tilde{\beta}$ essentially interpolates through the handful of points that lie almost exactly on the true line. That is exactly why its (nominal) variance is so small — it is using the answer to construct the weights.

Exercise 7.27

Write $T_i = \mathbf{1}\{|X_i| \leq c\}$ for the trimming indicator and assume $\mathbb{P}[|X| \leq c] > 0$ with

$$Q_c \equiv \mathbb{E}[XX'T] = \mathbb{E}[XX'\mathbf{1}\{|X| \leq c\}] > 0,$$

i.e. the retained region is rich enough to identify all k coefficients.

(a) Substituting $Y_i = X_i'\beta + e_i$,

$$\tilde{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' T_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i e_i T_i \right).$$

By the WLLN the first average converges to Q_c . For the second, note that T_i is a function of X_i , so by the conditioning theorem

$$\mathbb{E}[XeT] = \mathbb{E}[\mathbf{1}\{|X| \leq c\} X \mathbb{E}[e | X]] = \mathbf{0},$$

and $n^{-1} \sum_i X_i e_i T_i \xrightarrow{p} \mathbf{0}$. Hence $\tilde{\beta} \xrightarrow{p} \beta$ by the continuous mapping theorem.

Note carefully which assumption is doing the work. Trimming on X is a form of non-random sample selection, and it is harmless here *only* because the selection rule depends on X alone and because $\mathbb{E}[e | X] = 0$ holds for every x , including the retained ones. Under the weaker projection condition $\mathbb{E}[Xe] = \mathbf{0}$ the trimmed moment $\mathbb{E}[XeT]$ need not vanish, and $\tilde{\beta}$ would converge to the projection coefficient *within the trimmed population*, generally a different parameter. (Trimming on Y , or on the residual, would break consistency even under $\mathbb{E}[e | X] = 0$.)

(b) The summands $X_i e_i T_i$ are i.i.d. with mean zero and variance

$$\Omega_c = \mathbb{E}[X X' e^2 \mathbf{1}\{|X| \leq c\}],$$

so the CLT and Slutsky's theorem give

$$\sqrt{n}(\tilde{\beta} - \beta) \xrightarrow{d} N(\mathbf{0}, Q_c^{-1} \Omega_c Q_c^{-1}).$$

The moment conditions are unusually weak: on the retained set $\|X\| \leq c$ is bounded, so Q_c and Ω_c are finite as soon as $\mathbb{E}[e^2] < \infty$ — no moment assumptions on X are needed at all. This is the theoretical attraction of trimming: it buys robustness to heavy-tailed regressors, and the trimmed estimator is asymptotically normal even in cases where $\mathbb{E}[X^4] = \infty$ and the untrimmed THEOREM 7.3 does not apply. The price is efficiency, since $Q_c \leq Q_{XX}$ discards information, and the arbitrariness of c .

Exercise 7.28

The sub-sample is `race = 1`, `female = 0`, `hisp = 1`, with $n = 4,230$; standard errors below are HC1.

(a)

	coefficient	s.e. (HC1)
education ($\hat{\beta}_1$)	0.090449	0.002916
experience ($\hat{\beta}_2$)	0.035380	0.002585
experience ² /100 ($\hat{\beta}_3$)	-0.046506	0.005307
intercept ($\hat{\beta}_4$)	1.185209	0.046100

($s^2 = 0.32936$, $s = 0.5739$.)

(b) Differentiate the regression function. The return to a year of education is $\partial m / \partial \text{education} = \beta_1$. The return to a year of experience is

$$\frac{\partial m}{\partial \text{experience}} = \beta_2 + \frac{2\beta_3}{100} \text{experience},$$

so at experience = 10 it equals $\beta_2 + 0.2\beta_3$. Hence

$$\theta = \frac{\beta_1}{\beta_2 + 0.2\beta_3}, \quad \hat{\theta} = \frac{0.090449}{0.035380 + 0.2(-0.046506)} = \frac{0.090449}{0.026079} = 3.4683.$$

One extra year of schooling is worth about 3.5 years of experience for a worker with ten years of experience.

(c) Apply the delta method. Write $d = \beta_2 + 0.2\beta_3$ for the denominator. The gradient of $\theta = \beta_1/d$ is

$$R = \frac{\partial \theta}{\partial \beta} = \begin{pmatrix} 1/d \\ -\beta_1/d^2 \\ -0.2\beta_1/d^2 \\ 0 \end{pmatrix}, \quad \text{so} \quad s(\hat{\theta}) = \sqrt{\hat{R}' \hat{V}_{\hat{\beta}} \hat{R}},$$

with $\hat{R}$ evaluated at $\hat{\beta}$ and $\hat{V}_{\hat{\beta}}$ the robust covariance matrix estimator. Numerically

$$s(\hat{\theta}) = 0.22684.$$

(HC0 gives 0.22673 and HC3 gives 0.22735; the choice is immaterial at this sample size.)

(d) With $z_{0.95} = 1.645$,

$$\hat{\theta} \pm 1.645 s(\hat{\theta}) = 3.4683 \pm 0.3731 = \boxed{[3.095, 3.841]}$$

(e) Let $a = (12, 20, 4, 1)'$, recalling that the third regressor is $\text{experience}^2/100 = 400/100 = 4$. Then

$$\hat{m} = a'\hat{\beta} = 2.7922, \quad s(\hat{m}) = \sqrt{a'\hat{V}_{\hat{\beta}}a} = 0.01167,$$

so the 95% confidence interval for the regression function is

$$2.7922 \pm 1.96(0.01167) = \boxed{[2.769, 2.815]}$$

In wage units this is $[e^{2.769}, e^{2.815}] = [\$15.95, \$16.69]$ per hour. The interval is tight because it concerns the conditional *mean* at a well-populated point of the covariate space.

(f) Now let $a = (16, 5, 0.25, 1)'$. The point forecast is $a'\hat{\beta} = 2.7977$. The forecast error variance adds the irreducible error variance to the estimation variance (Exercise 7.10):

$$s_f = \sqrt{s^2 + a'\hat{V}_{\hat{\beta}}a} = \sqrt{0.32936 + 0.00039} = 0.57423.$$

With $z_{0.90} = 1.2816$ the 80% forecast interval for log wage is

$$2.7977 \pm 1.2816(0.57423) = \boxed{[2.062, 3.534]}$$

and, applying the exponential to both endpoints (legitimate by Exercise 5.9, since $\exp$ is increasing), the 80% interval for the wage is

$$[e^{2.062}, e^{3.534}] = \boxed{[\$7.86, \$34.25] \text{ per hour.}}$$

The contrast between (e) and (f) is the lesson. The confidence interval for the regression function has half-width 0.023; the forecast interval has half-width 0.736, thirty times wider. Almost all of s_f comes from $s^2 = 0.329$, the variation in log wages *within* education–experience cells, and only 0.0004 from the uncertainty about β . More data would shrink the former interval to nothing and leave the latter essentially unchanged. Predicting the average wage of a group is easy; predicting one person's wage is not.

Remark. The forecast interval relies on approximate normality of the regression error, which the moment conditions do not supply. Its coverage should be treated as approximate, and the back-transformed wage interval is an interval for the *median* wage of such an individual, not the mean — by Jensen, $\mathbb{E}[\exp(e)] > 1$ when $\mathbb{E}[e] = 0$.

Code (R).

```
dat <- read.table("cps09mar.txt")
names(dat) <- c("age", "female", "hisp", "education", "earnings", "hours",
               "week", "union", "uncov", "region", "race", "marital")
d <- subset(dat, race==1 & female==0 & hisp==1)
```

```

y <- log(d$earnings/(d$hours*d$week))
ex <- d$age - d$education - 6
X <- cbind(d$education, ex, ex^2/100, 1)
n <- nrow(X); k <- ncol(X)
XXi <- solve(crossprod(X)); b <- XXi %*% crossprod(X,y); e <- y - X %*% b
V <- (n/(n-k)) * XXi %*% crossprod(X, X*as.vector(e^2)) %*% XXi # HC1
den <- b[2] + 0.2*b[3]
th <- b[1]/den
R <- c(1/den, -b[1]/den^2, -0.2*b[1]/den^2, 0)
se <- sqrt(t(R) %*% V %*% R)
c(th, se, th - 1.645*se, th + 1.645*se) # (b)-(d)
a <- c(12,20,4,1); m <- sum(a*b); sm <- sqrt(t(a) %*% V %*% a) # (e)
c(m, sm, m - 1.96*sm, m + 1.96*sm)
a <- c(16,5,0.25,1); f <- sum(a*b) # (f)
s2 <- sum(e^2)/(n-k); sf <- sqrt(s2 + t(a) %*% V %*% a)
exp(c(f - 1.2816*sf, f + 1.2816*sf))

```

Chapter 8

Restricted Estimation

Solutions to all 22 exercises of Chapter 8 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercise 8.19 is empirical; the numbers were computed from the `cps09mar` data set on the textbook website.

The restriction is $R'\beta = c$ with R a $k \times q$ matrix of full rank q (Assumption 8.1). $\hat{\beta} = (X'X)^{-1}X'Y$ is unrestricted OLS and $\hat{Q}_{XX} = n^{-1}X'X$. The CLS estimator (8.8) is

$$\tilde{\beta}_{\text{cls}} = \hat{\beta} - (X'X)^{-1}R[R'(X'X)^{-1}R]^{-1}(R'\hat{\beta} - c),$$

and the minimum distance estimator with weight $\widehat{W}$ is (8.23),

$$\tilde{\beta}_{\text{md}} = \hat{\beta} - \widehat{W}^{-1}R(R'\widehat{W}^{-1}R)^{-1}(R'\hat{\beta} - c).$$

Throughout we write

$$P = X(X'X)^{-1}X', \quad A = (X'X)^{-1}R[R'(X'X)^{-1}R]^{-1}R'(X'X)^{-1},$$

and note two identities used repeatedly: A is symmetric, and

$$A(X'X)A = A. \tag{1}$$

(Multiply out: the middle $R'(X'X)^{-1}(X'X)(X'X)^{-1}R$ cancels one bracketed inverse.)

Exercise 8.1

Definition (8.3) minimizes $\text{SSE}(\beta) = \sum_i (Y_i - X'_{1i}\beta_1 - X'_{2i}\beta_2)^2$ over the constraint set $\{\beta : \beta_2 = \mathbf{0}\}$. On that set the criterion does not involve β_2 at all:

$$\min_{\beta_1, \beta_2 = \mathbf{0}} \sum_{i=1}^n (Y_i - X'_{1i}\beta_1 - X'_{2i}\beta_2)^2 = \min_{\beta_1} \sum_{i=1}^n (Y_i - X'_{1i}\beta_1)^2.$$

The right-hand side is the least squares problem for the regression of Y on X_1 , whose unique solution (when $\sum_i X_{1i}X'_{1i} > 0$) is

$$\tilde{\beta}_1 = \left(\sum_{i=1}^n X_{1i}X'_{1i} \right)^{-1} \left(\sum_{i=1}^n X_{1i}Y_i \right), \quad \tilde{\beta}_2 = \mathbf{0}.$$

So imposing an exclusion restriction by CLS is the same as simply dropping the excluded regressors. (Equivalently, (8.8) with $R' = [\mathbf{0} \quad \mathbf{I}_{k_2}]$ and $c = \mathbf{0}$ reduces to this formula after partitioned inversion.)

Exercise 8.2

On the constraint set $\{\beta_1 = c\}$ the criterion becomes

$$\sum_{i=1}^n (Y_i - X'_{1i}c - X'_{2i}\beta_2)^2 = \sum_{i=1}^n (Y_i^* - X'_{2i}\beta_2)^2, \quad Y_i^* = Y_i - X'_{1i}c,$$

a function of β_2 alone. This is the least squares criterion for the regression of Y^* on X_2 , so

$$\tilde{\beta}_1 = c, \quad \tilde{\beta}_2 = \left(\sum_{i=1}^n X_{2i} X'_{2i} \right)^{-1} \left(\sum_{i=1}^n X_{2i} (Y_i - X'_{1i} c) \right).$$

In practice: move the known part of the regression function to the left-hand side (an “offset”) and regress on what is left.

Exercise 8.3

Substitute the constraint $\beta_2 = -\beta_1$ into the regression function:

$$X'_1 \beta_1 + X'_2 \beta_2 = X'_1 \beta_1 - X'_2 \beta_1 = (X_1 - X_2)' \beta_1.$$

Minimizing SSE over the constraint set is therefore least squares of Y on $Z = X_1 - X_2$:

$$\tilde{\beta}_1 = \left(\sum_{i=1}^n Z_i Z'_i \right)^{-1} \left(\sum_{i=1}^n Z_i Y_i \right), \quad Z_i = X_{1i} - X_{2i}, \quad \tilde{\beta}_2 = -\tilde{\beta}_1.$$

This is the general recipe of Section 8.3: whenever a linear restriction can be solved for some coefficients in terms of others, CLS is OLS on the correspondingly reparameterized regressors.

Exercise 8.4

(a) By Exercise 8.1 the CLS estimator is OLS of Y on a constant alone,

$$\tilde{\alpha}_{\text{cls}} = \bar{Y}.$$

(b) Stack $\theta = (\alpha, \beta)'$, with unrestricted OLS estimate $\hat{\theta} = (\hat{\alpha}, \hat{\beta})'$ and robust covariance matrix estimate partitioned as

$$\hat{V}_\theta = \begin{bmatrix} \hat{V}_{\alpha\alpha} & \hat{V}_{\alpha\beta} \\ \hat{V}_{\beta\alpha} & \hat{V}_{\beta\beta} \end{bmatrix}.$$

The restriction is $R'\theta = \mathbf{0}$ with $R' = [\mathbf{0} \quad \mathbf{I}_k]$. Then $R'\hat{\theta} = \hat{\beta}$, $\hat{V}_\theta R = (\hat{V}'_{\alpha\beta}, \hat{V}'_{\beta\beta})'$ and $R'\hat{V}_\theta R = \hat{V}_{\beta\beta}$, so (8.25) gives

$$\tilde{\theta}_{\text{emd}} = \begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} - \begin{pmatrix} \hat{V}_{\alpha\beta} \\ \hat{V}_{\beta\beta} \end{pmatrix} \hat{V}_{\beta\beta}^{-1} \hat{\beta} \quad \implies \quad \boxed{\tilde{\alpha}_{\text{emd}} = \hat{\alpha} - \hat{V}_{\alpha\beta} \hat{V}_{\beta\beta}^{-1} \hat{\beta}, \quad \tilde{\beta}_{\text{emd}} = \mathbf{0},}$$

where $\hat{\alpha} = \bar{Y} - \bar{X}'\hat{\beta}$ is the unrestricted OLS intercept. This is (8.30) applied with the intercept as the retained coefficient.

Relation to part (a). The EMD estimator “adjusts” $\hat{\alpha}$ by the estimated regression of $\hat{\alpha}$ on $\hat{\beta}$. Under homoskedasticity $V_\theta = \sigma^2 Q^{-1}$ with $Q = \mathbb{E}[(1, X')'(1, X')]$, and the partitioned inverse formula gives $[Q^{-1}]_{\alpha\beta}[Q^{-1}]_{\beta\beta}^{-1} = -Q_{11}^{-1}Q_{12} = -\mathbb{E}[X]'$. With the sample analogue, $\tilde{\alpha}_{\text{emd}} = \hat{\alpha} + \bar{X}'\hat{\beta} = \bar{Y}$: the two estimators coincide, as THEOREM 8.9 predicts (CLS is efficient under homoskedasticity). Under heteroskedasticity they differ, and EMD exploits the information in $\mathbb{E}[Xe] = \mathbf{0}$ that $\bar{Y}$ ignores.

Exercise 8.5

Pre-multiply (8.8) by R' :

$$R'\tilde{\beta}_{\text{cls}} = R'\hat{\beta} - \underbrace{R'(X'X)^{-1}R[R'(X'X)^{-1}R]^{-1}}_{=\mathbf{I}_q}(R'\hat{\beta} - c) = R'\hat{\beta} - R'\hat{\beta} + c = c.$$

The bracketed matrix is invertible because $(X'X)^{-1} > 0$ and R has full column rank q .

Exercise 8.6

(1) Since $\hat{\beta} - \beta = (X'X)^{-1}X'e$ and $R'\beta = c$,

$$R'\hat{\beta} - c = R'(\hat{\beta} - \beta) + (R'\beta - c) = R'(X'X)^{-1}X'e.$$

(2) Subtract β from (8.8) and substitute part (1):

$$\begin{aligned}\tilde{\beta}_{\text{cls}} - \beta &= (X'X)^{-1}X'e - (X'X)^{-1}R[R'(X'X)^{-1}R]^{-1}R'(X'X)^{-1}X'e \\ &= ((X'X)^{-1}X' - AX')e.\end{aligned}$$

(3) Using $Y - X\beta = e$ and part (2),

$$\tilde{e} = Y - X\tilde{\beta}_{\text{cls}} = e - X(\tilde{\beta}_{\text{cls}} - \beta) = e - (P - XAX')e = (\mathbf{I}_n - P + XAX')e.$$

(4) Write $M = \mathbf{I}_n - P$ and $N = XAX'$. Both are symmetric (A is symmetric), so $M + N$ is symmetric. For idempotency:

- $MM = M$ (Exercise 3.8);
- $MN = (\mathbf{I}_n - P)XAX' = (X - X)AX' = \mathbf{0}$, and $NM = (MN)' = \mathbf{0}$;
- $NN = X(AX'XA)X' = XAX' = N$ by (1).

Hence $(M + N)^2 = M + \mathbf{0} + \mathbf{0} + N = M + N$.

(5) By linearity and the cyclic property of the trace,

$$\text{tr}(\mathbf{I}_n - P + XAX') = n - k + \text{tr}(AX'X)$$

and

$$\text{tr}(AX'X) = \text{tr}\left((X'X)^{-1}R[R'(X'X)^{-1}R]^{-1}R'\right) = \text{tr}\left([R'(X'X)^{-1}R]^{-1}R'(X'X)^{-1}R\right) = \text{tr}\mathbf{I}_q = q.$$

So the trace is $n - k + q$: imposing q restrictions gives back q degrees of freedom.

Exercise 8.7

By THEOREM 8.1.2, $\tilde{\beta}_{\text{cls}} - \beta = Be$ with $B = (X'X)^{-1}X' - AX'$ a function of X only. Hence

$$\mathbb{E}[\tilde{\beta}_{\text{cls}} - \beta \mid X] = B\mathbb{E}[e \mid X] = \mathbf{0}.$$

Note where the restriction was used: in Theorem 8.1.1, to replace $R'\hat{\beta} - c$ by $R'(\hat{\beta} - \beta)$. If the restriction is false, $R'\hat{\beta} - c$ has mean $R'\beta - c \neq \mathbf{0}$ and CLS is biased.

Exercise 8.8

With $B = (X'X)^{-1}X' - AX'$ as in Exercise 8.7 and $\text{var}[e | X] = \sigma^2 \mathbf{I}_n$,

$$\text{var}[\tilde{\beta}_{\text{cls}} | X] = B \text{var}[e | X] B' = \sigma^2 BB'.$$

Multiply out, using $X'X(X'X)^{-1} = \mathbf{I}_k$ and (1):

$$\begin{aligned} BB' &= ((X'X)^{-1}X' - AX')(X(X'X)^{-1} - XA) \\ &= (X'X)^{-1} - A - A + AX'XA \\ &= (X'X)^{-1} - A. \end{aligned}$$

Therefore $\text{var}[\tilde{\beta}_{\text{cls}} | X] = \sigma^2((X'X)^{-1} - A)$, which is the stated formula. Since $A \geq 0$, $V_{\tilde{\beta}}^0 \leq \sigma^2(X'X)^{-1}$: under homoskedasticity, imposing a true restriction can only reduce variance.

Exercise 8.9

By (8.17), which uses THEOREM 8.1.3–4, $(n - k + q)s_{\text{cls}}^2 = e'(\mathbf{I}_n - P + XAX')e$. By the trace trick, homoskedasticity, and THEOREM 8.1.5,

$$\mathbb{E}[e'(\mathbf{I}_n - P + XAX')e | X] = \sigma^2 \text{tr}(\mathbf{I}_n - P + XAX') = \sigma^2(n - k + q).$$

Dividing by $n - k + q$ gives $\mathbb{E}[s_{\text{cls}}^2 | X] = \sigma^2$. Consequently $\hat{V}_{\tilde{\beta}}^0 = ((X'X)^{-1} - A)s_{\text{cls}}^2$ satisfies $\mathbb{E}[\hat{V}_{\tilde{\beta}}^0 | X] = V_{\tilde{\beta}}^0$, the second claim of the theorem.

Exercise 8.10

The criterion is $J(\beta, \widehat{W}) = n(\widehat{\beta} - \beta)' \widehat{W}(\widehat{\beta} - \beta)$ and the Lagrangian $\mathcal{L}(\beta, \lambda) = \frac{1}{2}J(\beta, \widehat{W}) + \lambda'(R'\beta - c)$.

First-order conditions. Since $\partial J / \partial \beta = -2n\widehat{W}(\widehat{\beta} - \beta)$,

$$\frac{\partial \mathcal{L}}{\partial \beta} = -n\widehat{W}(\widehat{\beta} - \tilde{\beta}) + R\tilde{\lambda} = \mathbf{0}, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = R'\tilde{\beta} - c = \mathbf{0}.$$

The first gives $\tilde{\beta} = \widehat{\beta} - n^{-1}\widehat{W}^{-1}R\tilde{\lambda}$. Substituting into the second,

$$R'\widehat{\beta} - n^{-1}R'\widehat{W}^{-1}R\tilde{\lambda} = c \implies \tilde{\lambda}_{\text{md}} = n(R'\widehat{W}^{-1}R)^{-1}(R'\widehat{\beta} - c),$$

which is (8.22). Substituting back,

$$\tilde{\beta}_{\text{md}} = \widehat{\beta} - \widehat{W}^{-1}R(R'\widehat{W}^{-1}R)^{-1}(R'\widehat{\beta} - c),$$

which is (8.23). The quadratic criterion is strictly convex ($\widehat{W} > 0$) and the constraint is linear, so this critical point is the unique minimum.

The CLS special case. Set $\widehat{W} = \widehat{Q}_{XX} = n^{-1}X'X$, so $\widehat{W}^{-1} = n(X'X)^{-1}$. The factors of n cancel:

$$\tilde{\beta}_{\text{md}} = \widehat{\beta} - n(X'X)^{-1}R(nR'(X'X)^{-1}R)^{-1}(R'\widehat{\beta} - c) = \tilde{\beta}_{\text{cls}}$$

by (8.8). The same conclusion follows from (8.21): $\text{SSE}(\beta) = n\hat{\sigma}^2 + J^0(\beta)$, so SSE and J^0 differ by a term free of β and have the same constrained minimizer.

Exercise 8.11

Every ingredient of (8.23) has a probability limit:

- $\widehat{\beta} \xrightarrow{p} \beta$ (THEOREM 7.1, under Assumption 7.1);
- $\widehat{W} \xrightarrow{p} W > 0$ (Assumption 8.2), so $\widehat{W}^{-1} \xrightarrow{p} W^{-1}$ by the continuous mapping theorem (inversion is continuous at a non-singular matrix);
- $R'\widehat{W}^{-1}R \xrightarrow{p} R'W^{-1}R$, which is positive definite because $W^{-1} > 0$ and $\text{rank } R = q$; hence $(R'\widehat{W}^{-1}R)^{-1} \xrightarrow{p} (R'W^{-1}R)^{-1}$;
- $R'\widehat{\beta} - c \xrightarrow{p} R'\beta - c = \mathbf{0}$ by Assumption 8.1.

By the continuous mapping theorem,

$$\widetilde{\beta}_{\text{md}} \xrightarrow{p} \beta - W^{-1}R(R'W^{-1}R)^{-1}\mathbf{0} = \beta.$$

Consistency holds for *any* positive definite limiting weight, which is why CLS, EMD and every other member of the class are consistent when the restriction is true.

Exercise 8.12

Because $R'\beta = c$, we have $R'\widehat{\beta} - c = R'(\widehat{\beta} - \beta)$, and (8.23) can be written as

$$\sqrt{n}(\widetilde{\beta}_{\text{md}} - \beta) = \widehat{H} \sqrt{n}(\widehat{\beta} - \beta), \quad \widehat{H} = \mathbf{I}_k - \widehat{W}^{-1}R(R'\widehat{W}^{-1}R)^{-1}R'.$$

As in Exercise 8.11, $\widehat{H} \xrightarrow{p} H = \mathbf{I}_k - W^{-1}R(R'W^{-1}R)^{-1}R'$, and by THEOREM 7.3, $\sqrt{n}(\widehat{\beta} - \beta) \xrightarrow{d} N(0, V_\beta)$ with $V_\beta = Q_{XX}^{-1}\Omega Q_{XX}^{-1}$. By Slutsky's theorem,

$$\sqrt{n}(\widetilde{\beta}_{\text{md}} - \beta) \xrightarrow{d} N(0, HV_\beta H').$$

It remains to multiply out. Writing $G = (R'W^{-1}R)^{-1}$ (symmetric) and using the symmetry of W ,

$$\begin{aligned} HV_\beta H' &= (\mathbf{I}_k - W^{-1}RGR')V_\beta(\mathbf{I}_k - RGR'W^{-1}) \\ &= V_\beta - W^{-1}RGR'V_\beta - V_\beta RGR'W^{-1} + W^{-1}RGR'V_\beta RGR'W^{-1}, \end{aligned}$$

which is exactly (8.24).

Exercise 8.13

By Exercise 8.10, $\widetilde{\beta}_{\text{cls}} = \widetilde{\beta}_{\text{md}}$ with $\widehat{W} = \widehat{Q}_{XX}$. Under Assumption 7.2, $\widehat{Q}_{XX} \xrightarrow{p} Q_{XX} > 0$ by the WLLN, so Assumption 8.2 holds with $W = Q_{XX}$. THEOREM 8.7 then gives $\sqrt{n}(\widetilde{\beta}_{\text{cls}} - \beta) \xrightarrow{d} N(0, V_\beta(Q_{XX}))$, and substituting $W = Q_{XX}$ into (8.24) gives

$$\begin{aligned} V_{\text{cls}} &= V_\beta - Q_{XX}^{-1}R(R'Q_{XX}^{-1}R)^{-1}R'V_\beta - V_\beta R(R'Q_{XX}^{-1}R)^{-1}R'Q_{XX}^{-1} \\ &\quad + Q_{XX}^{-1}R(R'Q_{XX}^{-1}R)^{-1}R'V_\beta R(R'Q_{XX}^{-1}R)^{-1}R'Q_{XX}^{-1}, \end{aligned}$$

which is the statement of Theorem 8.8. Under homoskedasticity $V_\beta = \sigma^2 Q_{XX}^{-1}$ and this collapses to $\sigma^2(Q_{XX}^{-1} - Q_{XX}^{-1}R(R'Q_{XX}^{-1}R)^{-1}R'Q_{XX}^{-1})$, the asymptotic version of Theorem 8.3.

Exercise 8.14

With $W = V_\beta^{-1}$ we have $W^{-1} = V_\beta$, and $G = (R'W^{-1}R)^{-1} = (R'V_\beta R)^{-1}$. Substituting into (8.24),

$$\begin{aligned} V_\beta(V_\beta^{-1}) &= V_\beta - V_\beta RGR'V_\beta - V_\beta RGR'V_\beta + V_\beta \underbrace{RG R'V_\beta R GR'V_\beta}_{=G^{-1}} \\ &= V_\beta - 2V_\beta RGR'V_\beta + V_\beta RGR'V_\beta \\ &= V_\beta - V_\beta R(R'V_\beta R)^{-1}R'V_\beta, \end{aligned}$$

which is (8.26).

Exercise 8.15

By (8.26),

$$V_\beta - V_{\beta, \text{emd}} = V_\beta R(R'V_\beta R)^{-1}R'V_\beta.$$

Since $V_\beta > 0$ and $\text{rank } R = q$, $R'V_\beta R$ is positive definite and so is its inverse. For any $a \in \mathbb{R}^k$, setting $d = R'V_\beta a$,

$$a'(V_\beta - V_{\beta, \text{emd}})a = d'(R'V_\beta R)^{-1}d \geq 0.$$

So the difference is positive semi-definite and $V_{\beta, \text{emd}} \leq V_\beta$. The difference is non-zero (in fact of rank q), so imposing correct restrictions by EMD strictly improves efficiency in q directions — in contrast to CLS, which under heteroskedasticity need not improve on OLS (Exercise 8.17).

Exercise 8.16

The model is $Y = X_1'\beta_1 + X_2'\beta_2 + e$ with $\mathbb{E}[Xe] = \mathbf{0}$ and the true restriction $\beta_2 = \mathbf{0}$.

(8.29). By Exercise 8.1, $\tilde{\beta}_1 = (\sum_i X_{1i}X_{1i}')^{-1} \sum_i X_{1i}Y_i$. Because $\beta_2 = \mathbf{0}$, $Y = X_1'\beta_1 + e$ with $\mathbb{E}[X_1e] = \mathbf{0}$, so

$$\sqrt{n}(\tilde{\beta}_1 - \beta_1) = \left(\frac{1}{n} \sum_i X_{1i}X_{1i}' \right)^{-1} \frac{1}{\sqrt{n}} \sum_i X_{1i}e_i \xrightarrow{d} N(0, Q_{11}^{-1}\Omega_{11}Q_{11}^{-1})$$

by the WLLN, the CLT and Slutsky, with $Q_{11} = \mathbb{E}[X_1X_1']$ and $\Omega_{11} = \mathbb{E}[X_1X_1'e^2]$.

(8.30). The restriction is $R'\beta = \mathbf{0}$ with $R' = [\mathbf{0} \quad \mathbf{I}_{k_2}]$. Then $R'\hat{\beta} = \hat{\beta}_2$, $\hat{V}_\beta R = (\hat{V}_{12}', \hat{V}_{22})'$ and $R'\hat{V}_\beta R = \hat{V}_{22}$. By (8.25),

$$\begin{pmatrix} \bar{\beta}_1 \\ \bar{\beta}_2 \end{pmatrix} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} - \begin{pmatrix} \hat{V}_{12} \\ \hat{V}_{22} \end{pmatrix} \hat{V}_{22}^{-1} \hat{\beta}_2 = \begin{pmatrix} \hat{\beta}_1 - \hat{V}_{12} \hat{V}_{22}^{-1} \hat{\beta}_2 \\ \mathbf{0} \end{pmatrix}.$$

(8.31). By THEOREM 8.9 the asymptotic variance is the upper-left block of $V_\beta - V_\beta R(R'V_\beta R)^{-1}R'V_\beta$. Here $V_\beta R(R'V_\beta R)^{-1}R'V_\beta = (V_{12}', V_{22})'V_{22}^{-1}(V_{21}, V_{22})$, whose upper-left block is $V_{12}V_{22}^{-1}V_{21}$. Hence $\text{avar}[\bar{\beta}_1] = V_{11} - V_{12}V_{22}^{-1}V_{21}$.

Interpretation: $\bar{\beta}_1$ is $\hat{\beta}_1$ minus its linear projection on $\hat{\beta}_2$, using the known fact that $\hat{\beta}_2$ should be centred at zero. The variance formula is the corresponding residual variance — a control-variate argument.

Exercise 8.17

The design of Section 7.4 (Exercise 7.4) has

$$Q_{XX} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \quad \Omega = \begin{pmatrix} 1 & \frac{7}{8} \\ \frac{7}{8} & 1 \end{pmatrix}.$$

(8.32). $Q_{11 \cdot 2} = Q_{11} - Q_{12}Q_{22}^{-1}Q_{21} = 1 - \frac{1}{4} = \frac{3}{4}$. By (7.14), with all blocks scalars,

$$\text{avar}[\hat{\beta}_1] = \left(\frac{4}{3}\right)^2 \left(1 - \frac{1}{2} \cdot \frac{7}{8} - \frac{7}{8} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 \cdot \frac{1}{2}\right) = \frac{16}{9} \cdot \frac{6}{16} = \frac{2}{3}.$$

(8.33). By (8.29), $\text{avar}[\tilde{\beta}_1] = Q_{11}^{-1}\Omega_{11}Q_{11}^{-1} = 1$.

(8.34). We need the full $V_\beta = Q_{XX}^{-1}\Omega Q_{XX}^{-1}$. Here $Q_{XX}^{-1} = \frac{4}{3} \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$, and

$$Q_{XX}^{-1}\Omega = \frac{4}{3} \begin{pmatrix} \frac{9}{16} & \frac{3}{8} \\ \frac{3}{8} & \frac{9}{16} \end{pmatrix}, \quad V_\beta = \frac{16}{9} \begin{pmatrix} \frac{3}{8} & \frac{3}{32} \\ \frac{3}{32} & \frac{3}{8} \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} \end{pmatrix}.$$

(Consistently, $V_{11} = \frac{2}{3}$ agrees with (8.32).) By (8.31),

$$\text{avar}[\bar{\beta}_1] = V_{11} - \frac{V_{12}^2}{V_{22}} = \frac{2}{3} - \frac{1/36}{2/3} = \frac{2}{3} - \frac{1}{24} = \frac{5}{8}.$$

So $\text{avar}[\bar{\beta}_1] = \frac{5}{8} < \text{avar}[\hat{\beta}_1] = \frac{2}{3} < \text{avar}[\tilde{\beta}_1] = 1$. Imposing the *true* restriction $\beta_2 = 0$ by least squares *raises* the variance by 50%; imposing it efficiently lowers it. The reason is that X_2 , though irrelevant for the mean, is informative about the heteroskedastic error: $\mathbb{E}[X_2e] = 0$ is a valid moment condition that CLS throws away.

Exercise 8.18

Stack $\theta = (\beta_1', \beta_2')'$. By independence of the samples, $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$ with $V = \text{diag}(V_{\beta_1}, V_{\beta_2})$. The restriction is $R'\theta = \mathbf{0}$ with $R' = [\mathbf{I}_k \quad -\mathbf{I}_k]$.

(a) Here $R'\hat{\theta} = \hat{\beta}_1 - \hat{\beta}_2$, $R'\hat{V}R = \hat{V}_{\beta_1} + \hat{V}_{\beta_2} \equiv \hat{S}$ and $\hat{V}R = (\hat{V}_{\beta_1}', -\hat{V}_{\beta_2}')'$. By (8.25),

$$\tilde{\theta} = \begin{pmatrix} \hat{\beta}_1 - \hat{V}_{\beta_1}\hat{S}^{-1}(\hat{\beta}_1 - \hat{\beta}_2) \\ \hat{\beta}_2 + \hat{V}_{\beta_2}\hat{S}^{-1}(\hat{\beta}_1 - \hat{\beta}_2) \end{pmatrix}.$$

Using $\mathbf{I}_k - \hat{V}_{\beta_1}\hat{S}^{-1} = \hat{V}_{\beta_2}\hat{S}^{-1}$, both blocks equal $\hat{V}_{\beta_2}\hat{S}^{-1}\hat{\beta}_1 + \hat{V}_{\beta_1}\hat{S}^{-1}\hat{\beta}_2$, as they must. Using the identity $\hat{V}_{\beta_2}(\hat{V}_{\beta_1} + \hat{V}_{\beta_2})^{-1} = (\hat{V}_{\beta_1}^{-1} + \hat{V}_{\beta_2}^{-1})^{-1}\hat{V}_{\beta_1}^{-1}$ (verify by writing $\hat{V}_{\beta_1}^{-1} + \hat{V}_{\beta_2}^{-1} = \hat{V}_{\beta_1}^{-1}(\hat{V}_{\beta_1} + \hat{V}_{\beta_2})\hat{V}_{\beta_2}^{-1}$), the estimator is the *precision-weighted average*

$$\boxed{\tilde{\beta} = (\hat{V}_{\beta_1}^{-1} + \hat{V}_{\beta_2}^{-1})^{-1}(\hat{V}_{\beta_1}^{-1}\hat{\beta}_1 + \hat{V}_{\beta_2}^{-1}\hat{\beta}_2)}.$$

(b) Write $\Lambda = (V_{\beta_1}^{-1} + V_{\beta_2}^{-1})^{-1}$. Then $\sqrt{n}(\tilde{\beta} - \beta) = \Lambda(V_{\beta_1}^{-1}\sqrt{n}(\hat{\beta}_1 - \beta) + V_{\beta_2}^{-1}\sqrt{n}(\hat{\beta}_2 - \beta)) + o_p(1)$, and by independence

$$\sqrt{n}(\tilde{\beta} - \beta) \xrightarrow{d} N\left(0, \Lambda(V_{\beta_1}^{-1} + V_{\beta_2}^{-1})\Lambda\right) = N\left(0, (V_{\beta_1}^{-1} + V_{\beta_2}^{-1})^{-1}\right).$$

The precisions add, exactly as for pooling independent unbiased estimates.

(c) With unequal sizes the finite-sample variances are V_{β_j}/n_j , so the weights must use those. Let $\hat{S}_j = \hat{V}_{\beta_j}/n_j$ (the estimated covariance matrix of $\hat{\beta}_j$ itself, as reported by software). Then

$$\tilde{\beta} = (\hat{S}_1^{-1} + \hat{S}_2^{-1})^{-1}(\hat{S}_1^{-1}\hat{\beta}_1 + \hat{S}_2^{-1}\hat{\beta}_2), \quad \widehat{\text{var}}[\tilde{\beta}] = (\hat{S}_1^{-1} + \hat{S}_2^{-1})^{-1}.$$

Formally, with $n = n_1 + n_2$ and $n_1/n \rightarrow \pi \in (0, 1)$, $\sqrt{n}(\tilde{\beta} - \beta) \xrightarrow{d} N(0, (\pi V_{\beta_1}^{-1} + (1 - \pi)V_{\beta_2}^{-1})^{-1})$. The larger sample automatically receives more weight.

Exercise 8.19

The sample has $n = 4,230$. Marital codes 1–6 enter as dummies with never-married (code 7, 1,014 men) as the base. Standard errors are heteroskedasticity robust: HC1 for OLS, the sample analogue of THEOREM 8.8 (with the $n - k + q$ adjusted $\tilde{\Omega}$ of Section 8.7) for CLS, and the sample analogue of (8.26) for EMD.

	(a) OLS		(b) CLS		(c) EMD	
	coef.	s.e.	coef.	s.e.	coef.	s.e.
education	0.0874	0.0029	0.0875	0.0029	0.0876	0.0029
experience	0.0284	0.0028	0.0285	0.0028	0.0286	0.0028
experience ² /100	−0.0371	0.0055	−0.0372	0.0055	−0.0374	0.0055
married ₁ ($n = 2,734$)	0.1818	0.0250	0.1815	0.0250	0.1814	0.0250
married ₂ ($n = 1$)	−0.4790	—	−0.4786	—	−0.4786	—
married ₃ ($n = 119$)	−0.0388	0.0555	−0.0392	0.0555	−0.0392	0.0555
widowed ($n = 11$)	0.2372	0.1736	0.1815	0.0250	0.1814	0.0250
divorced ($n = 242$)	0.0744	0.0452	0.0560	0.0382	0.0514	0.0379
separated ($n = 109$)	0.0173	0.0529	0.0560	0.0382	0.0514	0.0379
intercept	1.1913	0.0462	1.1889	0.0460	1.1880	0.0460
SSE	1366.33		1366.60		1366.61	

(a) Note the cell sizes. Marital code 2 (married, Armed Forces spouse present) contains a *single* observation, so $\hat{\beta}_5$ is determined entirely by that one person's wage: its leverage is $h_{ii} = 1$, its residual is exactly zero, and no heteroskedasticity-robust standard error is meaningful for that coefficient (HC2 and HC3 are undefined; HC0/HC1 report a spurious number). This is the sparse-dummy problem of Section 4.16, and the entry is left blank. Widowed has only 11 observations, which is why its standard error is seven times that of married₁.

(b) Imposing $\beta_4 = \beta_7$ pools the 11 widowed men with the 2,734 married₁ men, so the common coefficient is essentially the married₁ estimate and the widowed standard error falls from 0.174 to 0.025. Pooling divorced and separated gives 0.056 (0.038), between the two unrestricted values. The other coefficients barely move. SSE rises by only 0.28, as it must (CLS minimizes SSE on a smaller set).

(c) EMD differs only slightly from CLS here — most visibly the pooled divorced/separated coefficient (0.051 versus 0.056) — and delivers weakly smaller standard errors, as THEOREM 8.9 predicts.

Its SSE is slightly larger than the CLS SSE, again as it must be: CLS is the SSE minimizer, EMD is not trying to be.

(d) The slope in experience is $\partial m / \partial \text{exp} = \beta_2 + 2\beta_3 \text{exp} / 100$, linear in experience. A linear function is non-negative on $[0, 50]$ iff it is non-negative at both endpoints: $\beta_2 \geq 0$ at $\text{exp} = 0$ and $\beta_2 + \beta_3 \geq 0$ at $\text{exp} = 50$. With $\beta_3 < 0$ (concavity) the binding condition is

$$\beta_2 + \beta_3 \geq 0.$$

At the CLS estimates $\tilde{\beta}_2 + \tilde{\beta}_3 = 0.0285 - 0.0372 = -0.0087 < 0$: the fitted profile peaks at about 38 years and turns down afterwards, violating the constraint.

(e) Because the inequality is violated at the equality-constrained estimate and the least squares criterion is convex, the inequality-constrained minimizer lies on the boundary $\beta_2 + \beta_3 = 0$. So the solution is CLS with the three equalities $\beta_4 = \beta_7$, $\beta_8 = \beta_9$, $\beta_2 + \beta_3 = 0$:

	coef.		s.e.		coef.		s.e.
education	0.0889	0.0029	married ₃	−0.0320	0.0554		
experience	0.0216	0.0016	widowed (= β_4)	0.1902	0.0247		
experience ² /100	−0.0216	0.0016	divorced	0.0638	0.0379		
married ₁	0.1902	0.0247	separated (= β_8)	0.0638	0.0379		
married ₂	−0.4899	—	intercept	1.2188	0.0447		

The experience profile is now $0.0216(\text{exp} - \text{exp}^2/100)$, which rises until exactly 50 years and is flat there. SSE rises to 1369.89. The standard errors are those of the equality-constrained estimator and should be read with care: the sampling distribution of an *inequality*-constrained estimator is non-normal near the boundary (it is a mixture of the constrained and unconstrained distributions), so these are conditional on the constraint binding, which it does here by a wide margin.

Exercise 8.20

(a) The conditions $\mathbb{E}[X^j e] = 0$ define m as the linear projection of Y on $Z(X) = (1, X, \dots, X^p)'$: the best p th-order polynomial approximation to Y (equivalently to the CEF, Exercise 2.19) in mean square. It equals $\mathbb{E}[Y | X = x]$ only if the CEF happens to be such a polynomial. Accordingly $g(x)$ is the slope of the best polynomial approximation at x — an approximation to the marginal effect $\partial \mathbb{E}[Y | X = x] / \partial x$, not in general equal to it.

(b) Estimate $\beta = (\beta_0, \dots, \beta_p)'$ by OLS of Y on $Z(X)$ and set

$$\hat{g}(x) = \hat{\beta}_1 + 2\hat{\beta}_2 x + \dots + p\hat{\beta}_p x^{p-1} = a(x)' \hat{\beta}, \quad a(x) = (0, 1, 2x, \dots, px^{p-1})'.$$

(c) For fixed x , $a(x)$ is non-random and $\hat{g}(x)$ is linear in $\hat{\beta}$, so by THEOREM 7.3

$$\sqrt{n}(\hat{g}(x) - g(x)) = a(x)' \sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, a(x)' V_\beta a(x)),$$

$V_\beta = Q_{ZZ}^{-1} \mathbb{E}[ZZ'e^2] Q_{ZZ}^{-1}$, provided $Q_{ZZ} > 0$ and $\mathbb{E}[X^{4p} e^2] < \infty$, $\mathbb{E}[X^{4p}] < \infty$.

(d) With $\hat{V}_{\hat{\beta}}$ a robust estimator of $\text{var}[\hat{\beta}]$,

$$s(\hat{g}(x)) = \sqrt{a(x)' \hat{V}_{\hat{\beta}} a(x)}, \quad \hat{g}(x) \pm 1.96 s(\hat{g}(x)).$$

(e) With $p = 2$, $m''(x) = 2\beta_2$, so concavity is the single linear inequality $\beta_2 \leq 0$. Inequality-constrained least squares (Section 8.13): if $\hat{\beta}_2 \leq 0$, the constraint is slack and the estimate is unrestricted OLS; if $\hat{\beta}_2 > 0$, the constrained minimizer sets $\tilde{\beta}_2 = 0$, i.e. OLS of Y on $(1, X)$. In either case $\tilde{g}(x) = \tilde{\beta}_1 + 2\tilde{\beta}_2 x$ (a constant slope in the boundary case).

(f) $m'(u) = \beta_1 + 2\beta_2 u$ is linear in u , so it is non-negative on $[x_L, x_U]$ iff it is non-negative at both endpoints:

$$\beta_1 + 2\beta_2 x_L \geq 0, \quad \beta_1 + 2\beta_2 x_U \geq 0.$$

Minimize SSE subject to these two linear inequalities (a quadratic program). If $\hat{\beta}$ satisfies both, keep OLS. If one is violated, impose it as an equality and re-estimate by CLS (checking that the other still holds); if both are violated at the solution, impose both. Then $\tilde{g}(x) = \tilde{\beta}_1 + 2\tilde{\beta}_2 x$.

Exercise 8.21

(a) **No — the ordering is the reverse, and it holds in every sample.** Residual variances measure in-sample fit, not sampling precision. OLS minimizes $\sum_i (Y_i - X_i' b)^2$ over all b , and CLS minimizes it over the constraint set, which contains $\bar{\beta}$. Hence algebraically

$$\hat{\sigma}^2 \leq \tilde{\sigma}^2 \leq \bar{\sigma}^2.$$

Moreover all three converge to the same σ^2 under the restriction, and the differences are $O_p(n^{-1})$. Efficiency is not fit (compare Exercise 4.21).

(b) Since $\hat{e}_i - \tilde{e}_i = X_i'(\tilde{\beta} - \hat{\beta})$,

$$\sum_{i=1}^n (\hat{e}_i - \tilde{e}_i)^2 = n(\tilde{\beta} - \hat{\beta})' \hat{Q}_{XX} (\tilde{\beta} - \hat{\beta}).$$

By (8.9), $\tilde{\beta} - \hat{\beta} = -\hat{Q}_{XX}^{-1} R(R' \hat{Q}_{XX}^{-1} R)^{-1} (R' \hat{\beta} - c)$, so

$$T_n = \frac{n(R' \hat{\beta} - c)' (R' \hat{Q}_{XX}^{-1} R)^{-1} (R' \hat{\beta} - c)}{\hat{\sigma}^2}.$$

Under $R' \beta = c$, $\sqrt{n}(R' \hat{\beta} - c) \xrightarrow{d} Z \sim N(0, R' V_\beta R)$, while $\hat{Q}_{XX} \xrightarrow{p} Q_{XX}$ and $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$. Hence

$$T_n \xrightarrow{d} Z' (\sigma^2 R' Q_{XX}^{-1} R)^{-1} Z \stackrel{d}{=} \sum_{j=1}^q \lambda_j \chi_{1,j}^2,$$

independent χ_1^2 variables weighted by the eigenvalues λ_j of $(\sigma^2 R' Q_{XX}^{-1} R)^{-1} R' V_\beta R$. In general this is *not* χ_q^2 .

(c) Yes. Under homoskedasticity $V_\beta = \sigma^2 Q_{XX}^{-1}$, so $R' V_\beta R = \sigma^2 R' Q_{XX}^{-1} R$, all $\lambda_j = 1$, and

$$T_n \xrightarrow{d} \chi_q^2.$$

T_n is then the homoskedastic Wald statistic (q times the F statistic of Section 9.14). Under heteroskedasticity it has the wrong asymptotic distribution and should not be compared with χ_q^2 critical values.

Exercise 8.22

(a) The ratio restriction is the linear restriction $\beta_1 = 2\beta_2$ (with $\beta_2 \neq 0$). Substituting,

$$X_1\beta_1 + X_2\beta_2 = \beta_2(2X_1 + X_2) = \beta_2 Z, \quad Z = 2X_1 + X_2.$$

So CLS is least squares of Y on the single constructed regressor Z :

$$\boxed{\tilde{\beta}_2 = \frac{\sum_{i=1}^n (2X_{1i} + X_{2i})Y_i}{\sum_{i=1}^n (2X_{1i} + X_{2i})^2}, \quad \tilde{\beta}_1 = 2\tilde{\beta}_2.}$$

(b) Under the restriction $Y = \beta_2 Z + e$ and $\mathbb{E}[Ze] = 2\mathbb{E}[X_1e] + \mathbb{E}[X_2e] = 0$. Hence

$$\sqrt{n}(\tilde{\beta}_2 - \beta_2) = \left(\frac{1}{n} \sum_i Z_i^2 \right)^{-1} \frac{1}{\sqrt{n}} \sum_i Z_i e_i \xrightarrow{d} N\left(0, \frac{\mathbb{E}[Z^2 e^2]}{(\mathbb{E}[Z^2])^2}\right),$$

and since $\tilde{\beta}_1 - \beta_1 = 2(\tilde{\beta}_2 - \beta_2)$,

$$\boxed{\sqrt{n}(\tilde{\beta}_1 - \beta_1) \xrightarrow{d} N\left(0, \frac{4\mathbb{E}[(2X_1 + X_2)^2 e^2]}{(\mathbb{E}[(2X_1 + X_2)^2])^2}\right),}$$

assuming $\mathbb{E}[(2X_1 + X_2)^2] > 0$ and the fourth moments are finite. (Because the problem has no intercept, X_1 and X_2 enter only through Z .)

Chapter 9

Hypothesis Testing

Solutions to all 29 exercises of Chapter 9 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 9.24–9.29 are computational; the numbers were produced from the data sets on the textbook website (cps09mar, Invest1993, Nerlove1963, MRW1992).

Notation: $T(\theta_0) = (\hat{\theta} - \theta_0)/s(\hat{\theta})$ is the t -statistic (9.1); for $\theta = r(\beta) \in \mathbb{R}^q$ the Wald statistic is $W = (\hat{\theta} - \theta_0)' \hat{V}_{\hat{\theta}}^{-1} (\hat{\theta} - \theta_0)$ with $\hat{V}_{\hat{\theta}} = \hat{R}' \hat{V}_{\hat{\beta}} \hat{R}$ (9.6). Under H_0 , $T \xrightarrow{d} N(0, 1)$ (THEOREM 9.1) and $W \xrightarrow{d} \chi_q^2$ (THEOREM 9.2). Unless stated otherwise $\hat{V}_{\hat{\beta}}$ is a heteroskedasticity-robust covariance matrix estimator.

Exercise 9.1

By definition $\bar{R}^2 = 1 - s^2/\tilde{\sigma}_Y^2$ with $\tilde{\sigma}_Y^2 = (n-1)^{-1} \sum_i (Y_i - \bar{Y})^2$, which does not depend on the regressors. So $\bar{R}^2$ increases iff s^2 decreases.

Let the short model have k regressors, with SSE_S and $s_S^2 = \text{SSE}_S/(n-k)$, and the long model $k+1$ regressors, with SSE_L and $s_L^2 = \text{SSE}_L/(n-k-1)$. Let x denote the added column and M the annihilator of the original X .

Step 1: the fall in SSE. By Frisch–Waugh–Lovell, $\hat{\beta}_{k+1} = (x'Mx)^{-1} x'MY$ and the long residual is $MY - Mx\hat{\beta}_{k+1}$, so

$$\text{SSE}_L = Y'MY - \frac{(x'MY)^2}{x'Mx} = \text{SSE}_S - \hat{\beta}_{k+1}^2 x'Mx.$$

By the partitioned inverse formula $[(X'X)^{-1}]_{k+1,k+1} = (x'Mx)^{-1}$ (with X now the long regressor matrix), so

$$\text{SSE}_S - \text{SSE}_L = \frac{\hat{\beta}_{k+1}^2}{[(X'X)^{-1}]_{k+1,k+1}} = T_{k+1}^2 s_L^2.$$

Step 2: compare the variance estimates. Using $\text{SSE}_L = (n-k-1)s_L^2$,

$$s_S^2 = \frac{\text{SSE}_L + T_{k+1}^2 s_L^2}{n-k} = s_L^2 \frac{n-k-1 + T_{k+1}^2}{n-k}.$$

Hence

$$s_L^2 < s_S^2 \iff n-k < n-k-1 + T_{k+1}^2 \iff T_{k+1}^2 > 1,$$

i.e. $\bar{R}^2$ rises iff $|T_{k+1}| > 1$ (with equality of the two $\bar{R}^2$ iff $|T_{k+1}| = 1$).

So “maximize $\bar{R}^2$ ” is a model-selection rule equivalent to including a variable whenever its t -ratio exceeds *one* — a far more lenient threshold than the conventional 1.96.

Exercise 9.2

(a) Each estimator satisfies $\sqrt{n}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, V_j)$ with $V_j = Q_{jj}^{-1}\Omega_j Q_{jj}^{-1}$, $Q_{jj} = \mathbb{E}[X_j X_j']$, $\Omega_j = \mathbb{E}[X_j X_j' e_j^2]$. Independence of the samples makes the two limits independent, so

$$\sqrt{n}((\hat{\beta}_2 - \hat{\beta}_1) - (\beta_2 - \beta_1)) \xrightarrow{d} N(0, V_1 + V_2).$$

(b) With robust estimators $\hat{V}_1, \hat{V}_2$ of V_1, V_2 ,

$$W = n(\hat{\beta}_2 - \hat{\beta}_1)'(\hat{V}_1 + \hat{V}_2)^{-1}(\hat{\beta}_2 - \hat{\beta}_1).$$

(If the covariance matrices are reported already divided by n , drop the leading n .)

(c) Under H_0 , $\sqrt{n}(\hat{\beta}_2 - \hat{\beta}_1) \xrightarrow{d} N(0, V_1 + V_2)$ and $\hat{V}_1 + \hat{V}_2 \xrightarrow{p} V_1 + V_2 > 0$, so $W \xrightarrow{d} \chi_k^2$. Reject at level α if W exceeds the $1 - \alpha$ quantile of χ_k^2 . This is the robust (unpooled) Chow test; unlike the classical version it does not require equal error variances in the two samples.

Exercise 9.3

(a) $|Z|$ has a continuous distribution, so by the continuous mapping theorem and the definition of convergence in distribution,

$$\mathbb{P}[|T| < c_1] + \mathbb{P}[|T| > c_2] \rightarrow \mathbb{P}[|Z| < c_1] + \mathbb{P}[|Z| > c_2] = \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha.$$

(b) **No.** Size is only half of a test's job; the other half is power, and this test spends half its size in the wrong place. A small $|T|$ means the estimate is close to $\theta = 0$ — the outcome *most* consistent with the null — yet the test rejects there.

Formally, under local alternatives $\theta_n = h/\sqrt{n}$ we have $T \xrightarrow{d} Z + \delta$ with $\delta \neq 0$. The proposed test puts $\alpha/2$ of its size in the region $\{|T| < c_1\}$, whose probability *falls* as $|\delta|$ grows, and only $\alpha/2$ in the upper tail, so its upper critical value is $c_2 = z_{1-\alpha/4}$ rather than $z_{1-\alpha/2}$. A direct calculation shows its power $\mathbb{P}[|Z + \delta| < c_1] + \mathbb{P}[|Z + \delta| > c_2]$ is below the power $\mathbb{P}[|Z + \delta| > z_{1-\alpha/2}]$ of the standard test for *every* $\delta \neq 0$ (for $\alpha = 0.05$: $c_1 = 0.031$, $c_2 = 2.241$ versus 1.960). The standard test dominates it. Rejection regions should be placed where the alternative puts relatively more probability than the null — here, the upper tail of $|T|$ only.

Exercise 9.4

(a) $W \xrightarrow{d} \chi_q^2$, a continuous distribution, so the rejection probability converges to $\mathbb{P}[\chi_q^2 < c_1] + \mathbb{P}[\chi_q^2 > c_2] = \alpha$.

(b) **No**, for the same reason as Exercise 9.3. W is a squared distance between $\hat{\theta}$ and $\mathbf{0}$. Under alternatives W converges to a non-central $\chi_q^2(\lambda)$, which is stochastically *larger* than χ_q^2 ; the lower tail becomes *less* likely, not more. Rejecting for very small W — estimates unusually close to the null value — wastes $\alpha/2$ of size where no alternative has more mass than the null. Its power is below that of “reject if $W > \chi_{q,1-\alpha}^2$ ” at every alternative (as a numerical check against the non-central χ_q^2 confirms for any q), so it is dominated. A small W is at most a signal of a misspecified variance estimate, never evidence against H_0 (compare Exercise 9.19).

Exercise 9.5

Wald test. The hypothesis is $R'\beta = \mathbf{0}$ with $R' = [\mathbf{I}_q \quad -\mathbf{I}_q]$ (plus zero columns for any other regressors). With $\widehat{V}_{\widehat{\beta}}$ partitioned conformably,

$$W = (\widehat{\beta}_1 - \widehat{\beta}_2)'(\widehat{V}_{11} - \widehat{V}_{12} - \widehat{V}_{21} + \widehat{V}_{22})^{-1}(\widehat{\beta}_1 - \widehat{\beta}_2) \xrightarrow{d} \chi_q^2$$

under H_0 (THEOREM 9.2). Reject if W exceeds the χ_q^2 critical value. The cross-covariance terms are essential; the two estimates come from the same regression and are correlated.

Equivalent reparameterization. Write $X_1'\beta_1 + X_2'\beta_2 = X_1'(\beta_1 - \beta_2) + (X_1 + X_2)'\beta_2$. Regress Y on X_1 and $X_1 + X_2$; the coefficient on X_1 is $\delta = \beta_1 - \beta_2$, and the Wald test of $\delta = \mathbf{0}$ is numerically identical to W (Exercise 4.19).

Exercise 9.6

No; the reasoning ignores multiple testing and conflates significance with importance.

Multiple testing. If none of the 20 regressors mattered and the t -ratios were roughly independent, each would exceed 2.5 in absolute value with probability $2\mathbb{P}[Z > 2.5] \approx 0.012$, and *at least one* would do so with probability $1 - (1 - 0.012)^{20} \approx 0.22$. Finding one “significant” variable among 20 is therefore unremarkable. A Bonferroni correction for 20 tests at familywise level 5% requires $|t| > 3.02$; 2.5 does not survive it.

Selection after the fact. The variable was singled out *because* its t -ratio was large, so its nominal p -value is not valid for that selected hypothesis. Its estimated coefficient is also biased away from zero by the selection.

Other deficiencies. (i) “Insignificant” does not mean “zero effect”: the other 19 coefficients may be imprecisely estimated rather than small. (ii) A t -ratio measures precision relative to the estimate, not predictive importance — “key predictor” is a claim about magnitude, which requires looking at the coefficients and the variation in the regressors. (iii) The regressors may be correlated, so individual t -tests say little about joint relevance; a joint Wald test of the 19 others is the appropriate check. (iv) A single regression cannot establish causality.

Exercise 9.7

The hypothesis is about the CEF at $x = 40$:

$$\theta = \mathbb{E}[Y \mid X = 40] = 40\beta_1 + 1600\beta_2 = R'\beta, \quad R = (40, 1600)', \quad H_0 : \theta = 20.$$

- (1) Estimate β by OLS of Y on (X, X^2) and compute a robust covariance matrix estimate $\widehat{V}_{\widehat{\beta}}$.
- (2) Compute $\widehat{\theta} = 40\widehat{\beta}_1 + 1600\widehat{\beta}_2$ and $s(\widehat{\theta}) = \sqrt{R'\widehat{V}_{\widehat{\beta}}R} = \sqrt{1600\widehat{V}_{11} + 2(64000)\widehat{V}_{12} + 1600^2\widehat{V}_{22}}$.
- (3) Form $T = (\widehat{\theta} - 20)/s(\widehat{\theta})$; under H_0 , $T \xrightarrow{d} N(0, 1)$.
- (4) Reject at 5% if $|T| > 1.96$; report the p -value $2(1 - \Phi(|T|))$ and the confidence interval $\widehat{\theta} \pm 1.96 s(\widehat{\theta})$.

Equivalently, re-centre the regressors at 40 — regress Y on $(X - 40)$ and $(X - 40)^2$ with an intercept — and the intercept is $\widehat{\theta}$ with its standard error reported directly.

Remark. As written the model has no intercept, which forces $\mathbb{E}[Y \mid X = 0] = 0$. That restriction is economically unmotivated; in practice one would include an intercept, in which case $R = (1, 40, 1600)'$. The testing procedure is otherwise unchanged.

Exercise 9.8

It is the same test (assuming X_1 and X_2 have the same dimension, so that $X_2 - X_1$ is defined). Rewrite

$$X_1' \beta_1 + X_2' \beta_2 = X_1' (\beta_1 + \beta_2) + (X_2 - X_1)' \beta_2,$$

so $\gamma_1 = \beta_1 + \beta_2$ and $\gamma_2 = \beta_2$. The paper's regressors are a non-singular linear transformation of yours, $Z = XC$ with $C = \begin{bmatrix} \mathbf{I} & -\mathbf{I} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$ (in the notation of Exercise 3.23). So $\hat{\gamma} = C^{-1} \hat{\beta}$ gives $\hat{\gamma}_2 = \hat{\beta}_2$ exactly, the residuals are identical, and by Exercise 4.19 the covariance matrix block for $\hat{\gamma}_2$ equals that for $\hat{\beta}_2$. The Wald statistics, and therefore the test decisions, are identical.

Exercise 9.9

They are not necessarily in conflict. Hypothesis tests make errors with known frequencies: under H_0 each study rejects with probability α , and under H_1 each fails to reject with probability $1 - \text{power}$, which can be large when the effect is modest or the sample small. Two independent studies of a true but moderate effect will disagree quite often.

A productive reading:

- “Failing to reject” is not “accepting H_0 ”. Compare the point estimates and confidence intervals rather than the binary decisions. If the estimates are similar and the second interval is simply wider, the studies agree and the second is less precise.
- If the intervals are close to non-overlapping, examine differences in samples, populations or specifications — heterogeneity is a substantive finding.
- Combine the evidence. With independent estimates $\hat{\theta}_1, \hat{\theta}_2$ and standard errors s_1, s_2 , the precision-weighted estimate (Exercise 8.18) has variance $(s_1^{-2} + s_2^{-2})^{-1}$ and is more informative than either test alone. Meta-analysis formalizes this.
- Be alert to publication bias: rejections are more likely to be reported, so the first result may overstate the evidence.

Exercise 9.10

Recall $V = \mathbb{E}[e^4] - \sigma^4$ (Exercise 7.8), estimated by $\hat{V} = n^{-1} \sum_i \hat{e}_i^4 - \hat{\sigma}^4$.

(a) $T_a = \frac{\sqrt{n}(\hat{\sigma}^2 - 1)}{\sqrt{\hat{V}}}$, asymptotically $N(0, 1)$ under H_0 .

(b) $g(u) = \sqrt{u}$ has $g'(\sigma^2) = 1/(2\sigma)$, so

$$\sqrt{n}(\hat{\sigma} - \sigma) \xrightarrow{d} N\left(0, \frac{V}{4\sigma^2}\right).$$

(c) $T_c = \frac{\sqrt{n}(\hat{\sigma} - 1)}{\sqrt{\hat{V}/(4\hat{\sigma}^2)}} = \frac{2\hat{\sigma}\sqrt{n}(\hat{\sigma} - 1)}{\sqrt{\hat{V}}}.$

(d) *The hypotheses are the same* ($\sigma > 0$, so $\sigma^2 = 1 \iff \sigma = 1$). *The tests are different.* Dividing,

$$\frac{T_a}{T_c} = \frac{\hat{\sigma}^2 - 1}{2\hat{\sigma}(\hat{\sigma} - 1)} = \frac{\hat{\sigma} + 1}{2\hat{\sigma}},$$

which converges to 1 under H_0 (so the tests are asymptotically equivalent there) but differs from 1 in any finite sample. If $\hat{\sigma} > 1$ then $|T_a| < |T_c|$; if $\hat{\sigma} < 1$ then $|T_a| > |T_c|$.

Contradiction. Suppose $\hat{\sigma} = 1.5$ and $T_c = 2.10$. Then $T_a = 2.10 \times \frac{2.5}{3} = 1.75$: the test on σ rejects at 5%, the test on σ^2 does not. Conversely with $\hat{\sigma} = 0.8$ and $T_c = -1.90$, $T_a = -2.14$: now only the σ^2 test rejects. The lesson is that Wald-type tests are *not invariant to reparameterization* of the null (Section 9.17); criterion-based statistics do not have this defect.

Exercise 9.11

No. With $m = \dots + \beta_2 \text{exp} + \beta_3 \text{exp}^2/100 + \dots$, experience has no effect on mean wages iff $\partial m / \partial \text{exp} = \beta_2 + 2\beta_3 \text{exp}/100 = 0$ for all experience levels, i.e. iff

$$H_0 : \beta_2 = 0 \text{ and } \beta_3 = 0.$$

The single t -test on β_2 tests only that the slope is zero *at* $\text{exp} = 0$; it can fail to reject even when experience matters strongly (and conversely the t -ratios can be individually insignificant when the two terms are jointly highly significant, because they are strongly collinear).

The appropriate method is the joint Wald test with $q = 2$: $W = \hat{\beta}'_{23}(\hat{V}_{23})^{-1}\hat{\beta}_{23}$ using the robust 2×2 covariance block, compared with χ^2_2 (5% critical value 5.99), or equivalently $F = W/2$.

Exercise 9.12

No. Power is the probability of rejecting *when* H_0 *is false*. Larger samples raise power only if the alternative is true; if H_0 is true the rejection probability stays at α however much data she collects. The statement presumes the conclusion.

Further, the p -value is a random variable. Under H_0 it is (asymptotically) uniform on $[0, 1]$, so $p = 0.08$ is perfectly ordinary under the null. And the strategy “collect more data until the test rejects” is optional stopping: with repeated looks, the probability of eventually rejecting a true null approaches one, so the procedure has no valid size.

A correct statement is: the estimate is imprecise, the data are consistent with a range of effects including zero, and a larger (pre-specified) sample would narrow the confidence interval. The researcher should report that interval.

Exercise 9.13

It is a statement about consistency of tests against *fixed* alternatives. If $\theta \neq \theta_0$ by any amount δ , then $T = \sqrt{n}(\hat{\theta} - \theta_0)/\sqrt{\hat{V}} \approx \sqrt{n}\delta/\sqrt{V} \rightarrow \pm\infty$, so the rejection probability tends to one. Since point null hypotheses (a coefficient exactly zero, an elasticity exactly one) are almost never literally true, in very large samples even economically trivial deviations become “statistically significant”.

Comments:

- Taken literally the view is false: if H_0 is exactly true, the rejection rate stays at α for every n .

- The practical lesson is that statistical significance and economic significance are different. With large n one should focus on the size of the estimate and its confidence interval, and ask whether the deviation from θ_0 matters.
- Local-power analysis (Section 9.19) formalizes the trade-off: a test at sample size n can detect deviations of order $n^{-1/2}$, so “significance” becomes a statement about ever smaller effects.

Exercise 9.14

(a) With $s(\hat{\theta}) = \sqrt{R'\hat{V}_{\hat{\beta}}R}$,

$$\hat{C} = \left[R'\hat{\beta} - 1.96\sqrt{R'\hat{V}_{\hat{\beta}}R}, \quad R'\hat{\beta} + 1.96\sqrt{R'\hat{V}_{\hat{\beta}}R} \right].$$

(b) $\theta_0 \notin \hat{C} \iff |R'\hat{\beta} - \theta_0| > 1.96 s(\hat{\theta}) \iff |T(\theta_0)| > 1.96$. Under H_0 , THEOREM 9.1 gives $T(\theta_0) \xrightarrow{d} N(0, 1)$, so

$$\mathbb{P}[\theta_0 \notin \hat{C}] = \mathbb{P}[|T(\theta_0)| > 1.96] \rightarrow \mathbb{P}[|Z| > 1.96] = 0.05.$$

Confidence intervals are exactly the sets of null values that the corresponding test does not reject.

Exercise 9.15

The simulated rejection frequency $\hat{p}$ is a sample mean of B Bernoulli draws, so $\hat{p} \approx N(p, p(1-p)/B)$. Test $H_0 : p = 0.05$.

(a) With $B = 100$, s.e. = $\sqrt{0.05 \times 0.95/100} = 0.0218$ and

$$z = \frac{0.07 - 0.05}{0.0218} = 0.92,$$

far from significant ($p \approx 0.36$). A 95% interval for the true size is $0.07 \pm 1.96\sqrt{0.07 \times 0.93/100} = [0.02, 0.12]$. **Disagree:** the evidence is consistent with correct size; 100 replications are simply too few to detect a 2-point distortion.

(b) With $B = 1000$, s.e. = 0.0069 and $z = 2.90$ ($p \approx 0.004$). Now there is statistically significant evidence that the true size exceeds 5%, so the colleague is *right* that the test over-rejects. Whether 7% versus 5% is a *practically* serious distortion is a separate judgement.

Exercise 9.16

(a) $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n (e_{1i}^2 - e_{2i}^2)$.

(b) $d_i = e_{1i}^2 - e_{2i}^2$ are i.i.d. with mean θ (note: the two errors are from the *same* observations, so they are correlated, and it is their difference that is averaged). If $\mathbb{E}[e_1^4] < \infty$ and $\mathbb{E}[e_2^4] < \infty$,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V), \quad V = \text{var}[e_1^2 - e_2^2] = \mathbb{E}[(e_1^2 - e_2^2)^2] - \theta^2.$$

(c) $\hat{V} = \frac{1}{n} \sum_{i=1}^n (e_{1i}^2 - e_{2i}^2 - \hat{\theta})^2$.

(d) $T = \sqrt{n}\hat{\theta}/\sqrt{\hat{V}}$; reject H_0 if $|T| > z_{1-\alpha/2}$. If H_0 is rejected with $T < 0$, model 1 fits better. (With estimated residuals the same statistic applies to first order under standard conditions.)

(e) Acceptance means the data do not *distinguish* the fit of the two models at the chosen level — not that they fit equally well. The test may have low power, and the confidence interval for θ should be reported. A degenerate case also deserves a warning: if the two models are nearly equivalent (e.g. $e_1 \approx e_2$), $V \approx 0$ and the normal approximation breaks down.

Exercise 9.17

Excluding X_2 means removing *every* term in which it appears: the level, its square, and the interaction.

(a) $H_0 : \beta_2 = \beta_4 = \beta_5 = 0$ against H_1 : at least one of $\beta_2, \beta_4, \beta_5$ is non-zero.

(b) With $\theta = (\beta_2, \beta_4, \beta_5)' = R'\beta$ (R selects those coefficients) and a robust covariance estimate,

$$W = \hat{\theta}' (R' \hat{V}_{\hat{\beta}} R)^{-1} \hat{\theta}.$$

(c) Under H_0 , $W \xrightarrow{d} \chi_3^2$.

(d) Reject at 5% if $W > 7.815$ (equivalently if the p -value $1 - G_3(W) < 0.05$); otherwise the data are consistent with excluding X_2 . Testing $\beta_2 = 0$ alone would be wrong: X_2 also affects Y through X_2^2 and $X_1 X_2$.

Exercise 9.18

Let $e = Y - X'\beta$ be the population projection error, $\beta = Q_{XX}^{-1} \mathbb{E}[XY]$.

(a) $\gamma = (\mathbb{E}[ZZ'])^{-1} \mathbb{E}[Ze]$: the projection coefficient of the first-stage error on Z .

(b) Since $\hat{e}_i = e_i - X_i'(\hat{\beta} - \beta)$,

$$\frac{1}{n} \sum_i Z_i \hat{e}_i = \frac{1}{n} \sum_i Z_i e_i - \left(\frac{1}{n} \sum_i Z_i X_i' \right) (\hat{\beta} - \beta) \xrightarrow{p} \mathbb{E}[Ze],$$

so $\tilde{\gamma} \xrightarrow{p} Q_{ZZ}^{-1} \mathbb{E}[Ze] = \gamma$.

(c) Treating $\hat{e}_i$ as data,

$$W = \tilde{\gamma}' \tilde{V}^{-1} \tilde{\gamma}, \quad \tilde{V} = (Z'Z)^{-1} \left(\sum_i Z_i Z_i' \tilde{u}_i^2 \right) (Z'Z)^{-1}.$$

(d) Using $\sqrt{n}(\hat{\beta} - \beta) = Q_{XX}^{-1} n^{-1/2} \sum_i X_i e_i + o_p(1)$,

$$\frac{1}{\sqrt{n}} \sum_i Z_i \hat{e}_i = \frac{1}{\sqrt{n}} \sum_i (Z_i - \Gamma X_i) e_i + o_p(1), \quad \Gamma = \mathbb{E}[ZX'] Q_{XX}^{-1}. \quad (*)$$

If $\mathbb{E}[ZX'] = \mathbf{0}$ then $\Gamma = \mathbf{0}$, the first-stage estimation error is negligible, and under H_0 ($\mathbb{E}[Ze] = \mathbf{0}$) $\sqrt{n}\tilde{\gamma} \xrightarrow{d} N(0, Q_{ZZ}^{-1} \mathbb{E}[ZZ'e^2] Q_{ZZ}^{-1})$. Under H_0 , $\tilde{u}_i = \hat{e}_i + o_p(1)$ and $n\tilde{V}$ is consistent for that variance, so

$$W \xrightarrow{d} \chi_\ell^2.$$

(e) **Yes.** By (*) the correct asymptotic variance is $Q_{ZZ}^{-1} \mathbb{E}[(Z - \Gamma X)(Z - \Gamma X)' e^2] Q_{ZZ}^{-1}$, whereas $n\tilde{V}$ estimates $Q_{ZZ}^{-1} \mathbb{E}[ZZ'e^2] Q_{ZZ}^{-1}$. So W is not asymptotically χ_ℓ^2 . Under homoskedasticity the correct middle matrix is $\sigma^2(Q_{ZZ} - \mathbb{E}[ZX'] Q_{XX}^{-1} \mathbb{E}[XZ']) \leq \sigma^2 Q_{ZZ}$: the naive variance is too large, W is

too small, and the test is conservative with reduced power. The remedy is to run one regression of Y on (X, Z) and Wald-test the coefficients on Z — by FWL that residualizes Z on X correctly.

Exercise 9.19

(a) One: the hypothesis imposes a single restriction. (Neither k nor n matters.)

(b) The 1% quantile of χ_1^2 is 0.000157, so $W = 0.34$ is *not* below it. The p -value is $\mathbb{P}[\chi_1^2 > 0.34] = 0.56$.

More importantly, even if W were below the 1% quantile it would *not* be a reason to reject. The Wald test rejects for *large* values of W ; a small W means $\hat{\beta}_2$ is close to zero relative to its standard error, which is evidence in favour of H_0 (Exercise 9.4). Do not reject.

Exercise 9.20

Yes, via the F statistic (9.13), which needs only the two SSEs:

$$F = \frac{(\text{SSE}_S - \text{SSE}_L)/q}{\text{SSE}_L/(n - k)} = \frac{(106 - 100)/3}{100/42} = 0.84,$$

with $q = 8 - 5 = 3$ restrictions and $n - k = 42$. The 5% critical value of $F_{3,42}$ is 2.83 and the p -value is 0.48. **Do not reject $H_0 : \beta_2 = 0$.**

Assumptions. This F statistic is the Wald statistic built with the homoskedastic covariance matrix (Section 9.14). Its validity (asymptotically $qF \xrightarrow{d} \chi_3^2$) requires *conditional homoskedasticity*; the exact $F_{3,42}$ distribution further requires normal errors. A heteroskedasticity-robust Wald statistic cannot be recovered from the reported summaries.

A consistency check worth doing. Both regressions share Y , so they must imply the same TSS. But $106/(1 - 0.20) = 132.5$ while $100/(1 - 0.26) = 135.1$: the reported numbers are mutually inconsistent (beyond what rounding of R^2 to two decimals can explain). Computing F from the R^2 's instead gives $F = (0.06/3)/(0.74/42) = 1.14$ ($p = 0.35$). Both versions lead to the same conclusion, but the discrepancy should be queried.

Exercise 9.21

The hypothesis is non-linear. It can be written in several algebraically equivalent ways, which lead to *different* Wald statistics; the product form is preferable because it avoids dividing by coefficients that may be near zero:

$$H_0 : \theta = r(\beta) = \beta_1\beta_4 - \beta_2\beta_3 = 0.$$

- (1) Estimate β by OLS with robust covariance estimate $\hat{V}_{\hat{\beta}}$.
- (2) Compute $\hat{\theta} = \hat{\beta}_1\hat{\beta}_4 - \hat{\beta}_2\hat{\beta}_3$ and the gradient $\hat{R} = \partial r / \partial \beta|_{\hat{\beta}} = (\hat{\beta}_4, -\hat{\beta}_3, -\hat{\beta}_2, \hat{\beta}_1)'$.
- (3) Form $W = \hat{\theta}^2 / (\hat{R}'\hat{V}_{\hat{\beta}}\hat{R})$, which under H_0 converges to χ_1^2 (THEOREM 9.2); reject at 5% if $W > 3.84$.

Because the Wald statistic depends on how the non-linear restriction is written (Section 9.17), a minimum distance statistic J^* (9.10) — which is invariant — is a sensible robustness check. The delta method also requires $R \neq 0$ at the truth, which fails if all four coefficients are zero.

Exercise 9.22

This repeats Exercise 9.7. Set $\theta = 40\beta_1 + 1600\beta_2$, compute $\hat{\theta} = 40\hat{\beta}_1 + 1600\hat{\beta}_2$ and $s(\hat{\theta}) = \sqrt{R'\hat{V}_{\hat{\beta}}R}$ with $R = (40, 1600)'$ and a robust $\hat{V}_{\hat{\beta}}$, form $T = (\hat{\theta} - 20)/s(\hat{\theta})$, and reject at 5% if $|T| > 1.96$. (Include an intercept in practice, in which case $R = (1, 40, 1600)'$.)

Exercise 9.23

Incomplete, and as stated not supported. The simulated frequency has Monte Carlo standard error $\sqrt{0.05 \times 0.95/200} = 0.0154$ under the hypothesis that the true size is 5%, so

$$z = \frac{0.070 - 0.050}{0.0154} = 1.30, \quad p \approx 0.19.$$

A 95% Monte Carlo interval for the true size is $0.070 \pm 1.96\sqrt{0.07 \times 0.93/200} = [0.035, 0.105]$, which contains 0.05. With 200 replications the experiment cannot distinguish a correctly sized test from one with true size 7%. She should report the Monte Carlo standard error and increase B (e.g. $B = 5000$ gives a standard error of about 0.003) before drawing a conclusion.

Exercise 9.24

The simulation used HC1 standard errors, $s(\hat{\theta}) = \hat{\theta}s(\hat{\beta})$ (delta method), and the random seed 20260916; a second run with $B = 20,000$ confirms the conclusions.

(a) **No.** Changing α to $\alpha + a$ adds the constant a to every Y_i . Since the regression contains an intercept, OLS absorbs the shift entirely into $\hat{\alpha}$: $\hat{\beta}$ (a function of $Y_i - \bar{Y}$) and the residuals $\hat{e}_i$ are unchanged. The robust variance of $\hat{\beta}$ depends only on the regressors and residuals, so $s(\hat{\beta})$ is unchanged, and hence so are $\hat{\theta}$, $s(\hat{\theta})$, T_{β} and T_{θ} .

(b)

	true value	mean over $B = 1000$	Monte Carlo s.e.
$\hat{\beta}$	1.000	0.999	0.016
$\hat{\theta}$	2.718	3.069	0.051

$\hat{\beta}$ shows no evidence of bias (it is exactly unbiased here since $\mathbb{E}[e | X] = 0$). $\hat{\theta}$ is clearly biased *upward* by about 13% ($t \approx 6.8$). This is Jensen's inequality: if $\hat{\beta} \approx N(\beta, v)$ then $\mathbb{E}[e^{\hat{\beta}}] = e^{\beta+v/2}$, and with $v \approx \sigma^2/(n \text{var}(X)) = 12/50 = 0.24$ this predicts $e^{1.12} = 3.065$ — almost exactly the simulated value.

(c) Asymptotic theory predicts $\mathbb{P}[T > 1.645] \rightarrow 0.05$ for both statistics. The simulations give

$$\mathbb{P}[T_{\beta} > 1.645] = 0.060 \text{ (s.e. 0.008)}, \quad \mathbb{P}[T_{\theta} > 1.645] = 0.003 \text{ (s.e. 0.002)}.$$

The t -statistic for β is reasonably well approximated (slight over-rejection, 0.059 with $B = 20,000$). The delta-method t -statistic for θ is badly behaved: its upper tail has almost no mass, while its lower tail has far too much ($\mathbb{P}[T_{\theta} < -1.645] = 0.123$). Because $s(\hat{\theta}) = \hat{\theta}s(\hat{\beta})$ grows with $\hat{\theta}$, large overestimates are divided by large standard errors, producing a strongly left-skewed statistic. A

one-sided test based on T_θ would be severely distorted even though T_θ is asymptotically $N(0, 1)$. The remedy (Exercise 5.9) is to test on the β scale, using $\theta \leq \theta_0 \iff \beta \leq \log \theta_0$.

Exercise 9.25

Standard errors are HC1.

(a)–(b)

	coefficient	s.e.	t	95% interval
Q	0.00277	0.00121	2.28	[0.00039, 0.00514]
C	0.00449	0.01523	0.29	[−0.02536, 0.03434]
D	0.01233	0.00623	1.98	[0.00013, 0.02454]
intercept	0.10078	0.00542	18.59	[0.09015, 0.11141]

(c) Joint test of $\beta_C = \beta_D = 0$: $W = 3.96$, χ^2_2 p -value 0.138 — not rejected at 5%. Test of $\beta_Q = 0$: $t = 2.28$, $p = 0.023$ — rejected, with a positive estimate.

Taken at face value this is consistent with q theory: Q predicts investment with the right sign, and cash flow and debt add nothing jointly. Two cautions. The debt coefficient is individually borderline ($t = 1.98$), so the joint non-rejection partly reflects the very noisy cash-flow estimate diluting it. And the economic magnitude of the Q effect is small: a one-unit rise in Q raises the investment rate by 0.003 against a sample mean of 0.113.

(d) The six non-linear terms are jointly highly significant: $W = 75.7$, χ^2_6 $p < 10^{-13}$. The linear specification is rejected, and with it the justification for reading the tests in (c) as tests of the theory: in the quadratic model Q^2 is significant (−0.00014, s.e. 0.00004) and D becomes significant (0.036, s.e. 0.012). The evidence for “ Q alone” is fragile to functional form.

Exercise 9.26

Standard errors are HC1 for OLS, the Section 8.7 estimator for CLS, and (8.26) for EMD.

	(a) OLS		(c) CLS		(d) EMD	
	coef.	s.e.	coef.	s.e.	coef.	s.e.
intercept	−3.527	1.719	−4.691	0.815	−4.745	0.809
$\log Q$	0.720	0.033	0.721	0.033	0.720	0.033
$\log P_L$	0.436	0.246	0.593	0.169	0.581	0.168
$\log P_K$	−0.220	0.324	−0.007	0.156	0.009	0.153
$\log P_F$	0.427	0.076	0.415	0.073	0.410	0.073

(b) Linear homogeneity of degree one in input prices: multiplying all three prices by λ multiplies cost by λ . It is implied by cost minimization (the cost function of any cost-minimizing firm is homogeneous of degree one in prices), so it is a test of the maintained economic theory. Note that $\beta_2 = 0.72 < 1$ indicates increasing returns to scale.

(c)–(d) Imposing homogeneity sharply improves precision of the price coefficients — the standard error on $\log P_K$ halves — and removes the theoretically awkward negative capital-price coefficient. CLS and EMD are close.

(e) $\hat{\beta}_3 + \hat{\beta}_4 + \hat{\beta}_5 = 0.643$; the robust Wald statistic for the sum equal to one is $W = 0.646$, $p = 0.42$. **Not rejected.**

(f) The efficient minimum distance statistic (9.10) is $J^* = n(\hat{\beta} - \tilde{\beta}_{\text{emd}})' \hat{V}_{\beta}^{-1} (\hat{\beta} - \tilde{\beta}_{\text{emd}}) = 0.646$, $p = 0.42$ — *numerically identical* to the Wald statistic, exactly as Section 9.12 shows for linear hypotheses. (The minimum distance statistic with the CLS weight $\hat{Q}_{XX}/s^2$ is 0.574; that version is valid only under homoskedasticity.) Homogeneity is consistent with the data.

Exercise 9.27

On the 98 non-oil countries, regress $\log(Y_{85}) - \log(Y_{60})$ on $\log Y_{60}$, $\log(I/\text{GDP})$, $\log(n + g + \delta)$ (with $g + \delta = 0.05$), $\log(\text{School})$ and an intercept. The estimates replicate the first column of Table 8.1:

	coef.	s.e.		coef.	s.e.
$\log \text{GDP}_{1960}$	-0.288	0.054	$\log(n + g + \delta)$	-0.506	0.236
$\log(I/\text{GDP})$	0.524	0.107	$\log(\text{School})$	0.231	0.066
			intercept	3.022	0.737

The restriction is $\theta = \beta_2 + \beta_3 + \beta_4 = 0$. We have $\hat{\theta} = 0.249$ with robust standard error 0.273, so $T = 0.91$ and $W = T^2 = 0.84$, χ_1^2 p -value 0.36. **The Solow restriction is not rejected**, which supports the constrained estimates reported in Table 8.1. The confidence interval $0.249 \pm 1.96(0.273) = [-0.29, 0.78]$ is wide, however, so the data are also consistent with sizeable departures.

Exercise 9.28

Note. The regression with marital-status variables is the one in **Table 4.2** (Section 4.19); Table 4.1 lists standard errors for a simple regression. We use the Table 4.2 specification and sample restriction (education ≥ 12); our code reproduces Table 4.2 exactly on the full sample. Standard errors are HC2 as in that table.

(a) Within this sub-sample the race and ethnicity indicators are constant: Hispanic ($= 0$), Black ($= 1$), American Indian, Asian and mixed race ($= 0$). All five must be dropped — the Black dummy would be collinear with the intercept and the others are identically zero.

(b) Marital status enters through four coefficients: married female, married male, formerly married female, formerly married male. The hypothesis is that all four are zero: $q = 4$ restrictions.

(c) With $n = 4,636$ and $k = 11$:

	coef.	s.e.
married female	0.057	0.024
married male	0.168	0.029
formerly married female	-0.013	0.027
formerly married male	0.080	0.036

The robust Wald statistic is $W = 42.6$, asymptotically χ_4^2 under H_0 , with $p = 1.3 \times 10^{-8}$. (Equivalently $F = W/4 = 10.6$ with $p = 1.4 \times 10^{-8}$ from $F_{4,4625}$.)

(d) Reject decisively: marital status is associated with mean wages of Black workers. The effect is concentrated among men — married Black men earn about 17% more than never-married Black men with the same education and experience, formerly married men about 8% more — while for women the married premium is smaller (6%) and the formerly-married effect is nil. This mirrors the pattern in the full sample (Table 4.2). As always, these are conditional associations, not causal effects.

Exercise 9.29

As in Exercise 9.28 we use the Table 4.2 specification and sample (education ≥ 12), and read “non-Hispanic” as applying to both racial groups. Race dummies other than Black are dropped; the regressors are education $\times$ each of the four groups (replacing the common education term), experience and its square, female, Black, Black $\times$ female (so each group has its own intercept), the two union dummies and the four marital-status dummies. $n = 37,786$ (white men 19,311, white women 13,839, Black men 2,157, Black women 2,479). HC2 standard errors.

(a)

return to education	coef.	s.e.
white men	0.1137	0.0020
white women	0.1164	0.0022
Black men	0.1108	0.0061
Black women	0.1267	0.0050

(b) H_0 : the four slopes are equal, i.e. three restrictions ($\beta_{wm} = \beta_{wf} = \beta_{bm} = \beta_{bf}$). The robust Wald statistic is $W = 6.72$, asymptotically χ_3^2 , $p = 0.081$ ($F = 2.24$, $p = 0.081$ from $F_{3,37770}$).

(c) At the 5% level we do not reject a common return to education; at 10% we would. The point estimates are strikingly similar — 11 to 12.7 log points per year — and the one group that stands out, Black women, is estimated about 1.3 points higher than the others with a standard error of 0.5. The evidence for heterogeneity is weak; a common return of roughly 11.5% is a reasonable summary.

Chapter 10

Resampling Methods

Solutions to all 31 exercises of Chapter 10 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 10.28–10.31 are computational. Bootstrap results use $B = 10,000$ replications (5,000 for the cluster bootstrap) with fixed seeds, so the last digit of a bootstrap quantity can differ across runs.

Notation. $\hat{\theta}_{(-i)}$ is the leave-one-out estimator and $\bar{\theta} = n^{-1} \sum_i \hat{\theta}_{(-i)}$; the jackknife variance estimator is (10.1),

$$\hat{V}_{\hat{\theta}}^{\text{jack}} = \frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(-i)} - \bar{\theta})(\hat{\theta}_{(-i)} - \bar{\theta})'.$$

$\hat{\theta}_b^*$ ($b = 1, \dots, B$) are bootstrap estimates with sample variance $\hat{V}_{\hat{\theta}}^{\text{boot}}$ (10.7) and empirical quantiles q_{α}^* . The percentile interval is $[q_{\alpha/2}^*, q_{1-\alpha/2}^*]$ (10.8); the BC interval uses $p^* = B^{-1} \sum_b \mathbf{1}\{\hat{\theta}_b^* \leq \hat{\theta}\}$, $z_0 = \Phi^{-1}(p^*)$ and quantile levels $x(\alpha) = \Phi(z_{\alpha} + 2z_0)$ (10.22)–(10.25); BC_a uses $x(\alpha) = \Phi(z_0 + (z_{\alpha} + z_0)/(1 - \hat{a}(z_{\alpha} + z_0)))$ with the jackknife acceleration $\hat{a}$ of Section 10.18. The percentile- t interval is $[\hat{\theta} - s(\hat{\theta})q_{1-\alpha/2}^*, \hat{\theta} - s(\hat{\theta})q_{\alpha/2}^*]$ where q^* are quantiles of $T^* = (\hat{\theta}^* - \hat{\theta})/s(\hat{\theta}^*)$.

Exercise 10.1

$\hat{\mu}_r$ is the sample mean of $Z_i = Y_i^r$. For a sample mean, $\hat{\mu}_{r,(-i)} = \frac{n}{n-1}\hat{\mu}_r - \frac{1}{n-1}Z_i$, the average of the leave-one-out estimates is $\hat{\mu}_r$ itself, and $\hat{\mu}_{r,(-i)} - \hat{\mu}_r = (\hat{\mu}_r - Z_i)/(n-1)$. Substituting into (10.1), as in (10.3),

$$\hat{V}_{\hat{\mu}_r}^{\text{jack}} = \frac{n-1}{n} \sum_{i=1}^n \frac{(Y_i^r - \hat{\mu}_r)^2}{(n-1)^2} = \frac{1}{n(n-1)} \sum_{i=1}^n (Y_i^r - \hat{\mu}_r)^2,$$

which is the unbiased sample variance of Y^r divided by n — the conventional estimator of $\text{var}[\hat{\mu}_r]$ (compare Exercise 4.1(d)).

Exercise 10.2

Applying the formula for $\hat{\theta}$ to the sample without observation i gives $\hat{\theta}_{(-i)} = a + C\hat{\beta}_{(-i)}$. Averaging, $\bar{\theta} = a + C\bar{\beta}$, so $\hat{\theta}_{(-i)} - \bar{\theta} = C(\hat{\beta}_{(-i)} - \bar{\beta})$ — the constant a cancels. Therefore

$$\hat{V}_{\hat{\theta}}^{\text{jack}} = \frac{n-1}{n} \sum_{i=1}^n C(\hat{\beta}_{(-i)} - \bar{\beta})(\hat{\beta}_{(-i)} - \bar{\beta})'C' = C\hat{V}_{\hat{\beta}}^{\text{jack}}C'.$$

Exercise 10.3

The essential point is that the leave-one-out estimator must redo *both* steps; deleting observation i only from the second step would ignore the sampling variability of $\hat{A}$. For each $i = 1, \dots, n$:

- (1) Re-estimate the first stage without observation i : $\hat{A}_{(-i)} = (\sum_{j \neq i} Z_j Z_j')^{-1} (\sum_{j \neq i} Z_j X_j')$.
- (2) Construct the generated regressors with this first stage, $\widehat{W}_j^{(-i)} = \hat{A}_{(-i)}' Z_j$ for $j \neq i$.

(3) Re-estimate the second stage without observation i :

$$\widehat{\beta}_{(-i)} = \left(\sum_{j \neq i} \widehat{W}_j^{(-i)} \widehat{W}_j^{(-i)'} \right)^{-1} \left(\sum_{j \neq i} \widehat{W}_j^{(-i)} Y_j \right).$$

Then set $\bar{\beta} = n^{-1} \sum_i \widehat{\beta}_{(-i)}$ and apply (10.1). The n first-stage fits can be computed cheaply with the rank-one updating formulas of Section 3.20 (e.g. $(Z'Z - Z_i Z_i')^{-1}$ by Sherman–Morrison), but correctness only requires that the whole estimator be recomputed. The result is asymptotically equivalent to the robust 2SLS covariance matrix, and automatically accounts for the generated regressor.

Exercise 10.4

On bootstrap sample b , $\widehat{\theta}^*(b) = a + C\widehat{\beta}^*(b)$, so the bootstrap mean is $\bar{\theta}^* = a + C\bar{\beta}^*$ and $\widehat{\theta}^*(b) - \bar{\theta}^* = C(\widehat{\beta}^*(b) - \bar{\beta}^*)$. Substituting into (10.7),

$$\widehat{V}_{\theta}^{\text{boot}} = \frac{1}{B-1} \sum_{b=1}^B C(\widehat{\beta}^*(b) - \bar{\beta}^*)(\widehat{\beta}^*(b) - \bar{\beta}^*)' C' = C \widehat{V}_{\beta}^{\text{boot}} C'.$$

This holds draw by draw, for the same bootstrap samples; in particular bootstrap standard errors for linear combinations can be computed from the bootstrap covariance matrix without re-running the bootstrap.

Exercise 10.5

Each bootstrap draw of the transformed parameter is $\widehat{\phi}_b^* = a + c\widehat{\beta}_b^*$. Suppose first $c > 0$. The map $u \mapsto a + cu$ is strictly increasing, so it preserves the ordering of the B draws: the j th order statistic of $\{\widehat{\phi}_b^*\}$ is $a + c$ times the j th order statistic of $\{\widehat{\beta}_b^*\}$ plus a . Every empirical quantile estimator is a function (an order statistic, or a weighted average of two adjacent ones) that commutes with increasing affine maps, so $q_{\alpha}^*(\widehat{\phi}^*) = a + c q_{\alpha}^*(\widehat{\beta}^*)$ and the interval is $[a + cL, a + cU]$.

If $c < 0$ the map reverses the order, $q_{\alpha}^*(\widehat{\phi}^*) = a + c q_{1-\alpha}^*(\widehat{\beta}^*)$, and the percentile interval is $[a + cU, a + cL]$ — the same set with endpoints listed in increasing order. (If $c = 0$ the parameter is the constant a .) More generally the percentile interval is transformation-respecting for any monotone map (Section 10.15).

Exercise 10.6

Conditional on the data, $\widehat{\theta}$ and $s(\widehat{\theta}) > 0$ are constants, so $T_b^* = (\widehat{\theta}_b^* - \widehat{\theta})/s(\widehat{\theta})$ is a strictly increasing affine function of $\widehat{\theta}_b^*$. By the argument of Exercise 10.5, the quantiles satisfy

$$q_{\alpha}^*(T^*) = \frac{q_{\alpha}^*(\widehat{\theta}^*) - \widehat{\theta}}{s(\widehat{\theta})} \implies \widehat{\theta} + s(\widehat{\theta}) q_{\alpha}^*(T^*) = q_{\alpha}^*(\widehat{\theta}^*).$$

Hence $C = [q_{\alpha/2}^*(\widehat{\theta}^*), q_{1-\alpha/2}^*(\widehat{\theta}^*)]$, the percentile interval. Studentizing by a *fixed* standard error adds nothing; the benefit of the percentile- t method comes from recomputing $s(\widehat{\theta}^*)$ in each bootstrap sample. (Note also the endpoints here use $q_{\alpha/2}^*$ on the left, whereas the percentile- t interval uses $-q_{1-\alpha/2}^*$ on the left.)

Exercise 10.7

We use two bootstrap analogues of standard facts, both proved exactly as their classical counterparts: (i) if $Z_n^* \xrightarrow{d^*} \xi$ then $Z_n^* = O_{p^*}(1)$, so $n^{-1/2}Z_n^* \xrightarrow{p^*} 0$; (ii) bootstrap Slutsky: if $A_n^* \xrightarrow{p^*} A$ (a constant) and $Z_n^* \xrightarrow{d^*} \xi$ then $A_n^*Z_n^* \xrightarrow{d^*} A\xi$ (apply THEOREM 10.5 to the jointly convergent pair (A_n^*, Z_n^*)).

Step 1: $\hat{\mu}^*$ is consistent. By (i), $\hat{\mu}^* - \hat{\mu} = n^{-1/2}\sqrt{n}(\hat{\mu}^* - \hat{\mu}) \xrightarrow{p^*} 0$, and by THEOREM 10.1, $\hat{\mu} \xrightarrow{p} \mu$ implies $\hat{\mu} \xrightarrow{p^*} \mu$. Hence $\hat{\mu}^* \xrightarrow{p^*} \mu$.

Step 2: mean value expansion. Let $\mathcal{N}$ be the neighbourhood on which g is continuously differentiable. On the event that $\hat{\mu}$ and $\hat{\mu}^*$ lie in a convex subset of $\mathcal{N}$ — whose bootstrap probability tends to one in probability by Step 1 — the mean value theorem applied row by row gives

$$g(\hat{\mu}^*) - g(\hat{\mu}) = G(\zeta^*)'(\hat{\mu}^* - \hat{\mu}),$$

where each row of $G(\zeta^*)$ is evaluated at a point on the segment between $\hat{\mu}$ and $\hat{\mu}^*$. Those points converge to μ in bootstrap probability, so by continuity of G and THEOREM 10.3, $G(\zeta^*) \xrightarrow{p^*} G(\mu) = G$.

Step 3: conclude. Multiplying by $\sqrt{n}$ and using (ii),

$$\sqrt{n}(g(\hat{\mu}^*) - g(\hat{\mu})) = G(\zeta^*)'\sqrt{n}(\hat{\mu}^* - \hat{\mu}) \xrightarrow{d^*} G'\xi.$$

If $\xi \sim N(0, V)$ then $G'\xi \sim N(0, G'VG)$.

Exercise 10.8

Let $W_i = h(Y_i)$. The bootstrap draws Y_i^* are i.i.d. from the EDF of $\{Y_i\}$, so $W_i^* = h(Y_i^*)$ are i.i.d. from the EDF of $\{W_i\}$, and $\hat{\mu}^* = n^{-1} \sum_i W_i^*$ is the bootstrap sample mean of the W_i .

- (1) The W_i are i.i.d. with $\mathbb{E}\|W\|^2 = \mathbb{E}\|h(Y)\|^2 < \infty$ and variance V , so the bootstrap CLT (THEOREM 10.4) gives $\sqrt{n}(\hat{\mu}^* - \hat{\mu}) \xrightarrow{d^*} N(0, V)$. (Theorem 10.4 is stated for $V > 0$; if V is singular apply it to a basis of the non-degenerate directions.)
- (2) By the WLLN, $\hat{\mu} \xrightarrow{p} \mu$.
- (3) g is continuously differentiable near μ by assumption.

These are the hypotheses of THEOREM 10.6, which yields $\sqrt{n}(\hat{\theta}^* - \hat{\theta}) = \sqrt{n}(g(\hat{\mu}^*) - g(\hat{\mu})) \xrightarrow{d^*} N(0, G'VG) = N(0, V_\theta)$.

Exercise 10.9

Write $\hat{V}^* = n^{-1} \sum_i h(Y_i^*)h(Y_i^*)' - \hat{\mu}^*\hat{\mu}^{*'}.$

- The matrices $h(Y_i)h(Y_i)'$ are i.i.d. with $\mathbb{E}\|hh'\| \leq \mathbb{E}\|h\|^2 < \infty$, hence uniformly integrable. The bootstrap WLLN (THEOREM 10.2) gives $n^{-1} \sum_i h(Y_i^*)h(Y_i^*)' \xrightarrow{p^*} \mathbb{E}[h(Y)h(Y)']$.
- Likewise $\hat{\mu}^* \xrightarrow{p^*} \mu$.
- By the bootstrap continuous mapping theorem (THEOREM 10.3), $\hat{V}^* \xrightarrow{p^*} \mathbb{E}[hh'] - \mu\mu' = V$.
- $G(\cdot)$ is continuous at μ , so $\hat{G}^* = G(\hat{\mu}^*) \xrightarrow{p^*} G$.

One more application of THEOREM 10.3 to the continuous map $(G, V) \mapsto G'VG$ gives $\widehat{V}_\theta^* \xrightarrow{p^*} G'VG = V_\theta$. Consequently $s(\widehat{\theta}^*)$ computed on bootstrap samples is consistent, which is what the percentile- t method needs.

Exercise 10.10

(a) No. $u \mapsto 1/u$ is strictly convex on $(0, \infty)$, so if $\bar{Y} > 0$ almost surely and $\bar{Y}$ is non-degenerate, Jensen's inequality gives $\mathbb{E}[1/\bar{Y}] > 1/\mathbb{E}[\bar{Y}] = \theta$. If instead Y can be negative (or has positive density at zero), $\bar{Y}$ can be arbitrarily close to 0 and $\mathbb{E}[\widehat{\theta}]$ typically does not exist at all (e.g. for normal Y).

(b) When $Y > 0$ almost surely the bias is *upward*, by (a). More generally a second-order expansion gives $\widehat{\theta} \approx \theta - \theta^2(\bar{Y} - \mu) + \theta^3(\bar{Y} - \mu)^2$, so the leading bias term is $\theta^3\sigma^2/n = \sigma^2/(n\mu^3) > 0$: the approximate bias is upward whenever the expansion is meaningful. Without positivity the exact direction cannot be signed because the expectation may not exist.

(c) Yes, provided μ is well separated from zero relative to $\sigma/\sqrt{n}$. The percentile interval is valid when some monotone transformation of the estimator is approximately symmetric and pivotal (Section 10.16). Here $\widehat{\theta} = \psi(\bar{Y})$ with $\psi(u) = 1/u$ strictly monotone on $(0, \infty)$, and $\bar{Y}$ is approximately normal and symmetric about μ . Because the percentile interval is transformation-respecting, the percentile interval for θ is just $[1/q_{1-\alpha/2}^*(\bar{Y}^*), 1/q_{\alpha/2}^*(\bar{Y}^*)]$ — the image of a good interval for μ — and it correctly captures the skewness of $\widehat{\theta}$ that a symmetric delta-method interval would miss. If μ is close to zero (many bootstrap $\bar{Y}^*$ near or below zero) neither method is reliable.

Exercise 10.11

For the drawn index,

$$Y^* = X'_{i'}\widehat{\beta} + \widehat{e}_{i'} = X'_{i'}\widehat{\beta} + (Y_{i'} - X'_{i'}\widehat{\beta}) = Y_{i'}.$$

So each drawn vector is exactly $(Y^*, X^*) = (Y_{i'}, X_{i'})$, the i' th observed pair. Drawing n indices with replacement and forming these vectors is literally drawing n pairs with replacement from the data, which is the nonparametric bootstrap. With the same random indices the two procedures produce identical bootstrap samples and therefore identical $\widehat{\beta}^*$ and statistics. (The procedure differs from the *residual* bootstrap, which draws X and $\widehat{e}$ with *independent* indices.)

Exercise 10.12

For the transformed parameter the estimate is $g(\widehat{\theta})$ and the bootstrap draws are $g(\widehat{\theta}_b^*)$. Since g is strictly increasing, $\mathbf{1}\{g(\widehat{\theta}_b^*) \leq g(\widehat{\theta})\} = \mathbf{1}\{\widehat{\theta}_b^* \leq \widehat{\theta}\}$ for every b , so

$$p_g^* = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{g(\widehat{\theta}_b^*) \leq g(\widehat{\theta})\} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\widehat{\theta}_b^* \leq \widehat{\theta}\} = p^*.$$

Yes, it extends: $z_0^* = \Phi^{-1}(p^*)$ depends on the data only through p^* , so it is invariant too, and so are the adjusted levels $x(\alpha) = \Phi(z_\alpha + 2z_0^*)$. Combined with quantile equivariance this is why the BC interval is transformation-respecting. (For a strictly decreasing g , $p_g^* = 1 - p^*$ up to ties and z_0^* changes sign, as it should.)

Exercise 10.13

(The printed statement has “ b ” where c is intended.) Let $\phi = a + c\beta$ with $\hat{\phi} = a + c\hat{\beta}$ and $s(\hat{\phi}) = |c|s(\hat{\beta})$ — a standard error of a scaled estimator scales with $|c|$, both in the sample and in each bootstrap sample. Then

$$T_{\phi}^* = \frac{\hat{\phi}^* - \hat{\phi}}{s(\hat{\phi}^*)} = \frac{c(\hat{\beta}^* - \hat{\beta})}{|c|s(\hat{\beta}^*)} = \text{sign}(c) T_{\beta}^*.$$

Case $c > 0$. $T_{\phi}^* = T_{\beta}^*$, the quantiles coincide, and

$$C_{\phi}^{pt} = [a + c\hat{\beta} - c s(\hat{\beta}) q_{1-\alpha/2}^*, a + c\hat{\beta} - c s(\hat{\beta}) q_{\alpha/2}^*] = [a + cL, a + cU].$$

Case $c < 0$. $T_{\phi}^* = -T_{\beta}^*$, so $q_{\phi,p}^* = -q_{\beta,1-p}^*$ and

$$C_{\phi}^{pt} = [a + c\hat{\beta} + |c|s(\hat{\beta}) q_{\alpha/2}^*, a + c\hat{\beta} + |c|s(\hat{\beta}) q_{1-\alpha/2}^*] = [a + cU, a + cL],$$

the same set with ordered endpoints. So percentile- t intervals are equivariant to *affine* maps. They are not equivariant to general monotone non-linear maps, because the standard error does not transform that way — which is the sense in which Section 10.19 says they are not transformation-respecting.

Exercise 10.14

T^* is centred at the wrong value. In the bootstrap world the true parameter is $\hat{\theta}$, not 0, so $T^* = \hat{\theta}^*/s(\hat{\theta}^*) \approx (\hat{\theta}^* - \hat{\theta})/s(\hat{\theta}^*) + T_n$ has a distribution centred near T_n itself. Its $1 - \alpha$ quantile is roughly $T_n + z_{1-\alpha}$, so the rule “ $T_n > q_{1-\alpha}^*$ ” essentially never rejects, whatever the truth: the test has size near zero and power near zero.

Correct versions: bootstrap the centred statistic $T^* = (\hat{\theta}^* - \hat{\theta})/s(\hat{\theta}^*)$, whose distribution mimics that of T_n under H_0 , and reject if $T_n > q_{1-\alpha}^*$; or generate bootstrap samples that *impose* H_0 (a restricted bootstrap) and then use the uncentred statistic.

Exercise 10.15

(a) $C^{pc} = [0.75, 1.30]$.

(c) **Neither.** The BC interval requires p^* , the fraction of bootstrap draws below $\hat{\theta} = 1.2$, and then quantiles at the adjusted levels $x(0.025)$ and $x(0.975)$ — none of which is reported. The percentile- t interval requires the quantiles of $T^* = (\hat{\theta}^* - \hat{\theta})/s(\hat{\theta}^*)$, which need a standard error computed on *each* bootstrap sample; the quantiles of $\hat{\theta}^*$ and a single $s(\hat{\theta})$ do not determine them.

The reported numbers are nonetheless informative: the interval is very asymmetric about 1.2 (0.45 below, 0.1 above) and 1.2 sits near the top of the bootstrap distribution, while the normal interval would be $1.2 \pm 0.39 = [0.81, 1.59]$. This is strong evidence of bias or skewness — exactly the situation in which the BC correction matters, so it is worth recovering p^* from the bootstrap draws.

Exercise 10.16

(a) Hold the regressors fixed at their sample values. Draw $e_1^*, \dots, e_n^*$ i.i.d. $N(0, \hat{\sigma}^2)$ and set $Y_i^* = X_i' \hat{\beta} + e_i^*$. Given the data F_n , $X_i' \hat{\beta}$ is a constant, so $Y_i^* | F_n \sim N(X_i' \hat{\beta}, \hat{\sigma}^2)$, independently across i .

(b) $\hat{\beta}^* = (X'X)^{-1}X'Y^* = \hat{\beta} + (X'X)^{-1}X'e^*$ with $e^* \mid F_n \sim N(\mathbf{0}, \hat{\sigma}^2 \mathbf{I}_n)$. A linear function of a normal vector is normal, with mean $\hat{\beta}$ and variance $(X'X)^{-1}X'\hat{\sigma}^2 \mathbf{I}_n X(X'X)^{-1} = \hat{\sigma}^2(X'X)^{-1}$.

(c) The bootstrap world is *exactly* a normal regression model with parameters $(\hat{\beta}, \hat{\sigma}^2)$ and the same design X . So THEOREMS 5.6–5.7 apply verbatim: with $s^{*2} = \hat{e}^* \hat{e}^* / (n - k)$,

$$\frac{\hat{\beta}_j^* - \hat{\beta}_j}{\sqrt{\hat{\sigma}^2[(X'X)^{-1}]_{jj}}} \sim N(0, 1), \quad \frac{(n - k)s^{*2}}{\hat{\sigma}^2} \sim \chi_{n-k}^2,$$

independent of each other. Their ratio, $T^* = (\hat{\beta}_j^* - \hat{\beta}_j) / \sqrt{s^{*2}[(X'X)^{-1}]_{jj}}$, is therefore exactly t_{n-k} (the unknown $\hat{\sigma}^2$ cancels). So the parametric percentile- t bootstrap reproduces the classical exact t interval, and nothing is gained over it in this model.

Exercise 10.17

(a) $\hat{m}(x) = x'\hat{\beta}$, $s(\hat{m}(x)) = \sqrt{x'\hat{V}_{\hat{\beta}}x}$ with a robust $\hat{V}_{\hat{\beta}}$, and $C = \hat{m}(x) \pm 1.96 s(\hat{m}(x))$ (Exercise 7.18).

(b) For $b = 1, \dots, B$: draw n pairs (Y_i^*, X_i^*) with replacement, compute $\hat{\beta}_b^*$ by OLS and $\hat{m}_b^* = x'\hat{\beta}_b^*$ (with x held fixed). The interval is $[q_{0.025}^*, q_{0.975}^*]$ of $\{\hat{m}_b^*\}$.

(c) In the same loop also compute the robust $\hat{V}_b^*$ on the bootstrap sample and $T_b^* = (x'\hat{\beta}_b^* - x'\hat{\beta}) / \sqrt{x'\hat{V}_b^*x}$. With $q_{0.025}^*, q_{0.975}^*$ the quantiles of $\{T_b^*\}$, the interval is

$$[\hat{m}(x) - s(\hat{m}(x)) q_{0.975}^*, \hat{m}(x) + s(\hat{m}(x)) q_{0.025}^*].$$

Exercise 10.18

(a) $\hat{\mu}_3 = n^{-1} \sum_{i=1}^n \hat{e}_i^3$ with $\hat{e}_i = Y_i - X_i'\hat{\beta}$.

(b) For $b = 1, \dots, B$ (e.g. $B = 10,000$):

- (1) draw n pairs (Y_i^*, X_i^*) with replacement;
- (2) re-estimate $\hat{\beta}^*$ by OLS *on the bootstrap sample* and form the bootstrap residuals $\hat{e}_i^* = Y_i^* - X_i^{*'}\hat{\beta}^*$;
- (3) compute $\hat{\mu}_3^* = n^{-1} \sum_i \hat{e}_i^{*3}$.

The 90% percentile interval is $[q_{0.05}^*, q_{0.95}^*]$ of $\{\hat{\mu}_{3,b}^*\}$. Step 2 matters: reusing the original residuals would ignore the estimation of β . Validity requires $\mathbb{E}[e^6] < \infty$ so that $\hat{\mu}_3$ is asymptotically normal; with heavy-tailed errors third-moment estimates are erratic and the interval should be read with caution.

Exercise 10.19

Estimate $\hat{\sigma}^2 = n^{-1} \sum_i \hat{e}_i^2$. For $b = 1, \dots, B$: resample n pairs, re-estimate $\hat{\beta}^*$ on the bootstrap sample, compute $\hat{\sigma}_b^{2*} = n^{-1} \sum_i (Y_i^* - X_i^{*'}\hat{\beta}^*)^2$. The $1 - \alpha$ percentile interval is $[q_{\alpha/2}^*, q_{1-\alpha/2}^*]$ of $\{\hat{\sigma}_b^{2*}\}$. Asymptotic validity needs $\mathbb{E}[e^4] < \infty$ (Exercise 7.8). Since $\hat{\sigma}^2$ is biased downward by the factor $(n - k)/n$ and has a right-skewed distribution, the BC or BC_a interval is a sensible refinement; by transformation-respecting properties the square roots of the endpoints give an interval for σ .

Exercise 10.20

- (1) On the data compute $\hat{\beta}_2$, a robust $s(\hat{\beta}_2)$ and $T = \hat{\beta}_2/s(\hat{\beta}_2)$.
- (2) For $b = 1, \dots, B$: draw n observations $(Y_i^*, X_{1i}^*, X_{2i}^*)$ with replacement; compute $\hat{\beta}_2^*$ and its robust standard error $s(\hat{\beta}_2^*)$ on the bootstrap sample; store the *centred* statistic $T_b^* = (\hat{\beta}_2^* - \hat{\beta}_2)/s(\hat{\beta}_2^*)$.
- (3) Let $q_{1-\alpha}^*$ be the $1 - \alpha$ quantile of $\{|T_b^*|\}$. Reject H_0 if $|T| > q_{1-\alpha}^*$. The bootstrap p -value is $B^{-1} \sum_b \mathbf{1}\{|T_b^*| > |T|\}$.

Centring at $\hat{\beta}_2$ makes the bootstrap distribution mimic the null distribution of T even though the resampling does not impose H_0 (compare Exercises 10.14 and 10.22). This symmetric test is second-order accurate.

Exercise 10.21

Let $\hat{\delta} = \hat{\beta}_1 - \hat{\beta}_2$ and $\hat{V}_\delta = R' \hat{V}_{\hat{\beta}} R$ with $R' = [\mathbf{I}_k \quad -\mathbf{I}_k]$.

- (1) Compute $W = \hat{\delta}' \hat{V}_\delta^{-1} \hat{\delta}$ on the data.
- (2) For each bootstrap sample of pairs compute $\hat{\delta}^*$ and $\hat{V}_\delta^*$ and store $W_b^* = (\hat{\delta}^* - \hat{\delta})' \hat{V}_\delta^{*-1} (\hat{\delta}^* - \hat{\delta})$ — centred at $\hat{\delta}$, not at zero.
- (3) Reject at level α if $W > q_{1-\alpha}^*(W^*)$; the p -value is $B^{-1} \sum_b \mathbf{1}\{W_b^* > W\}$.

Equivalently one can reparameterize as in Exercise 9.5 and bootstrap the Wald test on the coefficient of X_1 in the regression on $(X_1, X_1 + X_2)$.

Exercise 10.22

No; the method is wrong. She has made the error of Exercise 10.14: T^* is centred at 0, but the true value in the bootstrap world is $\hat{\alpha} = 2$. Her T^* is approximately distributed as $|\mathbf{N}(2, 1)|$, whose 95% quantile is about 3.6 — exactly what she found. The number 3.5 says nothing about the null distribution of T . With this procedure $|T|$ would essentially never exceed the “critical value”, however large the true effect.

The correct statistic is $T^* = (\hat{\alpha}^* - \hat{\alpha})/s(\hat{\alpha}^*)$; its 95% absolute quantile would typically be near 1.96 in a well-behaved regression, and the test would then reject since $|T| = 2$ is at least comparable. Her claim that “the bootstrap is more reliable” is true only for a correctly implemented bootstrap.

Exercise 10.23

- (a) $\hat{\theta} = \hat{\beta}_1 \hat{\beta}_2$. By the delta method (Exercise 7.14),

$$s(\hat{\theta}) = \sqrt{\hat{\beta}_2^2 \hat{V}_{11} + 2\hat{\beta}_1 \hat{\beta}_2 \hat{V}_{12} + \hat{\beta}_1^2 \hat{V}_{22}}, \quad C = \hat{\theta} \pm 1.96 s(\hat{\theta}),$$

with $\hat{V}_{j\ell}$ the robust covariance estimates of $(\hat{\beta}_1, \hat{\beta}_2)$.

- (b) Resample triples (Y_i, X_{1i}, X_{2i}) with replacement B times, compute $\hat{\theta}_b^* = \hat{\beta}_{1b}^* \hat{\beta}_{2b}^*$, and report $[q_{0.025}^*, q_{0.975}^*]$.
- (c) In each bootstrap sample also compute $s(\hat{\theta}_b^*)$ by the formula in (a) applied to the bootstrap estimates and their robust covariance matrix, and store $T_b^* = (\hat{\theta}_b^* - \hat{\theta})/s(\hat{\theta}_b^*)$. The interval is $[\hat{\theta} - s(\hat{\theta})q_{0.975}^*, \hat{\theta} - s(\hat{\theta})q_{0.025}^*]$.

When both coefficients are near zero the delta-method standard error is unreliable (Exercise 7.14), and the percentile interval is then the safest of the three.

Exercise 10.24

- (1) Compute $\hat{\theta} = \hat{\beta}_1/\hat{\beta}_2$ and, with gradient $\hat{R} = (1/\hat{\beta}_2, -\hat{\beta}_1/\hat{\beta}_2^2)'$ and robust $\hat{V}$, $s(\hat{\theta}) = \sqrt{\hat{R}'\hat{V}\hat{R}}$.
- (2) For $b = 1, \dots, B$: resample (Y_i, X_{1i}, X_{2i}) ; compute $\hat{\beta}^*$, its robust covariance $\hat{V}^*$, $\hat{\theta}^* = \hat{\beta}_1^*/\hat{\beta}_2^*$, $s(\hat{\theta}^*) = \sqrt{\hat{R}^*\hat{V}^*\hat{R}^*}$ and $T_b^* = (\hat{\theta}^* - \hat{\theta})/s(\hat{\theta}^*)$.
- (3) The 95% interval is $[\hat{\theta} - s(\hat{\theta})q_{0.975}^*, \hat{\theta} + s(\hat{\theta})q_{0.025}^*]$.

Ratios are fragile when β_2 is near zero relative to its standard error: some bootstrap draws of $\hat{\beta}_2^*$ then fall near zero, producing extreme $\hat{\theta}^*$. Check the proportion of such draws, and if it is non-negligible prefer a Fieller-type interval (invert the test of $\beta_1 - \theta_0\beta_2 = 0$ over θ_0). Exercise 10.30 shows the problem in data.

Exercise 10.25

No. The nonparametric (pairs) bootstrap resamples (Y_i, X_i) *jointly* from the EDF, which consistently estimates the joint distribution of (Y, X) — including any dependence of the conditional variance of e on X . Its theory (Theorems 10.7, 10.18–10.19) requires only i.i.d. sampling and moment conditions, and the bootstrap variance of $\hat{\beta}^*$ is consistent for the *robust* sandwich $Q_{XX}^{-1}\Omega Q_{XX}^{-1}$. What heteroskedasticity does invalidate are bootstraps that break the link between errors and regressors: the residual bootstrap (drawing $\hat{e}^*$ independently of X^*), or a percentile- t that studentizes with the homoskedastic standard error. If one wants to hold X fixed, the wild bootstrap preserves heteroskedasticity.

Exercise 10.26

Not correct. Resampling pairs from the data does not impose the null hypothesis of linearity. If the true CEF is non-linear, every bootstrap sample inherits the non-linearity, so the pseudo-true γ in the bootstrap world is $\tilde{\gamma} \neq 0$ and R^* is large — the critical value inflates exactly when H_0 is false, and the test loses its power. Under H_0 it is only accidentally close to right.

Two valid modifications:

- (1) *Centre the statistic*: keep the pairs bootstrap but store $R^* = ((\tilde{\gamma}^* - \tilde{\gamma})/s(\tilde{\gamma}^*))^2$, re-estimating both stages (including $\hat{Y}_i^*$) on each bootstrap sample.
- (2) *Impose the null* (preferred for specification tests): hold X_i fixed and generate $Y_i^* = X_i'\hat{\beta} + \hat{e}_i\eta_i$ with i.i.d. Rademacher η_i (a wild bootstrap, which also preserves heteroskedasticity). These data satisfy linearity exactly. Compute R^* on each sample exactly as on the data (re-estimating the first stage) and reject if R exceeds the 0.95 quantile of R^* .

Exercise 10.27

No. With $E[Xe] \neq 0$, $\hat{\beta} \xrightarrow{p} \beta^* = Q_{XX}^{-1}E[XY] = \beta + Q_{XX}^{-1}E[Xe] \neq \beta$. The bootstrap distribution of $\hat{\beta}^*$ approximates the sampling distribution of $\hat{\beta}$ around its own probability limit: it is centred at $\hat{\beta}$, which estimates β^* . The data contain no information about the distance $\beta - \beta^*$, because the observable distribution of (Y, X) identifies only the projection coefficient.

The BC adjustment corrects for *finite-sample median bias* of order n^{-1} — the discrepancy between the median of $\hat{\beta}$ and its limit — which the bootstrap can estimate. It cannot correct an *asymptotic* bias that does not vanish. The BC interval will cover β^* at roughly the nominal rate and β with probability tending to zero as n grows, since the interval shrinks around the wrong point. Endogeneity must be handled by identification (instruments, Chapter 12), not by resampling.

Exercise 10.28

Asymptotic standard errors are HC1; the jackknife is (10.1); the bootstrap is the pairs bootstrap with $B = 10,000$.

(a)

	coef.	asymptotic	jackknife	bootstrap
intercept	-3.527	1.719	1.788	1.713
$\log Q$	0.720	0.033	0.034	0.033
$\log P_L$	0.436	0.246	0.253	0.247
$\log P_K$	-0.220	0.324	0.336	0.323
$\log P_F$	0.427	0.076	0.078	0.077

The three methods agree closely. The jackknife is uniformly the largest, consistent with (10.5): it is approximately $\frac{n-1}{n}$ times HC3, typically the largest of the robust estimators.

(b) $\hat{\theta} = 0.643$ with standard errors 0.444 (asymptotic), 0.463 (jackknife) and 0.444 (bootstrap).

(c) Percentile 95% interval $[-0.221, 1.526]$; BC_a interval $[-0.165, 1.607]$ ($\hat{z}_0 = 0.025$, $\hat{a} = 0.026$). Both contain 1, agreeing with the Wald test of Exercise 9.26. The small positive z_0 and a shift the BC_a interval slightly to the right; the distribution of $\hat{\theta}$ is close to symmetric, so the refinements matter little here.

Exercise 10.29

$n = 98$, HC1 asymptotic standard errors, $B = 10,000$.

(a)

	coef.	asymptotic	jackknife	bootstrap
$\log \text{GDP}_{1960}$	-0.288	0.054	0.057	0.055
$\log(I/\text{GDP})$	0.524	0.107	0.112	0.106
$\log(n + g + \delta)$	-0.506	0.236	0.245	0.240
$\log(\text{School})$	0.231	0.066	0.069	0.066
intercept	3.022	0.737	0.756	0.740

(b) $\hat{\theta} = 0.249$; standard errors 0.273 (asymptotic), 0.281 (jackknife), 0.274 (bootstrap).

(c) Percentile $[-0.284, 0.803]$; BC $[-0.302, 0.784]$ ($\hat{z}_0 = -0.028$). Both contain zero, so the Solow restriction is again not rejected; the intervals are very close to the asymptotic interval $0.249 \pm 0.534 = [-0.285, 0.783]$. In this well-behaved linear problem all methods agree.

Exercise 10.30

The regression gives $\hat{\beta} = (0.0852, 0.0511, -0.1084, 1.1803)$ and

$$\hat{\theta} = \frac{\hat{\beta}_1}{\hat{\beta}_2 + 0.2\hat{\beta}_3} = \frac{0.0852}{0.0294} = 2.90.$$

(a) Standard errors: asymptotic (delta method, HC1) 0.760; jackknife 0.823; bootstrap ($B = 10,000$) 1.362.

(b) $\hat{\theta}$ is a ratio whose denominator $\hat{\beta}_2 + 0.2\hat{\beta}_3 = 0.029$ has standard error 0.0078 — a t -ratio of only 3.8. In some bootstrap samples the denominator is small, producing very large $\hat{\theta}^*$. The bootstrap distribution is strongly right-skewed: its quartiles are 2.49 and 3.56 (an interquartile range consistent with a standard deviation near 0.8), but the 95th and 99th percentiles are 4.94 and 6.59, and a few draws are far larger. The bootstrap variance is dominated by these tail draws. (For a ratio whose denominator has positive density at zero the exact variance is infinite, so $\hat{V}^{\text{boot}}$ is not even estimating a finite quantity; Sections 10.14–10.15 use exactly this ratio example to show that untrimmed bootstrap standard errors are unreliable.) The delta method and the jackknife are local linearizations that ignore the tails, so they are smaller and similar to each other. The discrepancy is a warning that a symmetric ± 1.96 standard-error interval is inappropriate here.

(c) The BC percentile interval is $[1.76, 5.31]$ ($\hat{z}_0 = -0.064$), compared with the percentile interval $[1.81, 5.63]$ and the delta-method interval $[1.41, 4.39]$. The bootstrap intervals are asymmetric about $\hat{\theta} = 2.90$, extending much further to the right, which correctly reflects the skewness of the ratio.

Exercise 10.31

Same sample as Exercise 4.26: $n = 5,269$ students in $G = 111$ schools. The pairs cluster bootstrap resamples entire schools with replacement ($B = 5,000$); the BC_a acceleration uses the delete-one-school jackknife, the natural cluster analogue.

	coef.	cluster asy.	cluster jack.	cluster boot.	BC_a 95% interval
tracking	0.1736	0.0767	0.0775	0.0762	[0.027, 0.321]
age at test	-0.0411	0.0134	0.0136	0.0130	[-0.065, -0.013]
girl	0.0817	0.0287	0.0289	0.0284	[0.025, 0.135]
contract teacher	0.1810	0.0377	0.0380	0.0374	[0.108, 0.255]
baseline percentile	0.0174	0.0007	0.0007	0.0007	[0.0160, 0.0188]
intercept	-0.7404	0.1305	0.1323	0.1287	[-0.995, -0.487]

The cluster bootstrap standard errors are essentially identical to the analytic cluster-robust ones (4.55)–(4.56) and the cluster jackknife, as Section 10.30 reports for the baseline specification. The bias and acceleration constants are tiny ($|\hat{z}_0| \leq 0.013$, $|\hat{a}| \leq 0.02$), so the BC_a intervals are close to $\hat{\beta} \pm 1.96 s$. All intervals exclude zero; the tracking effect is positive but imprecisely estimated, with an interval from about 0.03 to 0.32 standard deviations. Resampling *schools* rather than students is essential: an individual-level bootstrap would reproduce the misleadingly small standard errors of the non-clustered formula.

Chapter 11

Multivariate Regression

Solutions to all 15 exercises of Chapter 11 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022).

Notation follows Chapter 11. There are m equations $Y_j = X_j' \beta_j + e_j$. For observation i , Y_i and e_i are $m \times 1$ and $\bar{X}_i$ is the $m \times k$ matrix with the row vectors X_{ji}' on its block diagonal, so the system is $Y_i = \bar{X}_i \beta + e_i$ and $\hat{\beta} = (\sum_i \bar{X}_i' \bar{X}_i)^{-1} (\sum_i \bar{X}_i' Y_i)$. Also $\Omega = \mathbb{E}[\bar{X}' e e' \bar{X}]$, $Q = \mathbb{E}[\bar{X}' \bar{X}] = \text{diag}\{\mathbb{E}[X_j X_j']\}$ and $V_\beta = Q^{-1} \Omega Q^{-1}$. With common regressors $X_j = X$, $\bar{X}_i = \mathbf{I}_m \otimes X_i'$. The Kronecker product rules used below are the mixed-product rule $(A \otimes B)(C \otimes D) = AC \otimes BD$, $(A \otimes B)' = A' \otimes B'$ and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$ (Appendix A.21).

Exercise 11.1

$\bar{X}$ is a function of the regressors X , so by the law of iterated expectations and the conditioning theorem,

$$\Omega = \mathbb{E}[\bar{X}' e e' \bar{X}] = \mathbb{E}[\bar{X}' \mathbb{E}[e e' | X] \bar{X}] = \mathbb{E}[\bar{X}' \Sigma \bar{X}].$$

Exercise 11.2

With common regressors $\bar{X} = \mathbf{I}_m \otimes X'$, so $\bar{X}' = \mathbf{I}_m \otimes X$. Writing the $m \times 1$ vector as $e = e \otimes 1$ and applying the mixed-product rule,

$$\bar{X}' e = (\mathbf{I}_m \otimes X)(e \otimes 1) = (\mathbf{I}_m e) \otimes (X \cdot 1) = e \otimes X.$$

Hence, again by the mixed-product rule,

$$\bar{X}' e e' \bar{X} = (e \otimes X)(e \otimes X)' = (e \otimes X)(e' \otimes X') = e e' \otimes X X',$$

and taking expectations gives $\Omega = \mathbb{E}[e e' \otimes X X']$. In words: the (j, ℓ) block of Ω is $\mathbb{E}[e_j e_\ell X X']$.

Exercise 11.3

Start from Exercise 11.2. The Kronecker product is linear in its first argument and $X X'$ is a function of X , so

$$\Omega = \mathbb{E}[e e' \otimes X X'] = \mathbb{E}[\mathbb{E}[e e' | X] \otimes X X'] = \mathbb{E}[\Sigma \otimes X X'] = \Sigma \otimes \mathbb{E}[X X'].$$

Exercise 11.4

Substituting $Y_i = \bar{X}_i \beta + e_i$,

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n \bar{X}_i' \bar{X}_i \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \bar{X}_i' e_i \right).$$

Denominator. $\bar{X}_i' \bar{X}_i = \text{diag}\{X_{1i}X'_{1i}, \dots, X_{mi}X'_{mi}\}$, so by the WLLN (equation by equation) $n^{-1} \sum_i \bar{X}_i' \bar{X}_i \xrightarrow{p} Q$, which is positive definite because each block $\mathbb{E}[X_j X'_j] > 0$ under Assumption 7.2.

Numerator. $\bar{X}_i' e_i = (X_{1i}e'_{1i}, \dots, X_{mi}e'_{mi})'$ is i.i.d. with mean zero by (11.9). Its covariance matrix Ω is finite: each block $\mathbb{E}[X_j X'_j e_j e'_j]$ is bounded by Cauchy–Schwarz using the fourth moments in Assumption 7.2. The multivariate CLT gives $n^{-1/2} \sum_i \bar{X}_i' e_i \xrightarrow{d} N(0, \Omega)$.

Combine. By the continuous mapping theorem and Slutsky, $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} Q^{-1}N(0, \Omega) = N(0, Q^{-1}\Omega Q^{-1})$.

The content beyond single-equation theory is the *joint* limit: the off-diagonal blocks of V_β , namely $Q_{jj}^{-1} \mathbb{E}[X_j X'_j e_j e'_j] Q_{\ell\ell}^{-1}$, are generally non-zero, because the errors are correlated across equations.

Exercise 11.5

$\bar{X}' \bar{X} = (\mathbf{I}_m \otimes X)(\mathbf{I}_m \otimes X') = \mathbf{I}_m \mathbf{I}_m \otimes X X' = \mathbf{I}_m \otimes X X'$, and taking expectations $Q = \mathbf{I}_m \otimes \mathbb{E}[X X']$.

Exercise 11.6

Write $Q_{XX} = \mathbb{E}[X X']$. By Exercises 11.5 and 11.3, $Q^{-1} = \mathbf{I}_m \otimes Q_{XX}^{-1}$ and $\Omega = \Sigma \otimes Q_{XX}$. By the mixed-product rule,

$$V_\beta = (\mathbf{I}_m \otimes Q_{XX}^{-1})(\Sigma \otimes Q_{XX})(\mathbf{I}_m \otimes Q_{XX}^{-1}) = \Sigma \otimes Q_{XX}^{-1} Q_{XX} Q_{XX}^{-1} = \Sigma \otimes Q_{XX}^{-1}.$$

So $\text{acov}(\hat{\beta}_j, \hat{\beta}_\ell) = \sigma_{j\ell} Q_{XX}^{-1}$: the coefficient estimates of two equations are correlated exactly in proportion to the correlation of their errors.

Exercise 11.7

Assumption 7.3 states that $r(\cdot)$ is continuously differentiable at the true β with R of full rank. THEOREM 11.1 gives $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V_\beta)$. The delta method (THEOREM 6.8) applied to $\hat{\theta} = r(\hat{\beta})$ yields $\sqrt{n}(r(\hat{\beta}) - r(\beta)) \xrightarrow{d} R'N(0, V_\beta) = N(0, R'V_\beta R)$. Because $\hat{\beta}$ is estimated *jointly*, V_β contains the cross-equation covariances and R may load on coefficients from several equations — which is exactly what single-equation theory cannot deliver.

Exercise 11.8

Write $\hat{Q} = n^{-1} \sum_i \bar{X}_i' \bar{X}_i \xrightarrow{p} Q$ and $\delta = \hat{\beta} - \beta = O_p(n^{-1/2})$. Then $\hat{e}_i = e_i - \bar{X}_i \delta$.

Robust estimator. Expanding,

$$\frac{1}{n} \sum_i \bar{X}_i' \hat{e}_i \hat{e}_i' \bar{X}_i = \frac{1}{n} \sum_i \bar{X}_i' e_i e_i' \bar{X}_i - \frac{1}{n} \sum_i \bar{X}_i' e_i \delta' \bar{X}_i \bar{X}_i - \frac{1}{n} \sum_i \bar{X}_i' \bar{X}_i \delta e_i' \bar{X}_i + \frac{1}{n} \sum_i \bar{X}_i' \bar{X}_i \delta \delta' \bar{X}_i \bar{X}_i.$$

The first term converges to Ω by the WLLN, since $\mathbb{E}[\bar{X}' e e' \bar{X}] \leq \mathbb{E}[\bar{X}' \bar{X}] \mathbb{E}[e e'] \leq (\mathbb{E}[\bar{X}' \bar{X}]^4 \mathbb{E}[e e']^4)^{1/2} < \infty$. For the second (and symmetrically the third),

$$\left\| \frac{1}{n} \sum_i \bar{X}_i' e_i \delta' \bar{X}_i \bar{X}_i \right\| \leq \|\delta\| \frac{1}{n} \sum_i \|\bar{X}_i\|^3 \|e_i\| = o_p(1) O_p(1),$$

using Hölder's inequality, $\mathbb{E}[\bar{X}' \bar{X}]^3 \mathbb{E}[e e'] \leq (\mathbb{E}[\bar{X}' \bar{X}]^4 \mathbb{E}[e e']^4)^{1/4} < \infty$. The last term is bounded by $\|\delta\|^2 n^{-1} \sum_i \|\bar{X}_i\|^4 = o_p(1)$. Hence the middle matrix converges to Ω , and by the continuous

mapping theorem

$$n\hat{V}_{\hat{\beta}} = \hat{Q}^{-1} \left(\frac{1}{n} \sum_i \bar{X}_i' \hat{e}_i \hat{e}_i' \bar{X}_i \right) \hat{Q}^{-1} \xrightarrow{p} Q^{-1} \Omega Q^{-1} = V_{\beta}.$$

Homoskedastic estimator. The same expansion without the $\bar{X}_i'$ factors shows $\hat{\Sigma} = n^{-1} \sum_i \hat{e}_i \hat{e}_i' = n^{-1} \sum_i e_i e_i' + o_p(1) \xrightarrow{p} \Sigma$. Then

$$\frac{1}{n} \sum_i \bar{X}_i' \hat{\Sigma} \bar{X}_i = \frac{1}{n} \sum_i \bar{X}_i' \Sigma \bar{X}_i + \frac{1}{n} \sum_i \bar{X}_i' (\hat{\Sigma} - \Sigma) \bar{X}_i,$$

where the first term converges to $\mathbb{E}[\bar{X}' \Sigma \bar{X}]$ and the second is bounded by $\|\hat{\Sigma} - \Sigma\| n^{-1} \sum_i \|\bar{X}_i\|^2 = o_p(1)$. So $n\hat{V}_{\hat{\beta}}^0 \xrightarrow{p} V_{\beta}^0$, which equals V_{β} under (11.8) by Exercise 11.1.

Exercise 11.9

Let $\Sigma^{-1/2}$ be the symmetric inverse square root, so $\Sigma^{-1/2} \Sigma^{-1/2} = \Sigma^{-1}$ and $\Sigma^{-1/2} \Sigma \Sigma^{-1/2} = \mathbf{I}_m$. Premultiplying $Y_i = \bar{X}_i \beta + e_i$ by $\Sigma^{-1/2}$ gives

$$\tilde{Y}_i = \tilde{X}_i \beta + \tilde{e}_i, \quad \tilde{Y}_i = \Sigma^{-1/2} Y_i, \quad \tilde{X}_i = \Sigma^{-1/2} \bar{X}_i, \quad \tilde{e}_i = \Sigma^{-1/2} e_i,$$

with $\mathbb{E}[\tilde{e}_i | X] = 0$ and $\mathbb{E}[\tilde{e}_i \tilde{e}_i' | X] = \Sigma^{-1/2} \Sigma \Sigma^{-1/2} = \mathbf{I}_m$: the transformed errors are uncorrelated across equations and homoskedastic. Systems least squares on the transformed model is

$$\begin{aligned} \left(\sum_i \tilde{X}_i' \tilde{X}_i \right)^{-1} \left(\sum_i \tilde{X}_i' \tilde{Y}_i \right) &= \left(\sum_i \bar{X}_i' \Sigma^{-1/2} \Sigma^{-1/2} \bar{X}_i \right)^{-1} \left(\sum_i \bar{X}_i' \Sigma^{-1/2} \Sigma^{-1/2} Y_i \right) \\ &= \left(\sum_i \bar{X}_i' \Sigma^{-1} \bar{X}_i \right)^{-1} \left(\sum_i \bar{X}_i' \Sigma^{-1} Y_i \right), \end{aligned}$$

which is (11.16).

Exercise 11.10

Stack all n observations: $\mathbf{Y} = \bar{\mathbf{X}} \beta + \mathbf{e}$ with $\mathbf{e}$ of dimension nm . By independence across i and (11.8), $\mathbb{E}[\mathbf{e} \mathbf{e}' | X]$ is block diagonal with blocks Σ , i.e. $\mathbf{I}_n \otimes \Sigma$. Its inverse square root is $(\mathbf{I}_n \otimes \Sigma)^{-1/2} = \mathbf{I}_n \otimes \Sigma^{-1/2}$, since $(\mathbf{I}_n \otimes \Sigma^{-1/2})(\mathbf{I}_n \otimes \Sigma^{-1/2}) = \mathbf{I}_n \otimes \Sigma^{-1}$. Premultiplying by $\mathbf{I}_n \otimes \Sigma^{-1/2}$ makes the error covariance $(\mathbf{I}_n \otimes \Sigma^{-1/2})(\mathbf{I}_n \otimes \Sigma)(\mathbf{I}_n \otimes \Sigma^{-1/2}) = \mathbf{I}_n \otimes \mathbf{I}_m = \mathbf{I}_{nm}$. Least squares on the transformed system gives

$$\hat{\beta}_{\text{gls}} = (\bar{\mathbf{X}}' (\mathbf{I}_n \otimes \Sigma^{-1}) \bar{\mathbf{X}})^{-1} (\bar{\mathbf{X}}' (\mathbf{I}_n \otimes \Sigma^{-1}) \mathbf{Y}),$$

which is (11.17). Writing out the block-diagonal products recovers (11.16), as the text shows.

Exercise 11.11

Throughout, the model is (11.15): $\mathbb{E}[e | X] = 0$ where X collects *all* regressors. Substituting $Y_i = \bar{X}_i \beta + e_i$ into (11.18),

$$\sqrt{n}(\hat{\beta}_{\text{sur}} - \beta) = \left(\frac{1}{n} \sum_i \bar{X}_i' \hat{\Sigma}^{-1} \bar{X}_i \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_i \bar{X}_i' \hat{\Sigma}^{-1} e_i \right).$$

Denominator. By Exercise 11.8, $\hat{\Sigma} \xrightarrow{p} \Sigma$, hence $\hat{\Sigma}^{-1} \xrightarrow{p} \Sigma^{-1}$, and the argument used there gives $n^{-1} \sum_i \bar{X}'_i \hat{\Sigma}^{-1} \bar{X}_i \xrightarrow{p} \mathbb{E}[\bar{X}' \Sigma^{-1} \bar{X}]$, which is positive definite.

Numerator. Split it as

$$\frac{1}{\sqrt{n}} \sum_i \bar{X}'_i \Sigma^{-1} e_i + \frac{1}{\sqrt{n}} \sum_i \bar{X}'_i (\hat{\Sigma}^{-1} - \Sigma^{-1}) e_i.$$

The second term is a finite sum $\sum_{j,\ell} [\hat{\Sigma}^{-1} - \Sigma^{-1}]_{j\ell} n^{-1/2} \sum_i \bar{X}'_{ij} e_{\ell i}$, where $\bar{X}_{ij}$ is the j th row of $\bar{X}_i$. Each $n^{-1/2} \sum_i \bar{X}'_{ij} e_{\ell i}$ is $O_p(1)$ by the CLT, because $\mathbb{E}[X_j e_\ell] = \mathbb{E}[X_j \mathbb{E}[e_\ell | X]] = 0$ — this *cross-equation* orthogonality is why SUR needs $\mathbb{E}[e | X] = 0$ rather than only the equation-by-equation condition (11.9). Each coefficient is $o_p(1)$, so the second term is $o_p(1)$. The first term has i.i.d. mean-zero summands with covariance $\mathbb{E}[\bar{X}' \Sigma^{-1} e e' \Sigma^{-1} \bar{X}] = \mathbb{E}[\bar{X}' \Sigma^{-1} \mathbb{E}[e e' | X] \Sigma^{-1} \bar{X}] = \mathbb{E}[\bar{X}' \Sigma^{-1} \bar{X}]$ by (11.8), so it converges to $N(0, \mathbb{E}[\bar{X}' \Sigma^{-1} \bar{X}])$.

Combine. With $A = \mathbb{E}[\bar{X}' \Sigma^{-1} \bar{X}]$, Slutsky gives the limit $N(0, A^{-1} A A^{-1}) = N(0, A^{-1}) = N(0, V_\beta^*)$. Estimating Σ has no effect on the first-order distribution.

Exercise 11.12

Write $A = \mathbb{E}[\bar{X}' \bar{X}]$, $B = \mathbb{E}[\bar{X}' \Sigma^{-1} \bar{X}]$ and $C = \mathbb{E}[\bar{X}' \Sigma \bar{X}]$, all symmetric positive definite.

Step 1: reduction. It suffices to show $AB^{-1}A \leq C$. Indeed, pre- and post-multiplying an inequality between symmetric matrices by the same matrix A^{-1} (a congruence) preserves it, giving $B^{-1} \leq A^{-1} C A^{-1}$, which is the claim.

Step 2: transformation. Let $U = \Sigma^{1/2} \bar{X}$ and $V = \Sigma^{-1/2} \bar{X}$. (The printed hint writes $\Sigma^{1/2}$ for both; the second must be $\Sigma^{-1/2}$.) Then

$$\mathbb{E}[U'U] = C, \quad \mathbb{E}[V'V] = B, \quad \mathbb{E}[U'V] = \mathbb{E}[\bar{X}' \Sigma^{1/2} \Sigma^{-1/2} \bar{X}] = A.$$

Step 3: matrix Cauchy–Schwarz. For $\Gamma = B^{-1} \mathbb{E}[V'U] = B^{-1}A$,

$$0 \leq \mathbb{E}[(U - V\Gamma)'(U - V\Gamma)] = C - A\Gamma - \Gamma'A + \Gamma' B \Gamma = C - AB^{-1}A,$$

which is (B.33) in this setting. Hence $AB^{-1}A \leq C$ and, by Step 1, $V_\beta^* \leq V_\beta$.

Equality holds when U is a linear transformation of V , i.e. when $\Sigma \bar{X} = \bar{X} \Gamma$ for some Γ — for instance when Σ is diagonal and the regressors differ, or when the regressors are common across equations (Section 11.8), in which cases SUR and least squares coincide.

Exercise 11.13

Write $n \hat{V}_\beta = (n^{-1} \sum_i \bar{X}'_i \hat{\Sigma}^{-1} \bar{X}_i)^{-1}$. In the proof of Exercise 11.11 we showed that

$$n^{-1} \sum_i \bar{X}'_i \hat{\Sigma}^{-1} \bar{X}_i \xrightarrow{p} \mathbb{E}[\bar{X}' \Sigma^{-1} \bar{X}] > 0,$$

using $\hat{\Sigma} \xrightarrow{p} \Sigma$, the WLLN, and the bound

$$\left\| n^{-1} \sum_i \bar{X}'_i (\hat{\Sigma}^{-1} - \Sigma^{-1}) \bar{X}_i \right\| \leq \|\hat{\Sigma}^{-1} - \Sigma^{-1}\| n^{-1} \sum_i \|\bar{X}_i\|^2 = o_p(1).$$

Matrix inversion is continuous at a non-singular matrix, so $n\hat{V}_{\hat{\beta}} \xrightarrow{p} (\mathbb{E}[\bar{X}'\Sigma^{-1}\bar{X}])^{-1} = V_{\beta}^*$. (The printed statement writes V_{β} ; the limit is the SUR variance V_{β}^* .) Note that this estimator relies on homoskedasticity (11.8); without it the sandwich form must be used.

Exercise 11.14

Write $\hat{Q}_{ZZ} = n^{-1} \sum_i Z_i Z_i'$, $\hat{Q}_{ZY} = n^{-1} \sum_i Z_i Y_i$, and $u = X - \Gamma'Z$ (so $\mathbb{E}[u | Z] = 0$). Since $\hat{\pi}_i = \hat{\Gamma}'Z_i$,

$$\hat{\beta} = \left(\sum_i \hat{\pi}_i \hat{\pi}_i' \right)^{-1} \sum_i \hat{\pi}_i Y_i = (\hat{\Gamma}' \hat{Q}_{ZZ} \hat{\Gamma})^{-1} \hat{\Gamma}' \hat{Q}_{ZY}.$$

(a) $\hat{\Gamma} = \hat{Q}_{ZZ}^{-1} n^{-1} \sum_i Z_i X_i' \xrightarrow{p} Q_{ZZ}^{-1} \mathbb{E}[ZX'] = \Gamma$, because the CEF of X given Z is linear. Also $\mathbb{E}[ZY] = \mathbb{E}[Z(Z'\Gamma\beta + e)] = Q_{ZZ}\Gamma\beta$, using $\mathbb{E}[Ze] = 0$. Hence

$$\hat{\beta} \xrightarrow{p} (\Gamma' Q_{ZZ} \Gamma)^{-1} \Gamma' Q_{ZZ} \Gamma \beta = \beta,$$

provided $\Gamma' Q_{ZZ} \Gamma > 0$, i.e. $\text{rank } \Gamma = k$ (the instruments are relevant for all components of X). This estimator is two-stage least squares in the special case where the structural equation is written in terms of π (compare Chapter 12).

(b) If $\beta = 0$ then $Y_i = e_i$ and

$$\sqrt{n} \hat{\beta} = (\hat{\Gamma}' \hat{Q}_{ZZ} \hat{\Gamma})^{-1} \hat{\Gamma}' \frac{1}{\sqrt{n}} \sum_i Z_i e_i \xrightarrow{d} N\left(0, (\Gamma' Q_{ZZ} \Gamma)^{-1} \Gamma' \Omega (\Gamma' Q_{ZZ} \Gamma)^{-1}\right),$$

with $\Omega = \mathbb{E}[ZZ'e^2]$, by the CLT (mean zero since $\mathbb{E}[e | Z] = 0$) and Slutsky.

(c) For general β , $Y_i - \hat{\pi}_i' \hat{\beta} = e_i - Z_i'(\hat{\Gamma} - \Gamma)\beta$, and $\hat{Q}_{ZZ} \sqrt{n}(\hat{\Gamma} - \Gamma) = n^{-1/2} \sum_i Z_i u_i'$. One finds

$$\sqrt{n}(\hat{\beta} - \beta) = (\Gamma' Q_{ZZ} \Gamma)^{-1} \Gamma' \frac{1}{\sqrt{n}} \sum_i Z_i (e_i - u_i' \beta) + o_p(1),$$

so the first-stage error $u_i' \beta$ enters the asymptotic variance, which involves $\mathbb{E}[ZZ'(e - u' \beta)^2]$. The second-stage residuals $Y_i - \hat{\pi}_i' \hat{\beta}$ do not contain $u_i' \beta$, so the naive standard errors from the second-stage regression are wrong. When $\beta = 0$ the generated-regressor term vanishes: the first stage has no first-order effect and the naive second-stage variance formula is valid.

(d) Under $H_0 : \beta = 0$ use the robust Wald statistic from the second-stage regression,

$$W = n \hat{\beta}' \hat{V}^{-1} \hat{\beta}, \quad \hat{V} = (\hat{\Gamma}' \hat{Q}_{ZZ} \hat{\Gamma})^{-1} \hat{\Gamma}' \left(\frac{1}{n} \sum_i Z_i Z_i' \hat{e}_i^2 \right) \hat{\Gamma} (\hat{\Gamma}' \hat{Q}_{ZZ} \hat{\Gamma})^{-1},$$

with $\hat{e}_i = Y_i - \hat{\pi}_i' \hat{\beta}$ (or simply $\hat{e}_i = Y_i$, imposing the null). By (b) and the consistency of $\hat{V}$ under H_0 , $W \xrightarrow{d} \chi_k^2$; reject if W exceeds the χ_k^2 critical value.

Exercise 11.15

(a) The regressors are common, so SUR coincides with least squares (Section 11.8). Use equation-by-equation OLS, $\hat{\beta}_j = (\sum_i X_i X_i')^{-1} \sum_i X_i Y_{ji}$, $j = 1, 2$.

(b) By THEOREM 11.1 with $\bar{X} = \mathbf{I}_2 \otimes X'$ and $Q_{XX} = \mathbb{E}[XX']$,

$$\sqrt{n} \begin{pmatrix} \hat{\beta}_1 - \beta_1 \\ \hat{\beta}_2 - \beta_2 \end{pmatrix} \xrightarrow{d} N \left(0, \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix} \right), \quad V_{j\ell} = Q_{XX}^{-1} \mathbb{E}[XX' e_j e_\ell] Q_{XX}^{-1}.$$

The off-diagonal blocks are non-zero whenever the errors are correlated; under homoskedasticity $V = \Sigma \otimes Q_{XX}^{-1}$ (Exercise 11.6).

(c) Let $\hat{\delta} = \hat{\beta}_1 - \hat{\beta}_2$, with robust estimates $\hat{V}_{j\ell}$ built from $n^{-1} \sum_i X_i X_i' \hat{e}_{ji} \hat{e}_{\ell i}$. The Wald statistic

$$W = n \hat{\delta}' (\hat{V}_{11} - \hat{V}_{12} - \hat{V}_{21} + \hat{V}_{22})^{-1} \hat{\delta} \xrightarrow{d} \chi_k^2$$

under H_0 . The cross-covariance terms cannot be omitted. A convenient equivalent implementation: since $\hat{\delta}$ is exactly the OLS coefficient of $Y_1 - Y_2$ on X (common regressors), with residual $\hat{e}_1 - \hat{e}_2$, the robust Wald test that all coefficients in the regression of $Y_1 - Y_2$ on X are zero is numerically identical to W .

Chapter 12

Instrumental Variables

Solutions to all 27 exercises of Chapter 12 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 12.22–12.27 use the AJR2001, Card1995 and AK1991 data sets. The code reproduces the book’s own equations (12.86)–(12.92) and Tables 12.1–12.2 before extending them, which pins down the variable definitions.

Notation: $Y = X'\beta + e$ with instruments Z ($\ell \geq k$) satisfying $\mathbb{E}[Ze] = \mathbf{0}$. Partition $X = (X_1, X_2)$ into exogenous and endogenous parts and $Z = (Z_1, Z_2)$ with $Z_1 = X_1$. $P_Z = Z(Z'Z)^{-1}Z'$, $\hat{\beta}_{2sls} = (X'P_ZX)^{-1}X'P_ZY$, and $Q_{XZ} = \mathbb{E}[XZ']$, $Q_{ZZ} = \mathbb{E}[ZZ']$, $\Omega = \mathbb{E}[ZZ'e^2]$.

Exercise 12.1

As stated (no intercept),

$$\hat{\beta} = \frac{\sum_i D_i Y_i}{\sum_i D_i Z_i} = \frac{\sum_{i:D_i=1} Y_i}{\sum_{i:D_i=1} Z_i} = \frac{\bar{Y}_1}{\bar{Z}_1},$$

the ratio of the means of Y and Z in the $D = 1$ group.

With an intercept, $Y = \alpha + Z\beta + e$ and instruments $(1, D)$, the slope is the *Wald estimator* of Section 12.11,

$$\hat{\beta} = \frac{\widehat{\text{cov}}(D, Y)}{\widehat{\text{cov}}(D, Z)} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{Z}_1 - \bar{Z}_0},$$

the difference in mean outcomes between the $D = 1$ and $D = 0$ groups divided by the difference in mean treatment — the form most often used in practice.

Exercise 12.2

GLS is $\tilde{\beta} = (\sum_i \sigma^{-2}(X_i) X_i X_i')^{-1} (\sum_i \sigma^{-2}(X_i) X_i Y_i)$. Setting

$$Z_i = \frac{X_i}{\sigma^2(X_i)},$$

this is exactly $(\sum_i Z_i X_i')^{-1} (\sum_i Z_i Y_i)$, the IV estimator (12.24). Z is a valid instrument: $\mathbb{E}[Ze] = \mathbb{E}[\sigma^{-2}(X)X \mathbb{E}[e | X]] = \mathbf{0}$, and it is relevant since $\mathbb{E}[ZX'] = \mathbb{E}[XX'/\sigma^2(X)] > 0$. More generally, every linear estimator with weights that are functions of X is an IV estimator with the corresponding function of X as instrument.

Exercise 12.3

No — never. OLS minimizes $\sum_i (Y_i - X_i'b)^2$ over all b , so for any other estimator, IV included, $\sum_i \tilde{e}_i^2 \geq \sum_i \hat{e}_i^2$ in every sample (with equality only if $\tilde{\beta} = \hat{\beta}$). In large samples $\hat{\beta}$ converges to the projection coefficient, which is the population minimizer of mean-squared error, while $\tilde{\beta}$ converges to the structural β , so the gap is $O_p(1)$ when X is endogenous. IV is not designed to fit; it is designed to recover a causal coefficient. (Compare Exercises 4.21 and 8.21.)

Exercise 12.4

Substituting $u = X - \Gamma'Z$ into $\mathbb{E}[Zu'] = \mathbf{0}$:

$$\mathbb{E}[ZX'] - \mathbb{E}[ZZ']\Gamma = \mathbf{0} \implies \Gamma = (\mathbb{E}[ZZ'])^{-1}\mathbb{E}[ZX'].$$

Replacing expectations by sample averages, $\hat{\Gamma} = (n^{-1} \sum_i Z_i Z_i')^{-1} (n^{-1} \sum_i Z_i X_i') = (Z'Z)^{-1}(Z'X)$, where Z is $n \times \ell$ and X is $n \times k$. Column by column this is OLS of each element of X on Z (multivariate regression, Chapter 11).

Exercise 12.5

Substituting the first stage into the structural equation gives the reduced form for Y :

$$Y = Z'\Gamma\beta + (u'\beta + e) = Z'\lambda + v, \quad \lambda = \Gamma\beta, \quad \mathbb{E}[Zv] = \mathbf{0}.$$

The observable distribution identifies the reduced-form parameters (λ, Γ) , so β is identified iff $\lambda = \Gamma b$ has the unique solution $b = \beta$. If $\text{rank}(\Gamma) < k$ there is $c \neq \mathbf{0}$ with $\Gamma c = \mathbf{0}$, and then $\Gamma(\beta + tc) = \Gamma\beta = \lambda$ for every $t \in \mathbb{R}$: all $\beta + tc$ generate the same reduced form and hence the same distribution of observables. They are observationally equivalent, so β is not identified. Equivalently $\mathbb{E}[ZX'] = Q_{ZZ}\Gamma$ has rank less than k and the moment equations $\mathbb{E}[ZX']b = \mathbb{E}[ZY]$ have a continuum of solutions.

Exercise 12.6

By the WLLN under Assumption 12.2, $\hat{Q}_{ZZ} \xrightarrow{p} Q_{ZZ}$ and $\hat{Q}_{XZ} \xrightarrow{p} Q_{XZ}$, and $\hat{\beta}_{2\text{sls}} \xrightarrow{p} \beta$ (THEOREM 12.1). It remains to show $\hat{\Omega} = n^{-1} \sum_i Z_i Z_i' \hat{e}_i^2 \xrightarrow{p} \Omega$. With $\delta = \hat{\beta} - \beta$, $\hat{e}_i = e_i - X_i' \delta$ and

$$\hat{\Omega} = \frac{1}{n} \sum_i Z_i Z_i' e_i^2 - \frac{2}{n} \sum_i Z_i Z_i' e_i X_i' \delta + \frac{1}{n} \sum_i Z_i Z_i' (X_i' \delta)^2.$$

- The first term converges to Ω by the WLLN, since $\mathbb{E}\|Z\|^2 e^2 \leq (\mathbb{E}\|Z\|^4 \mathbb{E}[e^4])^{1/2} < \infty$ ($\mathbb{E}[e^4] < \infty$ follows from $\mathbb{E}[Y^4] < \infty$ and $\mathbb{E}\|X\|^4 < \infty$ by Minkowski).
- The second is bounded in norm by $2\|\delta\| n^{-1} \sum_i \|Z_i\|^2 |e_i| \|X_i\| = o_p(1)O_p(1)$, since by Hölder $\mathbb{E}\|Z\|^2 |e| \|X\| \leq (\mathbb{E}\|Z\|^4)^{1/2} (\mathbb{E}[e^4])^{1/4} (\mathbb{E}\|X\|^4)^{1/4} < \infty$.
- The third is bounded by $\|\delta\|^2 n^{-1} \sum_i \|Z_i\|^2 \|X_i\|^2 = o_p(1)$, since $\mathbb{E}\|Z\|^2 \|X\|^2 \leq (\mathbb{E}\|Z\|^4 \mathbb{E}\|X\|^4)^{1/2}$.

So $\hat{\Omega} \xrightarrow{p} \Omega$. Moreover (12.40) is a continuous function of $\hat{Q}_{XZ}$, $\hat{Q}_{ZZ}$ and $\hat{\Omega}$ at their limits, where $Q_{XZ}Q_{ZZ}^{-1}Q_{ZX} > 0$ by the rank condition. By the continuous mapping theorem $\hat{V}_\beta \xrightarrow{p} V_\beta$.

Exercise 12.7

(a) By the conditioning theorem, $\mathbb{E}[Xe] = \mathbb{E}[X \mathbb{E}[e | X]] = 0$ and likewise $\mathbb{E}[X^2 e] = 0$. So $\mathbb{E}[Ze] = \mathbf{0}$ (exogeneity) and $\mathbb{E}[ZX] = (\mathbb{E}[X^2], \mathbb{E}[X^3])'$ has rank one whenever $\mathbb{E}[X^2] > 0$ (relevance). Z is a valid instrument vector.

(b) $\hat{\beta}_{2\text{sls}} = (X'P_Z X)^{-1} X'P_Z Y$. Because X is itself a column of Z , it lies in the column space of Z and $P_Z X = X$. Hence

$$\hat{\beta}_{2\text{sls}} = (X'X)^{-1} X'Y = \hat{\beta}_{\text{ols}}$$

exactly. Adding further exogenous instruments to a regressor that instruments itself changes nothing for 2SLS: the first-stage fitted value is the regressor. (Efficient GMM, Chapter 13, would differ under heteroskedasticity, because it can exploit $\mathbb{E}[X^2 e] = 0$.)

Exercise 12.8

Equate demand and supply and solve for the equilibrium price (assuming $a_1 \neq b_1$):

$$P = \frac{b_0 - a_0}{a_1 - b_1} - \frac{a_2}{a_1 - b_1}Y + \frac{b_2}{a_1 - b_1}W + \frac{e_2 - e_1}{a_1 - b_1},$$

and Q follows from either equation. So the reduced forms are linear in $(1, Y, W)$.

- *Demand* excludes W , which can instrument P . Relevance requires the W coefficient in the price equation to be non-zero, i.e. $b_2 \neq 0$. Then $a_1 = \partial Q / \partial W / \partial P / \partial W$ (a ratio of reduced-form coefficients).
- *Supply* excludes Y , which can instrument P . Relevance requires $a_2 \neq 0$, and then $b_1 = \partial Q / \partial Y / \partial P / \partial Y$.

Each equation has one endogenous regressor and one excluded exogenous variable, so each is *just identified* provided $a_1 \neq b_1$, $b_2 \neq 0$ (for demand) and $a_2 \neq 0$ (for supply), together with the exogeneity conditions $\mathbb{E}[(1, Y, W)'e_j] = \mathbf{0}$. The intercepts and the coefficients on Y and W are then recovered too. The economic content: a demand curve is traced out by shifts in supply (W), and vice versa.

Exercise 12.9

(a) $\hat{\beta}_{iv} = \beta + (Z'X)^{-1}Z'e$, and $(Z'X)^{-1}Z'$ is a function of (X, Z) . By random sampling $\mathbb{E}[e_i | X, Z] = \mathbb{E}[e_i | X_i, Z_i] = 0$, so $\mathbb{E}[\hat{\beta}_{iv} | X, Z] = \beta$: IV is conditionally unbiased (on the event that $Z'X$ is invertible). The unconditional statement additionally requires the expectation to exist, which is not guaranteed since $(Z'X)^{-1}$ can be arbitrarily large.

(b) With $D = \text{diag}\{\sigma_1^2, \dots, \sigma_n^2\}$, $\sigma_i^2 = \mathbb{E}[e_i^2 | X_i, Z_i]$,

$$\text{var}[\hat{\beta}_{iv} | X, Z] = (Z'X)^{-1}Z'DZ(X'Z)^{-1},$$

which is $\sigma^2(Z'X)^{-1}Z'Z(X'Z)^{-1}$ under homoskedasticity. Because X is exogenous here, OLS is also unbiased with variance $\sigma^2(X'X)^{-1}$, which is no larger (Gauss–Markov): using an instrument when it is not needed costs efficiency.

Exercise 12.10

(The printed equation reads $Y = Z'\beta + u'\gamma + \nu$; substituting $e = u'\gamma + \nu$ into the structural equation gives $Y = X'\beta + u'\gamma + \nu$, which is the regression estimated.)

(a) $\mathbb{E}[X\nu] = \Gamma'\mathbb{E}[Z\nu] + \mathbb{E}[u\nu]$. The second term is zero by construction. For the first, $\nu = e - u'\gamma$, so $\mathbb{E}[Z\nu] = \mathbb{E}[Ze] - \mathbb{E}[Zu']\gamma = \mathbf{0}$. Hence $\mathbb{E}[X\nu] = \mathbf{0}$: once u is controlled for, the remaining error is uncorrelated with X .

(b) Let $W = (X', u')'$ and $\theta = (\beta', \gamma')'$. Then $Y = W'\theta + \nu$ with $\mathbb{E}[W\nu] = \mathbf{0}$ by (a) and the definition of ν . This is a linear projection model, so under the usual moment conditions

$$\sqrt{n} \begin{pmatrix} \hat{\beta} - \beta \\ \hat{\gamma} - \gamma \end{pmatrix} \xrightarrow{d} N(\mathbf{0}, Q_{WW}^{-1} \mathbb{E}[WW'\nu^2] Q_{WW}^{-1}), \quad Q_{WW} = \begin{pmatrix} \mathbb{E}[XX'] & \mathbb{E}[Xu'] \\ \mathbb{E}[uX'] & \mathbb{E}[uu'] \end{pmatrix}.$$

$Q_{WW} > 0$ requires that X is not collinear with u , i.e. $\Gamma'Z$ has a non-degenerate covariance — the rank condition. If u must be estimated by the first-stage residual, the generated-regressor correction of Section 12.27 applies; the point estimate $\hat{\beta}$ then equals 2SLS (Section 12.28).

Exercise 12.11

(a) Endogenous. X^2 is a function of the endogenous X ; there is no reason for $\mathbb{E}[X^2e] = 0$ when $\mathbb{E}[Xe] \neq 0$.

(b) There are $k = 3$ coefficients (with 1 exogenous and X, X^2 endogenous) and $\ell = 3$ instruments $(1, Z, Z^2)$, all valid because $\mathbb{E}[e | Z] = 0$. The order condition holds with equality: the model is *just identified*.

(c) Squaring (12.94) and using independence of u and Z with $\mathbb{E}[u] = 0$, $\sigma_u^2 = \mathbb{E}[u^2]$:

$$X^2 = (\gamma_0^2 + \sigma_u^2) + 2\gamma_0\gamma_1 Z + \gamma_1^2 Z^2 + v, \quad v = 2(\gamma_0 + \gamma_1 Z)u + u^2 - \sigma_u^2, \quad \mathbb{E}[v | Z] = 0.$$

The reduced-form coefficient matrix of $(1, X, X^2)$ on $(1, Z, Z^2)$ is

$$\Gamma = \begin{pmatrix} 1 & \gamma_0 & \gamma_0^2 + \sigma_u^2 \\ 0 & \gamma_1 & 2\gamma_0\gamma_1 \\ 0 & 0 & \gamma_1^2 \end{pmatrix}, \quad \det \Gamma = \gamma_1^3.$$

The rank condition holds iff $\gamma_1 \neq 0$. So the quadratic structural equation is identified exactly when the instrument is relevant for X itself; Z^2 then automatically serves as the instrument for X^2 .

Exercise 12.12

(a) With $\hat{\gamma} = \sum_i Z_i X_i / \sum_i Z_i^2$,

$$\hat{\beta} = \frac{\sum_i \hat{X}_i^2 Y_i}{\sum_i \hat{X}_i^4} = \frac{\sum_i Z_i^2 Y_i}{\hat{\gamma}^2 \sum_i Z_i^4}.$$

(b) $\hat{\gamma} \xrightarrow{p} \gamma$ and $n^{-1} \sum_i Z_i^4 \xrightarrow{p} \mathbb{E}[Z^4]$. Since $X^2 = \gamma^2 Z^2 + 2\gamma Z u + u^2$,

$$\frac{1}{n} \sum_i Z_i^2 Y_i \xrightarrow{p} \beta \mathbb{E}[Z^2 X^2] + \mathbb{E}[Z^2 e] = \beta(\gamma^2 \mathbb{E}[Z^4] + 2\gamma \mathbb{E}[Z^3 u] + \mathbb{E}[Z^2 u^2]) + \mathbb{E}[Z^2 e],$$

so

$$\hat{\beta} \xrightarrow{p} \beta + \beta \frac{2\gamma \mathbb{E}[Z^3 u] + \mathbb{E}[Z^2 u^2]}{\gamma^2 \mathbb{E}[Z^4]} + \frac{\mathbb{E}[Z^2 e]}{\gamma^2 \mathbb{E}[Z^4]}.$$

(c) **Not in general.** The stated assumptions do not imply $\mathbb{E}[Z^2 e] = 0$ or $\mathbb{E}[Z^3 u] = 0$. Even strengthening them to $\mathbb{E}[e | Z] = \mathbb{E}[u | Z] = 0$, which kills those terms,

$$\hat{\beta} \xrightarrow{p} \beta \left(1 + \frac{\mathbb{E}[Z^2 u^2]}{\gamma^2 \mathbb{E}[Z^4]} \right),$$

and $\mathbb{E}[Z^2 u^2] > 0$ unless $u \equiv 0$. The estimator is biased away from zero because $(\mathbb{E}[X | Z])^2 \neq \mathbb{E}[X^2 | Z]$: plugging fitted values into a non-linear function ignores $\text{var}[X | Z]$. The only reasonable condition giving consistency is $\beta = 0$ (with $\mathbb{E}[e | Z] = 0$), which makes $\hat{\beta}$ usable for testing $H_0 : \beta = 0$ but not for estimation. The correct approach is IV with Z^2 as the instrument for X^2 , $\tilde{\beta} = \sum_i Z_i^2 Y_i / \sum_i Z_i^2 X_i^2$, consistent when $\mathbb{E}[Z^2 e] = 0$ (Exercise 12.11).

Exercise 12.13

Substitute the reduced form for the endogenous regressors, $Y_2 = \Gamma'_{12} Z_1 + \Gamma'_{22} Z_2 + u_2$, into the structural equation:

$$Y_1 = Z_1'(\beta_1 + \Gamma_{12}\beta_2) + Z_2'(\Gamma_{22}\beta_2) + (e + u_2'\beta_2),$$

so $\lambda_2 = \Gamma_{22}\beta_2$. Under H_0 , $\lambda_2 = \mathbf{0}$. Conversely, if the rank condition $\text{rank}(\Gamma_{22}) = k_2$ holds, $\lambda_2 = \mathbf{0}$ implies $\beta_2 = \mathbf{0}$. So $H_0 : \beta_2 = \mathbf{0}$ is equivalent to $H_0 : \lambda_2 = \mathbf{0}$.

Test. Estimate (12.95) by OLS with a robust covariance matrix and compute

$$W = \hat{\lambda}_2'(\hat{V}_{\hat{\lambda}_2})^{-1}\hat{\lambda}_2 \xrightarrow{d} \chi_{\ell_2}^2$$

under H_0 (THEOREM 9.2); reject for large W . This is the Anderson–Rubin test evaluated at $\beta_2 = \mathbf{0}$. Its great virtue is that its null distribution does not depend on instrument strength at all: it is valid even with weak instruments (power, of course, does depend on Γ_{22}).

Exercise 12.14

Write $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Y}_2, \mathbf{e}$ for the stacked data and $P_2 = \mathbf{Z}_2(\mathbf{Z}_2' \mathbf{Z}_2)^{-1} \mathbf{Z}_2'$. We need $\ell_2 \geq k_1$ and $Q_{12} = \mathbb{E}[Z_1 Z_2']$ of rank k_1 .

(a) $\hat{\beta}_1 = (\mathbf{Z}_1' P_2 \mathbf{Z}_1)^{-1} \mathbf{Z}_1' P_2 \mathbf{Y}_1$ and $\mathbf{Y}_1 = \mathbf{Z}_1 \beta_1 + \mathbf{Y}_2 \beta_2 + \mathbf{e}$, so

$$b_{1n} = (\mathbf{Z}_1' P_2 \mathbf{Z}_1)^{-1} \mathbf{Z}_1' P_2 \mathbf{Y}_2 \beta_2, \quad r_{1n} = (\mathbf{Z}_1' P_2 \mathbf{Z}_1)^{-1} \mathbf{Z}_1' P_2 \mathbf{e}.$$

(b) Writing each block as sample moments,

$$r_{1n} = (\hat{Q}_{12} \hat{Q}_{22}^{-1} \hat{Q}_{21})^{-1} \hat{Q}_{12} \hat{Q}_{22}^{-1} \left(\frac{1}{n} \sum_i Z_{2i} e_i \right) \xrightarrow{p} (Q_{12} Q_{22}^{-1} Q_{21})^{-1} Q_{12} Q_{22}^{-1} \mathbb{E}[Z_2 e] = \mathbf{0}.$$

(c) Similarly, with $Q_{2Y} = \mathbb{E}[Z_2 Y_2']$,

$$b_{1n} \xrightarrow{p} b_1 = (Q_{12} Q_{22}^{-1} Q_{21})^{-1} Q_{12} Q_{22}^{-1} Q_{2Y} \beta_2, \quad \hat{\beta}_1 \xrightarrow{p} \beta_1 + b_1.$$

(d) Yes: b_1 is an omitted-variable bias — the instrumented Z_1 picks up the part of the omitted $Y_2 \beta_2$ that is predictable from Z_2 . There is no bias iff $Q_{12} Q_{22}^{-1} Q_{2Y} \beta_2 = \mathbf{0}$; sufficient conditions are $\beta_2 = \mathbf{0}$ (nothing omitted) or $\mathbb{E}[Z_2 Y_2'] = \mathbf{0}$ (the instruments do not predict the omitted variable). Instrumenting does not cure omitted variables: exclusion restrictions must hold relative to the *true* model.

(e) $\sqrt{n} r_{1n} = A_n n^{-1/2} \sum_i Z_{2i} e_i$ with $A_n \xrightarrow{p} A = (Q_{12} Q_{22}^{-1} Q_{21})^{-1} Q_{12} Q_{22}^{-1}$. By the CLT with $\Omega_2 = \mathbb{E}[Z_2 Z_2' e^2]$,

$$\sqrt{n}(\hat{\beta}_1 - \beta_1 - b_{1n}) \xrightarrow{d} N(\mathbf{0}, A \Omega_2 A').$$

Exercise 12.15

- (a) No. Z already appears as a regressor, so it cannot also serve as the excluded instrument for Y_2 . There are two coefficients and one instrument: $\ell = 1 < k = 2$, under-identified.
- (b) Yes, in principle: $\ell = 2 = k$, just identified, and Z^2 is exogenous since $\mathbb{E}[Z^2 e] = \mathbb{E}[Z^2 \mathbb{E}[e | Z]] = 0$.
- (c) That Z^2 is excluded from the structural equation: the direct effect of Z on Y_1 is exactly linear.
- (d) The reduced form $Y_2 = \pi_1 Z + \pi_2 Z^2 + u$ must have $\pi_2 \neq 0$: $\mathbb{E}[Y_2 | Z]$ must be non-linear in Z .
- (e) Generally not. Identification comes purely from functional form — linearity of the structural equation in Z together with non-linearity of the first stage — neither of which is typically implied by economic reasoning. A tiny curvature in the true structural relation would be misattributed to β_2 , and if the first stage is nearly linear the instrument is weak.

Exercise 12.16

- (a) $Y = (\gamma Z + u_2)\beta + e = Z(\gamma\beta) + (\beta u_2 + e)$, so $\lambda = \gamma\beta$, $u_1 = \beta u_2 + e$, and $\beta = \lambda/\gamma$ when $\gamma \neq 0$. $\mathbb{E}[Zu_1] = \beta\mathbb{E}[Zu_2] + \mathbb{E}[Ze] = 0$.
- (b) Both are OLS on the same regressor, so

$$\sqrt{n}(\hat{\theta} - \theta) = \left(\frac{1}{n} \sum_i Z_i^2 \right)^{-1} \frac{1}{\sqrt{n}} \sum_i Z_i u_i.$$

- (c) $\mathbb{E}[Zu] = (\mathbb{E}[Zu_1], \mathbb{E}[Zu_2])' = \mathbf{0}$ by (a) and assumption.
- (d) By the WLLN and CLT, with $\Omega_u = \mathbb{E}[Z^2 uu']$,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(\mathbf{0}, \Omega_u / (\mathbb{E}[Z^2])^2).$$

- (e) The gradient of λ/γ is $R = (1/\gamma, -\lambda/\gamma^2)' = \gamma^{-1}(1, -\beta)'$, so by the delta method

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N\left(0, \frac{(1, -\beta) \Omega_u (1, -\beta)'}{\gamma^2 (\mathbb{E}[Z^2])^2}\right).$$

- (f) Yes. $(1, -\beta)u = u_1 - \beta u_2 = e$, so $(1, -\beta)\Omega_u(1, -\beta)' = \mathbb{E}[Z^2 e^2]$, and the variance is

$$\mathbb{E}[Z^2 e^2] / (\gamma \mathbb{E}[Z^2])^2.$$

In the scalar just-identified case Theorem 12.2 gives $V_\beta = \Omega / Q_{ZX}^2 = \mathbb{E}[Z^2 e^2] / (\mathbb{E}[ZX])^2$, and $\mathbb{E}[ZX] = \gamma \mathbb{E}[Z^2]$. The two coincide, as they must, since ILS and IV are numerically identical here (Section 12.9).

Exercise 12.17

No. With $X = \hat{X} + \hat{u}$ (first-stage residual $\hat{u}$),

$$Y_i - \hat{X}_i' \hat{\beta} = (Y_i - X_i' \hat{\beta}) + \hat{u}_i' \hat{\beta},$$

so the second-stage residual contains the first-stage error scaled by the endogenous coefficients, and $n^{-1} \sum_i (Y_i - \hat{X}_i' \hat{\beta})^2 \xrightarrow{p} \mathbb{E}[(e + u'\beta)^2] \neq \sigma^2$ in general. The appropriate estimator uses the *original*

regressors,

$$\tilde{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - X_i' \hat{\beta}_{2\text{sls}})^2 \xrightarrow{p} \sigma^2,$$

possibly with a $n - k$ divisor. The same substitution is required inside robust covariance matrices (Exercise 12.20).

Exercise 12.18

(a) In each sample compute the IV estimator and its robust asymptotic covariance matrix,

$$\hat{\beta}_j = (\mathbf{Z}_j' \mathbf{X}_j)^{-1} \mathbf{Z}_j' \mathbf{Y}_j, \quad \hat{V}_j = n(\mathbf{Z}_j' \mathbf{X}_j)^{-1} \left(\sum_i Z_{ji} Z_{ji}' \hat{e}_{ji}^2 \right) (\mathbf{X}_j' \mathbf{Z}_j)^{-1},$$

with $\hat{e}_{ji} = Y_{ji} - X_{ji}' \hat{\beta}_j$. The Wald statistic is

$$W = n(\hat{\beta}_1 - \hat{\beta}_2)' (\hat{V}_1 + \hat{V}_2)^{-1} (\hat{\beta}_1 - \hat{\beta}_2).$$

(b) By Theorem 12.2 in each sample, $\sqrt{n}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, V_j)$, independently across samples. Under H_0 , $\sqrt{n}(\hat{\beta}_1 - \hat{\beta}_2) \xrightarrow{d} N(0, V_1 + V_2)$, and with $\hat{V}_j \xrightarrow{p} V_j$ (Exercise 12.6), $W \xrightarrow{d} \chi_k^2$.

(c) Estimate both equations, form W , and reject at level α if W exceeds the $1 - \alpha$ quantile of χ_k^2 . Equivalently pool the samples, interact every regressor and instrument with a sample dummy, and Wald-test the interaction coefficients.

Exercise 12.19

(a) Let Z be the vector of state dummies.

- *Exogeneity/exclusion*: $\mathbb{E}[Ze] = \mathbf{0}$ — state of residence affects Y *only through* X . States may not differ in any other determinant of Y (prices, labour markets, policies, culture), and households must not sort across states on unobservables related to Y .
- *Relevance*: mean X must differ across states (rank condition), and strongly enough to avoid weak instruments.
- *Sampling*: observations within a state share state-level shocks, so inference should cluster by state, and the effective number of observations for the first stage is the number of states.

These are strong assumptions; state-level instruments are rarely plausibly excluded.

(b) With an intercept there are (number of states -1) excluded instruments for one endogenous regressor — about 49 or 50 in U.S. data — so the model is heavily *overidentified*. Many-weak-instrument concerns (Section 12.37) then apply.

Exercise 12.20

The point estimates are correct: OLS of Y on $\hat{X} = P_Z X$ gives $(X' P_Z X)^{-1} X' P_Z Y = \hat{\beta}_{2\text{sls}}$.

The standard errors are not. The second-stage regression computes residuals $Y_i - \hat{X}_i' \hat{\beta}$ instead of $Y_i - X_i' \hat{\beta}$ (Exercise 12.17), so both the homoskedastic $s^2(\hat{X}' \hat{X})^{-1}$ and the robust sandwich use the wrong e_i . The standard errors are inconsistent, and can be too large or too small. Use a proper IV routine, or recompute the covariance matrix with the residuals $Y_i - X_i' \hat{\beta}$.

Exercise 12.21

This is Exercise 12.2 with $k = 1$: $\tilde{\beta}_{\text{gls}} = \sum_i \sigma^{-2}(X_i) X_i Y_i / \sum_i \sigma^{-2}(X_i) X_i^2$ is the IV estimator $\sum_i Z_i Y_i / \sum_i Z_i X_i$ with instrument $Z_i = X_i / \sigma^2(X_i)$.

Exercise 12.22

(a)–(b) Standard errors: homoskedastic, [robust HC1].

equation	coefficient	estimate	homoskedastic	robust
(12.86) OLS log GDP on risk	risk	0.516	0.063	[0.051]
(12.87) first stage risk on log mort	log mortality	−0.613	0.127	[0.152]
(12.88) 2SLS	risk	0.929	0.156	[0.173]

The 2SLS estimate 0.93 differs by 0.01 from the reported 0.94; the others round to the reported values. The reported standard errors (0.06), (0.13), (0.16) match the *homoskedastic* formulas.

(c) Reduced form for log GDP: $\hat{\lambda} = -0.570$. ILS: $\hat{\lambda}/\hat{\gamma} = -0.570 / -0.613 = 0.929$ — identical to 2SLS, as it must be in a just-identified model (Section 12.9).

(d) Regressing log GDP on the fitted $\widehat{\text{risk}}$ gives 0.929 — identical.

(e) Regressing log GDP on risk and the first-stage residual $\hat{u}$ gives risk coefficient 0.929 — identical (Section 12.28) — and $\hat{u}$ coefficient -0.569 with $t = -4.69$, reproducing the endogeneity test in Section 12.40.

(f) OLS (robust s.e.): risk 0.377 (0.066), latitude 1.383 (0.635), africa -0.723 (0.170). Both added variables are significant, so by least squares latitude and being in Africa predict GDP even conditional on institutions.

(g) 2SLS: risk 0.800 (0.281), latitude -0.055 (1.156), africa -0.348 (0.340). Once institutions are instrumented, latitude and africa become small and insignificant — the reversal the authors emphasize: their OLS significance reflected their correlation with institutions (whose OLS coefficient is attenuated), not a direct effect. Note that the instrument is weaker here (first-stage $t = -2.59$, $F \approx 6.7$), so these estimates are imprecise.

(h) With mortality in levels the first-stage coefficient is -0.00079 (homoskedastic s.e. 0.00038, $t = -2.06$) and $R^2 = 0.064$, versus $t = -4.83$ and $R^2 = 0.274$ with logs. Settler mortality is extremely skewed (from 8.5 to 2,940 per 1,000, median 78); a few extreme colonies dominate the levels regression while the relationship with institutions is concave. The log specification fits far better and gives a strong instrument; in levels it would be weak ($F \approx 4.3$).

(i) Reduced form for risk (robust s.e.): log mort -2.646 (0.737), $(\log \text{mort})^2$ 0.210 (0.082). The fitted relationship is convex, falling until log mort ≈ 6.3 (mortality ≈ 545) and flattening beyond: very high mortality adds little further deterioration of institutions. 2SLS with both instruments gives risk 0.772 (homoskedastic s.e. 0.115, robust 0.100), smaller than with one instrument.

(j) The (homoskedastic) first-stage F for the two excluded instruments is 18.4. From Table 12.4 with $\ell_2 = 2$: for 2SLS the critical values are 19.9 (10% size) and 11.6 (15%), so we reject that 2SLS

t -tests have size above 15% but cannot reject size above 10%; for LIML the 10% value is 8.7, so the instruments are strong for LIML.

(k) With two instruments and one endogenous regressor we can test the single over-identifying restriction. The Sargan statistic is 5.14, $\chi^2_1 p = 0.023$: rejected at 5%. The two instruments point to different values of the risk coefficient. This does not tell us which instrument is invalid — or whether the problem is heterogeneous effects or the linear structural specification rather than invalidity — but it is a warning.

(l) LIML with both instruments: risk 0.837 (homoskedastic s.e. 0.131, robust 0.133), $\hat{\kappa} = 1.082$. LIML lies between the just-identified 0.93 and 2SLS 0.77, consistent with 2SLS being pulled towards OLS in a moderately weak over-identified design.

Exercise 12.23

Asymptotic standard errors for the risk coefficient 0.929: 0.156 (homoskedastic), 0.173 (robust). Two runs of the pairs bootstrap with $B = 10,000$ and different seeds give

$$s_{\text{boot}} = 0.741 \quad \text{and} \quad s_{\text{boot}} = 0.477,$$

three to five times the asymptotic values and 55% apart from each other. The bootstrap distributions themselves are very similar in the middle — interquartile range /1.349 equals 0.178 and 0.177, very close to the asymptotic standard error — but contain a few absurd draws (from -30.8 to 54.9) generated by samples in which the first-stage covariance is near zero.

The model is just identified, so by Theorem 12.7 the 2SLS estimator has no finite variance (Section 12.24): the bootstrap is estimating an infinite quantity, and the answer is governed by whichever extreme draws occur. Bootstrap standard errors for IV should not be reported; percentile or BC intervals, which depend only on quantiles, are reliable.

Exercise 12.24

Experience = age – education – 6. (The Stata file stores age and education as 8-bit integers; they must be converted to floating point before squaring, or experience^2 silently overflows.) Robust standard errors in parentheses.

(a) Reduced form for education: experience -0.413 (0.032), $\text{experience}^2/100$ 0.093 (0.171), Black -1.006 (0.088), south -0.267 (0.079), urban 0.400 (0.085), public 0.430 (0.086), private 0.123 (0.101); robust $F = 13.87$ — exactly Table 12.2. 2SLS(a): education 0.161 (0.041), experience 0.119 (0.018), $\text{experience}^2/100$ -0.231 (0.037), Black -0.102 (0.044), south -0.095 (0.022), urban 0.116 (0.026), Sargan 0.82 — exactly Table 12.1 (up to the $n/(n-k)$ factor in one standard error).

(b) *nearc2* has coefficient 0.068 (0.074), $t = 0.91$. It adds no explanatory power and lowers the first-stage F from 13.5 to 9.3; 2SLS moves to 0.171 (0.041). Not useful: an irrelevant instrument only weakens the first stage.

(c) New coefficients: public -28.93 (2.83), public $\times$ age 1.483 (0.201), public $\times$ age $^2/100$ -1.554 (0.351); private 0.136 (0.105). The implied effect of growing up near a public college ranges from -2.3 years of schooling at age 24 to $+3.5$ at age 34 — implausibly large, and the interactions are

“significant” with a joint Wald statistic over 2,000. The explanation is mechanical. Experience is in the regression and, by construction, $\text{education} = \text{age} - \text{experience} - 6$. Holding experience fixed, age *is* education; the interactions let the regression reproduce that identity for the half of the sample (mean of public = 0.49) with public = 1. The “effects” are an accounting identity, not a proximity effect.

(d) 2SLS with the expanded set: education 0.083 (0.006), experience 0.087 (0.007), $\text{experience}^2/100$ -0.225 (0.032), Black -0.181 (0.018), south -0.122 (0.015), urban 0.157 (0.015). The return to schooling falls from 0.161 to 0.083, close to OLS (0.074), with a standard error seven times smaller. This is not a gain in precision: because the new instruments nearly reproduce education through the identity above — while experience, itself a function of the endogenous education, is treated as exogenous — 2SLS collapses towards OLS.

(e) First-stage $F = 384$ (homoskedastic; 588 robust) with $\ell_2 = 4$, far above every Stock–Yogo critical value in Table 12.4. By that test the instruments are strong — a reminder that the test checks relevance, not validity.

(f) Control function test: adding the first-stage residual gives coefficient -0.013 with $t = -1.72$ ($p \approx 0.09$): exogeneity of education is not rejected at 5%. With the original instruments (public, private) the same test gives $t = -2.38$ and rejects. The Sargan statistic for the expanded set is 10.98 (χ^2_3 , $p = 0.012$), rejecting the over-identifying restrictions — consistent with the invalidity of the added instruments.

(g) LIML: education 0.083 (0.006), $\hat{\kappa} = 1.004$; all coefficients agree with 2SLS to three decimals. With a first stage this strong LIML and 2SLS coincide. (For comparison, LIML for 2SLS(a) is 0.164 (0.042), matching Table 12.1.) The lesson of the exercise is that the problem in (d) is invalid instruments, which neither LIML nor weak-instrument diagnostics can fix.

Exercise 12.25

IV(a): education 0.132, robust s.e. 0.049 (asymptotic 95% interval [0.037, 0.228]). With $B = 10,000$ pairs bootstrap replications:

$$C^{bc} = [0.042, 0.258] \quad (\hat{z}_0 = 0.009), \quad C^{pc} = [0.041, 0.256].$$

The bootstrap standard error is 0.110, more than twice the asymptotic value, while the interquartile range /1.349 is 0.049. The discrepancy is driven by a handful of extreme draws (up to 9.6).

No, the bootstrap standard error should not be reported. The model is just identified, so the IV estimator has no finite variance (Theorem 12.7, Section 12.24); its bootstrap standard error is unstable across runs and meaningless. The BC interval depends only on bootstrap quantiles and is the appropriate summary: it is slightly right-shifted relative to the asymptotic interval, reflecting right skewness of the IV estimator.

Exercise 12.26

Our code reproduces the full-sample results: (12.89) education 0.081 (0.016) with $F = 4.75$; (12.92) 0.099 (0.021) with $F = 30.5$ and reduced-form quarter-of-birth coefficients 0.050, 0.101, 0.142. Standard errors below are homoskedastic, as in the text; robust ones are essentially identical.

(a) The *black* dummy must be omitted: it equals one for every observation and is collinear with the intercept. Among Black men only 50 states of birth appear, and some state-by-quarter cells are collinear with the other regressors, so of the nominal instruments only 163 are linearly independent. 2SLS gives education 0.046 (0.019), SMSA 0.205, married 0.227; LIML gives -1.19 (1.11).

(b) First-stage $F = 0.86$ for the 163 excluded instruments ($p = 0.91$). There is *no* evidence that these instruments predict education at all. The 2SLS estimate 0.046 is simply pulled toward the OLS value (0.058) — the classic many-weak-instruments bias — and its small standard error is spurious; LIML's erratic estimate is the honest signal that the model is not identified in practice.

(c) With the 30 quarter $\times$ year instruments: $F = 1.25$ ($p = 0.16$). Still no significant first stage. 2SLS 0.040 (0.036), LIML -0.041 (0.091).

(d) With the three quarter-of-birth dummies: $F = 4.85$ ($p = 0.002$), and the reduced-form coefficients 0.032, 0.158, 0.179 rise with quarter as in the full sample. From Table 12.4 with $\ell_2 = 3$:

- 2SLS critical values are 22.3 (10%), 12.8 (15%), 9.5 (20%), 7.8 (25%): $F = 4.85$ exceeds none, so the instruments are *too weak for 2SLS* even at 25% maximal size;
- LIML critical values are 6.5 (10%), 4.4 (15%), 3.7 (20%): $F = 4.85$ exceeds the 15% value, so the instruments are *adequate for LIML* with size distortion bounded by about 15% (but not 10%).

(e) The defensible specification is the 3-instrument model estimated by LIML, with controls for SMSA, marriage, year of birth and region of residence:

$$\hat{\beta}_{\text{edu}} = 0.084 \text{ (0.058)}, \quad 95\% \text{ interval } [-0.029, 0.196]$$

(2SLS gives 0.083 (0.057)). The point estimate is below but not far from the full-sample 0.099 (0.021) and above OLS for Black men (0.058). The Black-men estimate is roughly three times less precise; its interval contains zero, the OLS value and the full-sample IV value. Nothing in these data suggests a meaningfully different return to education for Black men — but nothing rules one out either. The more elaborate instrument sets should not be used for this subsample.

Exercise 12.27

2SLS: education 0.0833, asymptotic standard error 0.0572 (homoskedastic and robust agree). Pairs bootstrap, $B = 5,000$: standard error 0.0603; interquartile range /1.349 0.0549;

$$C^{bc} = [-0.026, 0.214] \quad (\hat{z}_0 = 0.044), \quad C^{pc} = [-0.032, 0.208],$$

compared with the asymptotic interval $0.083 \pm 1.96(0.057) = [-0.029, 0.195]$. The BC interval is shifted right and is longer on the right, reflecting the skewness caused by the weak first stage.

Reporting the bootstrap standard error. Here there are $\ell_2 - k_2 = 2$ over-identifying restrictions, so by Theorem 12.7 (under normality) the 2SLS estimator does have a finite variance, and the bootstrap standard error (0.060) is in fact close to the asymptotic value. It is therefore not meaningless as it was in Exercises 12.23 and 12.25. But with $F = 4.85$ the instruments are weak for 2SLS (Exercise 12.26), the sampling distribution is far from normal, and “estimate ± 1.96 standard errors” is unreliable whichever standard error is used. Section 12.24's advice applies: report the BC (or percentile) interval rather than a bootstrap standard error.

Chapter 13

Generalized Method of Moments

Solutions to all 28 exercises of Chapter 13 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 13.27–13.28 continue the empirical work of Exercises 12.22 and 12.24.

Notation: the linear moment model is $\mathbb{E}[Ze] = \mathbf{0}$ with $e = Y - X'\beta$, $Z \in \mathbb{R}^\ell$, $X \in \mathbb{R}^k$, $\ell \geq k$. $Q = \mathbb{E}[ZZ']$ ($\ell \times \ell$), $\Omega = \mathbb{E}[ZZ'e^2]$, $\bar{g}_n(\beta) = n^{-1}Z'(Y - X\beta)$, $J(\beta) = n\bar{g}_n(\beta)'W\bar{g}_n(\beta)$, and $\hat{\beta}_{\text{gmm}} = (X'ZWZ'X)^{-1}X'ZWZ'Y$, with asymptotic variance $V_\beta = (Q'WQ)^{-1}Q'W\Omega WQ(Q'WQ)^{-1}$. Efficient GMM uses $W = \Omega^{-1}$, giving $V_\beta = (Q'\Omega^{-1}Q)^{-1}$.

Exercise 13.1

The moment conditions are

$$\mathbb{E}[X(Y - X'\beta)] = \mathbf{0}, \quad \mathbb{E}[Z((Y - X'\beta)^2 - Z'\gamma)] = \mathbf{0},$$

$k + \dim \gamma$ equations in as many unknowns. The system is triangular: the first block involves only β . Its sample analogue gives OLS, $\hat{\beta} = (\sum_i X_i X_i')^{-1} \sum_i X_i Y_i$. Substituting into the second block, $\sum_i Z_i(\hat{e}_i^2 - Z_i'\hat{\gamma}) = \mathbf{0}$ with $\hat{e}_i = Y_i - X_i'\hat{\beta}$, so

$$\hat{\gamma} = \left(\sum_i Z_i Z_i' \right)^{-1} \sum_i Z_i \hat{e}_i^2,$$

the OLS regression of squared residuals on Z (the variance-function estimator of Exercise 7.21).

Exercise 13.2

With $W_n = (Z'Z)^{-1}$ the GMM estimator is 2SLS. Scaling each matrix by n ,

$$\sqrt{n}(\hat{\beta} - \beta) = \left[\left(\frac{1}{n} X'Z \right) \left(\frac{1}{n} Z'Z \right)^{-1} \left(\frac{1}{n} Z'X \right) \right]^{-1} \left(\frac{1}{n} X'Z \right) \left(\frac{1}{n} Z'Z \right)^{-1} \frac{1}{\sqrt{n}} Z'e.$$

By the WLLN $\frac{1}{n}Z'X \xrightarrow{p} Q$ and $\frac{1}{n}Z'Z \xrightarrow{p} M$; by the CLT $n^{-1/2}Z'e \xrightarrow{d} N(0, \Omega)$, and under homoskedasticity $\Omega = \mathbb{E}[ZZ'\mathbb{E}[e^2 | Z]] = \sigma^2 M$. By Slutsky,

$$\begin{aligned} \sqrt{n}(\hat{\beta} - \beta) &\xrightarrow{d} (Q'M^{-1}Q)^{-1}Q'M^{-1}N(0, \sigma^2 M) \\ &= N(0, \sigma^2(Q'M^{-1}Q)^{-1}Q'M^{-1}MM^{-1}Q(Q'M^{-1}Q)^{-1}) = N(0, \sigma^2(Q'M^{-1}Q)^{-1}). \end{aligned}$$

Since $W = M^{-1} \propto \Omega^{-1}$ here, 2SLS is efficient GMM under homoskedasticity.

Exercise 13.3

Let $\delta = \tilde{\beta} - \beta \xrightarrow{p} \mathbf{0}$. Then $\tilde{e}_i^2 = e_i^2 - 2e_i X_i' \delta + (X_i' \delta)^2$ and

$$\left\| \frac{1}{n} \sum_i Z_i Z_i' \tilde{e}_i^2 - \frac{1}{n} \sum_i Z_i Z_i' e_i^2 \right\| \leq 2\|\delta\| \frac{1}{n} \sum_i \|Z_i\|^2 \|e_i\| \|X_i\| + \|\delta\|^2 \frac{1}{n} \sum_i \|Z_i\|^2 \|X_i\|^2.$$

Under Assumption 12.2 the averages on the right are $O_p(1)$ (their expectations are finite by Hölder, exactly as in Exercise 12.6), so the difference is $o_p(1)$. By the WLLN $n^{-1} \sum_i Z_i Z_i' e_i^2 \xrightarrow{p} \Omega$. Hence

$n^{-1} \sum_i Z_i Z_i' \tilde{e}_i^2 \xrightarrow{p} \Omega$, and since $\Omega > 0$ and inversion is continuous there, $\widehat{W} \xrightarrow{p} \Omega^{-1}$. Note that only consistency of $\tilde{\beta}$ is required, not a rate.

Exercise 13.4

(a)

$$V_0 = (Q'\Omega^{-1}Q)^{-1}Q'\Omega^{-1}\Omega\Omega^{-1}Q(Q'\Omega^{-1}Q)^{-1} = (Q'\Omega^{-1}Q)^{-1}(Q'\Omega^{-1}Q)(Q'\Omega^{-1}Q)^{-1} = (Q'\Omega^{-1}Q)^{-1}.$$

(b) Take the $\ell \times k$ matrices

$$A = WQ(Q'WQ)^{-1}, \quad B = \Omega^{-1}Q(Q'\Omega^{-1}Q)^{-1}.$$

Then $A'\Omega A = (Q'WQ)^{-1}Q'W\Omega WQ(Q'WQ)^{-1} = V$ and $B'\Omega B = V_0$ by (a).

(c) $B'\Omega A = (Q'\Omega^{-1}Q)^{-1}Q'\Omega^{-1}\Omega WQ(Q'WQ)^{-1} = (Q'\Omega^{-1}Q)^{-1}(Q'WQ)(Q'WQ)^{-1} = (Q'\Omega^{-1}Q)^{-1}$, which equals $B'\Omega B = V_0$. Subtracting, $B'\Omega(A - B) = \mathbf{0}$, and by transposition $(A - B)'\Omega B = \mathbf{0}$.

(d) Writing $A = B + (A - B)$,

$$V = A'\Omega A = B'\Omega B + B'\Omega(A - B) + (A - B)'\Omega B + (A - B)'\Omega(A - B) = V_0 + (A - B)'\Omega(A - B).$$

Since $\Omega > 0$, $(A - B)'\Omega(A - B) \geq 0$, so $V \geq V_0$: $W = \Omega^{-1}$ is the efficient weight matrix.

Exercise 13.5

By the asymptotic normality of GMM (THEOREM 13.2) and the delta method (Assumption 7.3: r continuously differentiable with R of rank q), under H_0 , $\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} N(0, V_\theta)$ with $V_\theta = R'V_\beta R > 0$. The covariance estimator is consistent: $\widehat{V}_\beta \xrightarrow{p} V_\beta$ (THEOREM 13.5) and $\widehat{R} = R(\widehat{\beta}) \xrightarrow{p} R$ by continuity, so $\widehat{V}_\theta \xrightarrow{p} V_\theta$. By the continuous mapping theorem,

$$W = \sqrt{n}(\widehat{\theta} - \theta_0)'\widehat{V}_\theta^{-1}\sqrt{n}(\widehat{\theta} - \theta_0) \xrightarrow{d} \xi'V_\theta^{-1}\xi, \quad \xi \sim N(0, V_\theta),$$

and $\xi'V_\theta^{-1}\xi = \|V_\theta^{-1/2}\xi\|^2 \sim \chi_q^2$. Since χ_q^2 is continuous, $\mathbb{P}[W > c \mid H_0] \rightarrow 1 - G_q(c) = \alpha$.

Exercise 13.6

Let $\widehat{\beta} = (X'ZWZ'X)^{-1}X'ZWZ'Y$ be unconstrained GMM, so $X'ZWZ'(Y - X\widehat{\beta}) = \mathbf{0}$ (its first-order condition). Writing $Y - X\beta = (Y - X\widehat{\beta}) - X(\beta - \widehat{\beta})$ and expanding the criterion (up to the positive scale factor n^{-1}),

$$(Y - X\beta)'ZWZ'(Y - X\beta) = (Y - X\widehat{\beta})'ZWZ'(Y - X\widehat{\beta}) + (\beta - \widehat{\beta})'(X'ZWZ'X)(\beta - \widehat{\beta}),$$

the cross terms vanishing by the first-order condition. The first term does not involve β , so minimizing $J(\beta)$ subject to $R'\beta = c$ is minimizing the minimum-distance criterion $(\beta - \widehat{\beta})'\widehat{W}(\beta - \widehat{\beta})$ with $\widehat{W} = X'ZWZ'X$. By (8.23) (Lagrange multipliers, Exercise 8.10),

$$\widehat{\beta}_{\text{cgmm}} = \widehat{\beta} - (X'ZWZ'X)^{-1}R(R'(X'ZWZ'X)^{-1}R)^{-1}(R'\widehat{\beta} - c),$$

which is (13.16).

Exercise 13.7

With $W = \hat{\Omega}^{-1}$ and $\hat{Q} = n^{-1}Z'X$, $X'ZWZ'X = n^2\hat{Q}'\hat{\Omega}^{-1}\hat{Q} = n^2\hat{V}_\beta^{-1}$ where $\hat{V}_\beta = (\hat{Q}'\hat{\Omega}^{-1}\hat{Q})^{-1}$. Substituting into (13.16) the powers of n cancel:

$$(X'ZWZ'X)^{-1}R(R'(X'ZWZ'X)^{-1}R)^{-1} = n^{-2}\hat{V}_\beta R(n^{-2}R'\hat{V}_\beta R)^{-1} = \hat{V}_\beta R(R'\hat{V}_\beta R)^{-1},$$

so $\hat{\beta}_{\text{cgmm}} = \hat{\beta}_{\text{gmm}} - \hat{V}_\beta R(R'\hat{V}_\beta R)^{-1}(R'\hat{\beta}_{\text{gmm}} - c)$, which is (13.19).

Exercise 13.8

Under $R'\beta = c$, (13.16) becomes (13.17): $\sqrt{n}(\hat{\beta}_{\text{cgmm}} - \beta) = \hat{H}_n\sqrt{n}(\hat{\beta}_{\text{gmm}} - \beta)$ with $\hat{H}_n = \mathbf{I}_k - A_n^{-1}R(R'A_n^{-1}R)^{-1}R'$ and $A_n = n^{-2}X'ZW_nZ'X$ (the scaling cancels as in Exercise 13.7). Since $n^{-1}Z'X \xrightarrow{p} Q$ and $W_n \xrightarrow{p} W$, $A_n \xrightarrow{p} Q'WQ > 0$ and $\hat{H}_n \xrightarrow{p} H = \mathbf{I}_k - (Q'WQ)^{-1}R(R'(Q'WQ)^{-1}R)^{-1}R'$. With $\sqrt{n}(\hat{\beta}_{\text{gmm}} - \beta) \xrightarrow{d} N(0, V_\beta)$ (THEOREM 13.2) and Slutsky, $\sqrt{n}(\hat{\beta}_{\text{cgmm}} - \beta) \xrightarrow{d} N(0, HV_\beta H')$. Expanding $HV_\beta H'$ exactly as in Exercise 8.12, with W^{-1} there replaced by $(Q'WQ)^{-1}$, gives the four terms of (13.18).

Exercise 13.9

With the efficient weight, $W_n \xrightarrow{p} \Omega^{-1}$ (Exercise 13.3), so $Q'WQ = Q'\Omega^{-1}Q = V_\beta^{-1}$. Substituting $(Q'WQ)^{-1} = V_\beta$ into (13.18) and writing $G = (R'V_\beta R)^{-1}$,

$$V_{\text{cgmm}} = V_\beta - V_\beta RGR'V_\beta - V_\beta RGR'V_\beta + V_\beta RG(R'V_\beta R)GR'V_\beta = V_\beta - V_\beta R(R'V_\beta R)^{-1}R'V_\beta,$$

which is (13.20). This is the same algebra as Exercise 8.14: constrained efficient GMM has the efficient-minimum-distance variance.

Exercise 13.10

The moment function is $g_i(\beta) = Z_i(Y_i - m(X_i, \beta))$ with $\mathbb{E}[g_i(\beta)] = \mathbf{0}$ at the truth; identification requires $\ell \geq k$ and $G = \mathbb{E}[Z \partial m(X, \beta)/\partial \beta']$ of rank k .

- (1) *First step.* Minimize $J_1(\beta) = n\bar{g}_n(\beta)'(n^{-1}Z'Z)^{-1}\bar{g}_n(\beta)$ numerically (non-linear 2SLS) to obtain a consistent $\tilde{\beta}$.
- (2) *Weight.* $\hat{\Omega} = n^{-1}\sum_i Z_i Z_i' \tilde{e}_i^2$ with $\tilde{e}_i = Y_i - m(X_i, \tilde{\beta})$ (optionally centred).
- (3) *Second step.* $\hat{\beta} = \arg \min_\beta n\bar{g}_n(\beta)'\hat{\Omega}^{-1}\bar{g}_n(\beta)$, again by numerical optimization (Gauss-Newton works well). Optionally iterate steps 2-3 to convergence.
- (4) *Inference.* $\hat{V}_\beta = (\hat{G}'\hat{\Omega}^{-1}\hat{G})^{-1}$ with $\hat{G} = n^{-1}\sum_i Z_i \partial m(X_i, \hat{\beta})/\partial \beta'$ and $\hat{\Omega}$ recomputed at $\hat{\beta}$; standard errors are $\sqrt{n^{-1}[\hat{V}_\beta]_{jj}}$, and $J(\hat{\beta}) \xrightarrow{d} \chi_{\ell-k}^2$ tests the over-identifying restrictions.

Exercise 13.11

With $\hat{\Omega} = n^{-1}\sum_i Z_i Z_i' \hat{e}_i^2$ from a first-step estimate (OLS = 2SLS),

$$\hat{\beta}_{\text{gmm}} = \frac{X'Z\hat{\Omega}^{-1}Z'Y}{X'Z\hat{\Omega}^{-1}Z'X}, \quad Z'X = \begin{pmatrix} \sum X_i^2 \\ \sum X_i^3 \end{pmatrix}, \quad Z'Y = \begin{pmatrix} \sum X_i Y_i \\ \sum X_i^2 Y_i \end{pmatrix}.$$

Exercise 12.7 showed that 2SLS equals OLS exactly. Efficient GMM is *different* in general: it combines the two moment conditions $\mathbb{E}[Xe] = 0$ and $\mathbb{E}[X^2e] = 0$ with weights that depend on Ω .

Under conditional homoskedasticity $\Omega = \sigma^2 \mathbb{E}[ZZ']$, so efficient GMM is asymptotically equivalent to 2SLS and hence to OLS (it will differ slightly numerically because $\hat{\Omega}$ is estimated). Under heteroskedasticity it is asymptotically more efficient than OLS, since it exploits the extra valid moment $\mathbb{E}[X^2 e] = 0$ that OLS ignores; its variance is $(Q' \Omega^{-1} Q)^{-1} \leq \mathbb{E}[X^2 e^2] / (\mathbb{E}[X^2])^2$ by Exercise 13.4.

Exercise 13.12

(a) $X'(Y - X\beta) = X'Y - X'X\beta = X'X(\hat{\beta} - \beta)$. Hence

$$J(\beta) = n^{-1}(\beta - \hat{\beta})' X' X \hat{\Omega}^{-1} X' X (\beta - \hat{\beta}) = n(\beta - \hat{\beta})' \hat{Q}_{XX} \hat{\Omega}^{-1} \hat{Q}_{XX} (\beta - \hat{\beta}) = n(\beta - \hat{\beta})' \hat{V}_{\beta}^{-1} (\beta - \hat{\beta}),$$

with $\hat{V}_{\beta} = \hat{Q}_{XX}^{-1} \hat{\Omega} \hat{Q}_{XX}^{-1}$ the robust (HC0) asymptotic covariance matrix. So minimizing J subject to $r(\beta) = 0$ is efficient minimum distance (Section 8.8 / 8.14).

(b) For $R'\beta = c$, (8.25) gives $\hat{\beta} - \tilde{\beta} = \hat{V}_{\beta} R(R'\hat{V}_{\beta}R)^{-1}(R'\hat{\beta} - c)$, so

$$D = n(\hat{\beta} - \tilde{\beta})' \hat{V}_{\beta}^{-1} (\hat{\beta} - \tilde{\beta}) = n(R'\hat{\beta} - c)' (R'\hat{V}_{\beta}R)^{-1} (R'\hat{\beta} - c) = W.$$

For non-linear hypotheses the two statistics differ (and D is invariant to reparameterization).

Exercise 13.13

(a) $\Omega^{-1} > 0$ has a Cholesky factorization $\Omega^{-1} = CC'$ with C non-singular; then $\Omega = (CC')^{-1} = C'^{-1}C^{-1}$.

(b) $(C'\hat{\Omega}C)^{-1} = C^{-1}\hat{\Omega}^{-1}C'^{-1}$, so $n(C'\bar{g}_n)'(C'\hat{\Omega}C)^{-1}C'\bar{g}_n = n\bar{g}_n'CC^{-1}\hat{\Omega}^{-1}C'^{-1}C'\bar{g}_n = n\bar{g}_n'\hat{\Omega}^{-1}\bar{g}_n = J$.

(c) $\bar{g}_n(\hat{\beta}) = \frac{1}{n}Z'e - \frac{1}{n}Z'X(\hat{\beta} - \beta)$ and $\hat{\beta} - \beta = (\frac{1}{n}X'Z\hat{\Omega}^{-1}\frac{1}{n}Z'X)^{-1}\frac{1}{n}X'Z\hat{\Omega}^{-1}\bar{g}_n(\beta)$. Therefore $\bar{g}_n(\hat{\beta}) = [\mathbf{I}_{\ell} - \frac{1}{n}Z'X(\cdot\cdot\cdot)^{-1}\frac{1}{n}X'Z\hat{\Omega}^{-1}]\bar{g}_n(\beta)$, and premultiplying by C' and inserting $C'^{-1}C'$, $C'\bar{g}_n(\hat{\beta}) = D_n C'\bar{g}_n(\beta)$ with D_n as stated.

(d) $\frac{1}{n}Z'X \xrightarrow{p} Q$ and $\hat{\Omega}^{-1} \xrightarrow{p} \Omega^{-1} = CC'$, so $\hat{\Omega}^{-1}C'^{-1} \xrightarrow{p} C$ and $Q'\Omega^{-1}Q = (C'Q)'(C'Q) = R'R$ with $R = C'Q$. Hence $D_n \xrightarrow{p} \mathbf{I}_{\ell} - C'Q(R'R)^{-1}Q'C = \mathbf{I}_{\ell} - R(R'R)^{-1}R'$.

(e) $n^{1/2}C'\bar{g}_n(\beta) = C'n^{-1/2}Z'e \xrightarrow{d} C'N(0, \Omega) = N(0, C'C'^{-1}C^{-1}C) = N(0, \mathbf{I}_{\ell})$.

(f) By (b)–(e) and $C'\hat{\Omega}C \xrightarrow{p} C'\Omega C = \mathbf{I}_{\ell}$, $J = (D_n\sqrt{n}C'\bar{g}_n(\beta))'(C'\hat{\Omega}C)^{-1}(D_n\sqrt{n}C'\bar{g}_n(\beta)) \xrightarrow{d} u'P'u = u'Pu$, where $P = \mathbf{I}_{\ell} - R(R'R)^{-1}R'$ is symmetric idempotent.

(g) P is an orthogonal projection of rank $\text{tr } P = \ell - \text{tr}((R'R)^{-1}R'R) = \ell - k$. Write $P = H\Lambda H'$ with H orthogonal and Λ having $\ell - k$ ones and k zeros. Then $v = H'u \sim N(0, \mathbf{I}_{\ell})$ and $u'Pu = v'\Lambda v = \sum_{j=1}^{\ell-k} v_j^2 \sim \chi_{\ell-k}^2$.

Exercise 13.14

(a) Under H_0 , $\sqrt{n}m_n(\beta_0) = n^{-1/2}\sum_i Z_i e_i \xrightarrow{d} \xi \sim N(0, \Omega)$, so $J(\beta_0) \xrightarrow{d} \xi'W\xi = \sum_{j=1}^{\ell} \lambda_j \chi_{1,j}^2$, a weighted sum of independent χ_1^2 variables with weights the eigenvalues of $\Omega^{1/2}W\Omega^{1/2}$. No parameter is estimated.

(b) $W = \Omega^{-1}$ gives $J(\beta_0) \xrightarrow{d} \chi_{\ell}^2$ — degrees of freedom ℓ (not $\ell - k$), because β is fixed at β_0 rather than estimated.

- (c) Use the errors implied by the null: $\widehat{W}(\beta_0) = (n^{-1} \sum_i Z_i Z_i' (Y_i - X_i' \beta_0)^2)^{-1}$, consistent for Ω^{-1} under H_0 without any estimation.
- (d) Reject H_0 at level α if $J(\beta_0) = n m_n(\beta_0)' \widehat{W}(\beta_0) m_n(\beta_0)$ exceeds the $1 - \alpha$ quantile of χ_ℓ^2 .
- (e) Invert the test: $\widehat{C} = \{\beta_0 : J(\beta_0) \leq \chi_{\ell, 1-\alpha}^2\}$. Because $\widehat{W}(\beta_0)$ depends on β_0 , $J(\beta_0)$ is a ratio-of-quadratics function of β_0 , and the set is *not* an ellipse in general: it can be unbounded (when identification is weak), a union of disjoint pieces, or even empty (when the over-identifying restrictions fail badly, since $J(\beta) \geq \min_\beta J$ everywhere). In exchange, its coverage does not depend on instrument strength — it is the GMM form of the Anderson–Rubin confidence set.

Exercise 13.15

- (a) $\widehat{\beta} = (X'ZWZ'X)^{-1}X'ZWZ'Y$.
- (b) By (13.16) with $c = \mathbf{0}$, $\widetilde{\beta} = \widehat{\beta} - (X'ZWZ'X)^{-1}R(R'(X'ZWZ'X)^{-1}R)^{-1}R'\widehat{\beta}$.
- (c) With $W = \Omega^{-1}$, $V_\beta = (Q'\Omega^{-1}Q)^{-1}$ and, by Theorem 13.10,

$$\sqrt{n}(\widetilde{\beta} - \beta) \xrightarrow{d} N(0, V_\beta - V_\beta R(R'V_\beta R)^{-1}R'V_\beta).$$

Exercise 13.16

- (a) $\widehat{\beta} = (X'ZWZ'X)^{-1}X'ZWZ'Y$.
- (b) $\mathbb{E}[Ze] = \delta n^{-1/2} \mathbb{E}[Z] + \mathbb{E}[Z \mathbb{E}[u | Z]] = \delta n^{-1/2} \mu_Z \neq \mathbf{0}$ for $\delta \neq 0$.
- (c) With $\widehat{Q} = n^{-1}Z'X$,

$$\sqrt{n}(\widehat{\beta} - \beta) = (\widehat{Q}'W\widehat{Q})^{-1}\widehat{Q}'W \frac{1}{\sqrt{n}} \sum_i Z_i e_i = (\widehat{Q}'W\widehat{Q})^{-1}\widehat{Q}'W \left(\delta \bar{Z} + \frac{1}{\sqrt{n}} \sum_i Z_i u_i \right).$$

- (d) $\bar{Z} \xrightarrow{p} \mu_Z$, $\widehat{Q} \xrightarrow{p} Q$, and $n^{-1/2} \sum_i Z_i u_i \xrightarrow{d} N(0, \Omega_u)$ with $\Omega_u = \mathbb{E}[ZZ'u^2]$. Hence

$$\sqrt{n}(\widehat{\beta} - \beta) \xrightarrow{d} N\left((Q'WQ)^{-1}Q'W\mu_Z\delta, (Q'WQ)^{-1}Q'W\Omega_u WQ(Q'WQ)^{-1}\right).$$

Local misspecification of order $n^{-1/2}$ shifts the *mean* of the limit distribution — an asymptotic bias of the same order as the sampling error — while leaving the variance unchanged. This is the standard device for studying size distortion and the power of overidentification tests.

Exercise 13.17

It is feasible: two instruments for two coefficients, a just-identified model (GMM then equals IV and the weight matrix is irrelevant).

- *Exogeneity* holds automatically: $\mathbb{E}[Z^2 e] = \mathbb{E}[Z^2 \mathbb{E}[e | Z]] = 0$. The substantive assumption is the *exclusion* of Z^2 from the structural equation (the direct effect of Z is exactly linear).
- *Relevance* requires $\mathbb{E}[(Z, Z^2)'(Z, X)]$ to be non-singular: after partialling out Z , Z^2 must predict X , i.e. $\mathbb{E}[X | Z]$ must be non-linear in Z .

As in Exercise 12.15, identification comes entirely from functional form, which is rarely economically justified; the estimator is valid only under those two conditions and is fragile if either nearly fails.

Exercise 13.18

Stack the instruments $W_i = (X'_i, Q'_i)'$ (dropping any element of Q duplicated in X), $\ell = k + m$.

(1) First step: OLS $\hat{\beta}_1$ (consistent since $\mathbb{E}[Xe] = \mathbf{0}$), residuals $\hat{e}_i$.

(2) $\hat{\Omega} = n^{-1} \sum_i W_i W'_i \hat{e}_i^2$.

(3) $\hat{\beta} = (X'W\hat{\Omega}^{-1}W'X)^{-1}X'W\hat{\Omega}^{-1}W'Y$, with $\hat{V}_\beta = (\hat{Q}'\hat{\Omega}^{-1}\hat{Q})^{-1}$, $\hat{Q} = n^{-1}W'X$.

The model is over-identified by m and $J(\hat{\beta}) \xrightarrow{d} \chi_m^2$ tests the extra orthogonality conditions. Under homoskedasticity the efficient estimator is asymptotically equivalent to 2SLS with instruments (X, Q) , which equals OLS because X is in the instrument set (Exercise 12.7); the gain over OLS arises only under heteroskedasticity.

Exercise 13.19

The moments are $g_i(\mu) = (Y_i - \mu, X_i)'$ with $\ell = 2$, $k = 1$. Let $\Omega = \text{var}(g_i) = \begin{pmatrix} \sigma_Y^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_X^2 \end{pmatrix}$.

Efficient GMM minimizes $(\bar{Y} - \mu, \bar{X})\hat{\Omega}^{-1}(\bar{Y} - \mu, \bar{X})'$. Since $\Omega^{-1} \propto \begin{pmatrix} \sigma_X^2 & -\sigma_{XY} \\ -\sigma_{XY} & \sigma_Y^2 \end{pmatrix}$, the first-order condition in $a = \bar{Y} - \mu$ is $\sigma_X^2 a - \sigma_{XY} \bar{X} = 0$, so

$$\hat{\mu} = \bar{Y} - \hat{\beta} \bar{X}, \quad \hat{\beta} = \frac{\hat{\sigma}_{XY}}{\hat{\sigma}_X^2},$$

the sample mean of Y adjusted by the regression of Y on X times the sample mean of X (a control variate). Its asymptotic variance is $\sigma_Y^2 - \sigma_{XY}^2/\sigma_X^2 = \sigma_Y^2(1 - \rho_{XY}^2) \leq \sigma_Y^2$: knowledge of $\mathbb{E}[X]$ improves on $\bar{Y}$ whenever X and Y are correlated. (Use centred sample moments for $\hat{\Omega}$.)

Exercise 13.20

Compute unconstrained efficient two-step GMM $\hat{\beta}$ with $\hat{\Omega}$ and $\hat{V}_\beta = (\hat{Q}'\hat{\Omega}^{-1}\hat{Q})^{-1}$, then apply (13.19):

$$\tilde{\beta} = \hat{\beta} - \hat{V}_\beta R(R'\hat{V}_\beta R)^{-1}R'\hat{\beta},$$

with asymptotic variance $V_\beta - V_\beta R(R'V_\beta R)^{-1}R'V_\beta$ (Theorem 13.10). Equivalently, let $R_\perp$ be a $k \times (k - q)$ matrix whose columns span the null space of R' , write $\beta = R_\perp \theta$, estimate θ by efficient GMM in $Y = (R'_\perp X)\theta + e$ with instruments Z , and set $\tilde{\beta} = R_\perp \hat{\theta}$. The two are asymptotically equivalent; iterating the weight matrix with the constrained residuals is a natural refinement.

Exercise 13.21

Let $P_j = Z_j(Z'_j Z_j)^{-1}Z'_j$ and $A_j = X'P_j X$, so $\hat{\beta} = A_1^{-1}X'P_1 Y$ and $\tilde{\beta} = A_2^{-1}X'P_2 Y$. With the block-diagonal W ,

$$X'ZWZ'X = \lambda A_1 + (1 - \lambda)A_2, \quad X'ZWZ'Y = \lambda X'P_1 Y + (1 - \lambda)X'P_2 Y = \lambda A_1 \hat{\beta} + (1 - \lambda)A_2 \tilde{\beta}.$$

Therefore

$$\bar{\beta} = (\lambda A_1 + (1 - \lambda)A_2)^{-1}(\lambda A_1 \hat{\beta} + (1 - \lambda)A_2 \tilde{\beta}),$$

a matrix-weighted average with weights proportional to $\lambda X'P_1X$ and $(1 - \lambda)X'P_2X$: each 2SLS estimator is weighted by λ times the strength of its own first stage.

Exercise 13.22

- *Wald (1)* is valid: $W \xrightarrow{d} \chi_{k_1}^2$. It needs only the unrestricted estimate.
- *Distance (2)* is valid too. Just-identified GMM sets $\bar{g}_n(\hat{\beta}) = \mathbf{0}$, so $J(\hat{\beta}) = 0$ and $D = J(\hat{\beta})$, the minimized criterion under the restriction. With the same (efficient) weight matrix in both, D equals W exactly for this linear hypothesis (Exercise 13.12); with different weight estimates they are asymptotically equivalent. D has the advantage of invariance to reparameterization, which matters only for non-linear hypotheses.
- *Overidentification test (3)* is useless here. The unrestricted model has $\ell = k$, so $J(\hat{\beta}) = 0$ identically, with zero degrees of freedom: it cannot test anything about β_1 . (Computing J in the *restricted* model, where there are k_1 over-identifying restrictions, is exactly the distance statistic (2) — so the only sensible reading of (3) collapses into (2).)

Recommendation: use (1) or (2), which agree; reject (3).

Exercise 13.23

The moment conditions $\mathbb{E}[X(Y - X'Q\theta)] = \mathbf{0}$ are k equations in $m < k$ unknowns, over-identified by $k - m$. Efficient GMM with $\hat{\Omega} = n^{-1} \sum_i X_i X_i' \hat{e}_i^2$ (OLS residuals) is

$$\hat{\theta} = (Q'X'X\hat{\Omega}^{-1}X'XQ)^{-1}Q'X'X\hat{\Omega}^{-1}X'Y.$$

Using $X'Y = X'X\hat{\beta}_{ols}$ and $\hat{V}_{\beta} = \hat{Q}_{XX}^{-1}\hat{\Omega}\hat{Q}_{XX}^{-1}$, this is the efficient minimum distance projection

$$\hat{\theta} = (Q'\hat{V}_{\beta}^{-1}Q)^{-1}Q'\hat{V}_{\beta}^{-1}\hat{\beta}_{ols}, \quad \sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, (Q'V_{\beta}^{-1}Q)^{-1}),$$

(the GLS regression of $\hat{\beta}$ on Q). Under homoskedasticity it is asymptotically equivalent to OLS of Y on $Q'X$ (constrained least squares), but not otherwise.

Exercise 13.24

(a) $\bar{g}_n(\theta) = n^{-1} \sum_i X_i(Y_i - \theta) = \overline{XY} - \bar{X}\theta$. With $\hat{\Omega} = n^{-1} \sum_i X_i X_i'(Y_i - \tilde{\theta})^2$ from a first-step estimate $\tilde{\theta}$ (e.g. with identity weight, $\tilde{\theta} = \overline{X'XY}/\overline{X'X}$),

$$\hat{\theta} = \frac{\overline{X'}\hat{\Omega}^{-1}\overline{XY}}{\overline{X'}\hat{\Omega}^{-1}\overline{X}}.$$

(b) k moments for one parameter: over-identified when $k > 1$ (just identified when $k = 1$). Identification requires $\mathbb{E}[X] \neq \mathbf{0}$.

(c) $J = n\bar{g}_n(\hat{\theta})'\hat{\Omega}^{-1}\bar{g}_n(\hat{\theta}) \xrightarrow{d} \chi_{k-1}^2$. If X contains a constant, the constant moment identifies $\theta = \mathbb{E}[Y]$ and J tests whether the other elements of X are uncorrelated with Y .

Exercise 13.25

(a) The moment vector is $e = (e_1, \dots, e_n)'$ with covariance matrix $\sigma^2 \mathbf{I}_n$ under i.i.d. sampling, so the efficient weight (its inverse) is $\sigma^{-2} \mathbf{I}_n$.

(b) $J(\beta) = \sigma^{-2}(Y - X\beta)'(Y - X\beta)$ is minimized by OLS.

(c) $J(\hat{\beta}) = \sigma^{-2} \hat{e}' \hat{e} = n \hat{\sigma}^2 / \sigma^2$ (and $= n$ identically if σ^2 is replaced by $\hat{\sigma}^2$).

(d) $\ell = n$ moments and k parameters, so the asserted limit is χ_{n-k}^2 .

(e) No. The asymptotic theory requires a fixed number of moments, each estimated by an average over a growing sample. Here each “moment” uses a single observation, ℓ grows with n , and J is nothing but a scaled residual variance. (Under normal homoskedastic errors $\hat{e}' \hat{e} / \sigma^2 \sim \chi_{n-k}^2$ happens to hold exactly — but that is a statement about σ^2 , not a test of any over-identifying restriction; the statistic has no power against failures of $\mathbb{E}[Xe] = \mathbf{0}$.)

Exercise 13.26

(a) $\bar{X}' e = (X_1 e_1, \dots, X_n e_n)'$. By independence the cross blocks vanish, and conditioning on the regressors (which is what makes the hint relevant),

$$\Omega = \mathbb{E} \left[\bar{X}' e e' \bar{X} \mid X \right] = \text{diag} \{ \sigma^2 X_1 X_1', \dots, \sigma^2 X_n X_n' \},$$

each block of rank one. (Unconditionally $\Omega = \mathbf{I}_n \otimes \sigma^2 \mathbb{E}[X X']$.)

(b) Block by block, using $(aa')^- = aa'(a'a)^{-2}$, $W = \Omega^- = \text{diag} \{ \sigma^{-2} X_i X_i' (X_i' X_i)^{-2} \}$.

(c) $J(\beta) = \sum_i e_i(\beta) X_i' W_i X_i e_i(\beta) = \sigma^{-2} \sum_i e_i(\beta)^2 (X_i' X_i)^2 (X_i' X_i)^{-2} = \sigma^{-2} \sum_i (Y_i - X_i' \beta)^2$, minimized by OLS.

(d) $J(\hat{\beta}) = \sigma^{-2} \sum_i \hat{e}_i^2$.

(e) No. Theorem 13.14 would suggest χ_{nk-k}^2 , with mean near nk , whereas $J(\hat{\beta}) \approx n$. As in Exercise 13.25, the number of moments grows with the sample, each moment is “estimated” from one observation, and the weight matrix is singular; the conditions of the theorem fail and the approximation is meaningless.

Exercise 13.27

Robust standard errors in parentheses.

(a)

	2SLS	efficient GMM
risk	0.772 (0.100)	0.728 (0.090)
intercept	3.019 (0.682)	3.336 (0.613)

(b) $J = 3.75$ with one degree of freedom ($p = 0.053$), computed with the first-step weight matrix (4.02 if the weight is re-estimated at the GMM estimate).

(c) GMM moves the risk coefficient further down (from 0.77 to 0.73) and reduces the standard error by about 10%, as expected when the efficient weight accounts for heteroskedasticity. The J test is borderline: it just fails to reject at 5%, while the Sargan (homoskedastic) statistic of Exercise 12.22(k) rejected. Together with the just-identified estimate of 0.93, the evidence suggests the two instruments do not agree well, so the over-identified estimates should be interpreted cautiously.

Exercise 13.28

(a) 2SLS(a) specification (public, private): education 0.1615 (0.0405) by GMM versus 0.1611 (0.0405) by 2SLS; experience 0.1196 vs 0.1193, $\text{experience}^2/100$ -0.2315 vs -0.2305 , Black -0.101 , south -0.095 , urban 0.115 vs 0.116. No meaningful change: heteroskedasticity is mild and the model has only one over-identifying restriction.

(b) Expanded instrument set: education 0.0839 (0.0062) versus 0.0825 (0.0062) by 2SLS, with the other coefficients changing only in the third decimal. Again no meaningful change — and GMM cannot fix the invalidity of the added instruments discussed in Exercise 12.24.

(c) $J = 0.87$ with 1 d.f. ($p = 0.35$) for (a): no evidence against the over-identifying restriction. $J = 10.44$ with 3 d.f. ($p = 0.015$) for (b): the over-identifying restrictions are rejected, confirming that the age interactions are not valid excluded instruments.

Chapter 14

Time Series

Solutions to all 22 exercises of Chapter 14 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 14.18–14.22 use FRED-QD (1959Q1–2017Q4) and FRED-MD (1959m1–2017m12) from the textbook website. Growth rates are $100 \times (x_t/x_{t-1} - 1)$, as in the textbook’s own programs.

Key results used: transformations of strictly stationary ergodic processes are strictly stationary and ergodic (THEOREM 14.5); absolutely summable linear filters of stationary ergodic inputs are stationary ergodic (THEOREM 14.6); the Ergodic Theorem (THEOREM 14.9); and the MDS CLT (THEOREM 14.11). HC denotes heteroskedasticity-robust (HC1) standard errors; NW denotes Newey–West with Bartlett weights and $M = 5$.

Exercise 14.1

(The printed statement ends “ $\hat{\rho}(k) \xrightarrow{p} \gamma(k)$ ”; the limit is $\rho(k)$.) Expand

$$\hat{\gamma}(k) = \frac{1}{n} \sum_{t=k+1}^n Y_t Y_{t-k} - \bar{Y} \frac{1}{n} \sum_{t=k+1}^n Y_{t-k} - \bar{Y} \frac{1}{n} \sum_{t=k+1}^n Y_t + \frac{n-k}{n} \bar{Y}^2.$$

- $W_t = Y_t Y_{t-k}$ is a measurable function of $(Y_t, Y_{t-1}, \dots)$, so it is strictly stationary and ergodic (THEOREM 14.5), and $\mathbb{E}|W_t| \leq \mathbb{E}[Y_t^2] < \infty$ by Cauchy–Schwarz. By the Ergodic Theorem, $n^{-1} \sum_{t=k+1}^n Y_t Y_{t-k} \xrightarrow{p} \mathbb{E}[Y_t Y_{t-k}]$ (dropping k terms changes the average by $o_p(1)$).
- Likewise $\bar{Y} \xrightarrow{p} \mu$, and the partial averages over $t = k+1, \dots, n$ differ from $\bar{Y}$ by $O_p(n^{-1})$.

Hence $\hat{\gamma}(k) \xrightarrow{p} \mathbb{E}[Y_t Y_{t-k}] - \mu^2 - \mu^2 + \mu^2 = \text{cov}(Y_t, Y_{t-k}) = \gamma(k)$. Provided $\gamma(0) > 0$, the continuous mapping theorem gives $\hat{\rho}(k) \xrightarrow{p} \gamma(k)/\gamma(0) = \rho(k)$.

Exercise 14.2

X_{t-1} is $\mathcal{F}_{t-1} \subseteq \mathcal{F}_t$ -measurable and e_t is $\mathcal{F}_t$ -measurable, so u_t is adapted to $\mathcal{F}_t$. Assuming $\mathbb{E}|u_t| < \infty$, the conditioning theorem gives

$$\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = X_{t-1} \mathbb{E}[e_t \mid \mathcal{F}_{t-1}] = X_{t-1} \cdot 0 = 0.$$

This is why regression scores $X_{t-1}e_t$ in autoregressions are MDS, so the MDS CLT applies.

Exercise 14.3

e_t^2 is $\mathcal{F}_t$ -measurable and σ_t^2 is $\mathcal{F}_{t-1}$ -measurable, so u_t is adapted. It is integrable when $\mathbb{E}[e_t^2] < \infty$ (since $\mathbb{E}[\sigma_t^2] = \mathbb{E}[e_t^2]$). And $\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[e_t^2 \mid \mathcal{F}_{t-1}] - \sigma_t^2 = 0$.

Exercise 14.4

Assume, as throughout Section 14.11, that e_t is strictly stationary and ergodic. Then $u_t = e_t^2 - \sigma_t^2$ is strictly stationary and ergodic (THEOREM 14.5) and an MDS (Exercise 14.3). Its variance is

finite: by conditional Jensen $\sigma_t^4 \leq \mathbb{E}[e_t^4 | \mathcal{F}_{t-1}]$, so $\mathbb{E}[\sigma_t^4] \leq \mathbb{E}[e_t^4] < \infty$. The MDS CLT (THEOREM 14.11) gives the result with

$$v^2 = \mathbb{E}[u_t^2] = \mathbb{E}[e_t^4] - 2\mathbb{E}[e_t^2 \sigma_t^2] + \mathbb{E}[\sigma_t^4] = \mathbb{E}[e_t^4] - \mathbb{E}[\sigma_t^4],$$

using $\mathbb{E}[e_t^2 \sigma_t^2] = \mathbb{E}[\sigma_t^2 \mathbb{E}[e_t^2 | \mathcal{F}_{t-1}]] = \mathbb{E}[\sigma_t^4]$. Under conditional homoskedasticity ($\sigma_t^2 = \sigma^2$) this is $\mathbb{E}[e_t^4] - \sigma^4 = \text{var}(e_t^2)$, as in the i.i.d. case; with conditional heteroskedasticity v^2 is smaller than $\text{var}(e_t^2)$ by $\text{var}(\sigma_t^2)$.

Exercise 14.5

(a) Let $\mathcal{F}_t = \sigma(e_t, e_{t-1}, \dots; u_{t+1}, u_t, u_{t-1}, \dots)$. Then σ_t (a function of $u_t, u_{t-1}, \dots$) is $\mathcal{F}_{t-1}$ -measurable, Y_t is $\mathcal{F}_t$ -measurable, and e_t is independent of $\mathcal{F}_{t-1}$. Hence

$$\mathbb{E}[Y_t | \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[e_t] = 0.$$

Integrability holds since σ_t is log-normal under (b). By the LIE Y_t is then also an MDS with respect to its own history $\sigma(Y_{t-1}, Y_{t-2}, \dots) \subseteq \mathcal{F}_{t-1}$.

(b) With $|\beta| < 1$, back-substitution gives $\log \sigma_t^2 = \frac{\omega}{1-\beta} + \sum_{j=0}^{\infty} \beta^j u_{t-j}$. The coefficients are absolutely summable and u_t is i.i.d. with $\mathbb{E}|u_t| < \infty$, so $\log \sigma_t^2$ is a well-defined strictly stationary ergodic process (THEOREMS 14.4 AND 14.6). The pair (e_t, u_t) is i.i.d., hence stationary ergodic, and $Y_t = \exp(\frac{1}{2} \log \sigma_t^2) e_t = \phi(e_t, u_t, u_{t-1}, \dots)$ is a measurable function of its current and past values. By THEOREM 14.5, Y_t is strictly stationary and ergodic.

Exercise 14.6

Let $Y_t = \mu + e_t + \theta e_{t-1}$ with e_t white noise of variance σ^2 . Then $\gamma(0) = \text{var}(e_t + \theta e_{t-1}) = (1 + \theta^2)\sigma^2$ and $\gamma(1) = \mathbb{E}[(e_t + \theta e_{t-1})(e_{t-1} + \theta e_{t-2})] = \theta\sigma^2$, all other cross products having zero mean. So $\rho(1) = \theta/(1 + \theta^2)$. Note $|\rho(1)| \leq \frac{1}{2}$, with the bound attained at $\theta = \pm 1$.

Exercise 14.7

(The printed denominator has upper limit q ; for MA(∞) it is ∞ .) Let $Y_t = \mu + \sum_{j=0}^{\infty} \theta_j e_{t-j}$ with white noise e_t of variance σ^2 and $\sum_j \theta_j^2 < \infty$, so the series converges in mean square and expectations may be taken term by term. Then

$$\gamma(k) = \mathbb{E} \left[\sum_j \theta_j e_{t-j} \sum_l \theta_l e_{t-k-l} \right] = \sigma^2 \sum_{j,l} \theta_j \theta_l \mathbf{1}\{j = k+l\} = \sigma^2 \sum_{l=0}^{\infty} \theta_{l+k} \theta_l,$$

and $\gamma(0) = \sigma^2 \sum_j \theta_j^2$. Dividing gives the formula. For $\theta_j = 0$ beyond q it reduces to the MA(q) formula, which vanishes for $k > q$.

Exercise 14.8

$Y_t = \sum_{s=1}^t e_s$, so $\text{var}[Y_t] = t$. The variance grows linearly with t , so Y_t is not (covariance) stationary. Its first difference $\Delta Y_t = e_t$ is stationary: Y_t is integrated of order one.

Exercise 14.9

(a) Iterating, $Y_{t+j} = \alpha_1^j Y_t + \sum_{i=0}^{j-1} \alpha_1^i e_{t+j-i}$ and $Y_t = \alpha_1 Y_{t-1} + e_t$, so $b_j = \alpha_1^j$.

(b) $\hat{b}_j = \hat{\alpha}_1^j$.

(c) $\partial b_j / \partial \alpha_1 = j \alpha_1^{j-1}$, so $s(\hat{b}_j) = j |\hat{\alpha}_1|^{j-1} s(\hat{\alpha}_1)$ and the interval is $\hat{\alpha}_1^j \pm 1.96 j |\hat{\alpha}_1|^{j-1} s(\hat{\alpha}_1)$ for $j \geq 1$ ($b_0 = 1$ is known). For large j the delta method is poor — the interval shrinks to a point near 0 even though b_j is very uncertain when α_1 is near one; transforming the interval for α_1 (valid when $\alpha_1 > 0$, Exercise 5.9) is more reliable.

Exercise 14.10

(The printed model has $\alpha_2 Y_{t-1}$; an AR(2) has $\alpha_2 Y_{t-2}$.)

(a) The impulse responses satisfy $b_0 = 1$, $b_1 = \alpha_1$, $b_j = \alpha_1 b_{j-1} + \alpha_2 b_{j-2}$, so

$$b_1 = \alpha_1, \quad b_2 = \alpha_1^2 + \alpha_2, \quad b_3 = \alpha_1^3 + 2\alpha_1\alpha_2, \quad b_4 = \alpha_1^4 + 3\alpha_1^2\alpha_2 + \alpha_2^2.$$

(b) $\hat{b}_2 = \hat{\alpha}_1^2 + \hat{\alpha}_2$.

(c) The gradient with respect to (α_1, α_2) is $R = (2\alpha_1, 1)'$, so with $\hat{V}$ the covariance estimate of $(\hat{\alpha}_1, \hat{\alpha}_2)$,

$$s(\hat{b}_2) = \sqrt{4\hat{\alpha}_1^2 \hat{V}_{11} + 4\hat{\alpha}_1 \hat{V}_{12} + \hat{V}_{22}}, \quad \hat{b}_2 \pm 1.96 s(\hat{b}_2).$$

Exercise 14.11

(The printed second model reads $\alpha(L)Y_t = \mu + u_t$; the intended form is $Y_t = \mu + u_t$.) Applying $\alpha(L)$ to a constant gives $\alpha(L)\mu = (1 - \alpha_1 - \dots - \alpha_p)\mu = \alpha(1)\mu$. So from the second model $\alpha(L)Y_t = \alpha(1)\mu + \alpha(L)u_t = \alpha(1)\mu + e_t$, which is the first with $\alpha_0 = \alpha(1)\mu$. Conversely, if $\alpha(1) \neq 0$, set $\mu = \alpha_0/\alpha(1)$ and $u_t = Y_t - \mu$; then $\alpha(L)u_t = \alpha_0 + e_t - \alpha(1)\mu = e_t$. Hence

$$\mu = \frac{\alpha_0}{\alpha(1)} = \frac{\alpha_0}{1 - \alpha_1 - \dots - \alpha_p},$$

the unconditional mean of a stationary AR(p).

Exercise 14.12

Apply $\beta(L)$ to the first equation: $\beta(L)\alpha(L)Y_t = \beta(L)u_t = e_t$. So $\gamma(L) = \beta(L)\alpha(L)$, a lag polynomial of order $p + q$ (the product of the leading coefficients $\alpha_p\beta_q$ is generically non-zero), with $\gamma(0) = 1$. This is the logic of the test for omitted serial correlation in Section 14.37.

Exercise 14.13

(a) Y_t is covariance stationary with mean zero and

$$\gamma(0) = 2 + \theta^2, \quad \gamma(1) = \mathbb{E}[(u_t + \theta u_{t-1})(u_{t-1} + \theta u_{t-2})] = \theta, \quad \gamma(k) = 0 \quad (k \geq 2).$$

A covariance stationary process whose autocovariances vanish beyond lag one has (by the Wold decomposition) an MA(1) representation $Y_t = \eta_t + \psi\eta_{t-1}$ with white noise η_t , $\text{var}(\eta_t) = \sigma_\eta^2$, provided we can match the two non-zero autocovariances:

$$(1 + \psi^2)\sigma_\eta^2 = 2 + \theta^2, \quad \psi\sigma_\eta^2 = \theta.$$

(b) Dividing, $\psi/(1+\psi^2) = \rho \equiv \theta/(2+\theta^2)$. Since $|\rho| \leq 1/(2\sqrt{2}) < \frac{1}{2}$ the quadratic $\rho\psi^2 - \psi + \rho = 0$ has real roots; the invertible one ($|\psi| < 1$) is

$$\psi = \frac{1 - \sqrt{1 - 4\rho^2}}{2\rho}, \quad \rho = \frac{\theta}{2 + \theta^2}, \quad \sigma_\eta^2 = \theta/\psi$$

(and $\psi = 0$ when $\theta = 0$).

(c) $\theta = 1$ gives $\rho = 1/3$ and $\psi = \frac{3}{2}(1 - \sqrt{5}/3) = \frac{3-\sqrt{5}}{2} \approx 0.382$, with $\sigma_\eta^2 = \frac{3+\sqrt{5}}{2} \approx 2.618$. Check: $(1+\psi^2)\sigma_\eta^2 = 3 = \gamma(0)$. Adding white noise dampens the MA coefficient from 1 to 0.38.

Exercise 14.14

Apply $(1 - \alpha L)$: $(1 - \alpha L)Y_t = u_t + e_t - \alpha e_{t-1} \equiv v_t$. The process v_t has $\gamma_v(0) = \sigma_u^2 + (1 + \alpha^2)\sigma_e^2$, $\gamma_v(1) = -\alpha\sigma_e^2$ and $\gamma_v(k) = 0$ for $k \geq 2$, so by the argument of Exercise 14.13 $v_t = \eta_t + \psi\eta_{t-1}$ for white noise η_t , with ψ solving $\psi/(1+\psi^2) = -\alpha\sigma_e^2/(\sigma_u^2 + (1+\alpha^2)\sigma_e^2)$. Hence $(1 - \alpha L)Y_t = (1 + \psi L)\eta_t$: an ARMA(1,1) with the same AR root as X_t and an MA coefficient of opposite sign to α . Measurement error turns a pure AR into an ARMA.

Exercise 14.15

By THEOREM 14.21 the stationary solution is $Y_t = \mu + \sum_{j=0}^{\infty} \alpha_1^j e_{t-j}$ with $\mu = \alpha_0/(1 - \alpha_1)$. Each partial sum $S_N = \mu + \sum_{j=0}^N \alpha_1^j e_{t-j}$ is a linear combination of independent normals, hence $S_N \sim N(\mu, \sigma^2 \sum_{j=0}^N \alpha_1^{2j})$. $S_N \rightarrow Y_t$ almost surely and in mean square, so the characteristic functions converge: $\exp(is\mu - \frac{1}{2}s^2\sigma^2 \sum_{j \leq N} \alpha_1^{2j}) \rightarrow \exp(is\mu - \frac{1}{2}s^2\sigma^2/(1 - \alpha_1^2))$, which is the characteristic function of $N(\mu, \sigma^2/(1 - \alpha_1^2))$.

Exercise 14.16

By Exercise 14.15, $\mu = \alpha_0/(1 - \alpha_1)$, $\sigma_Y^2 = \sigma^2/(1 - \alpha_1^2)$, and since Y_t is normal $\kappa = 3\sigma_Y^4$.

The flaw: the fourth moment carries no information beyond the variance — κ is a known function of σ_Y^2 — so the three moment conditions pin down only two functions of the parameters, μ and σ_Y^2 . Formally the Jacobian of $(\alpha_0, \alpha_1, \sigma^2) \mapsto (\mu, \sigma_Y^2, \kappa)$ has rank two. Any (α_1, σ^2) with the same $\sigma^2/(1 - \alpha_1^2)$ (and α_0 adjusted to keep μ) produce identical moments, so the parameters are not identified. More fundamentally, *marginal* moments cannot identify *dynamics*: α_1 is identified only through serial dependence, e.g. $\mathbb{E}[(Y_t - \mu)(Y_{t-1} - \mu)] = \alpha_1\sigma_Y^2$. Counting moments and parameters is necessary but not sufficient for identification.

Exercise 14.17

Take logs: $\log Y_t = \alpha \log Y_{t-1} + (1 - \alpha) \log u_t$, an AR(1) in $\log Y_t$ with i.i.d. innovations $(1 - \alpha) \log u_t$.

(a) By THEOREM 14.21, if $|\alpha| < 1$ and $\mathbb{E}|\log u_t| < \infty$ then $\log Y_t$ is strictly stationary and ergodic, and so is $Y_t = \exp(\log Y_t)$ by THEOREM 14.5.

(b) $\log Y_t = (1 - \alpha) \sum_{j=0}^{\infty} \alpha^j \log u_{t-j}$, so

$$Y_t = \prod_{j=0}^{\infty} u_{t-j}^{(1-\alpha)\alpha^j},$$

a geometric weighted average of past shocks (the exponents sum to one).

Exercise 14.18**(a)–(c)** Sample 1960Q2–2017Q4, $n = 231$.

	coefficient	s.e. (HC)	s.e. (NW, $M = 5$)
intercept	0.491	0.148	0.143
Y_{t-1}	0.502	0.076	0.084
Y_{t-2}	0.168	0.070	0.069
Y_{t-3}	−0.026	0.063	0.065
Y_{t-4}	−0.068	0.053	0.052

The two sets of standard errors are very close: if the AR(4) is correctly specified the errors are an MDS and HC standard errors are valid.

(d) Investment growth is positively autocorrelated: the first two lags are significant (0.50 and 0.17) and the third and fourth are small and insignificant. The sum of the AR coefficients is 0.58, so shocks are moderately persistent, and the implied mean growth rate $0.491/(1 - 0.575) = 1.16\%$ per quarter matches the sample mean (1.19%). All roots of the AR polynomial lie outside the unit circle (the reciprocal roots have modulus 0.61 and 0.43): the model is stationary.

(e) With $b_0 = 1$ and $b_j = \sum_{i=1}^4 \hat{\alpha}_i b_{j-i}$:

j	1	2	3	4	5	6	7	8	9	10
b_j	0.502	0.420	0.269	0.124	0.062	0.016	−0.003	−0.009	−0.010	−0.007

A one-point shock to investment growth raises growth by 0.5 next quarter, 0.4 two quarters out, and dies out after about six quarters with a tiny undershoot. The cumulative response is about 2.4.

Exercise 14.19**(a)–(b)** $n = 231$ (1960Q2–2017Q4); first differences in dollars.

	coefficient	s.e. (HC)
intercept	0.149	0.447
ΔY_{t-1}	0.274	0.098
ΔY_{t-2}	−0.259	0.180
ΔY_{t-3}	0.026	0.073
ΔY_{t-4}	−0.072	0.120

(c) Wald statistic for $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$: $W = 10.47$, χ_4^2 $p = 0.033$ (HC). With Newey–West standard errors $W = 17.90$, $p = 0.001$.

(d) A random walk implies unpredictable changes; that is rejected at 5%. The main departure is momentum at one quarter (0.27): a rise in the real oil price tends to be followed by a further rise next quarter, partly offset at two quarters. Nevertheless predictability is modest — $R^2 \approx 0.11$, against a residual standard deviation of about \$7 per quarter — and the huge heteroskedasticity

of oil price changes (standard errors on the second lag are double those on the first) means the evidence rests on a few volatile episodes. The random walk is a reasonable first approximation but not literally true.

Exercise 14.20

All models use $n = 696$ observations (1960m1–2017m12) with the eight preceding months as initial conditions; $AIC(p) = n \log \hat{\sigma}^2(p) + 2p$ (Section 14.38).

(a)–(b)

p	1	2	3	4	5	6	7	8
AIC	–2399.5	–2407.2	–2457.0	–2478.5	–2497.4	–2501.5	–2501.9	–2499.9

(c) AR(7), although AR(6) is within 0.4.

(d) AR(7), HC standard errors:

	const	Y_{t-1}	Y_{t-2}	Y_{t-3}	Y_{t-4}	Y_{t-5}	Y_{t-6}	Y_{t-7}
coef.	0.084	0.982	0.192	–0.040	0.010	–0.066	–0.036	–0.057
s.e.	0.026	0.049	0.056	0.060	0.059	0.057	0.056	0.045

The AR coefficients sum to 0.986: unemployment is extremely persistent (near a unit root), with a positive second lag and small negative higher-order lags producing a hump-shaped response. The implied mean $0.084/(1 - 0.986) = 6.0\%$ matches the sample mean.

Exercise 14.21

Claims are measured in thousands; $n = 232$.

(a) DL regression, coefficients on $\text{claims}_{t-1}, \dots, \text{claims}_{t-4}$: 0.0094, 0.0017, 0.0010, 0.0058, with Newey–West s.e. 0.0019, 0.0017, 0.0015, 0.0018 (HC: 0.0021, 0.0035, 0.0030, 0.0016). Newey–West standard errors are appropriate: a distributed lag regression omits the dynamics of the dependent variable, so its error is strongly serially correlated (first-order autocorrelation of the residuals 0.93) and is not an MDS. (Even NW with $M = 5$ is likely too short for such persistence.)

(b) ADL (HC s.e.): unemployment lags 1.080 (0.099), –0.105 (0.173), 0.106 (0.174), –0.146 (0.089); claims lags 0.0057 (0.0014), –0.0038 (0.0019), –0.0003 (0.0015), –0.0006 (0.0008). The residuals are now essentially uncorrelated (autocorrelation 0.02), so HC standard errors are appropriate.

(c) Wald test that the four claims coefficients are zero: $W = 26.9$, χ_4^2 $p = 2 \times 10^{-5}$. Claims Granger-cause unemployment.

(d) Past initial claims help predict the unemployment rate even after conditioning on past unemployment. The effect is front-loaded: an extra 100,000 claims last quarter raises the predicted unemployment rate by about 0.57 percentage points, but the second lag largely offsets it, so the lagged *change* in claims is what matters; the sum of the claims coefficients is only 0.001. This is predictive (Granger) causality: claims are a timely leading indicator of layoffs, not a structural cause of unemployment.

Exercise 14.22

Housing starts in millions of units (annual rate); $n = 232$.

(a)–(b) DL coefficients on $\text{starts}_{t-1}, \dots, \text{starts}_{t-4}$: 2.46, -0.55 , -0.76 , -0.70 with NW s.e. 0.65, 0.77, 0.62, 0.47 (HC: 0.53, 0.68, 0.64, 0.49). As in Exercise 14.21, Newey–West standard errors are the safe choice for a DL regression, whose error need not be an MDS; here the residual autocorrelation is small (0.05) because GDP growth is itself only weakly persistent, so the two sets of standard errors are close.

(c) ADL (HC s.e.): GDP growth lags 0.032 (0.071), 0.157 (0.086); housing starts lags 2.52 (0.58), -1.00 (0.83), -0.87 (0.62), -0.33 (0.53).

(d) $W = 36.6$, χ_4^2 $p = 2 \times 10^{-7}$: housing starts Granger-cause GDP growth.

(e) Housing starts are a strong leading indicator. An increase of 100,000 starts last quarter predicts about 0.25 percentage points higher quarterly GDP growth, largely reversed over the following three quarters (sum of coefficients 0.32). So a *rise* in starts predicts faster growth. Once housing starts are included, lagged GDP growth adds little. As before, this is predictive content, not a causal estimate: starts respond quickly to the same interest-rate and credit conditions that later move GDP.

Chapter 15

Multivariate Time Series

Solutions to all 20 exercises of Chapter 15 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 15.14–15.20 use FRED-QD, FRED-MD and Kilian2009. The VAR code was checked against the book: it reproduces the orthogonalized and cumulative impulse responses underlying Figure 15.1 to six decimals. Shaded bands in the figures are 90% percentile intervals from a recursive-design residual bootstrap (500 replications; 100 for the short-run structural VARs); figures are stored in `figures/`.

Notation: a VAR(p) is $Y_t = a_0 + A_1 Y_{t-1} + \dots + A_p Y_{t-p} + e_t$ with $\text{var}(e_t) = \Sigma$. Its moving average representation (15.3) is $Y_t = \mu + \sum_{h \geq 0} \Theta_h e_{t-h}$ with $\Theta_0 = \mathbf{I}_m$. With B a square root of Σ ($BB' = \Sigma$) the orthogonalized impulse response is $\text{OIRF}(h) = \Theta_h B$; recursive VARs use $B = \text{chol}(\Sigma)$.

Exercise 15.1

A VAR(1) is strictly stationary (given $\mathbb{E}\|e_t\| < \infty$) iff all eigenvalues of A have modulus less than one.

- (a) Eigenvalues $0.7 \pm 0.2 = 0.9, 0.5$ (eigenvectors $(1, 1)', (1, -1)'$): **stationary**.
- (b) Eigenvalues $0.8 \pm 0.4 = 1.2, 0.4$: the root $1.2 > 1$ makes the process **explosive**, not stationary.
- (c) Eigenvalues $0.8 \pm 0.4i$ with modulus $\sqrt{0.64 + 0.16} = 0.894 < 1$: **stationary**, with damped oscillations.

Note that (b) and (c) have identical entries up to a sign, yet one is explosive and one stationary: stationarity is not visible from the size of the coefficients alone.

Exercise 15.2

Use the companion matrix $\begin{bmatrix} A_1 & A_2 \\ \mathbf{I}_2 & \mathbf{0} \end{bmatrix}$; stationarity requires all its eigenvalues to lie inside the unit circle. Both A_1 and A_2 have eigenvectors $(1, 1)'$ and $(1, -1)'$, so the system decouples:

- along $(1, 1)'$: $\lambda^2 - 0.5\lambda - 0.3 = 0$, roots 0.852 and -0.352 ;
- along $(1, -1)'$: $\lambda^2 - 0.1\lambda - 0.5 = 0$, roots 0.759 and -0.659 .

All four eigenvalues have modulus below one (largest 0.852), so Y_t is **strictly stationary**.

Exercise 15.3

From the first equation $u_{t-1} = Y_{t-1} - AY_{t-2}$. Substituting into $Y_t = AY_{t-1} + Bu_{t-1} + e_t$,

$$Y_t = (A + B)Y_{t-1} - BAY_{t-2} + e_t,$$

a VAR(2) with $A_1 = A + B$, $A_2 = -BA$ and error e_t . (Matrix order matters: $A_2 = -BA$, not $-AB$.)

Exercise 15.4

Each satisfies $Y_{it} = \alpha_{i0} + \alpha_{i1}Y_{i,t-1} + \cdots + \alpha_{ip}Y_{i,t-p} + e_{it}$. Stacking,

$$Y_t = a_0 + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + e_t, \quad a_0 = (\alpha_{10}, \dots, \alpha_{m0})', \quad A_j = \text{diag}(\alpha_{1j}, \dots, \alpha_{mj}),$$

with $e_t = (e_{1t}, \dots, e_{mt})'$ having diagonal covariance matrix $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_m^2)$ (independence). All coefficient matrices are diagonal: no series Granger-causes another, and orthogonalized and non-orthogonalized impulse responses coincide up to scale.

Exercise 15.5

Back-substituting, $Y_t = \sum_{h=0}^{\infty} A_1^h e_{t-h}$ (plus the mean), so $\Theta_h = A_1^h$.

Exercise 15.6

The recursion is $\Theta_0 = \mathbf{I}$, $\Theta_h = A_1 \Theta_{h-1} + A_2 \Theta_{h-2}$ ($\Theta_{-1} = \mathbf{0}$). Hence

$$\begin{aligned} \Theta_1 &= A_1, \\ \Theta_2 &= A_1^2 + A_2, \\ \Theta_3 &= A_1^3 + A_1 A_2 + A_2 A_1, \\ \Theta_4 &= A_1^4 + A_1^2 A_2 + A_1 A_2 A_1 + A_2 A_1^2 + A_2^2. \end{aligned}$$

The products do not commute in general, which is why terms such as $A_1 A_2$ and $A_2 A_1$ appear separately.

Exercise 15.7

Stack $\tilde{Y}_t = (Y_t', Y_{t-1}', \dots, Y_{t-p+1}')' (mp \times 1)$. Then

$$\tilde{Y}_t = \tilde{a}_0 + \mathbf{A} \tilde{Y}_{t-1} + \tilde{e}_t, \quad \mathbf{A} = \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ \mathbf{I}_m & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_m & & \mathbf{0} & \mathbf{0} \\ \vdots & & \ddots & & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_m & \mathbf{0} \end{bmatrix},$$

with $\tilde{a}_0 = (a_0', \mathbf{0}, \dots)'$ and $\tilde{e}_t = (e_t', \mathbf{0}, \dots)'$. The first block row is the VAR itself and the remaining rows are identities. By Exercise 15.5 applied to this VAR(1), the response of $\tilde{Y}_{t+h}$ to $\tilde{e}_t$ is $\mathbf{A}^h$. Selecting the first block with $J = [\mathbf{I}_m \ \mathbf{0} \ \cdots \ \mathbf{0}]$,

$$\text{IRF}(h) = \Theta_h = J \mathbf{A}^h J',$$

the upper-left $m \times m$ block of $\mathbf{A}^h$. Stationarity holds iff all eigenvalues of $\mathbf{A}$ are inside the unit circle (as used in Exercise 15.2).

Exercise 15.8

Lags of Y_2 do not enter the equation for Y_1 : the $(1, 2)$ elements vanish, $[A_1]_{12} = [A_2]_{12} = 0$. Both matrices are lower triangular.

Exercise 15.9

Also $[A_1]_{21} = [A_2]_{21} = 0$, so A_1 and A_2 are both diagonal. The two series may still be related *contemporaneously* through a non-diagonal Σ ; Granger non-causality restricts only the lag dynamics.

Exercise 15.10

Each equation has $mp + 1 = 8 \times 12 + 1 = 97$ coefficients (776 in the system, plus 36 distinct elements of Σ). There are 240 monthly observations, of which 12 are used as initial conditions, leaving $n = 228$ per equation — about 2.4 observations per coefficient. With $n - k = 131$ residual degrees of freedom per equation the estimates would be very imprecise, impulse responses (non-linear in the coefficients) worse still, and asymptotic approximations unreliable. It does not make sense as specified. Alternatives: select p by AIC; seasonally adjust or add seasonal dummies instead of twelve lags; reduce m ; or use shrinkage (Bayesian VARs) or factor methods (Section 15.30).

Exercise 15.11

Regress each residual on the preceding ones, without intercept (VAR residuals have mean zero):

$$\hat{e}_{1t} = \hat{v}_{1t}, \quad \hat{e}_{jt} = \sum_{k < j} \hat{\phi}_{jk} \hat{e}_{kt} + \hat{v}_{jt}, \quad j = 2, \dots, m.$$

Let Φ be the unit lower-triangular matrix with (j, k) entry $-\hat{\phi}_{jk}$, so $\hat{v}_t = \Phi \hat{e}_t$. Each $\hat{v}_{jt}$ is an OLS residual, orthogonal in sample to $\hat{e}_{1t}, \dots, \hat{e}_{j-1,t}$ and hence to $\hat{v}_{1t}, \dots, \hat{v}_{j-1,t}$ (which are linear in them). So the sample covariance of $\hat{v}_t$ is diagonal, $\hat{D} = \text{diag}(\hat{\sigma}_{v1}^2, \dots, \hat{\sigma}_{vm}^2)$, and

$$\Phi \hat{\Sigma} \Phi' = \hat{D} \implies \hat{\Sigma} = \Phi^{-1} \hat{D} \Phi^{-1'} = (\Phi^{-1} \hat{D}^{1/2}) (\Phi^{-1} \hat{D}^{1/2})'.$$

$\Phi^{-1} \hat{D}^{1/2}$ is lower triangular with positive diagonal, so by uniqueness of the Cholesky factor $\hat{B} = \Phi^{-1} \hat{D}^{1/2}$: the diagonal of $\hat{B}$ holds the residual standard deviations of the recursive regressions and the rest follows by inverting the unit-triangular Φ (forward substitution). (Use the same divisor n in $\hat{\Sigma}$ and $\hat{D}$.)

Exercise 15.12

(The printed $(2, 2)$ entry reads σ_1^2 ; it should be σ_2^2 .)

$$(a) \ B = \begin{bmatrix} \sigma_1 & 0 \\ \rho\sigma_2 & \sigma_2\sqrt{1-\rho^2} \end{bmatrix}; \text{ indeed } BB' = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \rho^2\sigma_2^2 + \sigma_2^2(1-\rho^2) \end{bmatrix}.$$

$$(b) \ B = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix}.$$

$$(c) \ R = \begin{bmatrix} \sqrt{1-\rho^2} & \rho \\ 0 & 1 \end{bmatrix}; \ RR' = \begin{bmatrix} 1-\rho^2+\rho^2 & \rho \\ \rho & 1 \end{bmatrix}.$$

$$(d) \ \text{With } \rho = 0.8, \ B = \begin{bmatrix} 1 & 0 \\ 0.8 & 0.6 \end{bmatrix} \text{ and } \text{OIRF}(h) = \Theta_h B = \begin{bmatrix} 1 & 0 \\ 1.8 & 0.6 \end{bmatrix}.$$

$$(e) \ R = \begin{bmatrix} 0.6 & 0.8 \\ 0 & 1 \end{bmatrix} \text{ and } \Theta_h R = \begin{bmatrix} 0.6 & 0.8 \\ 0.6 & 1.8 \end{bmatrix}.$$

(f) Very different. Under the first ordering, a shock to variable 1 moves Y_1 by 1 and Y_2 by 1.8, while the second shock moves Y_1 not at all. Under the reversed ordering the “first” shock moves Y_1 by only 0.6 and the second shock now moves Y_1 by 0.8. With $\rho = 0.8$ most of the innovation variance is common, and the ordering decides to which shock it is attributed. Both factorizations fit the data equally well; only an economic argument can choose between them.

Exercise 15.13

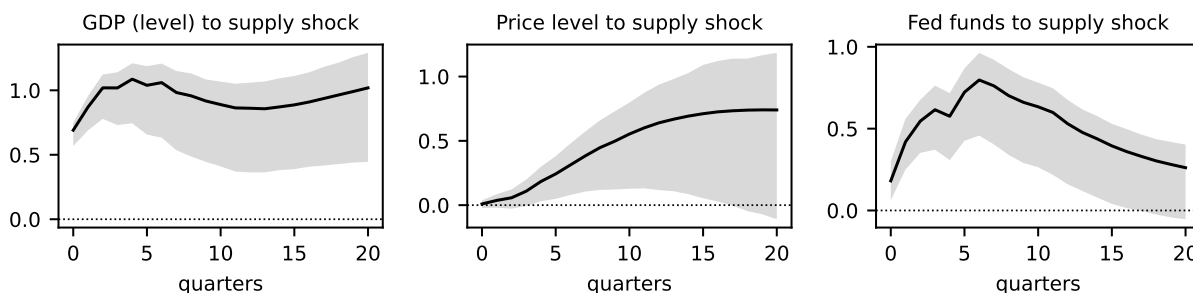
As one arbitrary statistical decomposition among many, not as causal effects. Recursive orthogonalization imposes exclusion restrictions (each variable does not respond within the period to shocks of variables ordered after it), and unless these are justified the “shocks” have no structural meaning. If the reduced-form errors are strongly correlated, other orderings can give very different responses (Exercise 15.12); if they are nearly uncorrelated the ordering matters little. With m variables there are $m!$ recursive orderings, and in general each produces different shocks and different responses; nothing in the data favours one over another, since all of them reproduce $\hat{\Sigma}$ exactly. A careful reader should ask for the residual correlation matrix and for robustness to alternative orderings, and should treat the reported responses as descriptive until an identification argument is supplied.

Exercise 15.14

Variables: $\Delta 100 \log(\text{GDP})$, $\Delta 100 \log(\text{deflator})$, fed funds rate; $n = 229$. The supply shock is the first Cholesky shock, one standard deviation (0.69% of GDP).

Why cumulate. GDP and prices enter as quarterly growth rates, so the VAR delivers responses of *growth*; the response of the *level* at horizon h is the sum of the growth responses up to h . The fed funds rate is already a level.

horizon (quarters)	0	1	2	4	8	12	20
GDP level (%)	0.69	0.87	1.02	1.09	0.96	0.86	1.02
price level (%)	0.01	0.04	0.06	0.18	0.45	0.64	0.74
fed funds (points)	0.18	0.42	0.55	0.58	0.70	0.53	0.26



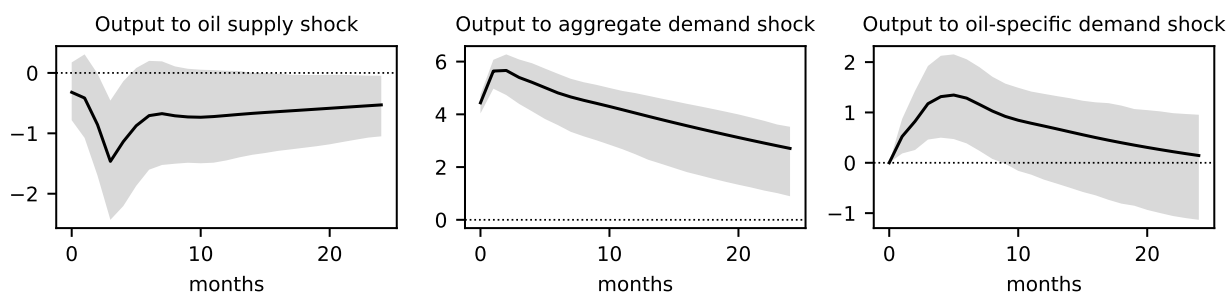
Comments. The shock raises output persistently — about 1% after a year and still 1% after five years (90% band at $h = 20$: $[0.44, 1.29]$) — consistent with a permanent supply improvement. The fed funds rate rises by up to 0.7 points and returns toward zero, a systematic policy tightening. But the price level *rises* steadily (0.45% at two years, band $[0.12, 0.66]$), whereas a favourable aggregate supply shock should lower prices. So the identified “supply” shock behaves like a demand shock,

which casts doubt on the recursive identification of Section 15.23 (the assumption that output does not respond to demand within the quarter).

Exercise 15.15

Monthly 1973m2–2007m12, $n = 415$; responses to one-standard-deviation shocks, 24 months.

months	0	1	3	6	12	18	24
oil supply shock	−0.32	−0.42	−1.47	−0.71	−0.71	−0.61	−0.53
aggregate demand shock	4.44	5.64	5.40	4.81	4.05	3.34	2.71
oil-specific demand shock	0.00	0.52	1.17	1.28	0.73	0.40	0.14



Comments. A supply disruption (lower oil production) lowers global real activity, with a trough of about -1.5 after three months and a persistent negative effect, marginally significant at longer horizons (90% band at 24 months $[-1.05, -0.05]$). An aggregate demand shock raises activity strongly and very persistently (band at 24 months $[0.89, 3.53]$) — by construction it is the innovation in the activity index itself. An oil-specific (precautionary) demand shock has no impact effect by the ordering, a small positive effect peaking at six months (band $[0.39, 2.06]$), and nothing afterwards. The main lesson matches Kilian: global activity is driven mostly by its own demand shocks, while oil supply shocks have modest and delayed real effects. (For completeness, the real oil price rises by 6.1 on impact after an oil-specific demand shock, by 0.4 after an aggregate demand shock rising to 4.8 after two years, and only briefly after a supply shock — the pattern of Figure 15.2.)

Exercise 15.16

(a) Loan growth $= 100\Delta \log(\text{realln})$. Permits start in 1960m1, which fixes the sample; with eight initial conditions $n = 688$ (1960m9–2017m12). Permits and starts are in thousands of units at annual rates.

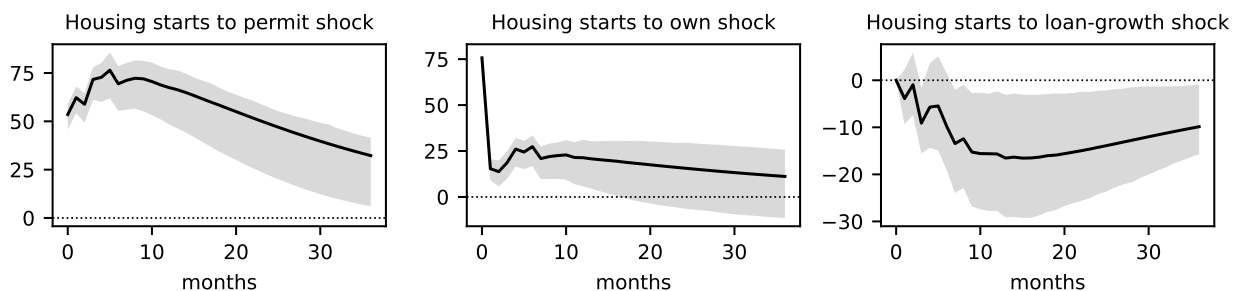
(b) $\text{AIC} = n \log |\hat{\Sigma}| + 2 \times (\text{number of coefficients})$, on the common sample:

p	1	2	3	4	5	6	7	8
AIC	11015.9	10876.1	10833.6	10779.1	10767.4	10739.8	10744.8	10758.2

AIC selects $p = 6$.

(c) Responses of housing starts (thousands, annual rate) to one-standard-deviation orthogonalized shocks:

months	0	1	3	6	12	24	36
permit shock	53.5	62.3	71.7	69.4	67.5	48.5	32.3
starts shock	75.7	15.4	18.8	27.4	21.3	15.7	11.2
loan shock	0.0	-3.9	-9.1	-9.8	-15.7	-14.3	-9.9



(d) A permit shock raises starts immediately (because the permit and start innovations are highly correlated, 0.58, and permits are ordered first) and the effect builds for three months and lasts for years: permits are the leading, persistent signal of construction. A shock to starts that is *not* predicted by permits is mostly transitory — it falls from 76 to 15 within a month — consistent with timing noise (weather, reporting) in monthly starts. A shock raising real estate loan growth is followed by *lower* starts, significantly so at longer horizons (band at 24 months $[-24, -2]$). One reading is that loan growth reflects purchases of the existing stock and tighter subsequent credit conditions; but this is a predictive pattern under a recursive ordering, not a structural credit-supply effect.

Exercise 15.17

Ordering (e_I, e_P, e_Y, e_R) for investment, prices, GDP and the fed funds rate.

(a)

$$A = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & a_{23} & 0 \\ a_{31} & a_{32} & 1 & 0 \\ a_{41} & a_{42} & a_{43} & 1 \end{bmatrix}, \quad \text{var}(\varepsilon_t) = \text{diag}(\sigma_I^2, \sigma_P^2, \sigma_Y^2, \sigma_R^2).$$

(b) Order condition: 6 free coefficients plus 4 variances = 10 = $m(m+1)/2$; exactly identified. Rank condition, equation by equation (Section 15.26): equation 1 has no free coefficient, so $\varepsilon_I = e_I - e_Y$ is identified. Equation 2 needs an instrument for e_Y uncorrelated with ε_P ; ε_I qualifies if $\text{cov}(e_Y, \varepsilon_I) \neq 0$. Solving the first three equations, $\text{cov}(e_Y, \varepsilon_I) \propto -a_{31}\sigma_I^2/\det(A_{11})$, so relevance requires $a_{31} \neq 0$. Equation 3 uses $(\varepsilon_I, \varepsilon_P)$ as instruments for (e_I, e_P) , relevant because the corresponding 2×2 block of A_{11}^{-1} has determinant $1/\det A_{11} \neq 0$. Equation 4 uses all three earlier shocks. So the model is identified provided $a_{31} \neq 0$ (GDP responds contemporaneously to investment) and $\det A_{11} = 1 + a_{31} - a_{23}a_{32} \neq 0$.

(c) Simultaneous: the price equation contains e_Y and the GDP equation contains e_P . Unlike the recursive example of Section 15.23, neither is predetermined within the quarter.

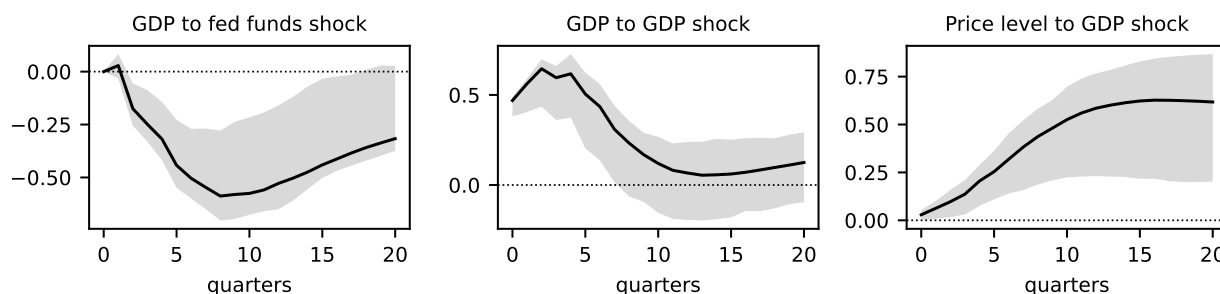
(d) AIC (VARs with trend, $p = 1, \dots, 8$) is minimized at $p = 6$, which justifies six lags. Solving $A\hat{\Sigma}A' = \text{diagonal}$ (the solution is unique):

$$\hat{A} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -0.062 & 0 \\ -0.154 & 0.403 & 1 & 0 \\ -0.043 & -0.590 & -0.096 & 1 \end{bmatrix}, \quad \hat{\sigma}_\varepsilon = (2.63, 0.22, 0.41, 0.65).$$

Written as structural equations: $e_P = 0.062e_Y + \varepsilon_P$ (prices rise slightly with output, Phillips-curve like); $e_Y = 0.154e_I - 0.403e_P + \varepsilon_Y$ (output rises with investment and falls with prices, a demand relation); $e_R = 0.043e_I + 0.590e_P + 0.096e_Y + \varepsilon_R$ (a Taylor-type rule responding mainly to prices). The signs are economically sensible.

(e) Variables are in log levels, so responses need no cumulation (units: % for GDP and prices).

quarters	0	1	2	4	8	12	20
GDP to fed funds shock	0.00	0.03	-0.18	-0.32	-0.59	-0.53	-0.32
GDP to GDP shock	0.47	0.56	0.65	0.62	0.23	0.07	0.13
prices to GDP shock	0.03	0.06	0.10	0.21	0.44	0.59	0.62



A monetary tightening (0.65 points) lowers GDP with a lag, by about 0.6% after two years (band at 12 quarters $[-0.65, -0.16]$) — the textbook contractionary effect. The GDP shock raises output temporarily (back near zero after three years) while the price level rises steadily (0.59% at 12 quarters, band $[0.23, 0.77]$): the GDP shock behaves like an aggregate demand shock.

Exercise 15.18

Ordering (supply = -oil, output, price).

(a) $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & a_{23} \\ a_{31} & a_{32} & 1 \end{bmatrix}$ with diagonal shock covariance.

(b) Order condition: $3 + 3 = 6 = m(m+1)/2$, exactly identified. Equation 1 has no free coefficients. Equation 2 (output) needs ε_1 as instrument for the price innovation; relevance requires $\text{cov}(e_3, \varepsilon_1) \propto -a_{31} \neq 0$ — the oil price must respond contemporaneously to supply shocks. Equation 3 uses $(\varepsilon_1, \varepsilon_2)$, which is relevant whenever $1 - a_{23}a_{32} \neq 0$. So identification requires $a_{31} \neq 0$.

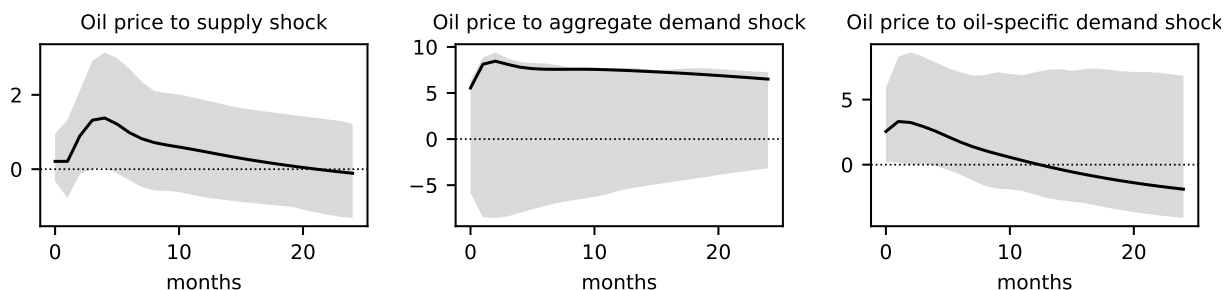
(c) Four lags, as in Exercise 15.15 and Section 15.24, for comparability (monthly data would support more). The unique solution is

$$\hat{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1.530 \\ -0.053 & -2.622 & 1 \end{bmatrix}, \quad \hat{\sigma}_\varepsilon = (20.0, 10.6, 12.8),$$

i.e. $e_{\text{output}} = -1.53e_{\text{price}} + \varepsilon_2$ and $e_{\text{price}} = 0.053e_{\text{supply}} + 2.62e_{\text{output}} + \varepsilon_3$. The implied simultaneous effects are large, but they should not be taken at face value: the estimate $\hat{a}_{31} = -0.053$ is close to zero and the correlation between the supply and price innovations is only 0.03. The identifying condition $a_{31} \neq 0$ is nearly violated, so the output equation is *weakly* identified — the structural VAR analogue of a weak instrument.

(d) Responses of the real oil price to one-standard-deviation structural shocks:

months	0	1	3	6	12	18	24
supply shock	0.21	0.21	1.32	0.98	0.48	0.14	-0.11
aggregate demand shock	5.54	8.12	8.11	7.58	7.48	7.07	6.51
oil-specific demand	2.55	3.32	2.94	1.74	0.09	-1.09	-1.88



Compared with the recursive VAR of Exercise 15.15 (impact responses 0.21, 0.40, 6.09), this model attributes most of the contemporaneous oil-price variation to aggregate demand rather than to oil-specific demand, and makes the demand effect large and persistent. But the bootstrap bands are extremely wide (at six months the 90% band for the aggregate-demand response is $[-7.3, 8.2]$), reflecting the weak identification in (c). The data cannot distinguish these structural stories; the recursive model's zero restriction on the response of activity to prices is doing a great deal of work in Kilian's results.

Exercise 15.19

Growth rates are $100\Delta \log$; $n = 231$.

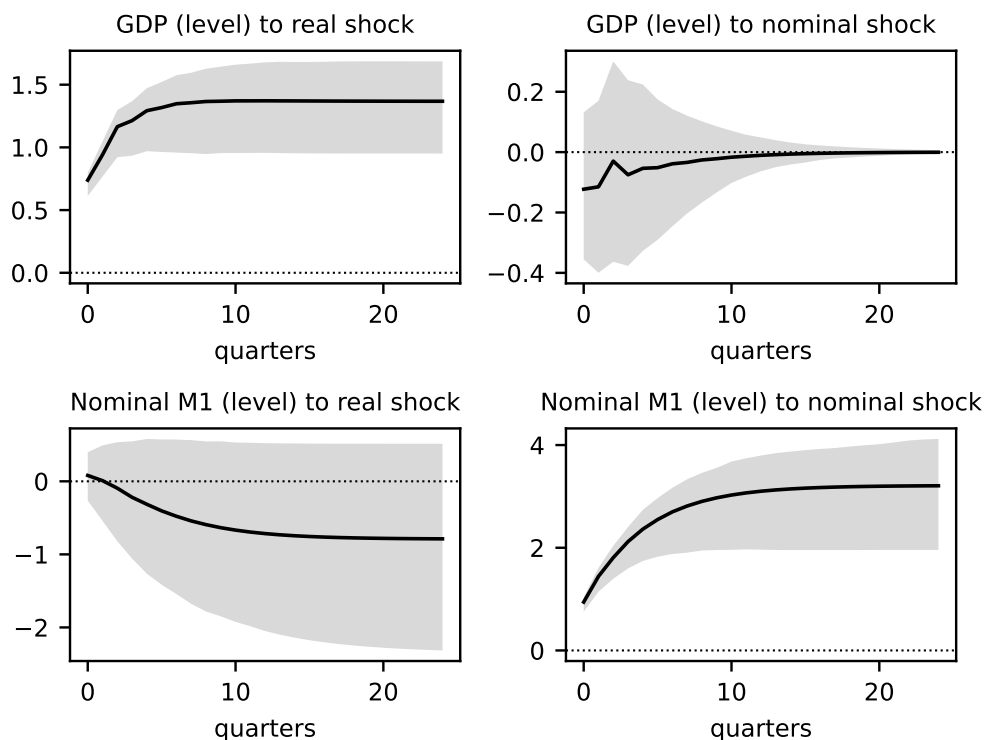
(a) Money growth coefficients (lags 1–4): 0.044 (0.057), 0.094 (0.066), -0.147 (0.054), 0.048 (0.051). Wald statistic 8.98, $\chi^2_4 p = 0.062$. Strict neutrality is not rejected at 5% (it would be at 10%): there is at most weak evidence that money growth helps predict GDP growth.

(b) Sum of the money coefficients 0.038 (0.039), $t = 0.97$, $p = 0.33$. Long-run neutrality is not rejected; the individual coefficients largely offset.

(c) Estimate a VAR(4) in $(\Delta\text{GDP}, \Delta\text{M1})$. Let $A(1) = \mathbf{I} - \hat{A}_1 - \dots - \hat{A}_4$ and define the long-run response matrix $C = A(1)^{-1}B$. Long-run neutrality says the second (nominal) shock has no long-run effect on the level of GDP: $C_{12} = 0$, so C is lower triangular and, as in Blanchard–Quah, $\hat{C} = \text{chol}(\hat{A}(1)^{-1}\hat{\Sigma}\hat{A}(1)^{-1'})$, $\hat{B} = \hat{A}(1)\hat{C}$. Estimates: $\hat{C} = \begin{bmatrix} 1.367 & 0 \\ -0.791 & 3.212 \end{bmatrix}$, $\hat{B} = \begin{bmatrix} 0.739 & -0.123 \\ 0.081 & 0.942 \end{bmatrix}$. The first shock is the “real” shock (the only one with permanent output effects), the second the “nominal” shock.

(d) Cumulated responses (levels, %):

quarters	0	1	2	4	8	12	24
GDP to real shock	0.74	0.94	1.17	1.29	1.37	1.37	1.37
GDP to nominal shock	-0.12	-0.12	-0.03	-0.05	-0.03	-0.01	0.00
M1 to real shock	0.08	0.01	-0.09	-0.31	-0.59	-0.72	-0.79
M1 to nominal shock	0.94	1.44	1.81	2.36	2.90	3.10	3.21



The real shock raises GDP permanently by about 1.4% and (insignificantly) lowers nominal money. The nominal shock permanently raises money by 3.2% but has only a tiny, short-lived and *negative* effect on output (band at four quarters $[-0.33, 0.22]$). There is no evidence here of short-run non-neutrality either, consistent with (a). Of course the long-run neutrality result is imposed, not estimated: the exercise shows what the data look like *given* that identifying assumption.

Exercise 15.20

(a) With ordering (hours, GDP, inflation) and shocks (labor supply, technology, demand),

$$C = \begin{bmatrix} c_{11} & 0 & 0 \\ c_{21} & c_{22} & 0 \\ c_{31} & c_{32} & c_{33} \end{bmatrix}.$$

(b) Yes, exactly: identification needs $m(m-1)/2 = 3$ restrictions and three zeros are imposed. Because C is lower triangular, $CC' = A(1)^{-1}\Sigma A(1)^{-1'}$ has the unique Cholesky solution (with positive diagonal).

(c) On a common sample ($n = 225$), AIC for $p = 1, \dots, 8$: -1075.0 , -1086.2 , -1083.6 , **-1086.6** , -1075.8 , -1067.0 , -1058.8 , -1070.5 . AIC selects $p = 4$.

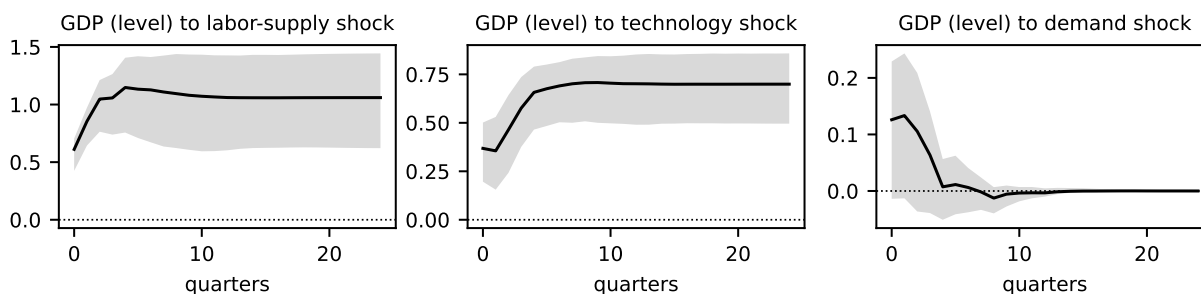
(d)

$$\hat{C} = \begin{bmatrix} 1.530 & 0 & 0 \\ 1.060 & 0.700 & 0 \\ 0.045 & 0.023 & 0.146 \end{bmatrix}.$$

Entries are long-run effects of one-standard-deviation shocks on the levels of hours and GDP (%) and on the *level* of inflation (the cumulated change). A labor-supply shock raises hours by 1.53% and GDP by 1.06% in the long run — an implied output elasticity of 0.69, close to labor's income share. A technology shock raises GDP by 0.70% with no long-run effect on hours. Demand shocks move only long-run inflation (0.15 points).

(e) Level of GDP (%):

quarters	0	1	2	4	8	12	24
labor-supply shock	0.61	0.85	1.05	1.15	1.10	1.06	1.06
technology shock	0.37	0.36	0.46	0.66	0.71	0.70	0.70
demand shock	0.13	0.13	0.11	0.01	-0.01	0.00	0.00



Both supply shocks have permanent, precisely estimated effects that are mostly in place within a year; labor-supply shocks are the larger source of long-run output movements. Demand shocks raise output briefly (0.13%) and die out within a year, and their effect is statistically indistinguishable from zero at every horizon beyond impact. The results mirror Shapiro and Watson's conclusion that permanent output movements are driven by supply-side shocks.

Chapter 16

Non-Stationary Time Series

Solutions to all 14 exercises of Chapter 16 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 16.12–16.14 use FRED-MD (1959m1–2017m12, 708 months). P-values are asymptotic, interpolated linearly from the book’s Tables 16.1 and 16.6–16.8 as described in Section 16.13. The code was checked against the book: it reproduces Table 16.2 (ADF and KPSS), the Engle–Granger statistic -4.0 , the VECM estimates of Tables 16.4–16.5 and the trace statistics 31.6, 2.8 and Table 16.9 (120, 68.3, 33.6, 10.8).

Notation: $W(r)$ is standard Brownian motion; $\Rightarrow$ and $\xrightarrow{d}$ both denote convergence in distribution, the former for processes on $[0, 1]$. “Case 1/2/3” refer to ADF regressions with no deterministic terms, an intercept, and an intercept and linear trend (the three ADF columns of Table 16.1).

Exercise 16.1

(a) $S_t = \sum_{i=1}^t e_i$, so $\mathbb{E}[S_t] = 0$ and $\text{var}[S_t] = t\sigma^2$.

(b) $Y_t = S_t/(\sigma\sqrt{t})$. Mean and variance are constant, but for $k \geq 0$

$$\text{cov}(Y_t, Y_{t+k}) = \frac{\text{cov}(S_t, S_t + e_{t+1} + \cdots + e_{t+k})}{\sigma^2 \sqrt{t(t+k)}} = \frac{t\sigma^2}{\sigma^2 \sqrt{t(t+k)}} = \sqrt{\frac{t}{t+k}},$$

which depends on t (e.g. $\text{cov}(Y_1, Y_2) = 0.71$ but $\text{cov}(Y_{100}, Y_{101}) = 0.995$). So Y_t is **not** covariance stationary, and hence not strictly stationary either. (With non-normal errors even the marginal distribution changes with t : $Y_1 = e_1/\sigma$ while Y_t is approximately normal for large t .) Standardizing the first two moments does not remove the non-stationarity, which lives in the dependence structure.

(c) By Theorem 16.1, $n^{-1/2}S_{\lfloor nr \rfloor} \Rightarrow \sigma W(r)$ on $[0, 1]$, and $\lfloor nr \rfloor/n \rightarrow r$ uniformly in r . Since

$$Y_{\lfloor nr \rfloor} = \frac{n^{-1/2}S_{\lfloor nr \rfloor}}{\sigma \sqrt{\lfloor nr \rfloor/n}}$$

and $(f, g) \mapsto f/\sqrt{g}$ is continuous on functions with $g \geq \delta/2 > 0$, the continuous mapping theorem gives

$$Y_{\lfloor nr \rfloor} \Rightarrow \frac{W(r)}{\sqrt{r}}, \quad r \in [\delta, 1].$$

For each fixed r the limit is $N(0, 1)$, but as a process it is not stationary: $\text{cov}(W(r)/\sqrt{r}, W(s)/\sqrt{s}) = \sqrt{r/s}$ for $r \leq s$. The restriction $r \geq \delta > 0$ is needed because the scaling $1/\sqrt{r}$ explodes at $r = 0$.

Exercise 16.2

Write $\Theta(z) = \Theta(1) + (1-z)\Theta^*(z)$ with $\Theta(1) = \mathbf{I} + \Theta_1 + \Theta_2$ and, by (16.4), $\Theta_j^* = -\sum_{\ell > j} \Theta_\ell$:

$$\Theta_0^* = -(\Theta_1 + \Theta_2), \quad \Theta_1^* = -\Theta_2, \quad \Theta_j^* = \mathbf{0} \quad (j \geq 2).$$

Check: $\Theta(1) + (1-z)(\Theta_0^* + \Theta_1^*z) = \Theta(1) + \Theta_0^* + (\Theta_1^* - \Theta_0^*)z - \Theta_1^*z^2 = \mathbf{I} + \Theta_1z + \Theta_2z^2$. Hence

$$\Delta Y_t = \xi_t + U_t - U_{t-1}, \quad \xi_t = (\mathbf{I} + \Theta_1 + \Theta_2)e_t, \quad U_t = -(\Theta_1 + \Theta_2)e_t - \Theta_2e_{t-1},$$

and summing, $Y_t = S_t + U_t + V_0$ with permanent component $S_t = (\mathbf{I} + \Theta_1 + \Theta_2) \sum_{i=1}^t e_i$, transitory MA(1) component U_t , and initial condition $V_0 = Y_0 - U_0$. For $\Theta_2 = \mathbf{0}$ this reduces to the book's MA(1) example.

Exercise 16.3

Assume, as in Theorem 16.4, that (e_t, u_t) is strictly stationary with finite variances and summable autocovariances, X_0 finite, and let $\omega_e^2 > 0$ be the long-run variance of e_t .

(a) $\Delta Y_t = e_t + u_t - u_{t-1}$ is stationary. Its long-run variance is the value at frequency zero of the spectral density. The filter $1 - L$ vanishes at frequency zero, so Δu_t contributes nothing, neither on its own nor through its cross-covariance with e_t :

$$\omega_{\Delta Y}^2 = \sum_{\ell=-\infty}^{\infty} \text{cov}(\Delta Y_t, \Delta Y_{t-\ell}) = \omega_e^2 > 0.$$

Hence ΔY_t is $I(0)$ and Y_t is $I(1)$. (Directly: Y_t has variance growing like $t\omega_e^2$, so it is not stationary.)

(b) $n^{-1/2}Y_{[nr]} = n^{-1/2}X_0 + n^{-1/2}\sum_{t \leq [nr]} e_t + n^{-1/2}u_{[nr]}$. The first term is $o_p(1)$. The second converges to $\omega_e W(r)$ by the FCLT (Theorem 16.4). For the third, by the union bound and strict stationarity,

$$\mathbb{P}\left[\max_{t \leq n} |u_t| > \varepsilon\sqrt{n}\right] \leq n \mathbb{P}[|u_1| > \varepsilon\sqrt{n}] \leq \varepsilon^{-2} \mathbb{E}[u_1^2 \mathbf{1}\{|u_1| > \varepsilon\sqrt{n}\}] \rightarrow 0,$$

so it is $o_p(1)$ uniformly in r . Therefore $n^{-1/2}Y_{[nr]} \Rightarrow \omega_e W(r)$, a Brownian motion with variance ω_e^2 : the stationary noise is asymptotically invisible at this scale.

Exercise 16.4

Let $\sigma^2 = \text{var}[e_t] > 0$.

(a) An i.i.d. sequence is strictly (hence weakly, given $\sigma^2 < \infty$) stationary. All autocovariances beyond lag zero vanish, so the long-run variance is $\sigma^2 > 0$, and Y_t is $I(0)$ by Definition 16.2.

(b) $X_t = e_t - e_{t-1}$ is an MA(1), a fixed function of the stationary pair (e_t, e_{t-1}) , so it is stationary. Its autocovariances are $\gamma(0) = 2\sigma^2$, $\gamma(\pm 1) = -\sigma^2$, and 0 otherwise, so its long-run variance is $\omega^2 = 2\sigma^2 - 2\sigma^2 = 0$ (equivalently $b(1)^2\sigma^2$ with $b(z) = 1 - z$). Definition 16.2 requires a positive long-run variance, so X_t is not $I(0)$. It is over-differenced ($I(-1)$): its partial sums $\sum_{t \leq n} X_t = e_n - e_0$ stay bounded in probability instead of growing like $\sqrt{n}$.

Exercise 16.5

Both series load on the random walk U_t , so both are $I(1)$ (assuming $\text{var}[e_t] > 0$). The combination $\beta'(Y_t, X_t)' = \beta_1 Y_t + \beta_2 X_t = (\beta_1 + 2\beta_2)U_t + \beta_1 v_t + \beta_2 w_t$ is stationary if and only if $\beta_1 + 2\beta_2 = 0$. So the cointegrating vector is

$$\beta = (2, -1)' \quad \text{or, normalized,} \quad \beta = (1, -\tfrac{1}{2})',$$

with equilibrium error $2Y_t - X_t = 2v_t - w_t$, i.i.d. and hence $I(0)$ provided $\text{var}[2v_t - w_t] > 0$. It is unique up to scale, since any other direction leaves U_t in the combination.

Exercise 16.6

The limits are consistent at the boundary. As $\alpha \rightarrow 1$ the stationary variance $1 - \alpha^2 \rightarrow 0$: the $\sqrt{n}$ -normalized estimator becomes degenerate, which signals that a faster rate applies. At $\alpha = 1$ indeed $\sqrt{n}(\hat{\alpha} - 1) = n^{-1/2} \cdot n(\hat{\alpha} - 1) = O_p(n^{-1/2}) \xrightarrow{p} 0$, matching the “ $N(0, 0)$ ” limit. With the correct rate n , Theorem 16.9 gives the non-degenerate Dickey–Fuller distribution $\int_0^1 W dW / \int_0^1 W^2$, which is skewed to the left (median below zero), so $\hat{\alpha}$ is biased downward.

The convergence is not uniform. For each fixed $|\alpha| < 1$ the normal approximation eventually works, but the sample size needed grows without bound as $\alpha \rightarrow 1$. What matters is $n(1 - \alpha)$, the number of “half-lives” in the sample. The unifying device is local-to-unity asymptotics, $\alpha_n = 1 + c/n$ with $c \leq 0$ fixed, under which

$$n(\hat{\alpha} - \alpha_n) \xrightarrow{d} \frac{\int_0^1 J_c dW}{\int_0^1 J_c^2}, \quad J_c(r) = \int_0^r e^{c(r-s)} dW(s),$$

an Ornstein–Uhlenbeck process. This family equals the Dickey–Fuller distribution at $c = 0$ and, suitably rescaled, tends to the normal as $c \rightarrow -\infty$. In practice, when $n(1 - \hat{\alpha})$ is not large (say $\hat{\alpha} = 0.95$ with $n = 100$) the $N(0, 1 - \alpha^2)$ approximation is poor: t -ratios are skewed and conventional confidence intervals under-cover. Unit-root distributions, or intervals obtained by inverting such tests (Stock 1991; Hansen’s 1999 grid bootstrap), should be used.

Exercise 16.7

Premultiply by β' : $\beta'Y_t - \beta'Y_{t-1} = \beta'\alpha\beta'Y_{t-1} + \beta'e_t$, i.e.

$$Z_t = (\mathbf{I}_r + \beta'\alpha)Z_{t-1} + \beta'e_t,$$

an AR(1) (a VAR(1) when $r > 1$) with coefficient $\mathbf{I}_r + \beta'\alpha$ and white-noise error $\beta'e_t$. Z_t is stationary if all eigenvalues of $\mathbf{I}_r + \beta'\alpha$ lie inside the unit circle. In the scalar case this means $-2 < \beta'\alpha < 0$: the adjustment coefficients must push the equilibrium error back toward zero without overshooting.

Exercise 16.8

Yes. The statistic is $t = (0.9 - 1)/0.05 = -2$, and they are implicitly comparing it with a normal critical value ($|t| > 1.96$). Under $H_0 : \alpha = 1$ the t -ratio does not have an asymptotic normal distribution but the Dickey–Fuller t distribution (Theorem 16.10), and the test is one-sided (reject for large negative values). The right critical value depends on the deterministic terms.

- As literally written (no intercept, Case 1): the 5% critical value is -1.94 and the interpolated p-value is $0.04 + 0.01 \times (2.03 - 2.00)/0.09 \approx 0.043$. The test barely rejects at 5%, but for the wrong reason.
- With an intercept, which is the usual specification since a no-intercept regression is appropriate only for data with a known zero mean (Case 2): the 5% critical value is -2.86 and the p-value is $0.20 + 0.10 \times (2.22 - 2.00)/0.25 \approx 0.29$. No rejection.

Further caveats: the regression should include enough lagged differences for e_t to be serially uncorrelated (the ADF test), and failing to reject would not establish a unit root.

Exercise 16.9

Yes. The arithmetic is right ($0.9 \pm 1.96 \times 0.04$), but the interval is justified by normal asymptotics, which fail near and at $\alpha = 1$ (Exercise 16.6). Whether $\alpha = 1$ is consistent with the data must be judged by the unit root distribution of $t = (0.9 - 1)/0.04 = -2.5$:

- no intercept (Case 1): p-value $\approx 0.01 + 0.01 \times (2.56 - 2.50)/0.25 = 0.012$, a rejection at 5%;
- intercept (Case 2): p-value $\approx 0.10 + 0.05 \times (2.57 - 2.50)/0.20 = 0.12$, no rejection;
- intercept and trend (Case 3): p-value ≈ 0.33 , no rejection.

In the realistic specifications $\alpha = 1$ is *not* rejected, so it belongs in any valid 95% confidence set. The normal interval has the wrong coverage, because the t -ratio is skewed and the correct interval is not symmetric about $\hat{\alpha}$. A valid set collects the values α_0 not rejected by a test with the right (local-to-unity) critical values, or comes from a grid bootstrap. And even a rejection only says that $\alpha = 1$ is unlikely, not that it is “inconsistent with the data”.

Exercise 16.10

Yes. The critical value -1.9 is the Case 1 value (-1.94 in Table 16.1): the software treats Z_t as a series with no estimated deterministic components. But the trend was estimated in the first step, and detrending by least squares beforehand is algebraically equivalent (asymptotically) to including the trend in the ADF regression. The statistic therefore follows the *detrended* Dickey–Fuller distribution (Theorem 16.14, Case 3), with 5% critical value -3.41 . Against the correct table,

$$p \approx 0.30 + 0.20 \times \frac{2.56 - 2.50}{2.56 - 2.18} \approx 0.33,$$

so the unit root is *not* rejected. (Even under Case 1 the p-value would be about 1.2%, not below 1%.) Estimating deterministic terms shifts the null distribution to the left; ignoring this makes stationarity far too easy to “find”.

Exercise 16.11

Yes, several.

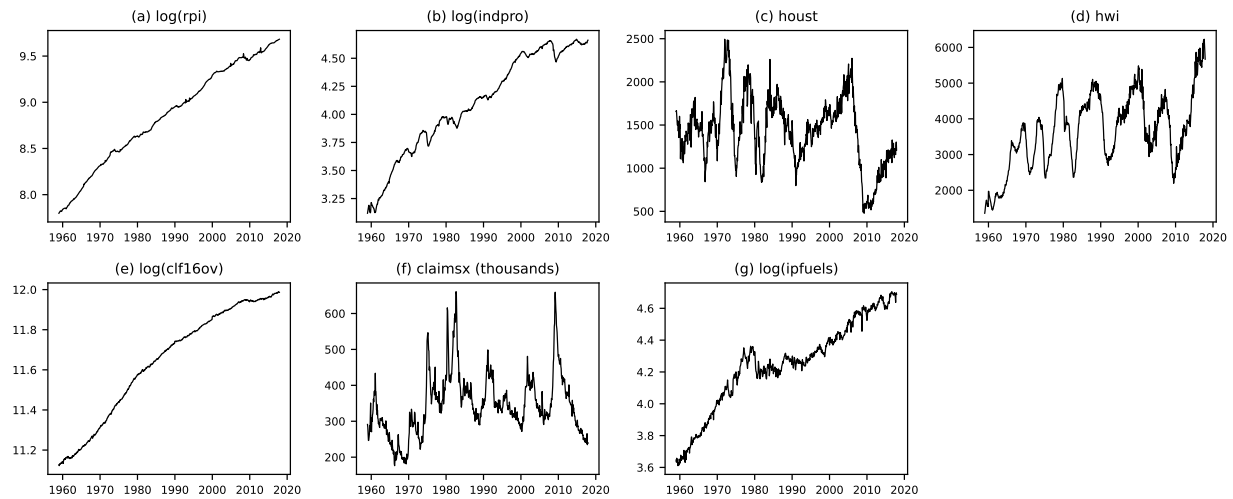
- (1) *Non-rejection is not acceptance.* With an intercept the p-value is about $0.20 + 0.10 \times 0.22/0.25 \approx 0.29$. The test fails to reject a unit root, which means only that the data do not provide strong evidence against it. Unit root tests have low power against persistent stationary alternatives, especially in short samples. The correct statement is “we cannot reject the hypothesis of a unit root.”
- (2) *A unit root is implausible for this variable.* A daily tweet count is non-negative and bounded in practice. A unit root process has variance growing linearly in time and eventually wanders arbitrarily far in both directions, which a count cannot do. Theory and context favour a stationary (possibly highly persistent) count process.
- (3) *Other features can masquerade as a unit root.* Level shifts (a new office, a campaign, a platform change), day-of-week effects and heteroskedasticity typical of count data all reduce the power

of the ADF test (Perron 1989). A count model (e.g. Poisson autoregression) with breaks would be more appropriate.

Consistent with Section 16.13, the test should be used to check a modeling presumption, not as a rule for declaring a series a unit root process.

Exercise 16.12

Method. The trend specification is chosen from the nature of the series (plotted below): a linear trend (Case 3) for the growing aggregates in (a), (b), (e), (g) and for *hwi*, which in this vintage (spliced with JOLTS data) trends upward; an intercept only (Case 2) for housing starts and initial claims, which fluctuate around a stable level. Aggregates growing exponentially are logged (the book lists *indpro*, *clf16ov* and *ipfuels* in levels; their level results are given in the notes). The order p minimizes AIC over $AR(p)$, $p \leq 12$, fit with the same deterministic terms on a common sample starting 1960m1. The ADF regression $\Delta Y_t = \mu(+\delta t) + \rho Y_{t-1} + \sum_{j=1}^{p-1} \gamma_j \Delta Y_{t-j} + e_t$ then uses all available observations. (Book typo: the FRED-MD mnemonic for initial claims is *claimsx*, not *claims*.)



series	deterministic	p	$\hat{\rho}$ (s.e.)	ADF	p-value
(a) $\log(rpi)$	trend	9	-0.0075 (0.0036)	-2.09	0.55
(b) $\log(indpro)$	trend	6	-0.0051 (0.0029)	-1.74	0.73
(c) <i>houst</i>	intercept	8	-0.0339 (0.0105)	-3.22	0.019
(d) <i>hwi</i>	trend	10	-0.0271 (0.0070)	-3.88	0.013
(e) $\log(clf16ov)$	trend	8	0.0023 (0.0018)	1.27	> 0.99
(f) <i>claimsx</i>	intercept	7	-0.0313 (0.0084)	-3.72	0.006
(g) $\log(ipfuels)$	trend	6	-0.0201 (0.0085)	-2.36	0.41

Notes on alternatives. In levels with trend: *indpro* $p = 5$, ADF = -3.02 ($p = 0.13$); *clf16ov* $p = 12$, ADF = 0.33 ($p > 0.99$); *ipfuels* $p = 3$, ADF = -2.64 ($p = 0.27$). Adding a trend to *houst* gives -3.47 (0.044), to *claimsx* -3.71 (0.022); *hwi* with intercept only gives -3.22 (0.019). The intercept-only test on $\log(clf16ov)$ gives -4.51 ($p < 0.001$). This is a spurious rejection from the

wrong specification: the concave growth path makes ΔY_t fall as Y_{t-1} rises, which looks like mean reversion when no trend is allowed.

Interpretation.

- Income, industrial production, the labor force and fuel production are consistent with unit roots (p-values 0.41–0.99), as expected for macroeconomic aggregates. For the labor force $\hat{\rho} > 0$: growth has decelerated since the 1990s, a shape a linear trend cannot capture.
- Housing starts and initial claims reject the unit root at 2% and 1%, consistent with cyclical, mean-reverting series, though reversion is slow: the autoregressive coefficients sum to about 0.97.
- *hwi* rejects at about 1%. But the series is a splice of the print help-wanted index and online vacancy data, and its behaviour changes markedly after 2010, so the rejection should be read cautiously (see also the KPSS results in Exercise 16.13).

Exercise 16.13

Same transformations and deterministic specifications as in Exercise 16.12: KPSS₂ (detrended) for trending series, KPSS₁ (demeaned) otherwise. Following Section 16.14, $M = \lfloor 3n^{1/3} \rfloor = 26$ ($n = 708$). This is large enough to control size for AR(1)-type persistence up to about 0.8. For sensitivity we also report $M = 6 \approx 4(n/100)^{1/4}$, a common short-lag rule.

series	test	KPSS ($M = 26$)	p-value	KPSS ($M = 6$)	p-value
(a) $\log(rpi)$	KPSS ₂	0.446	< 0.001	1.596	< 0.001
(b) $\log(indpro)$	KPSS ₂	0.354	< 0.001	1.219	< 0.001
(c) <i>houst</i>	KPSS ₁	0.455	0.052	1.454	< 0.001
(d) <i>hwi</i>	KPSS ₂	0.176	0.026	0.551	< 0.001
(e) $\log(clf16ov)$	KPSS ₂	0.649	< 0.001	2.423	< 0.001
(f) <i>claimsx</i>	KPSS ₁	0.423	0.064	1.283	< 0.001
(g) $\log(ipfuels)$	KPSS ₂	0.320	0.001	1.126	< 0.001

In levels: *indpro* 0.194 ($p = 0.018$), *clf16ov* 0.512 (< 0.001), *ipfuels* 0.278 (0.005).

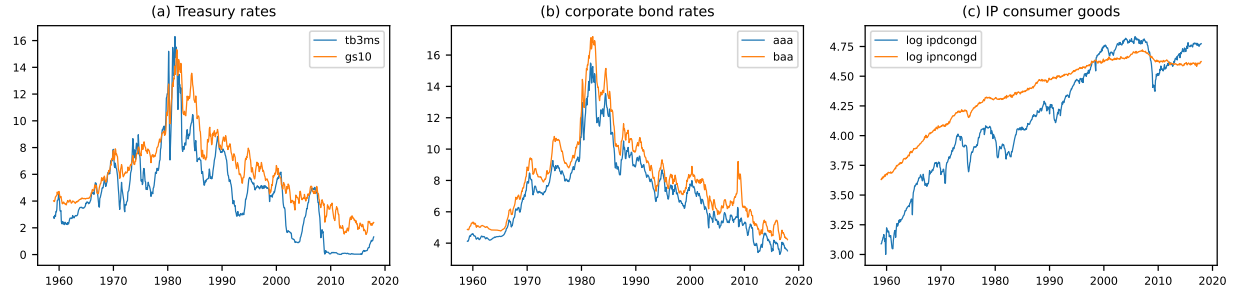
Choice of M matters. With $M = 6$ every series rejects stationarity decisively, but a short truncation underestimates the long-run variance of a persistent stationary series and over-rejects. Under the unit root alternative the statistic falls roughly like $1/M$, as the table illustrates. We base conclusions on $M = 26$.

Combining with the ADF results.

- (a), (b), (e), (g): KPSS rejects stationarity strongly and ADF does not reject a unit root. The two tests agree: these aggregates are best treated as $I(1)$.
- (c) housing starts and (f) initial claims: ADF rejects a unit root while KPSS₁ does not reject stationarity at 5% ($p = 0.05$ and 0.06 , borderline). Taken together the evidence favours stationary but very persistent series.
- (d) *hwi*: both tests reject their nulls (ADF $p = 0.013$, KPSS₂ $p = 0.026$). Neither “trend stationary” nor “unit root” describes the series well. The conflict is typical of a series with structural breaks (here the shift from print to online job advertising and the associated splice), and a break-robust analysis would be needed.

Exercise 16.14

The VAR order minimizes AIC over levels VARs with $p \leq 12$ (common sample from 1960m1). The trace statistic is $LR(r) = -n \sum_{j>r} \log(1 - \hat{\lambda}_j)$, compared with Tables 16.6–16.8 using $m - r$ degrees of freedom.



Trend specification. Interest rates are untrended, so we use Trend Model 2 (intercept restricted to the cointegrating relation). The IP series are trended and grow at different average rates (2.9% versus 1.7% a year), so any cointegrating relation must allow a linear trend. We report Trend Models 3 and 4.

pair	model	p	LR(0)	p-value	LR(1)	p-value
(a) <i>tb3ms</i> , <i>gs10</i>	2	10	34.2	< 0.001	2.45	0.69
(b) <i>aaa</i> , <i>baa</i>	2	8	25.4	0.009	1.70	0.83
(c) log IP durables, nondurables	3	3	38.8	< 0.001	4.37	0.037
(c) log IP durables, nondurables	4	3	48.0	< 0.001	7.83	0.27

(a) No cointegration is strongly rejected and $r = 1$ is not rejected ($p = 0.69$), so the rates are cointegrated with one cointegrating vector. The estimated relation is $tb3ms - 1.017 gs10 + 1.63$: the spread is stationary with the 3-month rate on average about 1.6 points below the 10-year rate, matching the quarterly results of Table 16.4 and Section 16.21. The conclusion is robust to p : with $p = 4$, $LR(0) = 33.3$ and $LR(1) = 2.77$; with $p = 12$, 29.6 and 3.20.

(b) $LR(0) = 25.4$ exceeds the 1% critical value 25.1 and $LR(1) = 1.70$ is far from significant. The AAA and BAA yields are cointegrated, with relation $aaa - 0.924 baa + 0.39$, so the default spread is stationary. The coefficient is somewhat below one. With $p = 4$: 27.1 and 1.54; with $p = 12$: 22.0 ($p = 0.03$) and 2.26. The rejection of no cointegration weakens with longer lags but remains significant at 5%.

(c) Both specifications reject no cointegration at any conventional level. Under Model 3, $LR(1) = 4.37$ also rejects $r = 1$ at 5%, which would imply rank 2, i.e. two stationary series. That contradicts the unit root evidence for industrial production (Exercise 16.12). The rejection is a symptom of misspecification: Model 3 excludes a trend from the cointegrating relation, although the two series trend at different rates. Under Model 4, which allows a trend in the relation, $LR(1) = 7.83$ ($p = 0.27$), and the results support $r = 1$: $\log(ipdcngd) - 1.39 \log(ipncngd)$ is stationary around a linear trend. With $p = 12$ the same pattern holds ($LR(0) = 31.9$, $p = 0.01$; $LR(1) = 9.18$, $p = 0.17$). The evidence for cointegration is weaker than for the interest rates, and the large swings in the

durables/nondurables gap visible in the plot (e.g. after 2007) suggest the equilibrium error is very persistent.

Chapter 17

Panel Data

Solutions to all 18 exercises of Chapter 17 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 17.15–17.18 use `AB1991` and `Invest1993`. The dynamic panel GMM estimators were programmed directly, following Stata `xtdpd` conventions. They reproduce Blundell and Bond's (1998) estimates in (17.114) exactly, and the book's Table 17.3 to within 0.002 (see the remark in Exercise 17.17). Robust standard errors are clustered by individual; two-step standard errors use Windmeijer's (2005) correction.

Notation: $\bar{Y}_i$ is the individual mean, $\dot{Y}_{it} = Y_{it} - \bar{Y}_i$ the within transformation, $M_i = \mathbf{I}_i - \mathbf{1}_i(\mathbf{1}_i'\mathbf{1}_i)^{-1}\mathbf{1}_i'$ and $P_i = \mathbf{I}_i - M_i$. N is the number of individuals and $n = \sum_i T_i$ the number of observations. Observations are independent across i throughout.

Exercise 17.1

Under Assumption 17.1, $\mathbb{E}[e_i | X_i] = 0$ and $\mathbb{E}[e_i e_i' | X_i] = \sigma_\varepsilon^2 \Omega_i$ with $\Omega_i = \mathbf{I}_i + \mathbf{1}_i \mathbf{1}_i' \sigma_u^2 / \sigma_\varepsilon^2$, and independence across i gives $\mathbb{E}[e_i e_j' | X] = \mathbf{0}$ for $i \neq j$.

(a) $\hat{\beta}_{\text{gls}} - \beta = A^{-1} \sum_i X_i' \Omega_i^{-1} e_i$ with $A = \sum_i X_i' \Omega_i^{-1} X_i$. Hence

$$V_{\text{gls}} = A^{-1} \left(\sum_i X_i' \Omega_i^{-1} \sigma_\varepsilon^2 \Omega_i \Omega_i^{-1} X_i \right) A^{-1} = \sigma_\varepsilon^2 A^{-1} = \left(\sum_{i=1}^N X_i' \Omega_i^{-1} X_i \right)^{-1} \sigma_\varepsilon^2.$$

Similarly $\hat{\beta}_{\text{pool}} - \beta = (\sum_i X_i' X_i)^{-1} \sum_i X_i' e_i$, so

$$V_{\text{pool}} = \left(\sum_{i=1}^N X_i' X_i \right)^{-1} \left(\sum_{i=1}^N X_i' \Omega_i X_i \right) \left(\sum_{i=1}^N X_i' X_i \right)^{-1} \sigma_\varepsilon^2.$$

(Book typos: the sums in (17.11)–(17.12) run over individuals $i = 1, \dots, N$, not n ; and (17.12) omits the factor σ_ε^2 . Both variances must carry it for (17.13) to compare like with like.)

(b) Stack $X = (X_1', \dots, X_N')'$ and $\Omega = \text{diag}(\Omega_i)$. Let $C = (X'X)^{-1}X' - (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}$, so $CX = \mathbf{0}$. Then

$$\frac{V_{\text{pool}}}{\sigma_\varepsilon^2} = [(X'\Omega^{-1}X)^{-1}X'\Omega^{-1} + C]\Omega[\Omega^{-1}X(X'\Omega^{-1}X)^{-1} + C'] = (X'\Omega^{-1}X)^{-1} + C\Omega C',$$

since the cross terms contain $CX = \mathbf{0}$. As $C\Omega C' \geq 0$, $V_{\text{gls}} \leq V_{\text{pool}}$, with equality iff $C\Omega C' = \mathbf{0}$, e.g. when $\sigma_u^2 = 0$.

Exercise 17.2

No. $\hat{\beta}_{\text{fe}} - \beta = (\sum_i \dot{X}_i' \dot{X}_i)^{-1} \sum_i \dot{X}_i' \varepsilon_i$ and $\dot{X}_i' \varepsilon_i = \sum_t \dot{X}_{it} \varepsilon_{it}$ with $\dot{X}_{it} = X_{it} - \bar{X}_i$. The individual mean $\bar{X}_i$ contains the regressors of *all* periods, so unbiasedness needs $\mathbb{E}[\varepsilon_{it} | X_{i1}, \dots, X_{iT}] = 0$: strict exogeneity (17.18). Contemporaneous exogeneity allows X_{is} , $s > t$, to respond to ε_{it} (feedback), which makes $\mathbb{E}[\bar{X}_i \varepsilon_{it}] \neq 0$.

Example. $T = 2$, $Y_{it} = X_{it}\beta + u_i + \varepsilon_{it}$ with $X_{it} = \varepsilon_{i,t-1}$ and ε_{it} i.i.d. $(0, \sigma^2)$. Then $\mathbb{E}[\varepsilon_{it} | X_{it}] = \mathbb{E}[\varepsilon_{it} | \varepsilon_{i,t-1}] = 0$. But with $T = 2$ the FE estimator is the regression of ΔY_{i2} on $\Delta X_{i2} = \varepsilon_{i1} - \varepsilon_{i0}$

(Exercise 17.6), and $\mathbb{E}[\Delta X_{i2} \Delta \varepsilon_{i2}] = \mathbb{E}[(\varepsilon_{i1} - \varepsilon_{i0})(\varepsilon_{i2} - \varepsilon_{i1})] = -\sigma^2$. So $\hat{\beta}_{fe} \xrightarrow{p} \beta - \sigma^2/(2\sigma^2) = \beta - \frac{1}{2}$: biased and inconsistent. The leading case is the dynamic panel with a lagged dependent variable (Nickell bias), which motivates Sections 17.36–17.42.

Exercise 17.3

The inequality holds *on average over t* in general, and for each t under an exchangeability condition, but not for every t in general.

Averaged version. Let $\mu_t = \mathbb{E}[X_{it}]$ and $\tilde{X}_{it} = X_{it} - \mu_t$. Then $\dot{X}_{it} - \mathbb{E}[\dot{X}_{it}] = \tilde{X}_{it} - \bar{\tilde{X}}_i$. The algebraic identity $\sum_t (\tilde{X}_{it} - \bar{\tilde{X}}_i)^2 = \sum_t \tilde{X}_{it}^2 - T \bar{\tilde{X}}_i^2$ gives, taking expectations,

$$\sum_{t=1}^T \text{var}[\dot{X}_{it}] = \sum_{t=1}^T \text{var}[X_{it}] - T \text{var}[\bar{X}_i] \leq \sum_{t=1}^T \text{var}[X_{it}].$$

This is the sense in which the within transformation “reduces the variance of the regressors”. It is the population version of $\sum_i \dot{X}_i' \dot{X}_i \leq \sum_i X_i' X_i$ used in (17.28).

Each t under exchangeability. If $\text{var}[X_{it}] = \sigma^2$ and $\text{cov}(X_{it}, X_{is}) = c$ for all $t \neq s$ (e.g. $X_{it} = a_i + v_{it}$ with i.i.d. v_{it}), then $\text{cov}(X_{it}, \bar{X}_i) = \text{var}[\bar{X}_i] = (\sigma^2 + (T-1)c)/T$ and

$$\text{var}[\dot{X}_{it}] = \sigma^2 - 2 \text{cov}(X_{it}, \bar{X}_i) + \text{var}[\bar{X}_i] = \frac{T-1}{T}(\sigma^2 - c) \leq \sigma^2,$$

because positive semi-definiteness requires $c \geq -\sigma^2/(T-1)$.

A counterexample for a single t . Take $T = 3$ with unit variances and $\text{cov}(X_1, X_2) = \text{cov}(X_2, X_3) = -0.5$, $\text{cov}(X_1, X_3) = 0.5$ (a valid covariance matrix, with determinant 0.5). Then $\dot{X}_{i2} = (-X_{i1} + 2X_{i2} - X_{i3})/3$ has variance $(1 + 4 + 1 + 2 + 1 + 2)/9 = 11/9 > 1$. So the claim should be read as the averaged (or exchangeable) statement.

Exercise 17.4

Since $M_i \mathbf{1}_i = \mathbf{0}$ and $\dot{X}_i' M_i = \dot{X}_i'$, $\hat{\beta}_{fe} - \beta = (\sum_i \dot{X}_i' \dot{X}_i)^{-1} \sum_i \dot{X}_i' \varepsilon_i$, a linear function of ε given X . With $\Sigma_i = \mathbb{E}[\varepsilon_i \varepsilon_i' | X_i]$ and $\mathbb{E}[\varepsilon_i \varepsilon_j' | X] = \mathbb{E}[\varepsilon_i | X_i] \mathbb{E}[\varepsilon_j' | X_j] = \mathbf{0}$ for $i \neq j$ (independence and (17.18)),

$$\begin{aligned} V_{fe} &= \left(\sum_i \dot{X}_i' \dot{X}_i \right)^{-1} \left(\sum_i \sum_j \dot{X}_i' \mathbb{E}[\varepsilon_i \varepsilon_j' | X] \dot{X}_j \right) \left(\sum_i \dot{X}_i' \dot{X}_i \right)^{-1} \\ &= \left(\sum_{i=1}^N \dot{X}_i' \dot{X}_i \right)^{-1} \left(\sum_{i=1}^N \dot{X}_i' \Sigma_i \dot{X}_i \right) \left(\sum_{i=1}^N \dot{X}_i' \dot{X}_i \right)^{-1}. \end{aligned}$$

Exercise 17.5

Under (17.25)–(17.26) and $u_i = 0$, $V_{\text{pool}} = \sigma_\varepsilon^2 (\sum_i X_i' X_i)^{-1}$ and $V_{fe}^0 = \sigma_\varepsilon^2 (\sum_i \dot{X}_i' \dot{X}_i)^{-1}$. Because $\mathbf{I}_i = M_i + P_i$ with P_i idempotent,

$$\sum_i X_i' X_i - \sum_i \dot{X}_i' \dot{X}_i = \sum_i X_i' P_i X_i = \sum_i T_i \bar{X}_i \bar{X}_i' \geq 0.$$

For positive definite matrices $A \geq B > 0$ implies $B^{-1} \geq A^{-1}$. With $A = \sum_i X_i' X_i$ and $B = \sum_i \dot{X}_i' \dot{X}_i$ (assumed invertible), $V_{fe}^0 \geq V_{pool}$. (The book's text refers to "Exercise 17.28"; it means this exercise and Exercise 17.3.)

Exercise 17.6

With $T = 2$, $\bar{X}_i = (X_{i1} + X_{i2})/2$, so $\dot{X}_{i1} = -\frac{1}{2}\Delta X_{i2}$ and $\dot{X}_{i2} = \frac{1}{2}\Delta X_{i2}$, and likewise for Y . Therefore

$$\sum_t \dot{X}_{it} \dot{X}_{it}' = \frac{1}{2} \Delta X_{i2} \Delta X_{i2}', \quad \sum_t \dot{X}_{it} \dot{Y}_{it} = \frac{1}{2} \Delta X_{i2} \Delta Y_{i2},$$

and $\hat{\beta}_{fe} = (\sum_i \frac{1}{2} \Delta X_{i2} \Delta X_{i2}')^{-1} \sum_i \frac{1}{2} \Delta X_{i2} \Delta Y_{i2} = \hat{\beta}_\Delta$. (In an unbalanced panel with $T_i \leq 2$, individuals with $T_i = 1$ contribute zero to both estimators, so the equality still holds.)

Exercise 17.7

In the random effects model the level error is $e_{it} = u_i + \varepsilon_{it}$ with $\mathbb{E}[e_{it}^2] = \sigma_u^2 + \sigma_\varepsilon^2$ (by (17.9)–(17.10) and (17.6)). $\hat{e}_{it}$ estimates e_{it} : $\hat{e}_{it} = e_{it} - X_{it}'(\hat{\beta}_{fe} - \beta)$ and $\hat{\beta}_{fe}$ is consistent under the random effects assumptions. Hence $\hat{\sigma}_e^2 \xrightarrow{p} \sigma_u^2 + \sigma_\varepsilon^2$ as $N \rightarrow \infty$, while $\hat{\sigma}_e^2 \xrightarrow{p} \sigma_\varepsilon^2$. A consistent estimator is

$$\tilde{\sigma}_u^2 = \max[0, \hat{\sigma}_e^2 - \hat{\sigma}_\varepsilon^2].$$

Remarks: if X_{it} contains no intercept but $\mathbb{E}[u_i] \neq 0$, use the demeaned residuals $\hat{e}_{it} - \bar{\hat{e}}$ in $\hat{\sigma}_e^2$. A finite-sample refinement divides by $n - k$ instead of n . Unlike (17.44), this estimator does not require the between regression, but it does rely on homoskedasticity ($\mathbb{E}[e_{it}^2]$ equal across i, t).

Exercise 17.8

In full-sample notation (17.34), $Y = X\beta + Du + \varepsilon$, $\dot{X} = M_D X$ and $\hat{\varepsilon} = \dot{Y} - \dot{X}\hat{\beta}_{fe} = (M_D - P_{\dot{X}})Y$ with $P_{\dot{X}} = \dot{X}(\dot{X}'\dot{X})^{-1}\dot{X}'$. Since $M_D D = \mathbf{0}$ and $(M_D - P_{\dot{X}})X = \dot{X} - \dot{X} = \mathbf{0}$, we get $\hat{\varepsilon} = Q\varepsilon$ with $Q = M_D - P_{\dot{X}}$. Q is symmetric and idempotent ($P_{\dot{X}}M_D = P_{\dot{X}}$), with $\text{tr} Q = (n - N) - k$. By (17.18), (17.25), (17.26) and independence across i , $\mathbb{E}[\varepsilon\varepsilon' | X] = \sigma_\varepsilon^2 \mathbf{I}_n$, so

$$\mathbb{E}[\hat{\varepsilon}'\hat{\varepsilon} | X] = \mathbb{E}[\text{tr}(Q\varepsilon\varepsilon') | X] = \sigma_\varepsilon^2 \text{tr} Q = \sigma_\varepsilon^2(n - N - k),$$

and $\mathbb{E}[\hat{\sigma}_\varepsilon^2 | X] = \sigma_\varepsilon^2$. (The sum in (17.37) should run over $i = 1, \dots, N$.)

Exercise 17.9

$\sqrt{N}(\hat{\beta}_\Delta - \beta) = (N^{-1} \sum_i \Delta X_i' \Delta X_i)^{-1} N^{-1/2} \sum_i \Delta X_i' \Delta \varepsilon_i$. The only moment condition used is $\mathbb{E}[\Delta X_i' \Delta \varepsilon_i] = \sum_{t=2}^T \mathbb{E}[\Delta X_{it} \Delta \varepsilon_{it}] = 0$, and $\mathbb{E}[\Delta X_{it} \Delta \varepsilon_{it}] = \mathbb{E}[X_{it} \varepsilon_{it}] - \mathbb{E}[X_{i,t-1} \varepsilon_{it}] - \mathbb{E}[X_{it} \varepsilon_{i,t-1}] + \mathbb{E}[X_{i,t-1} \varepsilon_{i,t-1}]$ involves only adjacent periods.

Theorem. Assume (1) $Y_{it} = X_{it}'\beta + u_i + \varepsilon_{it}$, $t = 1, \dots, T$, $T \geq 2$; (2) (ε_i, X_i) i.i.d. across i ; (3') $\mathbb{E}[X_{is} \varepsilon_{it}] = 0$ for $|s - t| \leq 1$; (4) $Q_\Delta = \mathbb{E}[\Delta X_i' \Delta X_i] > 0$; (5) $\mathbb{E}\varepsilon_{it}^4 < \infty$; (6) $\mathbb{E}\|X_{it}\|^4 < \infty$. Then as $N \rightarrow \infty$,

$$\sqrt{N}(\hat{\beta}_\Delta - \beta) \xrightarrow{d} N(0, Q_\Delta^{-1} \Omega_\Delta Q_\Delta^{-1}), \quad \Omega_\Delta = \mathbb{E}[\Delta X_i' \Delta \varepsilon_i \Delta \varepsilon_i' \Delta X_i].$$

Proof. By (2), (5), (6) and the WLLN, $N^{-1} \sum \Delta X_i' \Delta X_i \xrightarrow{p} Q_\Delta$. By (3'), $\Delta X_i' \Delta \varepsilon_i$ has mean zero; by (5)–(6) and Cauchy–Schwarz it has finite variance Ω_Δ . So the CLT applies, and the result follows from (4) and the CMT.

Assumption 3' is strictly weaker than strict exogeneity: X_{it} may be correlated with ε_{is} for $|s - t| \geq 2$, e.g. feedback from ε_{it} to $X_{i,t+2}$ and beyond. (The exact requirement is even weaker, $\sum_t \mathbb{E}[\Delta X_{it} \Delta \varepsilon_{it}] = 0$, but 3' is the natural primitive condition.) By contrast the within transformation uses all periods, so the FE estimator needs exogeneity at all leads and lags.

Exercise 17.10

With residuals built from the true β and a balanced panel, $\hat{\varepsilon}_{it} = \varepsilon_{it} - \bar{\varepsilon}_i$. By (17.26) the errors are conditionally uncorrelated across t , so $\mathbb{E}[\varepsilon_{it} \bar{\varepsilon}_i | X_i] = \sigma_{it}^2/T$ and $\mathbb{E}[\bar{\varepsilon}_i^2 | X_i] = T^{-2} \sum_s \sigma_{is}^2 = \bar{\sigma}_i^2/T$ with $\bar{\sigma}_i^2 = T^{-1} \sum_s \sigma_{is}^2$. Hence

$$\mathbb{E}[\hat{\varepsilon}_{it}^2 | X_i] = \sigma_{it}^2 - \frac{2\sigma_{it}^2}{T} + \frac{\bar{\sigma}_i^2}{T} = \left(\frac{T-2}{T}\right)\sigma_{it}^2 + \frac{\bar{\sigma}_i^2}{T}.$$

Exercise 17.11

(a) Summing (17.57) over t : $\sum_t \mathbb{E}[\hat{\varepsilon}_{it}^2 | X_i] = \frac{T-2}{T} T \bar{\sigma}_i^2 + \bar{\sigma}_i^2 = (T-1)\bar{\sigma}_i^2$, so $\mathbb{E}[\hat{\sigma}_i^2 | X_i] = \bar{\sigma}_i^2$ (the book's σ_i^2).

(b) Write $S = (\dot{X}' \dot{X})^{-1}$, $V_{fe} = S(\sum_{i,t} \dot{X}_{it} \dot{X}_{it}' \sigma_{it}^2)S$ and $B = S(\sum_i \dot{X}_i' \dot{X}_i \bar{\sigma}_i^2)S$. From the book's calculation (with $k = 0$), $\mathbb{E}[\hat{V}_{fe} | X] = \frac{T-2}{T-1} V_{fe} + \frac{1}{T-1} B$, and by (a) $\mathbb{E}[\hat{B}_{fe} | X] = B$. To remove the bias we need

$$\tilde{V}_{fe} = \frac{T-1}{T-2} \left(\hat{V}_{fe} - \frac{1}{T-1} \hat{B}_{fe} \right) = \frac{T-1}{T-2} \hat{V}_{fe} - \frac{1}{T-2} \hat{B}_{fe},$$

for then $\mathbb{E}[\tilde{V}_{fe} | X] = V_{fe} + \frac{1}{T-2} B - \frac{1}{T-2} B = V_{fe}$.

Remark. Book typo in (17.58). As printed, $\tilde{V}_{fe} = \frac{T-1}{T-2} \hat{V}_{fe} - \frac{1}{T-1} \hat{B}_{fe}$ has expectation $V_{fe} + \frac{1}{(T-1)(T-2)} B$ and is biased. The factor $\frac{T-1}{T-2}$ must multiply the whole difference, as in Stock and Watson (2008); equivalently, the coefficient on $\hat{B}_{fe}$ is $\frac{1}{T-2}$. This is also what the unbalanced formula (17.60) reduces to when $T_i \equiv T$ (Exercise 17.12).

Exercise 17.12

(a)–(b) Exercises 17.10–17.11(a) with T replaced by T_i and sums over the observed periods $t \in S_i$ (selection is conditioned on, as in Assumption 17.3): $\mathbb{E}[\hat{\varepsilon}_{it}^2 | X_i] = \frac{T_i-2}{T_i} \sigma_{it}^2 + \frac{\sigma_i^2}{T_i}$ and $\mathbb{E}[\hat{\sigma}_i^2 | X_i] = \sigma_i^2$, with $\sigma_i^2 = T_i^{-1} \sum_{t \in S_i} \sigma_{it}^2$.

(c) It suffices that $\mathbb{E}[\tilde{\Omega}_{fe} | X] = \sum_i \sum_{t \in S_i} \dot{X}_{it} \dot{X}_{it}' \sigma_{it}^2$.

- $T_i > 2$: $\mathbb{E}\left[\frac{T_i \hat{\varepsilon}_{it}^2 - \hat{\sigma}_i^2}{T_i - 2} | X_i\right] = \frac{(T_i - 2)\sigma_{it}^2 + \sigma_i^2 - \sigma_i^2}{T_i - 2} = \sigma_{it}^2$ for each t .
- $T_i = 2$: $\mathbb{E}\left[\frac{T_i \hat{\varepsilon}_{it}^2}{T_i - 1} | X_i\right] = 2(0 \cdot \sigma_{it}^2 + \sigma_i^2/2) = \sigma_i^2$, which is not σ_{it}^2 term by term. But with two periods $\dot{X}_{i1} = -\dot{X}_{i2}$, so $\dot{X}_{i1} \dot{X}_{i1}' = \dot{X}_{i2} \dot{X}_{i2}'$ and $\sum_t \dot{X}_{it} \dot{X}_{it}' \sigma_{it}^2 = \dot{X}_{i1} \dot{X}_{i1}' (\sigma_{i1}^2 + \sigma_{i2}^2) = \sum_t \dot{X}_{it} \dot{X}_{it}' \sigma_{it}^2$.
- $T_i = 1$: $\dot{X}_{i1} = 0$, so these individuals contribute nothing to either side.

Summing over i and pre- and post-multiplying by $(\dot{X}'\dot{X})^{-1}$ gives $\mathbb{E}[\tilde{V}_{fe} | X] = V_{fe}$. With $T_i \equiv T$, $\tilde{\Omega}_{fe} = \frac{T}{T-2} \sum \dot{X}_{it}\dot{X}'_{it}\hat{\varepsilon}_{it}^2 - \frac{1}{T-2} \sum_i \dot{X}'_i\dot{X}_i\hat{\sigma}_i^2$, i.e. $\tilde{V}_{fe} = \frac{T-1}{T-2}\hat{V}_{fe} - \frac{1}{T-2}\hat{B}_{fe}$, confirming the correction to (17.58) noted in Exercise 17.11.

Exercise 17.13

No. The within transformation must be applied to each regressor, and the regressor is X_{it}^2 . Its within transform is $(X^2)_{it} = X_{it}^2 - \overline{X^2}_i$, not the square of the demeaned variable, $\dot{X}_{it}^2 = (X_{it} - \overline{X}_i)^2$. Demeaning the model gives

$$\dot{Y}_{it} = \dot{X}_{it}\beta_1 + (X^2)_{it}\beta_2 + \dot{\varepsilon}_{it}, \quad (X^2)_{it} - \dot{X}_{it}^2 = 2\overline{X}_i\dot{X}_{it} - s_i^2, \quad s_i^2 = \overline{X^2}_i - \overline{X}_i^2.$$

Using $\dot{X}_{it}^2$ therefore omits the term $(2\overline{X}_i\dot{X}_{it} - s_i^2)\beta_2$, which is correlated with the included regressors. The estimator is inconsistent unless $\beta_2 = 0$. (Equivalently, $\dot{X}_{it}^2$ keeps an interaction between the individual mean $\overline{X}_i$ and the regressor.) The correct fixed effects estimator regresses $\dot{Y}_{it}$ on $\dot{X}_{it}$ and $(X^2)_{it}$: first create $Z_{it} = X_{it}^2$, then demean it. This is what `xtreg, fe` or the dummy-variable regression does automatically.

Exercise 17.14

Just identification means $k_1 = \ell_2$. The instrument matrix $\mathcal{Z} = (\dot{X}_1, \dot{X}_2, \overline{X}_1, Z_1)$ then has as many columns as the regressor matrix $\mathcal{X} = (X_1, X_2, Z_1, Z_2)$. 2SLS reduces to IV, which solves $\mathcal{Z}'(Y - \mathcal{X}\hat{\theta}) = 0$. Two facts do the work: a within-transformed variable is orthogonal to every time-invariant variable, $\dot{X}'_j\overline{X}_1 = \dot{X}'_jZ_1 = \dot{X}'_jZ_2 = 0$, because $\sum_t \dot{X}_{jit} = 0$ for each i ; and $\dot{X}'_jX_l = \dot{X}'_j\dot{X}_l$, $\dot{X}'_jY = \dot{X}'_j\dot{Y}$.

- The first two blocks of equations are $\dot{X}'_j(\dot{Y} - \dot{X}_1\hat{\beta}_1 - \dot{X}_2\hat{\beta}_2) = 0$, $j = 1, 2$: the normal equations of the within regression. So $(\hat{\beta}_1, \hat{\beta}_2) = \hat{\beta}_{fe}$.
- Define the estimated individual effects $\hat{u}_i = \overline{Y}_i - \overline{X}'_{1i}\hat{\beta}_1 - \overline{X}'_{2i}\hat{\beta}_2$. Because $\overline{X}_1$ and Z_1 are time invariant, $\overline{X}'_1(Y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2 - Z_1\gamma_1 - Z_2\gamma_2) = \sum_i T_i\overline{X}_{1i}(\hat{u}_i - Z'_{1i}\gamma_1 - Z'_{2i}\gamma_2)$, and similarly with Z_1 . The remaining equations are therefore

$$\sum_{i=1}^N T_i \begin{pmatrix} Z_{1i} \\ \overline{X}_{1i} \end{pmatrix} (\hat{u}_i - Z'_{1i}\hat{\gamma}_1 - Z'_{2i}\hat{\gamma}_2) = 0,$$

exactly the IV estimator in the regression of $\hat{u}_i$ on (Z_{1i}, Z_{2i}) with instruments $(Z_{1i}, \overline{X}_{1i})$: Z_1 instruments itself and $\overline{X}_1$ instruments Z_2 . It is computed at the observation level (weights T_i ; unweighted in a balanced panel).

Exercise 17.15

Year effects enter as time dummies. Arellano–Bond uses all lags $K_{i,t-2}, K_{i,t-3}, \dots$ as instruments for the differenced equation (751 observations, 1978–1984, 36 instruments). Blundell–Bond adds the level equations (891 observations) with instrument $\Delta K_{i,t-1}$ (44 instruments).

estimator	$\hat{\alpha}$	clustered s.e.
pooled OLS (levels, year dummies)	1.005	
fixed effects (within)	0.708	
(a) Arellano–Bond one-step	0.571	(0.174)
(b) Blundell–Bond one-step	1.048	(0.019)

(c) The estimates differ enormously, and the Arellano–Bond estimate is even *below* the within estimate, which is itself biased downward (Nickell bias). The explanation is weak identification (Section 17.40). Log capital is extremely persistent (pooled OLS gives 1.005). When α is near one, lagged levels are almost uncorrelated with subsequent differences. The Anderson–Hsiao first-stage coefficient of $\Delta K_{i,t-1}$ on $K_{i,t-2}$ is only 0.008, so the Arellano–Bond estimator is weakly identified: imprecise (s.e. 0.17) and biased toward the inconsistent least squares estimate of the differenced equation. The Blundell–Bond level moments use $\Delta K_{i,t-1}$ as an instrument for $K_{i,t-1}$ and remain informative near a unit root, hence the precise estimate.

The Blundell–Bond estimate 1.048 exceeds one, however, and the level moments rest on mean stationarity (17.101), which is dubious for a variable with a (near) unit root. The two-step Hansen statistics are $J = 39.9$ (df 28, $p = 0.07$) for Arellano–Bond and $J = 60.4$ (df 35, $p = 0.005$) for Blundell–Bond. Their difference, 20.5 on 7 df ($p \approx 0.005$), rejects the additional level moments. The data therefore say that α is close to one but cannot pin it down credibly: Arellano–Bond is weakly identified, and Blundell–Bond’s extra assumption appears to fail.

Exercise 17.16

One-step estimates with clustered standard errors:

	(a) AB, strictly exogenous	(b) AB, predetermined	(c) BB system
$N_{i,t-1}$	0.489 (0.113)	0.708 (0.084)	0.871 (0.044)
W_{it}	−0.467 (0.177)	−0.709 (0.117)	−0.782 (0.117)
$W_{i,t-1}$	0.217 (0.106)	0.500 (0.111)	0.512 (0.169)
K_{it}	0.313 (0.054)	0.466 (0.101)	0.472 (0.070)
$K_{i,t-1}$	−0.031 (0.066)	−0.215 (0.086)	−0.360 (0.072)
instruments	40	92	114

(a) Under strict exogeneity the differenced regressors ΔW_{it} , $\Delta W_{i,t-1}$, ΔK_{it} , $\Delta K_{i,t-1}$ serve as their own instruments, and $N_{i,t-2}$, $N_{i,t-3}$, ... instrument the lagged dependent variable.

(b) “ $W_{i,t-1}$, $K_{i,t-1}$ predetermined” means $\mathbb{E}[W_{i,t-1-s}\varepsilon_{it}] = 0$ for $s \geq 0$. For the differenced equation at t the valid instruments are then $W_{i,t-2}$, $W_{i,t-3}$, ... (and likewise for K and N): current wages and capital are allowed to be endogenous. This reproduces (17.114) *exactly*: 0.7075 (0.0842), −0.7088 (0.1171), 0.5000 (0.1113), 0.4660 (0.1010), −0.2151 (0.0859). Relative to (a), every coefficient is larger in magnitude: persistence rises from 0.49 to 0.71 and the wage and capital effects roughly double. Treating wages and capital as strictly exogenous ignores their feedback from employment shocks (e.g. a firm hit by a negative shock reduces both employment and investment).

Using their differences as instruments then biases the estimates toward the (downward-biased) differenced least squares estimates. The predetermined specification is more credible.

(c) Adding the level moments raises persistence to 0.87 and roughly halves the standard error of the lagged dependent variable, consistent with the weak-instrument argument of Section 17.40. The wage and capital coefficients are similar to (b).

(d) Read as a partial-adjustment labor demand equation, employment adjusts slowly: 71–87% of last year’s level carries over. A 10% wage increase lowers employment by about 7–8% on impact, partly reversed next year (positive $W_{i,t-1}$ coefficient); a 10% capital increase raises employment by about 4.7% on impact. The long-run elasticities (delta-method s.e.) are

	AB (b)	BB (c)
wage, $(\beta_W + \beta_{W,-1})/(1 - \alpha)$	−0.71 (0.42)	−2.09 (1.15)
capital, $(\beta_K + \beta_{K,-1})/(1 - \alpha)$	0.86 (0.22)	0.87 (0.16)

Labor and capital are complements, with a long-run capital elasticity near one (constant returns would imply one). The long-run wage elasticity is negative as theory predicts, but imprecise: $1/(1 - \hat{\alpha})$ amplifies the uncertainty when $\hat{\alpha}$ is close to one.

(e) Without clustering (classical one-step standard errors) the Blundell–Bond standard errors are

$$0.031, 0.089, 0.071, 0.057, 0.061 \quad \text{instead of} \quad 0.044, 0.117, 0.169, 0.070, 0.072,$$

i.e. too small by a factor of 1.2–1.4 for most coefficients and by 2.4 for $W_{i,t-1}$. The classical formula is invalid here for two reasons: the idiosyncratic errors are heteroskedastic across firms, and in the level equations the error $u_i + \varepsilon_{it}$ is correlated within firm by construction. Forgetting to cluster would overstate precision substantially, e.g. making the lagged wage effect look highly significant ($t = 7.2$ instead of 3.0).

Exercise 17.17

1962 firms, 1960–1991, 23,642 differenced observations. Following the exercise, no time effects are included. Adding them changes $\hat{\alpha}$ by less than 0.02: Arellano–Bond 0.592, Blundell–Bond 0.625 with all lags.

Remark. Replication and a numerical caveat. Very few firms are observed in 1960–1962 (6, 8 and 13). Instrument columns built from those years are supported by so few firms that the two-step weight matrix $\hat{\Omega} = \sum_i Z_i' \hat{u}_i \hat{u}_i' Z_i$ is singular and must be inverted with a generalized inverse. With a careless pseudo-inverse the two-step estimates and naive standard errors become erratic. With a scale-invariant generalized inverse (used here; results unchanged for tolerances 10^{-6} to 10^{-14}) the book’s Table 17.3 is reproduced to within 0.002: 0.3174 (0.0173) and 0.0297 (0.0112) versus 0.3191 (0.0172) and 0.0309 (0.0112). The remaining gap reflects Stata’s different choice of generalized inverse.

(a)–(b) With all available lags as instruments:

	$\hat{\alpha}$	Windmeijer s.e.	Hansen J (df, p)
(a) Arellano–Bond two-step	0.576	(0.056)	513 (422, 0.002)
(b) Blundell–Bond two-step	0.641	(0.046)	560 (452, < 0.001)

Leverage is moderately persistent: about 60% of a shock to debt/assets remains after one year.

(c) Estimates of α (first-lag coefficient), with robust / classical standard errors:

model	instrument lags	Arellano–Bond		Blundell–Bond	
		one-step	two-step	one-step	two-step
AR(1)	all	0.550 (0.060 0.013)	0.576 (0.056 0.010)	0.630 (0.053 0.008)	0.641 (0.046 0.006)
AR(1)	2–6	0.548 (0.077 0.014)	0.584 (0.063 0.013)	0.634 (0.058 0.008)	0.643 (0.044 0.008)
AR(1)	2–3	0.516 (0.087 0.015)	0.658 (0.052 0.027)	0.630 (0.058 0.008)	0.627 (0.038 0.012)
AR(2)	all	0.528 (0.051 0.014)	0.554 (0.048 0.010)	0.636 (0.040 0.009)	0.646 (0.034 0.005)
AR(2)	2–6	0.532 (0.069 0.016)	0.587 (0.055 0.015)	0.647 (0.048 0.009)	0.654 (0.035 0.007)
AR(2)	2–3	0.494 (0.089 0.018)	0.618 (0.053 0.031)	0.645 (0.049 0.010)	0.645 (0.033 0.011)

The AR(2) coefficient is small, 0.03–0.06, and mostly insignificant with robust standard errors (e.g. 0.044 (0.027) for two-step Arellano–Bond, 0.026 (0.046) for two-step Blundell–Bond), so AR(1) is adequate.

- *Coefficients.* The biggest difference is Arellano–Bond versus Blundell–Bond: the system estimates are typically 0.05–0.15 higher (the exception is two-step Arellano–Bond with lags 2–3), consistent with a downward weak-instrument bias of the difference estimator. Within Arellano–Bond the estimate is also sensitive to the step and the instrument set. With lags 2–3 the one-step estimate is 0.52 but the two-step estimate is 0.66. Blundell–Bond is remarkably stable (0.63–0.65) across all variations. AR(1) versus AR(2) hardly matters.
- *Standard errors.* By far the biggest difference is classical versus robust: classical standard errors are typically 3–8 times smaller, which would grossly overstate precision. Among robust standard errors the two-step (Windmeijer-corrected) ones are somewhat smaller than one-step, and the one-step robust standard errors rise as fewer instrument lags are used, as expected.
- The Hansen tests reject the overidentifying restrictions in every specification. This is a sign of heterogeneity or misspecification (e.g. remaining serial correlation in ε_{it}) that deserves further investigation, keeping in mind that J tests are unreliable with hundreds of instruments.

Exercise 17.18

All regressors are dated $t - 1$ and predetermined, so the differenced equation at t uses D, I, Q, CF dated $t - 2$ and earlier as instruments. The level equation (Blundell–Bond) uses their first differences dated $t - 1$.

(a)–(b) Two-step estimates with Windmeijer standard errors:

instrument lags	Arellano–Bond		Blundell–Bond	
	all	2–6	all	2–6
$D_{i,t-1}$	0.508 (0.065)	0.520 (0.082)	0.600 (0.057)	0.621 (0.060)
$I_{i,t-1}$	−0.095 (0.071)	−0.053 (0.066)	−0.041 (0.059)	−0.010 (0.053)
$Q_{i,t-1}$	0.0003 (0.0013)	0.0004 (0.0013)	0.0006 (0.0011)	0.0006 (0.0011)
$CF_{i,t-1}$	−0.043 (0.017)	−0.060 (0.026)	−0.037 (0.017)	−0.042 (0.021)
instruments	1692	560	1813	681
Hansen J (p)	1516 (1.00)	756 (< 0.001)	1666 (0.99)	903 (< 0.001)

Interpretation. Leverage is persistent ($\hat{\alpha} \approx 0.5$ – 0.6). Higher cash flow is followed by lower leverage, the one robust finding ($t \approx -2.0$ to -2.5 in every column). This is consistent with the pecking-order view that firms repay debt from internal funds. Lagged investment and Tobin’s Q have no significant effect on debt.

Too many instruments. With all lags there are 1692 (AB) and 1813 (BB) instruments for only 1962 firms. The 1692–1813-dimensional estimate of Ω is a sum of 1962 rank-one terms and is nearly singular. The two-step estimator then collapses to the one-step estimator (they agree to three decimals), the naive two-step standard errors are essentially zero, and the J statistic has p-value ≈ 1 , a telltale symptom of instrument proliferation rather than evidence of valid moments. Restricting to lags 2–6 gives a well-behaved two-step estimator and a J test with power (it rejects). Specifications (a) and (b) should therefore be reported with a limited lag depth.

(c) Highlights from the full grid (one- and two-step, AB and BB, lags all / 2–6 / 2–3):

- *Coefficients.* The investment coefficient is the most fragile. It ranges from -0.13 (Arellano–Bond one-step, lags 2–3) to $+0.035$ (Blundell–Bond two-step, lags 2–3), changing sign with the step and the lag depth. Its sign should not be interpreted. The persistence α again depends mainly on the estimator (Arellano–Bond 0.49 – 0.52 versus Blundell–Bond 0.60 – 0.62). The cash-flow coefficient is stable, between -0.04 and -0.07 .
- *Standard errors.* Classical versus robust is again the dominant factor: e.g. for $\hat{\alpha}$ (AB, lags 2–6) the one-step classical s.e. is 0.011 versus robust 0.082 . With too many instruments the uncorrected two-step standard errors collapse toward zero, which makes the Windmeijer correction (or one-step robust inference) essential.

Chapter 18

Difference in Differences

Solutions to all 8 exercises of Chapter 18 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 18.6–18.8 use CK1994, DS2004 and BMN2016. Before answering them we reproduced the chapter’s numbers. Tables 18.1–18.3, regressions (18.2) and (18.4), the regional tests ($p = 0.60$, $p = 0.23$), the police estimate -0.087 and (18.7) all match. (18.8) matches up to rounding except that we obtain $+0.005$ (0.005) rather than -0.005 for the *OffOutFlows* coefficient. Standard errors are clustered at the unit level (restaurant, city block, state), with Stata’s small-sample adjustment.

Exercise 18.1

The claim concerns variables that vary only at the *state-by-time* level, like Time_t and $D_{it} = \text{State}_i \text{Time}_t$. It is false for general individual-level variables: demeaning X_{it} at the state level leaves the individual’s own mean deviation, which is not orthogonal to that individual’s dummy.

(a) Let individual j belong to state $s(j)$ and let W_{st} be a state-by-time variable. Its state-demeaned version is $\widetilde{W}_{it} = W_{s(i)t} - \overline{W}_{s(i)}$, where $\overline{W}_s$ is the mean over all observations in state s . In a balanced panel every individual is observed in the same T periods, so $\overline{W}_s = T^{-1} \sum_t W_{st}$. The inner product with the dummy $d_{j,it} = \mathbf{1}\{i = j\}$ is

$$\sum_{i,t} d_{j,it} \widetilde{W}_{it} = \sum_{t=1}^T (W_{s(j)t} - \overline{W}_{s(j)}) = T\overline{W}_{s(j)} - T\overline{W}_{s(j)} = 0.$$

The same holds for variables demeaned at the state level and by time (two-way demeaning), since the time means are also state-by-time variables.

(b) No. If individual j is observed only in periods S_j , then $\overline{W}_s$ is a weighted average over periods, with weights proportional to the number of individuals observed in each period. In general $\sum_{t \in S_j} (W_{st} - \overline{W}_s) \neq 0$. In the unbalanced Card–Krueger sample (794 observations, some restaurants observed once) the state-dummy regression gives $\hat{\theta} = 2.754$ while restaurant fixed effects give 2.750: close, but not identical.

(c) By Frisch–Waugh–Lovell, $\hat{\theta} = \widetilde{D}'Y/\widetilde{D}'\widetilde{D}$, where $\widetilde{D}$ is the residual from projecting D on the other regressors. In the first regression these are the state and time dummies, so $\widetilde{D}$ is D two-way demeaned at the state and time level, a state-by-time variable. The individual dummies span the state dummies, so the second regression’s controls are the state dummies, the time dummies and the extra directions $M_S I$ (individual dummies residualized on state and time dummies). By (a), $\widetilde{D}$ is orthogonal to every individual dummy and hence to $M_S I$. Projecting further on $M_S I$ therefore leaves $\widetilde{D}$ unchanged, and both regressions produce the same $\widetilde{D}$ and the same $\hat{\theta}$ (and the same coefficient on Time_t). The extra restaurant dummies only absorb within-state heterogeneity in Y , which changes the residuals and hence the standard errors, not the estimate.

Exercise 18.2

The regressors $(1, \text{State}, D)$ take three distinct values, one for each of the groups: control state (both periods), treated state before, and treated state after. The regression is saturated in these three groups, so the fitted values equal the group means: $\hat{\beta}_0 = \bar{Y}_0$. (control state, pooled over both periods), $\hat{\beta}_0 + \hat{\beta}_1 = \bar{Y}_{10}$, and $\hat{\beta}_0 + \hat{\beta}_1 + \hat{\theta} = \bar{Y}_{11}$.

(a) $\hat{\theta} = \bar{Y}_{11} - \bar{Y}_{10}$, the change over time in the treated state's mean.

(b) The formula involves only treated-state observations. The control observations affect only $\hat{\beta}_0$ (and, through it, $\hat{\beta}_1$), never $\hat{\theta}$.

(c) No. It is a single *difference* (before–after) estimator with no control group.

(d) It estimates the treatment effect only if the outcome in the treated state would not have changed in the absence of treatment: no time effect ($\beta_2 = 0$), i.e. no common shocks, trends or seasonal differences between the two periods. In the Card–Krueger data this assumption is clearly violated. Pennsylvania employment fell by 2.3, so the before–after estimate 0.47 differs sharply from the difference-in-difference estimate 2.75.

Exercise 18.3

Use fixed-effects 2SLS (Section 17.28) with two-way effects. Eliminate u_i with the within transformation and include time dummies, or equivalently apply the two-way within transformation $\ddot{Y}_{it}, \ddot{D}_{it}, \ddot{Z}_{it}$. Then estimate

$$\hat{\theta} = \frac{\sum_{i,t} \ddot{Z}_{it} \ddot{Y}_{it}}{\sum_{i,t} \ddot{Z}_{it} \ddot{D}_{it}} \quad (\text{just identified; with several instruments use 2SLS or GMM}),$$

i.e. 2SLS of Y on D with unit and time fixed effects, instrumenting D by Z . In Stata: `xtivreg y (d = z) i.year, fe vce(cluster id)`. Requirements:

- *Exogeneity*: $\mathbb{E}[Z_{is}\varepsilon_{it}] = 0$ for all s, t , since the within transformation mixes periods.
- *Relevance*: $\ddot{Z}$ must be correlated with $\ddot{D}$, so Z must vary over time within units and differentially across units; a purely cross-sectional instrument is absorbed by u_i .

Standard errors should be clustered by unit. With a binary instrument and $T = 2$ the estimator is the ratio of two diff-in-diffs, $\hat{\theta} = \text{DiD}(Y)/\text{DiD}(D)$ (a “Wald” or fuzzy diff-in-diff estimator). With heterogeneous effects it estimates a LATE-type average for units whose treatment responds to Z .

Exercise 18.4

With $T = 2$, let d_j be the unit (region) dummies and Time_t the post-period dummy.

- *Treatment effects*. The treatment dummy is the sum of the treated units' post-period interactions, $D_{it} = \sum_{j \in \text{treated}} d_{j,i} \text{Time}_t$. Adding all N_2 interactions alongside D_{it} creates perfect collinearity. Dropping one makes the omitted unit the baseline: θ becomes its treatment effect and the $N_2 - 1$ interaction coefficients measure the other units' deviations from it. Homogeneity is the hypothesis that these $N_2 - 1$ coefficients are zero.

- *Control effects.* All N unit-by-post interactions sum to Time_t , and the treated ones sum to D_{it} . So the control units' interactions sum to $\text{Time}_t - D_{it}$, collinear with the included time effect and treatment dummy. Only $N_1 - 1$ can be added, measuring the other control units' time effects relative to the base control unit. Equal control effects is the hypothesis that these are zero.

In both cases the number of free parameters is the number of distinct unit-specific changes (N_2 or N_1) minus the one common effect already in the model.

Exercise 18.5

- (a) Difference in differences: $(16.72 - 15.23) - (18.10 - 16.42) = 1.49 - 1.68 = -0.19$. Wages grew by 0.19 less in Wisconsin than in Minnesota.
- (b) $\text{Act10} = \text{Wisconsin} \times \text{Post2010}$, so the regression is saturated in the four state-period cells. OLS fitted values equal the cell means, which are exactly the county averages given in the exercise. Hence $\hat{\alpha} = 16.42$, $\hat{\delta} = 18.10 - 16.42 = 1.68$, and $\hat{\beta} = -0.19$, the diff-in-diff estimate, whatever the number of counties per state.
- (c) $\hat{\gamma} = 15.23 - 16.42 = -1.19$, the pre-period Wisconsin–Minnesota difference.
(In the book's table the periods "2001–2010" and "2010–2020" overlap in 2010. The post period should be 2011–2020, matching the regressor definition.)

Exercise 18.6

- (a) Price is missing for 57 of 820 observations. Deleting them and their paired observations leaves a balanced panel of 356 restaurants (288 in New Jersey, 68 in Pennsylvania), 712 observations. The meal price averages \$3.33.
- (b) Average price of a meal (dollars):

	New Jersey	Pennsylvania	Difference
Before increase	3.359	3.088	0.271
After increase	3.425	3.054	0.371
Difference	0.066	−0.034	0.100

Meal prices rose by 6.6 cents in New Jersey and fell by 3.4 cents in Pennsylvania, a diff-in-diff increase of 10 cents, about 3% of the pre-period New Jersey price.

- (c) With standard errors clustered by restaurant,

$$\text{price}_{it} = \underset{(0.074)}{3.088} + \underset{(0.084)}{0.271} \text{State}_i - \underset{(0.037)}{0.034} \text{Time}_t + \underset{(0.044)}{0.100} D_{it}.$$

- (d) State fixed effects: $\hat{\theta} = 0.100$, $\text{Time} = -0.034$. With classical standard errors (as in `xtreg`, `fe` with the state as panel; clustering by two states is impossible) the standard errors are 0.121 and 0.109. Clustered by restaurant they are 0.044 and 0.037.
- (e) Restaurant fixed effects, clustered: $\hat{\theta} = 0.100$ (0.044), $\text{Time} = -0.034$ (0.037).
- (f) The point estimates are identical in (c), (d) and (e), as Exercise 18.1 predicts for a balanced panel. The standard errors are not. The classical state-FE standard errors ignore the strong within-restaurant correlation of prices over time, and the restaurant effects absorb most of the

price variation. With clustered standard errors, $t = 2.27$ and $p \approx 0.02$: the minimum wage increase raised meal prices by about 10 cents. In logs the effect is 2.9% (1.3%). Most of it comes from the entree price (7.5 cents, s.e. 4.2); fry and soda prices did not change differentially.

(g) Average meal price by region:

	South NJ	Central NJ	North NJ	PA 1	PA 2
Before increase	3.314	3.484	3.340	3.136	3.052
After increase	3.375	3.480	3.432	3.072	3.040
Difference	0.061	-0.004	0.092	-0.063	-0.012
restaurants	80	52	156	29	39

Prices rose in South and North New Jersey (6–9 cents) but not in Central New Jersey. They fell slightly in both Pennsylvania regions, more in PA 1 (Philadelphia suburbs).

(h) Adding $\text{Time} \times \text{South NJ}$ and $\text{Time} \times \text{North NJ}$ (Central NJ as the base) gives interaction coefficients 0.065 (0.071) and 0.096 (0.066). The joint Wald test is 2.19 with $p = 0.34$ (χ^2_2). Homogeneous treatment effects are not rejected, although the point estimates suggest a smaller effect in Central NJ.

(i) Adding $\text{Time} \times \text{PA 1}$ gives -0.051 (0.083), $p = 0.54$. Equal control effects are not rejected, supporting the use of Pennsylvania as a single control group.

Exercise 18.7

The sample has 876 blocks: 37 contain a protected institution, 161 are one block away and 678 further away. July is excluded, leaving 7008 block-months.

(a) Excluding the 37 protected blocks, average car thefts per block per month:

	One block away	More than one block	Difference
April–June	0.0921	0.0959	-0.0038
August–December	0.0963	0.1067	-0.0104
Difference	0.0042	0.0108	-0.0067

The diff-in-diff estimate is -0.007 thefts per block-month, less than a tenth of the same-block effect (-0.087).

(b) With block and month fixed effects and standard errors clustered by block:

$$\text{thefts}_{it} = \underbrace{-0.0883}_{(0.0303)} \text{Same}_i \times \text{Post}_t - \underbrace{0.0067}_{(0.0157)} \text{One}_i \times \text{Post}_t + u_i + v_t + \varepsilon_{it}.$$

The difference between the two effects is -0.082 (0.033). Adding a two-blocks-away treatment gives -0.0045 (0.0137), also negligible.

(c) No. The deterrence effect is large and significant on the protected block ($t = -2.9$, a 78% reduction) but statistically and economically negligible one block away ($t = -0.4$), and significantly smaller than the same-block effect. Police presence reduces car theft only in its immediate vicinity. There is also no evidence that theft was displaced to neighbouring blocks, which would show up as

a positive coefficient. This supports the chapter's caution against extrapolating the result to broad crime deterrence.

Exercise 18.8

We do both beverages. The regressions are state fixed-effects models of log per-capita sales on Sunday on- and off-premise sales hours, the unemployment rate, the cross-border outflow variables and year effects. The second specification adds state-specific linear trends. Standard errors are clustered by state (47 states, 1722 state-years, 1970–2007).

	log beer		log wine	
	(18.7)	(18.8) trends	(18.7)	(18.8) trends
OnHours	0.0011 (0.0028)	−0.0013 (0.0022)	0.0059 (0.0037)	0.0032 (0.0028)
OffHours	0.0030 (0.0019)	0.0005 (0.0020)	−0.0031 (0.0040)	0.0014 (0.0029)
UR	−0.0004 (0.0037)	−0.0070 (0.0026)	−0.0048 (0.0061)	−0.0039 (0.0049)
OnOutFlows	0.0044 (0.0049)	−0.0008 (0.0017)	−0.0030 (0.0048)	0.0030 (0.0027)
OffOutFlows	0.0003 (0.0047)	−0.0001 (0.0028)	0.0055 (0.0071)	0.0041 (0.0048)

For comparison, the liquor regressions give OnHours = 0.0109 (0.0030) without and 0.0002 (0.0020) with state trends.

Interpretation.

- Unlike liquor, neither beer nor wine sales respond significantly to Sunday on- or off-premise sales hours, with or without state trends. All hour coefficients are below 0.006 in absolute value (under 0.6% per additional allowed hour) and none is significant at 5%; the largest, wine on-premise, has $t = 1.6$. Extending Sunday sales hours seems to shift the timing of purchases rather than raise consumption. This is consistent with the book's reading of the off-premise liquor results and with rational consumers who plan purchases around known hours.
- The cross-border outflow variables are insignificant, so there is no evidence of cross-state shopping in response to neighbours' hours.
- Year effects are jointly highly significant for both beverages ($p < 0.001$), reflecting strong national trends (e.g. per-capita beer sales rose steadily through the 1970s). Adding state-specific trends mostly reduces the standard errors and makes the unemployment effect on beer significant (−0.7% per point): beer sales are pro-cyclical once state trends are controlled for.
- As with liquor, results that appear without state trends (the liquor OnHours effect) can vanish once trends are allowed. For beer and wine there is no such effect to begin with, so the conclusion that Sunday sales laws do not affect beer or wine consumption is robust to the trend specification.

Chapter 19

Nonparametric Regression

Solutions to all 11 exercises of Chapter 19 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 19.7–19.11 use DDK2011, cps09mar, Invest1993, RR2010 and FRED-QD. We follow the chapter’s methods and Hansen’s R code: Gaussian kernel, Fan–Gijbels rule-of-thumb bandwidth h_{rot} (19.9) from a quartic pilot, and leave-one-out cross-validation over $[h_{\text{rot}}/3, 3h_{\text{rot}}]$, whose minimizer is used unless it falls on the grid boundary. Pointwise 95% bands use (19.18) with leave-one-out prediction errors (clustered versions where noted). Replication checks: the chapter’s CPS example gives $h_{\text{rot}} = 5.14$ and $h_{\text{cv}} = 4.34$ (book: 5.14, 4.32). The DDK girls example gives conventional $h_{\text{cv}} = 12.2$ and clustered $h_{\text{cv}} = 6.2$ (book: 12.3, 6.2). Figures are in **figures/**.

Notation: $B(x)$ is the bias function ($B_{\text{nw}}(x) = \frac{1}{2}m''(x) + f(x)^{-1}f'(x)m'(x)$, $B_{\text{LL}}(x) = \frac{1}{2}m''(x)$), $R_K = \int K^2$, $\bar{B} = \int B(x)^2 f(x)w(x)dx$ and $\bar{\sigma}^2 = \int \sigma^2(x)w(x)dx$, so that $\text{AIMSE} = h^4\bar{B} + R_K\bar{\sigma}^2/(nh)$ (19.5).

Exercise 19.1

Let $Y^* = cY$ with $c = 100$. Then $m^*(x) = cm(x)$ and $e^* = ce$, so $\sigma^{2*}(x) = c^2\sigma^2(x)$. Both bias functions are linear in m and its derivatives, so $B^*(x) = cB(x)$ and $\bar{B}^* = c^2\bar{B}$, while $\bar{\sigma}^{2*} = c^2\bar{\sigma}^2$. Therefore

$$h_0^* = \left(\frac{R_K\bar{\sigma}^{2*}}{4\bar{B}^*}\right)^{1/5} n^{-1/5} = \left(\frac{R_K c^2 \bar{\sigma}^2}{4c^2 \bar{B}}\right)^{1/5} n^{-1/5} = h_0.$$

The bandwidth should not change. This makes sense: h is measured in units of X , and rescaling Y multiplies both the squared bias and the variance by c^2 , so the bias–variance trade-off, and the estimator itself up to the factor c , are unchanged. (Rescaling X by c would instead multiply h_0 by c .)

Exercise 19.2

Let $g(h) = h^4\bar{B} + a/h$ with $a = R_K\bar{\sigma}^2/n > 0$ and $\bar{B} > 0$. Then $g'(h) = 4h^3\bar{B} - a/h^2$, which is negative for small h , positive for large h , and zero only at $h^5 = a/(4\bar{B})$. Also $g''(h) = 12h^2\bar{B} + 2a/h^3 > 0$, so g is strictly convex on $h > 0$ and the stationary point is the unique global minimizer:

$$h_0 = \left(\frac{a}{4\bar{B}}\right)^{1/5} = \left(\frac{R_K\bar{\sigma}^2}{4\bar{B}}\right)^{1/5} n^{-1/5},$$

which is (19.6). Substituting, $g(h_0) = h_0^{-1}(h_0^5\bar{B} + a) = \frac{5}{4}a/h_0 = \frac{5}{4}4^{1/5}(R_K^4\bar{B}\bar{\sigma}^4)^{1/5}n^{-4/5} \approx 1.65(R_K^4\bar{B}\bar{\sigma}^4)^{1/5}n^{-4/5}$. This is the book’s expression. The printed $(R_K^4\bar{B}\bar{\sigma}^8)^{1/5}$ reads correctly with $\sigma^8 = (\bar{\sigma}^2)^4$.

Exercise 19.3

The asymptotic bias is $h^2B_{\text{LL}}(x) = \frac{1}{2}h^2m''(x)$. Where m is convex ($m'' > 0$) the estimator is biased *upward*; where m is concave ($m'' < 0$) it is biased *downward*; near inflection points ($m'' = 0$) the

leading bias vanishes. The bias grows with curvature and with h^2 , and does not depend on the slope.

This is intuitive. The local linear estimator fits a straight line to the data in a window around x , which amounts to replacing m by a local weighted average of a linear approximation. In a convex region the function lies above its tangent, so the fitted line in the window sits above $m(x)$ at the centre. By Jensen's inequality averaging a convex function lifts it. In a concave region the opposite happens. Smoothing therefore flattens peaks (downward bias at local maxima) and fills in troughs (upward bias at local minima), the "smoothing bias" of Figure 19.2.

Exercise 19.4

With $m' = \beta$ and $m'' = 0$,

$$B_{\text{nw}}(x) = \beta \frac{f'(x)}{f(x)} = \beta \frac{d}{dx} \log f(x).$$

For $\beta > 0$: $B(x) > 0$ where the density is increasing ($f'(x) > 0$, e.g. to the left of the mode of a unimodal density) and $B(x) < 0$ where it is decreasing (to the right of the mode). For $\beta < 0$ the signs reverse: negative bias where $f' > 0$ and positive where $f' < 0$.

Intuition. NW is a local *average* of Y . Where $f'(x) > 0$ the window around x contains more observations to the right of x than to the left. If m is increasing these observations have larger Y values, so the average overshoots $m(x)$; if m is decreasing it undershoots. (By contrast the local linear estimator has zero bias here, since it fits the line exactly; the design bias of NW is removed by the local slope term.)

Exercise 19.5

With $m' = m'' = 0$, $B_{\text{nw}}(x) = 0$ for all x , so $\bar{B} = 0$ and $\text{AIMSE}(h) = R_K \bar{\sigma}^2 / (nh)$. This is strictly decreasing in h , so there is no finite minimizer: formally $h_0 = (R_K \bar{\sigma}^2 / (4 \cdot 0))^{1/5} n^{-1/5} = \infty$. With no bias to trade off against variance, the optimal procedure uses all observations with equal weight. As $h \rightarrow \infty$ the NW estimator becomes the global sample mean $\bar{Y}$, the natural estimator of a constant regression function, unbiased with variance σ^2/n (the parametric rate instead of $(nh)^{-1}$). In practice cross-validation tends to select very large bandwidths when the regression is flat.

Exercise 19.6

With a common bandwidth and a product kernel (Section 19.23),

$$\text{AIMSE}(h) = h^4 \bar{B}_d + \frac{a_d}{nh^d}, \quad \bar{B}_d = \int_S \left(\sum_{j=1}^d B_j(x) \right)^2 f(x) w(x) dx, \quad a_d = R_K^d \int_S \sigma^2(x) w(x) dx.$$

Assume $\bar{B}_d > 0$. The derivative $4h^3 \bar{B}_d - d a_d / (nh^{d+1})$ is negative for small h , positive for large h , and vanishes at a single point, so the unique minimizer solves $h^{d+4} = d a_d / (4 \bar{B}_d n)$:

$$h_0 = \left(\frac{d a_d}{4 \bar{B}_d} \right)^{1/(4+d)} n^{-1/(4+d)} = c n^{-1/(4+d)}.$$

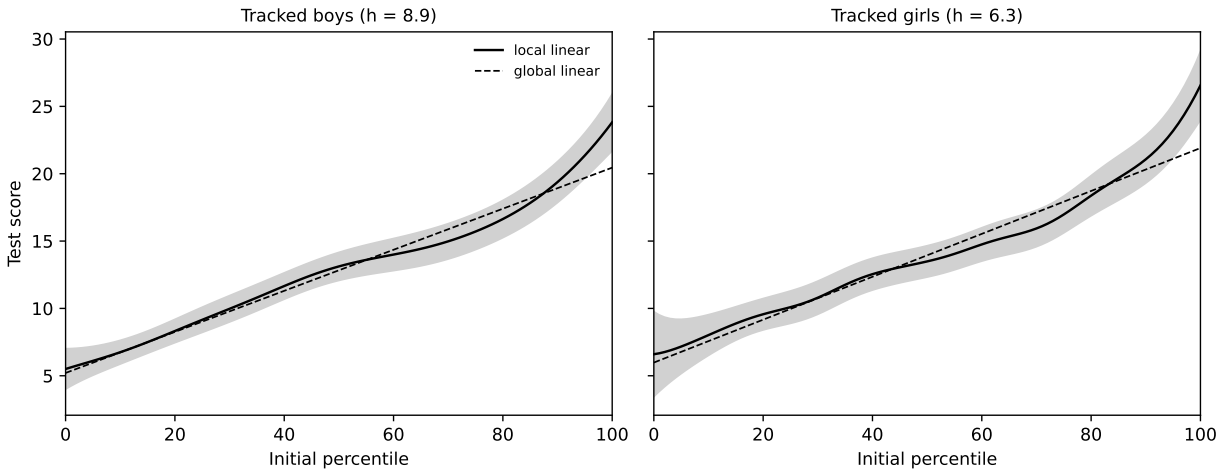
At h_0 , $a_d/(nh_0^d) = 4\bar{B}_d h_0^4/d$, so

$$\text{AIMSE}(h_0) = \left(1 + \frac{4}{d}\right) \bar{B}_d h_0^4 = \left(1 + \frac{4}{d}\right) \bar{B}_d c^4 n^{-4/(4+d)} = O(n^{-4/(4+d)}).$$

For $d = 1$ this is Theorem 19.3. As d grows, the variance term $(nh^d)^{-1}$ forces larger bandwidths and slower convergence: $n^{-4/5}$, $n^{-2/3}$, $n^{-4/7}$, ... for $d = 1, 2, 3$.

Exercise 19.7

Tracked boys: $n = 1473$ in 60 schools. As in Section 19.21 we use the raw total score, delete-school (clustered) cross-validation and clustered standard errors. Bandwidths: $h_{\text{rot}} = 5.4$, conventional $h_{\text{cv}} = 9.4$, clustered $h_{\text{cv}} = 8.9$ (used). For the girls the same procedure gives $h = 6.3$.



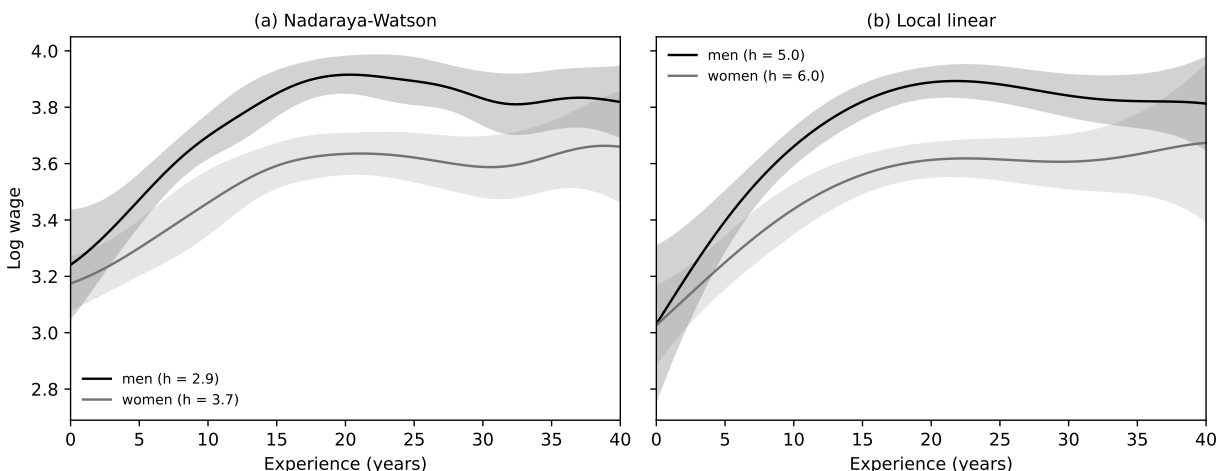
initial percentile	0	10	25	50	75	90	100
boys	5.5 (0.8)	6.7 (0.5)	9.2 (0.5)	13.1 (0.6)	15.7 (0.7)	19.4 (0.8)	23.8 (1.1)
girls	6.6 (1.7)	8.0 (0.8)	10.1 (0.6)	13.5 (0.6)	17.0 (0.7)	21.1 (0.9)	26.6 (1.4)

Similarities. For both groups the relationship is close to the global linear fit (boys $5.2 + 0.152x$, girls $6.0 + 0.159x$) up to about the 80th percentile, and then becomes convex: students starting at the top gain more than a linear control predicts. Clustered standard errors are about 1.5 times the unclustered ones in both samples.

Differences. The girls' curve lies somewhat above the boys' curve, especially at the top (26.6 versus 23.8 at the 100th percentile), and its upper-tail convexity is stronger (4.6 versus 3.4 points above the linear fit at 100). The boys' curve is smoother because clustered CV selects a larger bandwidth. The pointwise bands overlap almost everywhere, so the gender differences are not statistically decisive.

Exercise 19.8

Men $n = 1179$, women $n = 640$. Bandwidths (CV): NW 2.9 (men), 3.7 (women); LL 5.0 (men), 6.0 (women). The ROT bandwidths are 3.2 and 3.8.



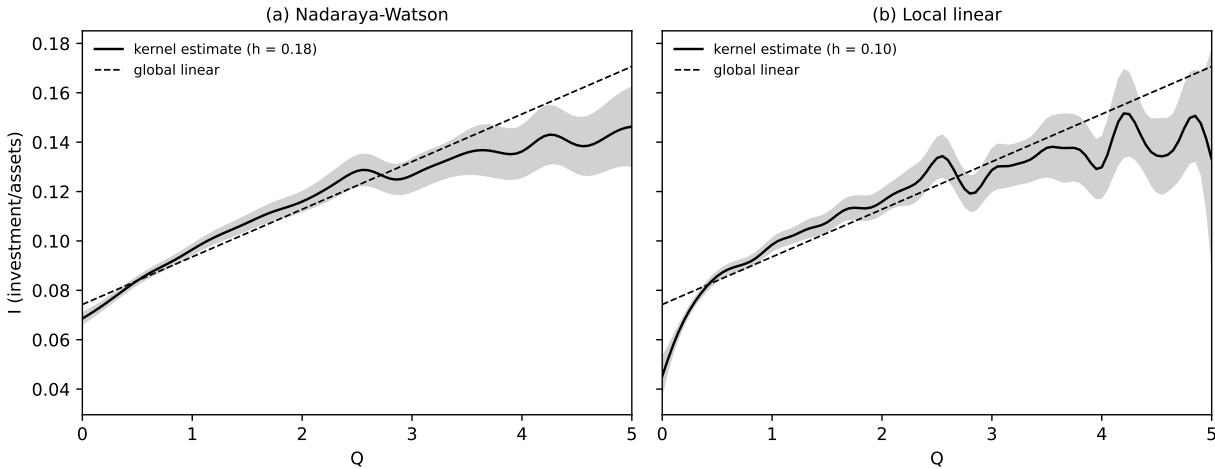
experience	0	5	10	20	30	35	40
men, NW	3.24 (0.10)	3.47 (0.05)	3.70 (0.04)	3.92 (0.03)	3.83 (0.05)	3.83 (0.05)	3.82 (0.07)
men, LL	3.03 (0.14)	3.40 (0.06)	3.66 (0.04)	3.89 (0.03)	3.84 (0.04)	3.82 (0.05)	3.81 (0.09)
women, NW	3.18 (0.05)	3.30 (0.05)	3.46 (0.06)	3.64 (0.04)	3.59 (0.06)	3.63 (0.06)	3.66 (0.10)
women, LL	3.03 (0.07)	3.25 (0.05)	3.44 (0.05)	3.61 (0.03)	3.61 (0.05)	3.63 (0.07)	3.67 (0.14)

(a) For both sexes wages rise steeply for the first 15–20 years (by roughly 70–85 log points for men and 45–60 for women, depending on the estimator) and then flatten. Women earn substantially less than men at every experience level beyond the first few years, with a gap of about 0.15–0.30 log points. For men the NW estimate falls by 0.10 log points between 20 and 40 years, but the pointwise bands at 20 (3.92 ± 0.07) and at 30–40 overlap heavily. For women there is no decline at all. The evidence therefore does not support a fall in expected wages after 20 years for this group: the profile is better described as flat after about 20 years.

(b) The local linear estimates are smoother (CV selects larger bandwidths) and differ from NW mainly at the boundaries. At zero experience LL is 0.15–0.2 lower, because NW is biased upward at a left boundary when the regression is increasing (Theorem 19.5). The LL bands are wider at both boundaries (e.g. 0.14 versus 0.10 at 0 for men, and 0.14 versus 0.10 at 40 for women). The conclusions are unchanged: rapid growth followed by a plateau, with no significant decline.

Exercise 19.9

$n = 26,247$ firm-years (1934 firms); the quartiles of Q are 0.40, 0.73 and 1.36. Because these are panel data, the bands use standard errors clustered by firm, which are 1.2–1.3 times the unclustered ones. Bandwidths: $h_{\text{rot}} = 0.18$. LL cross-validation selects 0.10. For NW the CV criterion keeps falling down to the smallest bandwidth considered ($h_{\text{rot}}/3$), driven by NW's boundary bias at $Q \approx 0$, so we use h_{rot} for NW.



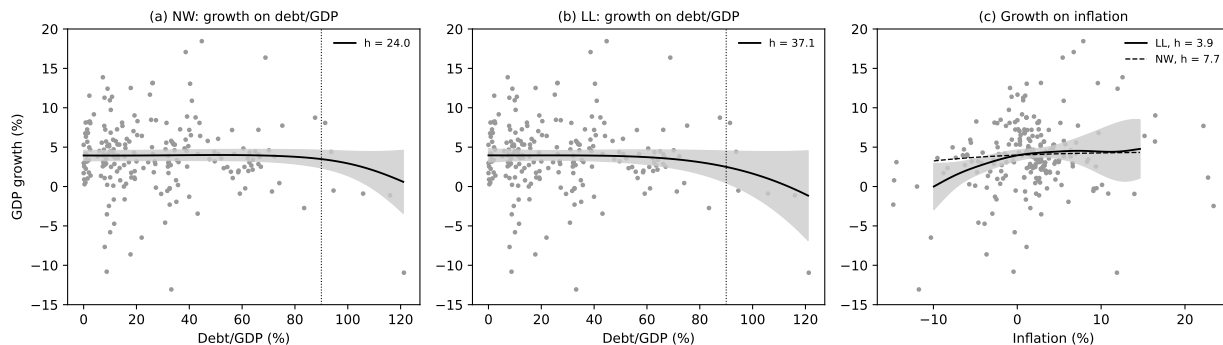
Estimated $\mathbb{E}[I | Q]$ (clustered s.e.):

Q	0	0.25	0.5	1	1.5	2	3	4	5
NW	0.069	0.076	0.084	0.097	0.107	0.116	0.127	0.136	0.146
LL	0.045	0.072	0.086	0.099	0.108	0.116	0.129	0.130	0.133
s.e. (LL)	(0.004)	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.005)	(0.007)	(0.023)

(c) Yes. The global linear fit is $I = 0.074 + 0.019Q$. The local linear estimate is steep for low Q (investment rises by 4 percentage points as Q goes from 0 to 0.5), has a slope close to 0.02 between 0.5 and 3, and flattens above 3. The linear fit lies outside the (tight) bands near $Q = 0$ and for Q above about 3.5. Investment is a concave function of Q , in contrast to the linear specification implied by the Q theory of investment. (The NW estimate misses the steep initial segment because of boundary bias. Its CV preference for ever-smaller bandwidths is a symptom of exactly this.)

Exercise 19.10

$n = 219$. Debt/GDP ranges from 0 to 121% but exceeds 90% in only six years (1944–1949). Bandwidths: $h_{\text{rot}} = 13.4$; CV selects 24.0 for NW and 37.1 for LL.



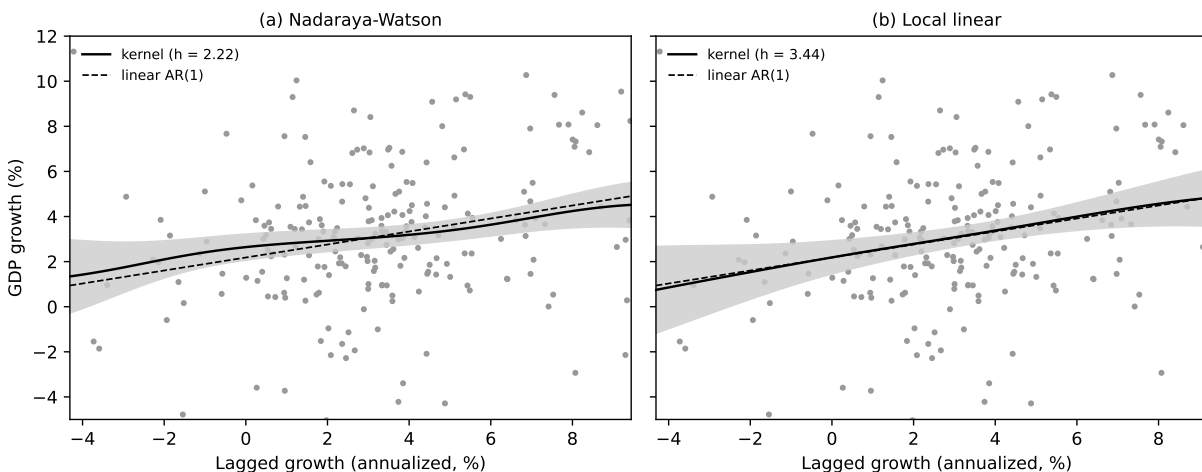
(a)–(b) Both estimates are flat at about 3.9–4.0% growth for debt ratios up to 50–60%. Above that they decline: NW 3.78 (0.46) at 80%, 3.48 (0.57) at 90%, 2.94 (0.79) at 100%; LL 3.09 (0.76) at 80%, 2.49 (1.03) at 90%, 1.64 (1.43) at 100%. The pointwise bands widen rapidly above 80%.

(c) There is weak evidence of a decline in growth at high debt levels, but no evidence of a break at 90%. The estimated decline is gradual (a smooth kernel fit cannot show a threshold, and there is no visible jump in the data), and the difference between growth at 30% and at 90% debt is well within sampling error. Most importantly, the high-debt observations are the six post-war years 1944–1949, when output fell as wartime production was unwound (average growth -0.2% versus 4.0% in all other years). The negative association at high debt reflects this single episode, whose causality runs from war to both debt and the post-war output contraction, not from debt to growth.

(d) Growth on inflation: NW ($h = 7.7$) rises gently from 3.2% at -10% inflation to 4.3% at 15% . LL (the CV minimizer is at the grid boundary, so $h_{\text{rot}} = 3.9$ is used) shows the shape more clearly: 2.5 (0.5) at -5% , 4.0 (0.3) at 0% , 4.2 at 2% , 4.5 at 5% . Deflation years are associated with markedly lower growth. Above about 5% inflation the estimate is flat and imprecise. The deflation–low growth association comes largely from 19th-century and early-20th-century depressions. It is a correlation in a nonstationary historical sample, not a causal effect of inflation.

Exercise 19.11

$n = 234$ quarterly observations (1959Q3–2017Q4), mean annualized growth 3.1% . Evaluation region: the 2.5–97.5% quantiles of Y_{t-1} , $[-4.3, 9.4]$. Bandwidths: $h_{\text{rot}} = 3.44$. CV selects 2.22 for NW, while for LL the CV criterion is minimized at the largest bandwidth considered ($3h_{\text{rot}}$), i.e. it prefers an essentially linear fit, so h_{rot} is used for the plot.



Estimates of $m(x)$ (s.e.): NW 2.10 (0.42) at $x = -2$, 2.64 (0.29) at 0 , 3.04 (0.23) at 3 , 3.64 (0.25) at 6 , 4.24 (0.38) at 8 ; LL 1.54 (0.62), 2.19 (0.37), 3.08 (0.22), 3.99 (0.31), 4.56 (0.50) at the same points. The linear AR(1) is $\hat{Y}_t = 2.18 + 0.288Y_{t-1}$.

(d) No. Both kernel estimates are close to the linear AR(1) line, which lies inside the 95% bands over the whole range. Cross-validation for the local linear estimator prefers the largest bandwidth, i.e. linearity. The NW estimate is slightly flatter in the middle and steeper at the extremes, but the deviations are small relative to the bands. With 234 observations there is no detectable nonlinearity in the conditional mean of U.S. GDP growth. Asymmetries between recessions and expansions, if present, are not captured by a nonlinear function of last quarter's growth alone.

Chapter 20

Series Regression

Solutions to all 18 exercises of Chapter 20 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). The empirical exercises use `cps09mar`, `RR2010`, `DDK2011`, `CHJ2004` and `AL1999`. Following Section 20.18, standard errors are HC3 (computed from leave-one-out prediction errors), or delete-cluster versions for clustered data. Cross-validation is the leave-one-out criterion $CV = n^{-1} \sum \hat{e}_i^2$ (Section 20.17). As a check, the Angrist–Lavy reading-score NPIV estimates of Table 20.1, the Wald statistic 12.7 ($p = 0.013$) and the effect -2.96 (1.21) are reproduced exactly. Figures are in `figures/`.

Exercise 20.1

For $X > 2$ both spline terms are active, so the slope is $2 + 5 - 3 = 4$. (For $X < 1$ it is 2; for $1 < X < 2$ it is 7.)

Exercise 20.2

m_K is continuous and piecewise linear with slopes β_1 on $(-\infty, \tau_1)$, $\beta_1 + \beta_2$ on (τ_1, τ_2) , $\beta_1 + \beta_2 + \beta_3$ on (τ_2, τ_3) and $\beta_1 + \beta_2 + \beta_3 + \beta_4$ on (τ_3, ∞) . A continuous piecewise-linear function is non-decreasing iff every slope is non-negative:

$$\beta_1 \geq 0, \quad \beta_1 + \beta_2 \geq 0, \quad \beta_1 + \beta_2 + \beta_3 \geq 0, \quad \beta_1 + \beta_2 + \beta_3 + \beta_4 \geq 0.$$

(On a bounded support $[a, b] \ni \tau_1, \tau_2, \tau_3$ the same four conditions apply.)

Exercise 20.3

A continuous piecewise-linear function is concave iff its slopes are non-increasing, i.e. each change of slope at a knot is non-positive:

$$\beta_2 \leq 0, \quad \beta_3 \leq 0, \quad \beta_4 \leq 0,$$

with β_0, β_1 unrestricted.

Exercise 20.4

(The book prints $\beta_2 x^3$; for a quadratic spline the term is $\beta_2 x^2$.) The function is continuously differentiable, and its second derivative is piecewise constant:

$$m_K''(x) = \begin{cases} 2\beta_2, & x < \tau_1, \\ 2(\beta_2 + \beta_3), & \tau_1 < x < \tau_2, \\ 2(\beta_2 + \beta_3 + \beta_4), & \tau_2 < x < \tau_3, \\ 2(\beta_2 + \beta_3 + \beta_4 + \beta_5), & x > \tau_3. \end{cases}$$

A C^1 function with piecewise continuous second derivative is concave iff $m'' \leq 0$ wherever it exists:

$$\beta_2 \leq 0, \quad \beta_2 + \beta_3 \leq 0, \quad \beta_2 + \beta_3 + \beta_4 \leq 0, \quad \beta_2 + \beta_3 + \beta_4 + \beta_5 \leq 0.$$

(With the literal cubic term, m'' is linear within each segment. Concavity then requires $m'' \leq 0$ at both endpoints of every segment of the support; on an unbounded support it forces $\beta_2 = 0$.)

Exercise 20.5

Take the linear spline regressor $(X - \tau)\mathbf{1}\{X \geq \tau\}$. If $\tau \geq \max_i X_i$ it equals zero for every observation, so the design matrix has a zero column and its coefficient is not identified. If $\tau \leq \min_i X_i$ it equals $X_i - \tau$ for every observation, an exact linear combination of the intercept and X_i : perfect collinearity again. The same holds for higher-order splines: below the support $(X - \tau)^p$ is a polynomial of degree p already spanned by $1, X, \dots, X^p$. In either case $\mathbf{X}'\mathbf{X}$ is singular and OLS cannot be computed (software drops the variable). The knot adds flexibility only by splitting the sample, so it must lie strictly inside the support, with enough observations on both sides to estimate the change in slope precisely.

Exercise 20.6

- (a) $\hat{m}'_K(x) = \hat{\beta}_1 + 2\hat{\beta}_2x + 3\hat{\beta}_3x^2 + \dots + p\hat{\beta}_px^{p-1}$.
- (b) Yes: $\hat{m}'_K(x) = a_K(x)'\hat{\beta}_K$ with $a_K(x) = (0, 1, 2x, \dots, px^{p-1})'$. The functional $\theta = a(m) = m'(x)$ is linear in m .
- (c) By Theorem 20.8, with $K = p + 1$ and $r_K = m - m_K$,

$$\frac{\sqrt{n}(\hat{m}'_K(x) - m'(x) + r'_K(x))}{V_K^{1/2}} \xrightarrow{d} N(0, 1), \quad V_K = a_K(x)'Q_K^{-1}\Omega_KQ_K^{-1}a_K(x).$$

The bias is the derivative of the approximation error, $r'_K(x)$.

- (d) With $\hat{V}_{\hat{\beta}}$ the HC3 covariance matrix (20.32), $s(\hat{m}'_K(x)) = \sqrt{a_K(x)'\hat{V}_{\hat{\beta}}a_K(x)}$, and a 95% point-wise interval is $\hat{m}'_K(x) \pm 1.96 s(\hat{m}'_K(x))$. As for $\hat{m}_K(x)$, the interval ignores the bias $r'_K(x)$: it is valid under undersmoothing, or as an interval for the pseudo-true derivative. Derivatives are estimated less precisely than levels, especially near the boundary, where $|a_K(x)|$ is large.

Exercise 20.7

$\text{CV}(K) = n^{-1} \sum_i \tilde{e}_i^2$ with $\tilde{e}_i = \hat{e}_i / (1 - h_{ii})$.

- *Rescaling Y to cY* multiplies every residual and prediction error by c and leaves the leverages unchanged. So $\text{CV}(K)$ is multiplied by c^2 for every K and the minimizing K is unchanged.
- *Rescaling X to cX* : for polynomials, $\{1, cX, \dots, (cX)^p\}$ spans the same space as $\{1, X, \dots, X^p\}$. The same is true for splines when the knots are rescaled with X ($(cX - c\tau)^p = c^p(X - \tau)^p$). Fitted values, residuals and leverages are unchanged, so the CV function and its minimizer are unchanged. (If spline knots are kept fixed in the original units while X is rescaled, the model itself changes.)

Scaling does matter numerically: rescaling X to $[0, 1]$ reduces the ill-conditioning of high-order polynomial designs, which is why the exercises below do it.

Exercise 20.8

Write $e_K = e + r_K(X)$ with $r_K(x) = m(x) - X_K(x)'\beta_K$ the approximation error.

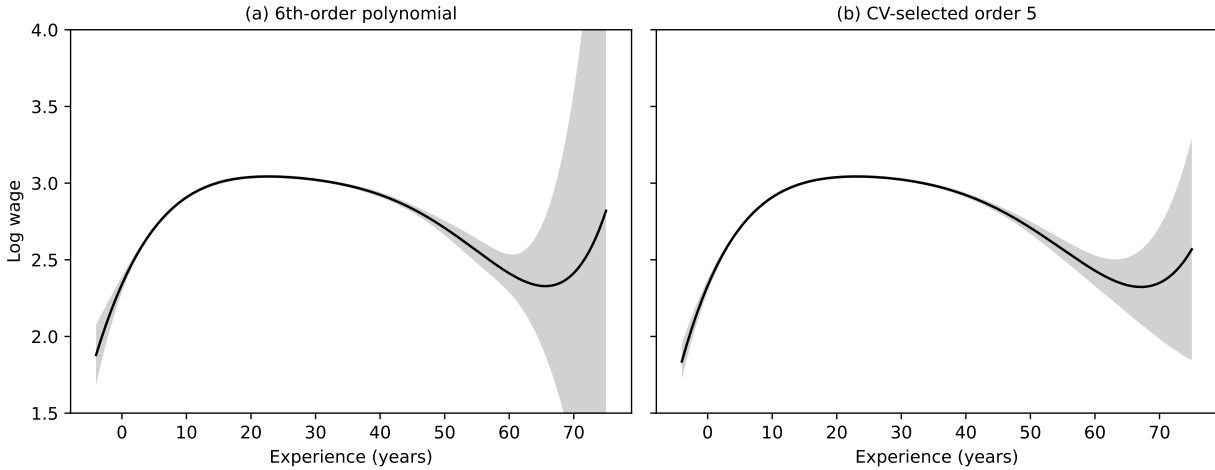
(a) In general no: $\mathbb{E}[e_K | Z] = \mathbb{E}[e | Z] + \mathbb{E}[r_K(X) | Z] = \mathbb{E}[r_K(X) | Z]$, which is non-zero because X depends on Z (relevance) and $r_K \neq 0$ unless m lies exactly in the span of X_K .

(b) Yes, if $\mathbb{E}[Z_K X'_K]$ is non-singular: define $\beta_K = \mathbb{E}[Z_K X'_K]^{-1} \mathbb{E}[Z_K Y]$. Then $\mathbb{E}[Z_K e_K] = \mathbb{E}[Z_K Y] - \mathbb{E}[Z_K X'_K] \beta_K = 0$ by construction. This is the just-identified IV projection coefficient; its approximation error need not be small in any L_2 sense.

(c) In general no. $\mathbb{E}[Z_L e_K] = \mathbb{E}[Z_L m(X)] - \mathbb{E}[Z_L X'_K] \beta_K$ is L equations in $K < L$ unknowns. They have a solution only if $\mathbb{E}[Z_L m(X)]$ lies in the column space of $\mathbb{E}[Z_L X'_K]$, e.g. if $r_K \equiv 0$. With a genuine approximation error the overidentifying restrictions fail. β_K can then only be defined as a pseudo-true GMM value that makes $\mathbb{E}[Z_L e_K]$ as small as possible, and overidentification tests will tend to reject for small K .

Exercise 20.9

$n = 50,742$. Experience (age – education – 6) ranges from -4 to 75 and is rescaled as $(x - \min)/(\max - \min)$.



Selected values of the 6th-order fit (s.e.): 2.34 (0.03) at 0 years, 2.91 (0.006) at 10, 3.04 (0.005) at 20, 3.02 (0.005) at 30, 2.92 (0.008) at 40, 2.71 (0.02) at 50, 2.41 (0.06) at 60, 2.33 (0.20) at 65 and 2.41 (0.61) at 70.

(c) Log wages rise steeply over the first ten years (by about 57 log points), flatten around 20–30 years, and decline after about 35 years, by roughly 60 log points by 60 years of experience. This is a concave profile, similar to but more flexible than the quadratic Mincer specification. Above about 60 years the estimate turns *up* again, but this is not credible. Only 3 of the 50,742 workers have more than 65 years of experience (and 25 more than 60), so the polynomial is determined almost entirely by its behaviour elsewhere. The pointwise band widens explosively (the s.e. at 70 is 0.6, i.e. ± 120 log points). The upturn is a boundary artefact of a global polynomial with a large regressor norm $\zeta_K(x)$ at the edge of the support (Section 20.18). No conclusion should be drawn for experience above about 60.

Exercise 20.10

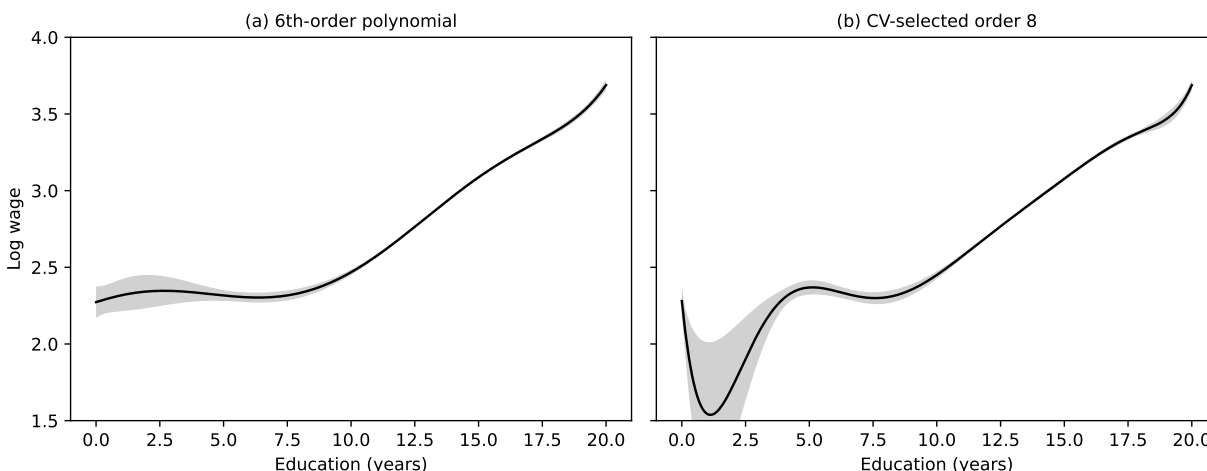
order	1	2	3	4	5	6	7	8
CV	0.45408	0.44027	0.43921	0.43891	0.43878	0.43886	0.43892	0.43878
AIC+41000	940.0	-627.9	-749.3	-785.3	-798.9	-797.4	-796.4	-798.0

(a) Both criteria select order 5 (order 8 is a near tie), a large improvement over the quadratic.

(b) The order-5 estimate is shown in panel (b) of the figure in Exercise 20.9. It is practically identical to the order-6 fit up to 60 years of experience. Its bands are narrower at the upper boundary (s.e. 0.09 at 65 and 0.19 at 70), and the spurious upturn is milder. The interpretation is unchanged: rapid early growth, a peak around 20–25 years (≈ 3.04), and a gradual decline.

Exercise 20.11

Education ranges from 0 to 20 years but takes only a few distinct values (0, 4, 6, 8, 9, ..., 14, 16, 18, 20).

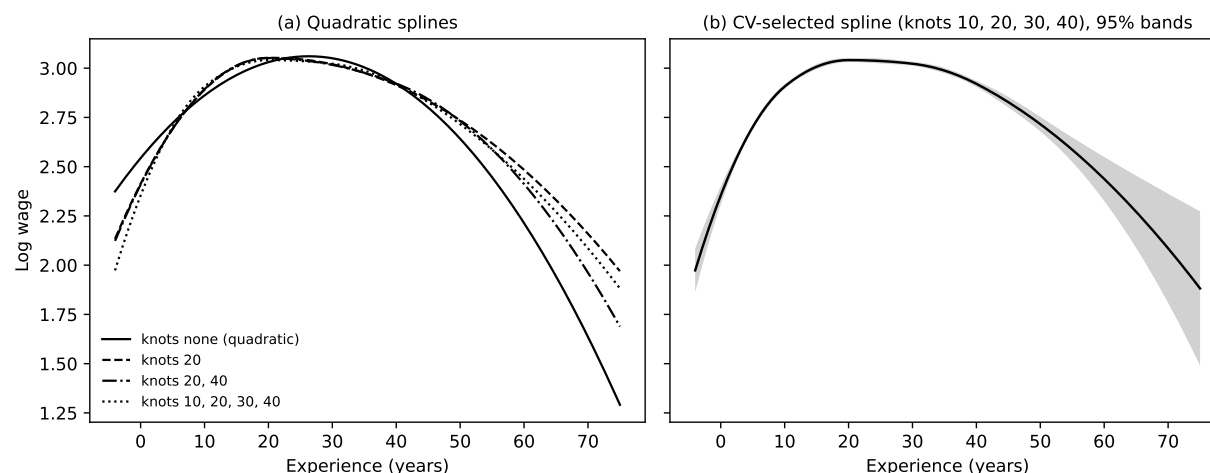


Fitted values (s.e.): 2.27 (0.05) at 0, 2.33 (0.03) at 4, 2.33 (0.02) at 8, 2.47 (0.01) at 10, 2.70 (0.004) at 12, 2.96 (0.004) at 14, 3.20 (0.005) at 16, 3.38 (0.008) at 18 and 3.69 (0.02) at 20. The return to schooling is essentially zero below 8 years and about 10–15% per year above 10 years. This is the convex profile seen in earlier chapters. Precision is lowest at low education, where there are few workers.

Exercise 20.12

CV $\times 10^5$ for orders 1–8: 36868, 36623, 36563, 36563, 36514, 36511, 36512, **36507**; AIC is also minimized at order 8. The criterion keeps falling, slightly, up to the largest order considered, though orders 5–8 are within 0.02% of each other. The order-8 estimate (panel (b) above) is very close to the order-6 fit: flat below 8 years and steep and convex above. With so few distinct values of education, high-order polynomials are close to fitting a separate mean for each education level.

Exercise 20.13



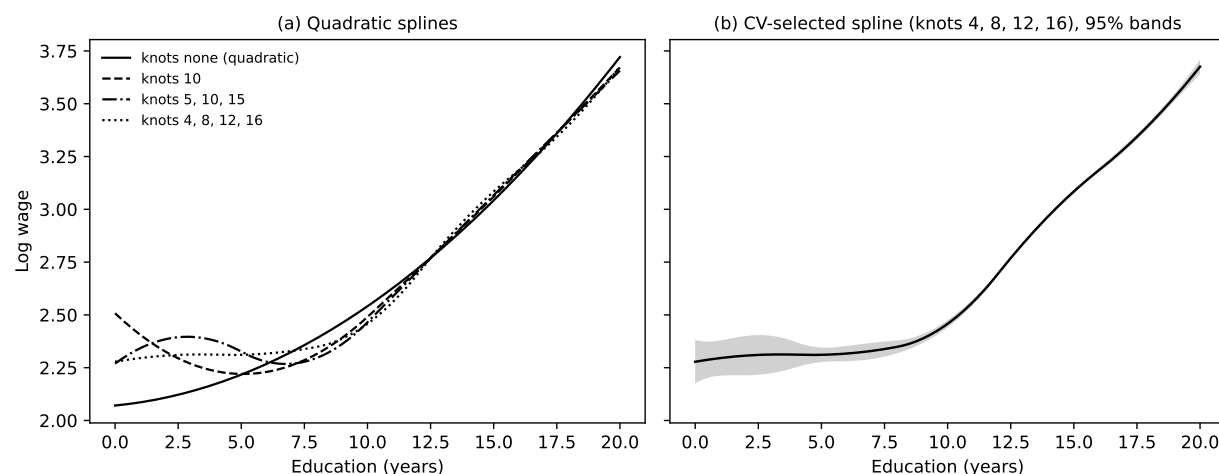
(a) All four agree between about 5 and 45 years: a peak of ≈ 3.04 – 3.05 around 20–25 years. The quadratic is too rigid. It overstates wages at zero experience (2.54 versus 2.35–2.41 for the splines) and forces an accelerating decline (to 1.64 at 70 years). The spline models allow a steep early rise followed by a flat middle section.

(b) CV: 0.44027 (quadratic), 0.43891 (knot 20), 0.43891 (knots 20, 40), **0.43882** (knots 10, 20, 30, 40). AIC (+41000): -627.9 , -784.2 , -784.9 , **-795.2** . Both select the four-knot spline.

(c) The four-knot spline (right panel) rises from 2.35 at 0 to 3.04 at 20 years, is flat to 30 years, and declines to 2.72 at 50 and 2.44 at 60. The bands are tight except at the boundaries.

(d) The order-5 polynomial has a marginally smaller CV (0.43878 versus 0.43882), but the difference is negligible. The spline is preferable for interpretation and extrapolation: it does not produce the implausible upturn beyond 60 years that the polynomials show, because each quadratic piece depends only locally on the data.

Exercise 20.14



(a) The quadratic is smooth and convex but misses the flat segment at low education; the one-knot spline overshoots at 0 years (2.51). The three- and four-knot splines capture a flat profile up to 8 years and a steep, roughly linear-convex rise afterwards.

(b) CV: 0.36623, 0.36559, 0.36535, **0.36516**; AIC (+51000): 30.0, −58.4, −91.6, **−117.6**. The four-knot spline is selected.

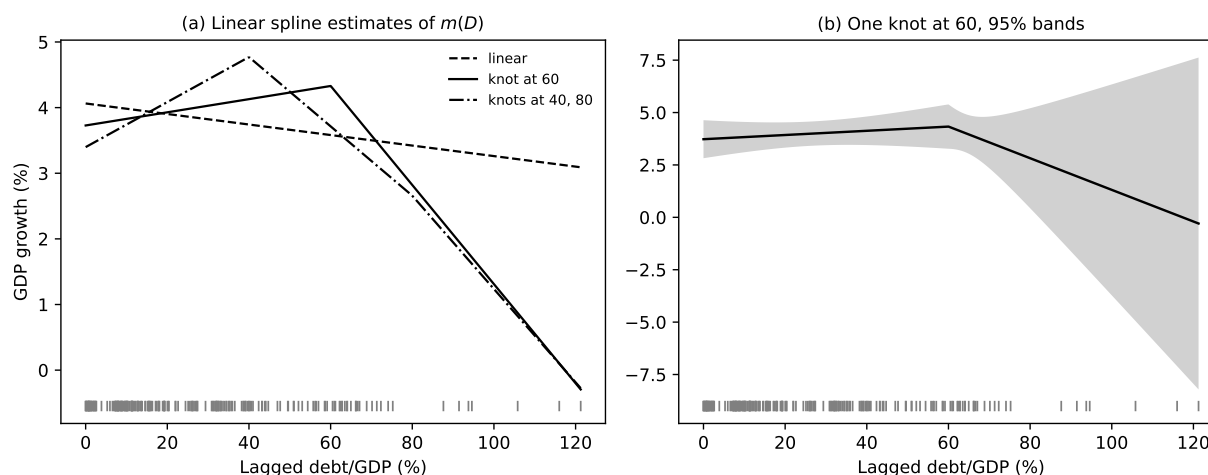
(c) Selected fit (s.e.): 2.31 (0.03) at 4, 2.35 (0.02) at 8, 2.70 (0.004) at 12, 2.97 (0.005) at 14, 3.19 (0.005) at 16, 3.40 (0.007) at 18 and 3.68 (0.02) at 20. Returns are near zero through primary school and about 11–14% per year above 10 years.

(d) The order-8 polynomial has a slightly lower CV (0.36507 versus 0.36516), so by CV the polynomial wins narrowly. The two fits are visually indistinguishable where the data are dense. Given only a dozen distinct education values, either is acceptable; a set of education dummies would be the natural saturated alternative.

Exercise 20.15

$n = 218$ (1792–2009). Estimates, with HC3 standard errors:

	$\hat{\alpha}$	slope of m , segment 1	segment 2	segment 3
(1) linear	0.300 (0.092)	−0.008 (0.014)		
(2) knot at 60	0.284 (0.093)	0.010 (0.014)	−0.076 (0.070)	
(3) knots at 40, 80	0.280 (0.094)	0.034 (0.025)	−0.053	−0.071
CV	18.29 (1), 18.41 (2), 18.60 (3); AIC: 631.1, 630.1, 630.0			



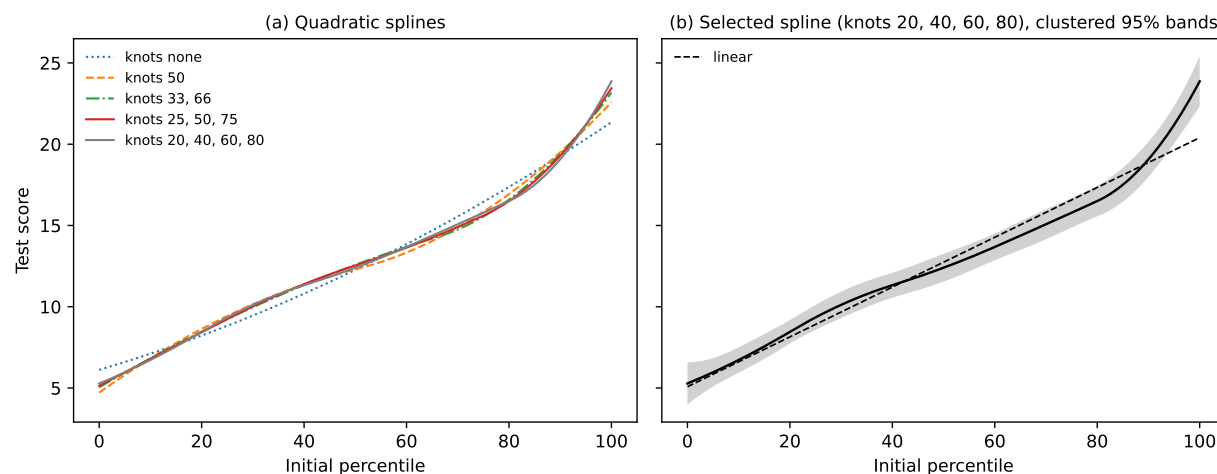
(c) CV prefers the linear model; AIC prefers the spline models by a trivial margin (about 1 point). With no decisive evidence, the simplest model is preferred.

(d) There is no significant evidence that debt affects subsequent growth. In the linear model the slope is −0.008 percentage points of growth per point of debt/GDP ($t = -0.6$). In the one-knot model the slope above 60% is −0.076 (0.070), negative but insignificant, and the band in panel (b) is very wide above 60. The apparent “kink” is driven by only seven observations with debt above 80%: 1944–1950, the post-war demobilization, when output fell as military spending was cut. That

episode is a special historical event rather than evidence of a general threshold at 90%. U.S. data alone cannot support the Reinhart–Rogoff claim.

Exercise 20.16

We use the raw total score (*totalscore*, as in Section 19.21); $n = 5304$ pupils in 111 schools.



(a) All specifications agree: scores rise with the initial percentile, roughly linearly up to about 80 and more steeply above. The pure quadratic differs slightly at the extremes.

(b) Leave-school-out CV: 63.244, 62.954, 62.932, 62.924, **62.923** (conventional CV and AIC give the same ranking up to ties). Any knot improves on the quadratic, and the models with three or four knots are essentially tied. We select the four-knot spline, the CV minimizer, noting that the three-knot model is practically equivalent.

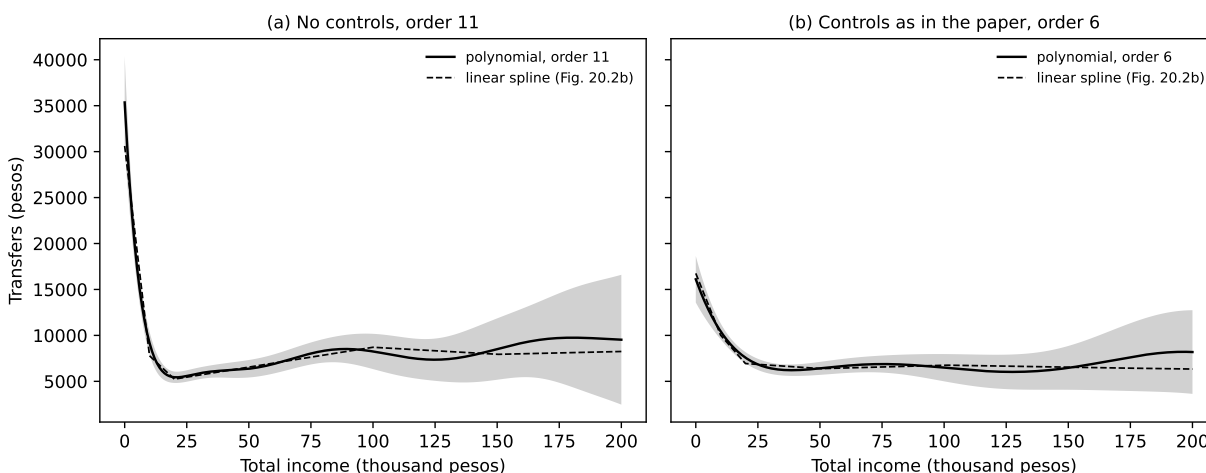
(c) Fitted scores (clustered s.e.): 5.3 (0.7) at percentile 0, 6.7 (0.4) at 10, 9.3 (0.4) at 25, 12.4 (0.4) at 50, 15.8 (0.5) at 75, 19.1 (0.6) at 90 and 23.9 (0.8) at 100. The slope is about 0.13 points per percentile in the middle of the distribution, rising to about 0.2 between the 75th and 90th percentiles and about 0.5 in the top decile. Students who start at the top of the distribution gain more than a linear control would predict, the same conclusion as the kernel analysis of Section 19.21. A linear specification of the percentile control, as in Duflo, Dupas and Kremer, is adequate except in the top quintile.

Exercise 20.17

$n = 8684$ households; income is rescaled to units of 100,000 pesos. Orders 1–12 were compared by CV.

- (a) No controls: CV is minimized at order 11 (orders 10 and 11 are virtually tied; order 12 and higher deteriorate sharply as the design becomes ill-conditioned).
- (b) Paper's 15 controls (education dummies, age, female, married, children, household size, employment status): CV selects order 6 (AIC 7).

- (c) A smaller demographic set (age, female, married, size, both-work, not-employed): CV selects order 7. Its best CV (3.333×10^8) is worse than with the full control set (3.060×10^8 ; no controls: 3.824×10^8), so the paper's controls are preferred by CV.



Estimated transfers (s.e.) with the paper's controls, at the mean of the controls: 16,106 (1290) at zero income, 10,443 (444) at 10,000, 7578 (309) at 20,000, 6428 (310) at 30,000, 6411 (361) at 50,000, 6485 (757) at 100,000 and 6467 (1216) at 150,000 pesos.

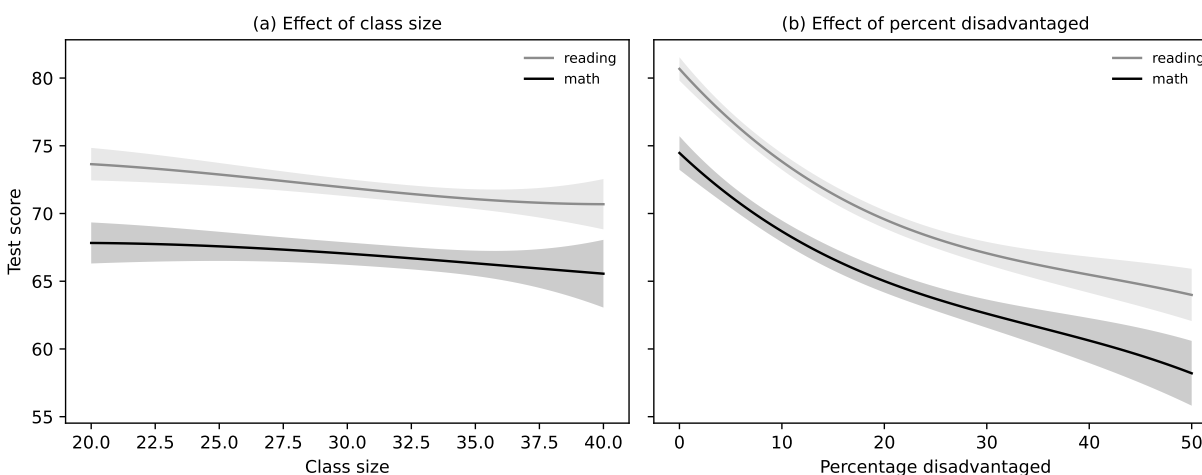
Comparison with Figure 20.2(b). With controls, the polynomial closely tracks the book's linear spline (dashed). Transfers fall steeply at low incomes, with average slopes of -0.57 between 0 and 10,000 pesos and -0.29 between 10,000 and 20,000 (the spline gives -0.7 and -0.3). They are then essentially flat from about 25,000 pesos onward. This is the crowding-out pattern predicted by altruistic transfer models, though with a slope smaller than the theoretical -1 . The polynomial is smoother around the knots and its bands widen at high incomes, where data are sparse. Without controls the initial decline is much steeper (average slope -2.6 below 10,000 pesos): part of what looks like crowding out is explained by household characteristics (education, employment status, household composition) that are correlated with very low income. The estimate above 150,000 pesos is imprecise in all specifications.

Exercise 20.18

Same specification as Table 20.1: cubics in $c = \text{classsize}/40$ and $d = \text{disadvantaged}/14$, the interaction cd , enrollment and a grade-4 dummy. Instruments are $p/40$, $(p/40)^2$, $(p/40)^3$ and $(p/40)d$, where p is the Maimonides prediction $p = \text{enrollment}/(1 + \lfloor (\text{enrollment} - 1)/40 \rfloor)$. Standard errors are clustered by school.

(Two book typos. (20.42) should read $1 + \lfloor (\text{enrollment} - 1)/40 \rfloor$ in the denominator, as in the book's own code. In the class-size effect $\theta = \frac{1}{2}\beta_1 + \frac{3}{4}\beta_2 + \frac{7}{8}\beta_3 + \frac{1}{2}\beta_7\bar{d}$, the last term involves the interaction coefficient β_7 , not β_4 . With $\bar{d} \approx 1$ this reproduces $\hat{\theta} = -2.96$ (1.21) for reading.)

	reading (Table 20.1)	math
c, c^2, c^3	34.2, -61.2, 29.0 (33.5), (53.0), (26.9)	22.7, -28.1, 10.0 (45.0), (71.7), (36.6)
cd	0.81 (1.77)	-2.68 (2.24)
d, d^2, d^3	-12.4, 3.33, -0.377 (1.7), (0.54), (0.078)	-7.95, 2.90, -0.387 (2.17), (0.70), (0.098)
enrollment	0.015 (0.007)	0.030 (0.008)
grade 4	-1.96 (0.16)	1.53 (0.22)
Wald, no class-size effect (χ_4^2)	12.7 ($p = 0.013$)	3.1 ($p = 0.54$)
θ : class size 20 $\rightarrow$ 40	-2.96 (1.21)	-2.27 (1.58)



Comparison. The estimated class-size effect on math is similar in magnitude to that on reading, a decline of about 2.3 points as class size rises from 20 to 40, again nearly linear. But it is estimated less precisely and is not statistically significant: the Wald test of no class-size effect has $p = 0.54$, and $\hat{\theta}$ has $t = -1.4$. The effect of the share of disadvantaged students is again large, nonlinear and precisely estimated: math scores fall from 74.5 at 0% to 65.0 at 20% and 58.2 at 50% disadvantaged, a decline of about 16 points, similar to that for reading. Fourth graders score higher than fifth graders in math but lower in reading; since the tests differ by grade, such level differences are not comparable. Overall, the nonparametric IV evidence for a class-size effect is substantially weaker for math than for reading.

Chapter 21

Regression Discontinuity

Solutions to all 9 exercises of Chapter 21 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 21.5–21.9 use LM2007 (Ludwig and Miller’s Head Start data, 2783 counties, cut-off $c = 59.1984$). Local linear RDD estimates use the normalized Triangular kernel $K(u) = (1 - |u|/\sqrt{6})\mathbf{1}\{|u| < \sqrt{6}\}/\sqrt{6}$, estimated separately on each side of the cut-off. Standard errors follow Section 21.5, using leave-one-out prediction errors. The baseline estimate of Table 21.1, -1.51 (0.71), is reproduced exactly.

Notation: $m(x) = \mathbb{E}[Y \mid X = x]$, $m_d(x) = \mathbb{E}[Y_d \mid X = x]$, $m(c\pm)$ the one-sided limits, and $\theta(x) = m_1(x) - m_0(x)$.

Exercise 21.1

Now $m(x) = m_1(x)\mathbf{1}\{x \leq c\} + m_0(x)\mathbf{1}\{x > c\}$. Under continuity of m_0 and m_1 at c ,

$$\theta = \theta(c) = m_1(c) - m_0(c) = m(c-) - m(c+).$$

Only the sign convention changes: the treated side is now the left side, so $\hat{\theta} = \hat{m}(c-) - \hat{m}(c+)$. Estimation is otherwise identical (separate local linear fits on each side, the same bandwidth choices and standard errors). In regression form (21.4) define $D = \mathbf{1}\{X \leq c\}$ and the interaction $(X - c)D$; then $\hat{\theta}$ is again the coefficient on D . Whether the point $X = c$ is assigned to the treated or untreated side is immaterial for a continuous running variable.

Exercise 21.2

There are two discontinuities. If m_0 and m_1 are continuous at c_1 and at c_2 ,

$$\theta(c_1) = m(c_1+) - m(c_1-), \quad \theta(c_2) = m(c_2-) - m(c_2+).$$

At c_1 units switch from untreated to treated, and at c_2 from treated to untreated. So the conditional ATEs at the two cut-offs are identified, and can be estimated by one-sided local linear regressions at each boundary. The hypothesis $\theta(c_1) = \theta(c_2)$ can be tested. Nothing more is identified without extrapolation: inside (c_1, c_2) only m_1 is observed and outside only m_0 , so $\theta(x)$ for other x requires assumptions about the unobserved potential-outcome functions.

Exercise 21.3

The observed outcome is $Y = Y_0\mathbf{1}\{X < c\} + Y_1\mathbf{1}\{X \geq c\}$. The indicators are functions of X , so conditional on $X = x$ they are constants and can be taken outside the expectation:

$$\begin{aligned} m(x) &= \mathbb{E}[Y \mid X = x] = \mathbf{1}\{x < c\}\mathbb{E}[Y_0 \mid X = x] + \mathbf{1}\{x \geq c\}\mathbb{E}[Y_1 \mid X = x] \\ &= m_0(x)\mathbf{1}\{x < c\} + m_1(x)\mathbf{1}\{x \geq c\}. \end{aligned}$$

Exercise 21.4

The regressors of (21.4), $(1, X, (X - c)D, D)$, span the same space as $((1 - D), (1 - D)(X - c), D, D(X - c))$: a separate intercept and slope on each side of the cut-off, expressed in deviations from c . A regression on this reparameterization decouples, so the fit is the same as two separate least squares regressions of Y on $(1, X - c)$: one on $c - h \leq X < c$ and one on $c \leq X \leq c + h$. In the original parameterization the left intercept at c is $\beta_0 + \beta_1 c$ and the right intercept is $\beta_0 + \beta_1 c + \theta$. Hence

$$\hat{\theta} = \hat{m}(c+) - \hat{m}(c-).$$

An unweighted least squares fit of $(1, X - c)$ on the observations with $|X - c| \leq h$ is exactly the local linear estimator at the boundary point $x = c$ with the (unnormalized) Rectangular kernel $K(u) = \frac{1}{2}\mathbf{1}\{|u| \leq 1\}$ and bandwidth h , computed from one side. So (21.4) is the RDD local linear estimator (21.2) with that kernel. The normalized (unit-variance) Rectangular kernel has support $[-\sqrt{3}, \sqrt{3}]$, so a normalized bandwidth h corresponds to the window $\pm h\sqrt{3}$. One caveat: the conventional (homoskedastic) regression standard errors from (21.4) impose a common error variance on both sides. Heteroskedasticity-robust standard errors coincide with those of the two separate regressions.

Exercise 21.5

Least squares on the window, White (HC0) standard errors as in the book's code:

window	n	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_3$	$\hat{\theta}$
$c \pm 7$	427	-4.13 (10.69)	0.130 (0.196)	0.004 (0.227)	-1.86 (1.06)
$c \pm 8$	482	-3.11 (9.13)	0.111 (0.168)	0.181 (0.235)	-2.20 (1.06)
$c \pm 13.8$	751	-1.01 (3.32)	0.074 (0.066)	0.044 (0.100)	-1.57 (0.74)
$c \pm 20$	1059	-0.01 (2.02)	0.056 (0.041)	0.009 (0.061)	-1.30 (0.61)

Remark. Book discrepancy. The text says (21.5) uses the window 59.1984 ± 13.8 with 482 observations. But the reported estimates, -3.11 (9.13), 0.11 (0.17), 0.18 (0.23), -2.20 (1.06), and the sample size 482 are exactly those of the window ± 8 (row 2). The window ± 13.8 contains 751 counties and gives $\hat{\theta} = -1.57$ (0.74), very close to the Triangular-kernel estimate of Table 21.1. (The book's R code computes $h_0 = h\sqrt{3}$ but then selects the sample with h .)

The estimated effect is negative for every window: grant-writing assistance lowered Head Start-related mortality among 5–9 year olds by 1.3–2.2 deaths per 100,000. Narrower windows give larger effects and larger standard errors: with ± 7 the effect is significant only at 10% ($p = 0.08$), while with ± 13.8 and ± 20 , $p = 0.03$.

Exercise 21.6

h (normalized)	h ($[-1, 1]$ Triangular)	$\hat{\theta}$	s.e.	p-value
4	9.8	-2.10	(1.02)	0.038
8	19.6	-1.51	(0.71)	0.035
12	29.4	-1.25	(0.58)	0.030

The baseline -1.51 (0.71) reproduces Table 21.1. The estimated untreated mortality at the cut-off is $\hat{m}(c-) = 3.31$, so the effect is a reduction of about 45%. Halving the bandwidth raises the estimated effect to -2.1 and inflates the standard error by about 40%. Increasing it to 12 shrinks both. This is the expected bias–variance pattern (Section 21.6): larger bandwidths straighten the fitted curves, and in particular the curvature above the cut-off, attenuating the jump. All three estimates are significant at 5%.

Exercise 21.7

Injuries are not a target of Head Start’s health services, so no discontinuity is expected.

h	4	8	12
$\hat{\theta}$ (s.e.)	0.25 (3.32)	1.62 (2.58)	1.17 (2.31)

(Simple regression, window ± 13.8 : 2.42 (2.71).) Relative to a baseline injury mortality of about 23 per 100,000 at the cut-off, the estimates are small, positive and far from significant at every bandwidth. There is no discontinuity in non-targeted child mortality. This supports the interpretation that the Head Start-related estimate reflects the program rather than a general discontinuity in child health or data quality at the cut-off.

Exercise 21.8

h	4	8	12
$\hat{\theta}$ (s.e.)	2.33 (5.89)	4.48 (4.35)	6.34 (3.84)

(Simple regression, ± 13.8 : 4.57 (4.57).) Adults were not exposed to Head Start, so their mortality from the same causes is a natural placebo. The estimates are positive, not negative, and insignificant ($p = 0.30$ at $h = 8$; $p = 0.10$ at $h = 12$). They are tiny relative to the adult rate of about 132 per 100,000 (3%). There is no evidence of a mortality reduction among adults, as expected if the child effect is due to Head Start.

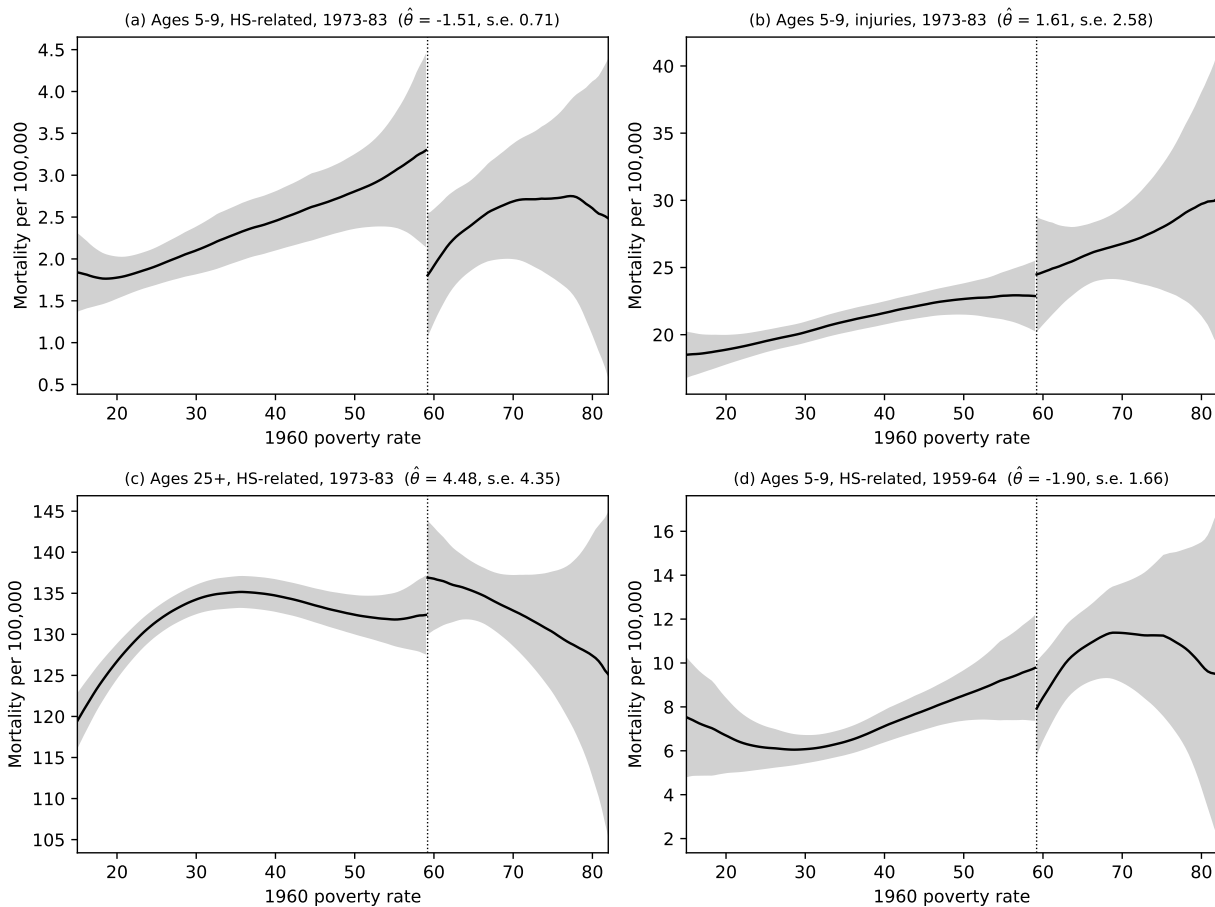
Exercise 21.9

h	4	8	12
$\hat{\theta}$ (s.e.)	-3.74 (1.74)	-1.90 (1.66)	-1.17 (1.52)
p-value	0.032	0.25	0.44

(Simple regression, ± 13.8 : -1.58 (1.83).) Before the program existed there should be no discontinuity. At the baseline bandwidth $h = 8$ the estimate is insignificant, as are those at $h = 12$ and in the simple regression. This is the evidence Ludwig and Miller use to argue that the post-program jump is causal. Two cautions are in order, however. The point estimates are negative and, in

absolute terms, as large as the post-program effect. At the small bandwidth $h = 4$ the pre-period estimate is significant ($p = 0.03$). Relative to the much higher 1959–64 mortality rate (≈ 9.8 per 100,000 at the cut-off versus 3.3 in 1973–83) the pre-period jump is proportionally smaller (about 20% versus 45%). But the placebo test is not sharp enough to rule out a pre-existing discontinuity. The post-program estimate should therefore be viewed as suggestive rather than conclusive. A difference-in-discontinuities comparison (post minus pre) would be a natural next step.

Figure. Local linear RDD fits with 95% pointwise bands ($h = 8$) for the four outcomes of Exercises 21.6–21.9.



Chapter 22

M-Estimators

Solutions to all 4 exercises of Chapter 22 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). All asymptotic covariance matrices follow Theorem 22.4: $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V)$ with $V = Q^{-1}\Omega Q^{-1}$, $Q = \mathbb{E}[\partial^2 \rho / \partial \theta \partial \theta']$ and $\Omega = \mathbb{E}[\psi \psi']$.

Notation: $e = Y - X'\theta_0$. For Exercises 22.2–22.4 we give V in general and also in the leading special case where e is independent of X , which lets the expectations factor.

Exercise 22.1

(a) By independence, $\mathbb{P}[Y \leq y \mid X = x] = \mathbb{P}[e \leq y - x'\theta \mid X = x] = F(y - x'\theta)$, where F is the distribution function of e . Differentiating in y gives the conditional density $f(y - x'\theta)$.

(b) The MLE is the m-estimator with the negative log-density:

$$\rho(Y, X, \theta) = -\log f(Y - X'\theta), \quad \psi(Y, X, \theta) = \frac{\partial \rho}{\partial \theta} = X \frac{f'(Y - X'\theta)}{f(Y - X'\theta)} = -X \phi(Y - X'\theta),$$

where $\phi(u) = -f'(u)/f(u)$ is the location score of f .

(c) At θ_0 , $\psi_i = -X_i \phi(e_i)$, and $\mathbb{E}[\phi(e)] = \int f'(u) du = 0$, so the scores have mean zero. With $\mathcal{I}_f = \mathbb{E}[\phi(e)^2] = \int (f')^2 / f$ (the Fisher information for location) and independence,

$$\Omega = \mathbb{E}[X X' \phi(e)^2] = \mathbb{E}[X X'] \mathcal{I}_f.$$

For the Hessian, $S(\theta) = \mathbb{E}[-\log f(e - X'(\theta - \theta_0))]$, so $Q = \mathbb{E}[X X'] \mathbb{E}[\phi'(e)]$ when f is twice differentiable. Integrating by parts,

$$\mathbb{E}[\phi'(e)] = \int \left(\frac{f'(u)^2}{f(u)} - f''(u) \right) du = \mathcal{I}_f$$

(the boundary terms vanish under the usual regularity conditions). The same limit holds with only one derivative, by differentiating $\mathbb{E}[X \phi(e - X'\delta)]$ under the integral. So $Q = \Omega$, the information matrix equality, and

$$V = Q^{-1} = (\mathbb{E}[X X'])^{-1} \frac{1}{\mathcal{I}_f}.$$

For normal errors $\mathcal{I}_f = 1/\sigma^2$ and $V = \sigma^2 \mathbb{E}[X X']^{-1}$, the OLS variance. For other densities $\mathcal{I}_f > 1/\text{var}[e]$ (Cramér–Rao), so the MLE improves on OLS. For example, Laplace errors with scale b have $\mathcal{I}_f = 1/b^2 = 2/\text{var}[e]$, halving the variance.

Exercise 22.2

(a) $\rho(Y, X, \theta) = g(Y - X'\theta)$ and, for differentiable g , $\psi(Y, X, \theta) = -X g'(Y - X'\theta)$.

(b) The estimand θ_0 is defined by the population first-order condition $\mathbb{E}[X g'(Y - X'\theta_0)] = 0$. This equals the regression coefficient only if $\mathbb{E}[g'(e) \mid X] = 0$, e.g. when e is symmetric about zero given X and g is even. With $e = Y - X'\theta_0$ and g twice differentiable,

$$Q = \mathbb{E}[X X' g''(e)], \quad \Omega = \mathbb{E}[X X' g'(e)^2], \quad V = Q^{-1} \Omega Q^{-1}.$$

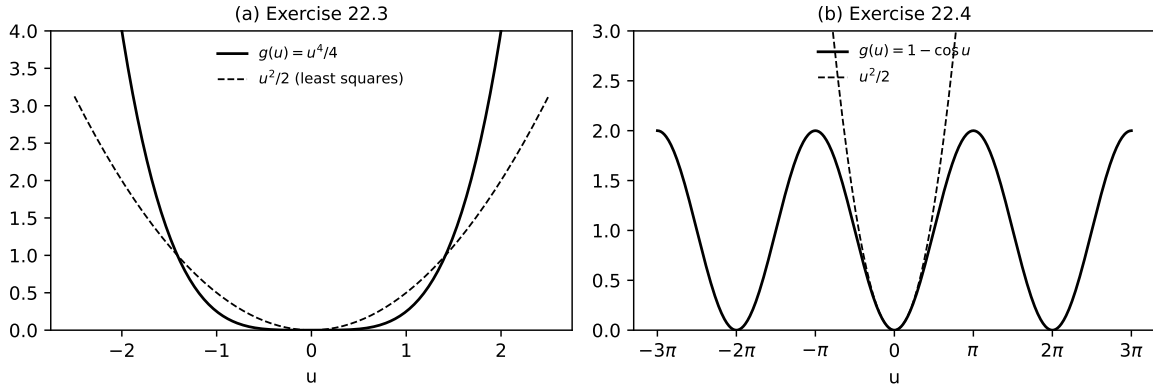
If e is independent of X ,

$$V = (\mathbb{E}[XX'])^{-1} \frac{\mathbb{E}[g'(e)^2]}{(\mathbb{E}[g''(e)])^2}.$$

(If g is not twice differentiable, as for LAD, Q is computed as $\partial^2 S(\theta)/\partial\theta\partial\theta'$ at θ_0 ; see Section 22.7.) Least squares, $g(u) = u^2/2$, gives $\mathbb{E}[g''^2]/(\mathbb{E}g'')^2 = \sigma^2$, as it should.

Exercise 22.3

(a) g is a flat-bottomed, strictly convex, even function (left panel below). It is a polynomial, so it is continuous and infinitely differentiable: $g'(u) = u^3$ and $g''(u) = 3u^2$. Compared with least squares it penalizes small residuals less and large residuals much more.



(b) $\rho(Y, X, \theta) = \frac{1}{4}(Y - X'\theta)^4$ and $\psi(Y, X, \theta) = -X(Y - X'\theta)^3$.

(c) With $e = Y - X'\theta_0$, where θ_0 solves $\mathbb{E}[Xe^3] = 0$,

$$Q = 3\mathbb{E}[XX'e^2], \quad \Omega = \mathbb{E}[XX'e^6], \quad V = \frac{1}{9}(\mathbb{E}[XX'e^2])^{-1}\mathbb{E}[XX'e^6](\mathbb{E}[XX'e^2])^{-1},$$

which requires $\mathbb{E}\|X\|^2 e^6 < \infty$. Under independence, $V = \mathbb{E}[XX']^{-1} \mathbb{E}[e^6]/(9\sigma^4)$ with $\sigma^2 = \mathbb{E}[e^2]$. For normal errors $\mathbb{E}[e^6] = 15\sigma^6$, so $V = \frac{5}{3}\sigma^2\mathbb{E}[XX']^{-1}$, 67% larger than OLS. For light-tailed (platykurtic) errors it can beat OLS. For uniform errors on $[-a, a]$, $\mathbb{E}e^6/(9\sigma^4) = (a^6/7)/(9a^4/9) = a^2/7$ versus $\sigma^2 = a^2/3$, so the fourth-power estimator is more than twice as efficient as OLS. The criterion is strictly convex, so the minimizer is unique and easy to compute.

Exercise 22.4

(a) g is even, bounded between 0 and 2, and periodic with period 2π . It behaves like $u^2/2$ near zero but levels off, so it is not convex (right panel above). It is continuous and infinitely differentiable: $g'(u) = \sin u$, $g''(u) = \cos u$.

(b) $\rho(Y, X, \theta) = 1 - \cos(Y - X'\theta)$ and $\psi(Y, X, \theta) = -X \sin(Y - X'\theta)$.

(c) With θ_0 solving $\mathbb{E}[X \sin(Y - X'\theta_0)] = 0$,

$$Q = \mathbb{E}[XX' \cos(e)], \quad \Omega = \mathbb{E}[XX' \sin^2(e)], \quad V = Q^{-1}\Omega Q^{-1},$$

and under independence $V = \mathbb{E}[XX']^{-1} \mathbb{E}[\sin^2 e]/(\mathbb{E}[\cos e])^2$. Theorem 22.4 requires $Q > 0$, i.e. $\mathbb{E}[\cos e] > 0$: the errors must be concentrated well within $(-\pi/2, \pi/2)$ relative to the period. For

$e \sim N(0, \sigma^2)$, $\mathbb{E} \cos e = e^{-\sigma^2/2}$ and $\mathbb{E} \sin^2 e = (1 - e^{-2\sigma^2})/2$, so

$$V = \mathbb{E}[X X']^{-1} \sinh(\sigma^2) \approx \sigma^2 \mathbb{E}[X X']^{-1} \quad \text{for small } \sigma^2,$$

nearly efficient for small errors but rapidly worse as σ grows.

Remarks. Because ψ is bounded ($|\sin| \leq 1$) and redescending, the estimator is robust to outliers, like Andrews' sine estimator. But periodicity creates identification problems. If X contains an intercept, the criterion is unchanged when the intercept is shifted by 2π , so θ is identified only up to that shift. More generally $S(\theta)$ has multiple local minima, so a restricted parameter space and good starting values (e.g. OLS) are needed. The data should be scaled so that the residuals are small relative to π .

Chapter 23

Nonlinear Least Squares

Solutions to all 10 exercises of Chapter 23 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 23.8–23.10 use PSS2017, RR2010 and Nerlove1963. As a check, the CES and regression-kink estimates of Table 23.1 are reproduced to the reported precision. Standard errors use (23.6) with the NLLS gradient $\hat{m}_{\theta_i}$ as the effective regressor: heteroskedasticity-robust, or clustered by country for the CES model.

Exercise 23.1

(a) The CEF is the constant $\exp(\theta)$, which is nonlinear in θ , so formally this is a nonlinear regression model. But it is nonlinear only because of the parameterization: with $\mu = \exp(\theta)$ it becomes the linear intercept-only model $Y = \mu + e$.

(b) Yes. Estimate μ by least squares, $\hat{\mu} = \bar{Y}$, and set $\hat{\theta} = \log \bar{Y}$ (defined when $\bar{Y} > 0$).

(c) It is the same. NLLS minimizes $\sum (Y_i - e^\theta)^2$, and the first-order condition $-2e^\theta \sum (Y_i - e^\theta) = 0$ gives $e^\theta = \bar{Y}$. Least squares is invariant to one-to-one reparameterization. If $\bar{Y} \leq 0$ the NLLS criterion has no minimizer (it decreases as $\theta \rightarrow -\infty$), matching the fact that $\log \bar{Y}$ is undefined.

Exercise 23.2

No, not in the sense of this chapter, although it looks like one. A nonlinear regression model has the form $Y = m(X, \theta) + e$ with $\mathbb{E}[e | X] = 0$ for a fixed dependent variable. Here the dependent variable itself depends on the parameter λ , and the model restricts $\mathbb{E}[Y^{(\lambda)} | X]$, not $\mathbb{E}[Y | X]$. The implied CEF of Y ,

$$\mathbb{E}[Y | X] = \mathbb{E}[(1 + \lambda(\beta_0 + \beta_1 X + e))^{1/\lambda} | X],$$

depends on the whole conditional distribution of e , not just on $(\lambda, \beta_0, \beta_1)$. Consequently NLLS on $\sum (Y_i^{(\lambda)} - \beta_0 - \beta_1 X_i)^2$ is not valid. The criterion is not comparable across λ , because the scale of $Y^{(\lambda)}$ changes with λ , and minimizing it over λ can simply shrink the dependent variable. The model is a transformation model. It is estimated by maximum likelihood (e.g. under normal e), whose log-likelihood includes the Jacobian term $(\lambda - 1) \sum \log Y_i$. Equivalently, one can use least squares on $Y^{(\lambda)}$ divided by $\bar{Y}^{\lambda-1}$ with $\bar{Y}$ the geometric mean.

Exercise 23.3

(a) No. For any $k \neq 0$, $(k\beta_1, k\beta_2, k\beta_3)$ gives the same regression function, so the parameters are identified only up to scale.

(b) The ratios are identified (provided $\beta_1 \neq 0$, $\beta_3 \neq 0$, and X takes at least two values). Normalize $\beta_1 = 1$ and write

$$Y = \frac{1}{\gamma_1 + \gamma_2 X} + e, \quad \gamma_1 = \beta_2/\beta_1, \quad \gamma_2 = \beta_3/\beta_1,$$

and estimate (γ_1, γ_2) by NLLS with robust standard errors. Equivalent normalizations are possible, e.g. $\beta_2 = 1$ giving $Y = \delta_1/(1 + \delta_2 X)$ with $\delta_1 = \beta_1/\beta_2$, $\delta_2 = \beta_3/\beta_2$. If $\beta_1 = 0$ the regression function is identically zero and neither ratio is identified.

Exercise 23.4

(a) Yes, provided $\beta_1 \neq 0$ and X is non-degenerate. If $m(x) = \beta_1 e^{\beta_2 x} = b_1 e^{b_2 x}$ at two values $x_1 \neq x_2$, dividing gives $e^{\beta_2(x_1 - x_2)} = e^{b_2(x_1 - x_2)}$, so $b_2 = \beta_2$ and then $b_1 = \beta_1$. If $\beta_1 = 0$, β_2 is not identified; if X is constant, only $\beta_1 e^{\beta_2 X}$ is.

(b) The linearized regressor is $m_\theta(X, \theta) = (e^{\beta_2 X}, \beta_1 X e^{\beta_2 X})'$. With $\hat{m}_{\theta i} = m_\theta(X_i, \hat{\theta})$ and NLLS residuals $\hat{e}_i$,

$$\hat{Q} = \frac{1}{n} \sum_{i=1}^n \hat{m}_{\theta i} \hat{m}_{\theta i}', \quad \hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \hat{m}_{\theta i} \hat{m}_{\theta i}' \hat{e}_i^2, \quad \hat{V}_{\hat{\theta}} = \frac{1}{n} \hat{Q}^{-1} \hat{\Omega} \hat{Q}^{-1},$$

with standard errors the square roots of the diagonal of $\hat{V}_{\hat{\theta}}$ (or $n^{-1} \hat{Q}^{-1} \hat{\sigma}^2$ under homoskedasticity).

Exercise 23.5

The conditional log-likelihood is

$$\ell_n(\theta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - m(X_i, \theta))^2.$$

For any σ^2 , maximizing over θ is equivalent to minimizing the sum of squared errors, so $\hat{\theta}_{\text{mle}} = \hat{\theta}_{\text{nlls}}$. Concentrating, $\partial \ell_n / \partial \sigma^2 = 0$ gives $\hat{\sigma}_{\text{mle}}^2 = n^{-1} \sum_i (Y_i - m(X_i, \hat{\theta}))^2$, the average squared NLLS residual.

Exercise 23.6

(a) The order condition is $\ell \geq k$. The local rank condition is that $G = \mathbb{E}[\partial g / \partial \theta'] = -\mathbb{E}[Z X' \exp(X' \theta_0)]$ has full column rank k .

(b) The moment function is $g_i(\theta) = Z_i(Y_i - \exp(X_i' \theta))$ with $\bar{g}_n(\theta) = n^{-1} \sum_i g_i(\theta)$. The GMM estimator is

$$\hat{\theta} = \arg \min_{\theta} \bar{g}_n(\theta)' \mathbf{W} \bar{g}_n(\theta),$$

computed numerically. A natural first step uses $\mathbf{W} = (n^{-1} \sum Z_i Z_i')^{-1}$ (nonlinear 2SLS). With first-step residuals $\tilde{e}_i$, the efficient two-step estimator uses $\mathbf{W} = \hat{\Omega}^{-1}$, $\hat{\Omega} = n^{-1} \sum_i Z_i Z_i' \tilde{e}_i^2$. If $\ell = k$ it simply solves $\bar{g}_n(\theta) = 0$.

(c) Let $\hat{G} = -n^{-1} \sum_i Z_i X_i' \exp(X_i' \hat{\theta})$ and $\hat{\Omega} = n^{-1} \sum_i Z_i Z_i' \hat{e}_i^2$ with $\hat{e}_i = Y_i - \exp(X_i' \hat{\theta})$. For efficient GMM $\hat{V} = (\hat{G}' \hat{\Omega}^{-1} \hat{G})^{-1}$; for a general weight matrix $\hat{V} = (\hat{G}' \mathbf{W} \hat{G})^{-1} \hat{G}' \mathbf{W} \hat{\Omega} \mathbf{W} \hat{G} (\hat{G}' \mathbf{W} \hat{G})^{-1}$. Standard errors are the square roots of the diagonal of $n^{-1} \hat{V}$.

Exercise 23.7

By the delta method $m(x, \hat{\theta})$ is asymptotically normal, with variance equal to the quadratic form of the gradient $m_{\theta}(x, \theta_0)$ in $\text{var}[\hat{\theta}]$. Hence

$$m(x, \hat{\theta}) \pm 1.96 \sqrt{m_{\theta}(x, \hat{\theta})' \hat{V} m_{\theta}(x, \hat{\theta})}, \quad m_{\theta}(x, \theta) = \frac{\partial}{\partial \theta} m(x, \theta),$$

with the derivative computed analytically or numerically. For strongly nonlinear m the interval may be improved by forming it on a transformed scale where m is closer to linear and mapping back (e.g. on $\log m$ when $m > 0$), or by a bootstrap.

Exercise 23.8

The alternative inputs are available for 338 country-years (26 countries). They are measured in currency units (around 10^{10}), so we express them in billions. This only shifts β by $\nu \log 10^9$ and helps the numerical optimization. NLLS uses a multi-start search. Standard errors are clustered by country, with $\sigma = 1/(1 - \rho)$ by the delta method.

	capacities (Table 23.1)	capital stocks
ρ	0.36 (0.29)	0.41 (0.25)
ν	1.05 (0.03)	1.04 (0.03)
α	0.39 (0.06)	0.32 (0.04)
β	1.66 (0.30)	9.02 (0.08) (inputs in billions)
$\sigma = 1/(1 - \rho)$	1.57 (0.71)	1.70 (0.73)
n	390	338

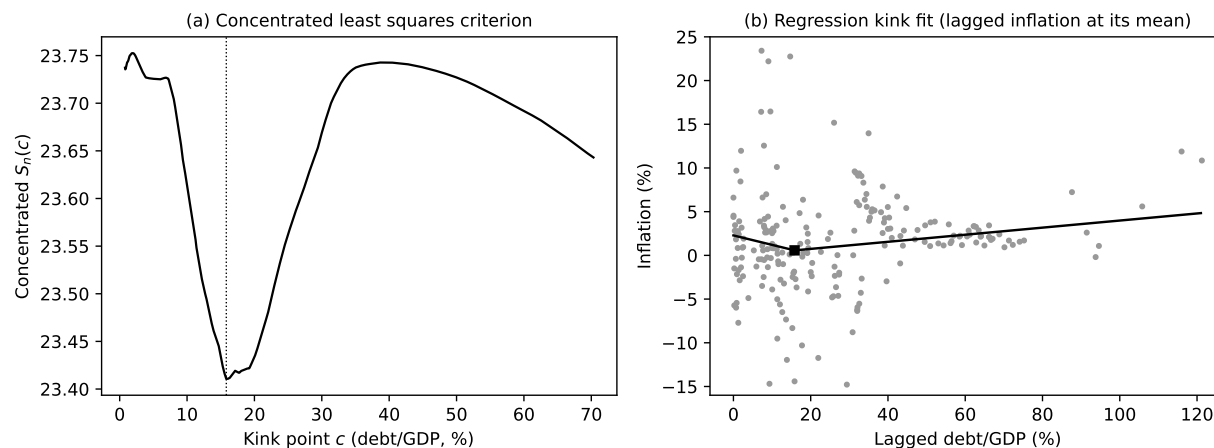
The capacity column reproduces Table 23.1, except the standard error of σ . The delta method gives $s(\hat{\sigma}) = s(\hat{\rho})/(1 - \hat{\rho})^2 = 0.288/0.405 = 0.71$. The book's 0.46 equals $s(\hat{\rho})/(1 - \hat{\rho})$, which omits a factor $(1 - \hat{\rho})^{-1}$.

Comparison. The results are robust. The substitution parameter is again positive ($\hat{\rho} = 0.41$, $\hat{\sigma} = 1.7$): clean and dirty generation are substitutes, with an elasticity of substitution above one. But the estimate is imprecise ($t = 1.6$ for $\rho = 0$), so Cobb–Douglas cannot be rejected. The scale elasticity is essentially unchanged at 1.04, mildly increasing returns. The main difference is the share parameter. With capital stocks, clean capital gets a smaller weight (0.32 versus 0.39), more precisely estimated. Dirty technology remains the dominant input.

Exercise 23.9

Model: $\pi_t = \beta_1(D_{t-1} - c)_- + \beta_2(D_{t-1} - c)_+ + \beta_3\pi_{t-1} + \beta_4 + e_t$, $n = 218$. We estimate it by concentrated least squares over c (a fine grid between the 5% and 95% quantiles of debt, then local refinement), with robust standard errors. As a check, the same code with $Y = \text{GDP growth}$ reproduces Table 23.1: 0.033 (0.025), -0.067 (0.047), 0.28 (0.09), 3.78 (0.68), $c = 43.9$ (11.9).

$$\hat{\pi}_t = \underset{(0.116)}{-0.108}(D_{t-1} - 15.8)_- + \underset{(9.5)}{0.041}(D_{t-1} - 15.8)_+ + \underset{(0.018)}{0.458}\pi_{t-1} - \underset{(0.095)}{0.16}.$$



Interpretation. Below a debt ratio of about 16% the slope is negative but insignificant. Above it, inflation rises with debt: each 10 points of debt/GDP are associated with 0.4 points higher inflation in the next year, significant at 5%, plus the persistence captured by $\hat{\beta}_3 = 0.46$. But the kink is poorly determined. The concentrated criterion (panel (a)) is very flat, and every c between about 9% and 70% gives a sum of squared errors within 1% of the minimum. The kink model reduces the sum of squared errors by only 1.4% relative to a linear AR(1) with debt (slope 0.019). Moreover, the standard error of $\hat{c}$ is not reliable because the regression kink estimator has a non-standard distribution when the kink is weakly identified. The evidence is consistent with a mild positive association between high debt and subsequent inflation, most visible in war and post-war episodes, rather than a well-defined threshold.

Exercise 23.10

(The text's "for values much above β_7 " should read "much above γ ".)

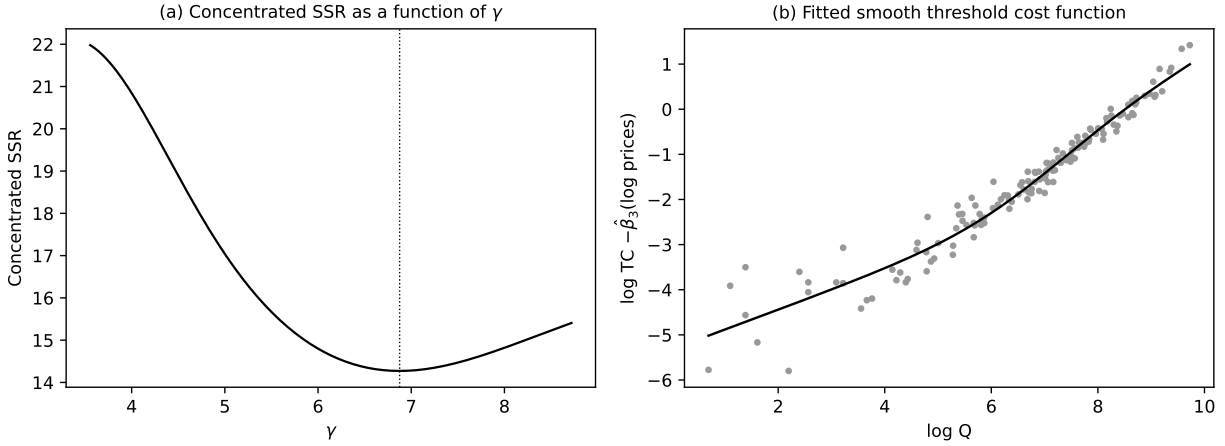
(a) $\log Q$ ranges from 0.69 to 9.72 ($n = 145$). Its 10th–15th smallest values are 3.09–3.76 and its 10th–15th largest 8.67–8.97. We restrict $\gamma \in [3.56, 8.72]$, the 13th smallest to the 13th largest observation, so that at least 13 firms lie on each side.

(b) A multi-start local optimizer over all five parameters with γ constrained to this range (25×5 starting values) converges to a single minimum:

$$\hat{\beta}_1 = -5.321, \quad \hat{\beta}_2 = 0.438, \quad \hat{\beta}_3 = 0.371, \quad \hat{\beta}_4 = 0.224, \quad \hat{\gamma} = 6.875, \quad \text{SSR} = 14.272,$$

against 22.114 for the linear model. If γ is left unrestricted the global minimum ($\text{SSR} = 12.94$) is at $\gamma = -0.78$, outside the range of the data. There the transition function is nearly constant except for the smallest firms, and the fit relies on huge offsetting coefficients ($\hat{\beta}_2 = 16.3$, $\hat{\beta}_4 = -15.4$). This degenerate solution is exactly what the restriction in (a) is designed to exclude.

(c) Concentrating out the linear coefficients by OLS for each γ on a grid of 4001 points gives the same estimates ($\hat{\gamma} = 6.875$, $\text{SSR} = 14.272$). The concentrated criterion has a single local minimum in the admissible range (panel (a)), so the two methods agree. Concentration is faster and more reliable, since it only requires a one-dimensional search.



(d) Heteroskedasticity-robust (homoskedastic) standard errors from (23.6):

$$\begin{aligned}\hat{\beta}_1 &= -5.32 \text{ (0.49)}, & \hat{\beta}_2 &= 0.438 \text{ (0.100)}, & \hat{\beta}_3 &= 0.371 \text{ (0.044)}, \\ \hat{\beta}_4 &= 0.224 \text{ (0.055)}, & \hat{\gamma} &= 6.88 \text{ (0.36)},\end{aligned}$$

with homoskedastic values 0.60, 0.048, 0.062, 0.029 and 0.32. The cost elasticity with respect to output is 0.44 for small firms and $0.44 + 0.22 = 0.66$ for large firms. The transition is centred at $\log Q = 6.9$ (about 970 million kWh), with 68 firms below and 77 above. Economies of scale are strong for small utilities (returns to scale $1/0.44 \approx 2.3$) and weaker, though still present, for large ones ($1/0.66 \approx 1.5$). This is the pattern Nerlove found by splitting the sample by size, here captured by a smooth parametric transition. The standard error of $\hat{\gamma}$ relies on smoothness of the logistic transition; it would not be valid for a sharp threshold.

Chapter 24

Quantile Regression

Solutions to all 16 exercises of Chapter 24 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Quantile regressions are computed exactly by linear programming (the Koenker–Bassett dual). The code reproduces the tracking coefficients of Table 24.2 (0.069, 0.136, 0.125, 0.185, 0.151). Standard errors are nonparametric bootstrap standard errors (500 replications; 1000 clustered by school for DDK), with 95% bias-corrected (BC) percentile intervals as recommended in Section 24.8.

Notation: $\rho_\tau(u) = u(\tau - \mathbf{1}\{\{\} u < 0\})$ is the check function, $\psi_\tau(u) = \tau - \mathbf{1}\{\{\} u < 0\}$, $F(\cdot | x)$ is the conditional distribution function of Y , and $q_\tau(x)$ is its τ th quantile.

Exercise 24.1

Continuity implies $F(m(x) | x) = \frac{1}{2}$ and $\mathbb{P}[Y = m(x) | X = x] = 0$. Hence

$$\begin{aligned} \mathbb{E}[\text{sgn}(e) | X = x] &= \mathbb{P}[Y > m(x) | X = x] - \mathbb{P}[Y < m(x) | X = x] \\ &= (1 - F(m(x) | x)) - F(m(x) | x) = \frac{1}{2} - \frac{1}{2} = 0. \end{aligned}$$

Exercise 24.2

Fix x , write $m = m(x)$ and $F(\cdot) = F(\cdot | x)$, and take $\theta > m$. Pointwise,

$$|y - \theta| - |y - m| = \begin{cases} \theta - m, & y \leq m, \\ \theta + m - 2y \geq -(\theta - m), & m < y < \theta, \\ -(\theta - m), & y \geq \theta, \end{cases}$$

so $|y - \theta| - |y - m| \geq (\theta - m)(\mathbf{1}\{\{\} y \leq m\} - \mathbf{1}\{\{\} y > m\})$. Taking conditional expectations, which are finite since $\mathbb{E}|Y| < \infty$,

$$\mathbb{E}[|Y - \theta| - |Y - m| | X = x] \geq (\theta - m)(F(m) - (1 - F(m))) = (\theta - m)(2F(m) - 1) = 0.$$

The case $\theta < m$ is symmetric. Hence m minimizes $\mathbb{E}[|Y - \theta| | X = x]$ (uniquely if F is strictly increasing at m).

Exercise 24.3

$\mathbb{E}[\psi(Y - \theta)] = \tau - \mathbb{P}[Y < \theta]$, so the condition says $\mathbb{P}[Y < \theta] = \tau$.

- If Y is continuously distributed, then $F(\theta) = \mathbb{P}[Y < \theta] = \tau$, so θ is a τ th quantile. If F is flat at level τ , every point of the flat interval is a solution, and all of them are τ -quantiles in the sense of minimizing $\mathbb{E}[\rho_\tau(Y - \theta)]$.
- In general any solution satisfies $\mathbb{P}[Y < \theta] \leq \tau \leq \mathbb{P}[Y \leq \theta]$, which is the defining property of a (generalized) τ -quantile. So when a solution exists it is a quantile.
- But with atoms a solution may not exist. If Y is Bernoulli($\frac{1}{2}$) and $\tau = 0.3$, $\mathbb{P}[Y < \theta] \in \{0, \frac{1}{2}, 1\}$ never equals 0.3, although the quantile (0) exists. The moment condition characterizes quantiles

only for continuous distributions, which is why Theorem 24.2 assumes continuity. The check-function minimization (24.12) always defines the quantile.

Exercise 24.4

- (a) By symmetry $\text{med}[e | X] = 0$ and, when $\mathbb{E}|e| < \infty$, $\mathbb{E}[e | X] = 0$. Hence $\mathbb{E}[Y | X] = \text{med}[Y | X] = X'\beta$.
- (b) The same β : the mean and median regressions coincide, so both estimators are consistent for it (OLS needs finite variance for $\sqrt{n}$ -asymptotics).
- (c) The choice is about efficiency and robustness. If e is independent of X with variance σ^2 and density f , the asymptotic variances are $\sigma^2\mathbb{E}[XX']^{-1}$ for OLS and $\{4f(0)^2\}^{-1}\mathbb{E}[XX']^{-1}$ for LAD.
- Normal errors: $\{4f(0)^2\}^{-1} = \pi\sigma^2/2$, so OLS is more efficient (LAD has 57% larger variance). Prefer OLS for near-normal or light-tailed errors.
 - Laplace errors: $\{4f(0)^2\}^{-1} = \sigma^2/2$, so LAD has half the variance. More generally prefer LAD for heavy-tailed errors or outliers, where OLS is inefficient. For Cauchy errors OLS is not even consistent while LAD works.

Also prefer LAD if the median is the object of interest or the data contain gross errors or top-coding. Prefer OLS if the conditional mean is of interest (e.g. for aggregation), since without symmetry the two estimate different coefficients.

Exercise 24.5

No. $R^2 = 1 - \sum \hat{e}_i^2 / \sum (Y_i - \bar{Y})^2$ measures fit in terms of squared errors, and OLS minimizes the sum of squared errors by construction. Therefore OLS has the highest R^2 among all linear fits, including LAD, in every sample. The comparison is guaranteed in advance and says nothing about which estimator is more accurate for β . By the same logic LAD would “win” on the sum of absolute errors. The choice should rest on the object of interest (conditional mean or median) and on the likely error distribution: efficiency, robustness to outliers, heavy tails (Exercise 24.4).

Exercise 24.6

$\mathbb{E}[\psi_\tau(e) | X = x] = \tau - \mathbb{P}[Y < q_\tau(x) | X = x] = \tau - F(q_\tau(x) | x) = \tau - \tau = 0$, using continuity: $F(q_\tau(x) | x) = \tau$, and $\mathbb{P}[Y = q_\tau(x) | X = x] = 0$.

Exercise 24.7

Fix x , let $q = q_\tau(x)$, $F = F(\cdot | x)$, and take $\theta > q$. Pointwise:

- $y \leq q$: both arguments are ≤ 0 , so $\rho_\tau(y - \theta) - \rho_\tau(y - q) = -(1 - \tau)(y - \theta) + (1 - \tau)(y - q) = (1 - \tau)(\theta - q)$;
- $y \geq \theta$: $\tau(y - \theta) - \tau(y - q) = -\tau(\theta - q)$;
- $q < y < \theta$: $-(1 - \tau)(y - \theta) - \tau(y - q) \geq -\tau(\theta - q)$, since the first term is positive and $y - q < \theta - q$.

Hence $\rho_\tau(y - \theta) - \rho_\tau(y - q) \geq (\theta - q)[(1 - \tau)\mathbf{1}\{y \leq q\} - \tau\mathbf{1}\{y > q\}]$, and taking expectations,

$$\mathbb{E}[\rho_\tau(Y - \theta) - \rho_\tau(Y - q) | X = x] \geq (\theta - q)[(1 - \tau)F(q) - \tau(1 - F(q))] = (\theta - q)(F(q) - \tau) = 0.$$

The case $\theta < q$ is symmetric. So $q_\tau(x)$ minimizes the expected check loss. For $\tau = \frac{1}{2}$ this is Exercise 24.2 ($\rho_{1/2}(u) = \frac{1}{2}|u|$).

Exercise 24.8

$Q_\tau[Y | X]$ takes only two values, $q_\tau(0)$ and $q_\tau(1)$. Any function of a binary variable is linear in it:

$$Q_\tau[Y | X] = q_\tau(0) + (q_\tau(1) - q_\tau(0))X.$$

Exercise 24.9

The pair takes four values, so the conditional quantile is saturated by an interaction model:

$$Q_\tau[Y | X_1, X_2] = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2,$$

with $\alpha = q_\tau(0, 0)$, $\beta_1 = q_\tau(1, 0) - q_\tau(0, 0)$, $\beta_2 = q_\tau(0, 1) - q_\tau(0, 0)$ and $\beta_3 = q_\tau(1, 1) - q_\tau(1, 0) - q_\tau(0, 1) + q_\tau(0, 0)$. Without the interaction the linear model is generally misspecified. Unlike the mean, quantiles are not additive, so even independent effects of X_1 and X_2 on Y do not in general produce $\beta_3 = 0$.

Exercise 24.10

First identify Q_τ as the Hessian of $M(\beta) = \mathbb{E}[\rho_\tau(Y - X'\beta)]$. Since $\mathbb{E}[\psi_\tau(Y - X'\beta) | X] = \tau - F(X'\beta | X)$, the gradient is $-\mathbb{E}[X(\tau - F(X'\beta | X))]$ and the Hessian is $\mathbb{E}[XX'f_{Y|X}(X'\beta | X)]$. At β_τ , with $e = Y - X'\beta_\tau$, $f_{Y|X}(X'\beta_\tau | X) = f_\tau(0 | X)$, so $Q_\tau = \mathbb{E}[XX'f_\tau(0 | X)]$ as in (24.18). If the error density at zero does not depend on x , $f_\tau(0 | x) = f_\tau(0)$ for all x , then it factors out of the expectation:

$$Q_\tau = \mathbb{E}[XX'f_\tau(0 | X)] = \mathbb{E}[XX']f_\tau(0),$$

which is (24.19).

Exercise 24.11

Under correct specification $Q_\tau[Y | X] = X'\beta_\tau$, so $\mathbb{P}[e < 0 | X] = \tau$ (continuous distribution). Since $\mathbf{1}\{\{\} e < 0\}^2 = \mathbf{1}\{\{\} e < 0\}$,

$$\mathbb{E}[\psi_\tau^2 | X] = \mathbb{E}[\tau^2 - 2\tau\mathbf{1}\{\{\} e < 0\} + \mathbf{1}\{\{\} e < 0\} | X] = \tau^2 - 2\tau^2 + \tau = \tau(1 - \tau).$$

By iterated expectations $\Omega_\tau = \mathbb{E}[XX'\mathbb{E}[\psi_\tau^2 | X]] = \tau(1 - \tau)\mathbb{E}[XX'] = \tau(1 - \tau)Q$. (Section 24.8's formula for $\hat{\Omega}_\tau$ should be scaled by $1/n$, not $1/h$.)

Exercise 24.12

With $h(0, x, u) = 0$, the causal effect is $C(x, u) = h(1, x, u)$. Assumption 24.1.3 is used in the proof of Theorem 24.5 only to interchange $h(d, x, \cdot)$ with the quantile operator, $Q_\tau[h(d, X, U) | X = x] = h(d, x, Q_\tau[U | X = x])$ for $d = 0, 1$.

- For $d = 1$, $h(1, x, u) = C(x, u)$ is increasing in u by Assumption 24.1.2, so the interchange holds without 24.1.3.
- For $d = 0$, $h(0, x, u) = 0$ is constant, so $Q_\tau[h(0, X, U) | X = x] = 0 = h(0, x, Q_\tau[U | X = x])$ trivially.

Hence, using Assumption 24.1.4 to replace $Q_\tau[h(d, X, U) \mid X = x]$ by $q_\tau(d, x) = Q_\tau[Y \mid D = d, X = x]$,

$$\begin{aligned} Q_\tau(x) &= Q_\tau[C(X, U) \mid X = x] = C(x, Q_\tau[U \mid X = x]) \\ &= Q_\tau[h(1, X, U) \mid X = x] - 0 = q_\tau(1, x) - q_\tau(0, x) = D_\tau(x). \end{aligned}$$

Monotonicity of the untreated response is automatic here, so only 24.1.1, 24.1.2 and 24.1.4 are needed. (The exercise's " $h(0, X_2, U)$ " should read $h(0, X, U)$.)

Exercise 24.13

$n = 3393$ (OLS slope 0.137). Bootstrap standard errors in parentheses:

τ	0.1	0.3	0.5	0.7	0.9
education	0.111 (0.011)	0.134 (0.006)	0.136 (0.006)	0.138 (0.009)	0.148 (0.012)
95% BC interval	[0.087, 0.128]	[0.125, 0.146]	[0.119, 0.145]	[0.123, 0.154]	[0.129, 0.169]
intercept	0.67 (0.14)	0.79 (0.07)	1.03 (0.08)	1.29 (0.11)	1.54 (0.15)

The return to a year of education rises with the quantile, from 11% at the 10th percentile to 15% at the 90th. The difference $q_{0.9} - q_{0.1}$ in slopes is 0.037 (paired bootstrap s.e. 0.015, 95% interval [0.015, 0.073]), so the increase is statistically significant. Education raises the whole conditional distribution of wages but stretches it: wage dispersion among Hispanic men is larger at higher education levels. The median return (13.6%) is close to the OLS estimate.

Exercise 24.14

$n = 2525$ (OLS slope 0.115).

τ	0.1	0.3	0.5	0.7	0.9
education	0.097 (0.010)	0.110 (0.008)	0.118 (0.003)	0.116 (0.005)	0.118 (0.008)
95% BC interval	[0.080, 0.120]	[0.101, 0.126]	[0.113, 0.128]	[0.105, 0.127]	[0.105, 0.139]
intercept	0.75 (0.14)	0.95 (0.11)	1.07 (0.05)	1.32 (0.08)	1.66 (0.11)

For women the returns are lower than for men at every quantile above the 10th (11–12% versus 13–15%). They are nearly constant above the 30th percentile. Only the 10th percentile shows a lower return (9.7%). The slope difference $q_{0.9} - q_{0.1} = 0.021$ (s.e. 0.012, interval $[-0.008, 0.042]$) is not significant. So for Hispanic women education shifts the wage distribution roughly uniformly (a location shift in log wages), with at most a smaller gain at the very bottom, whereas for men it also increases dispersion.

Exercise 24.15

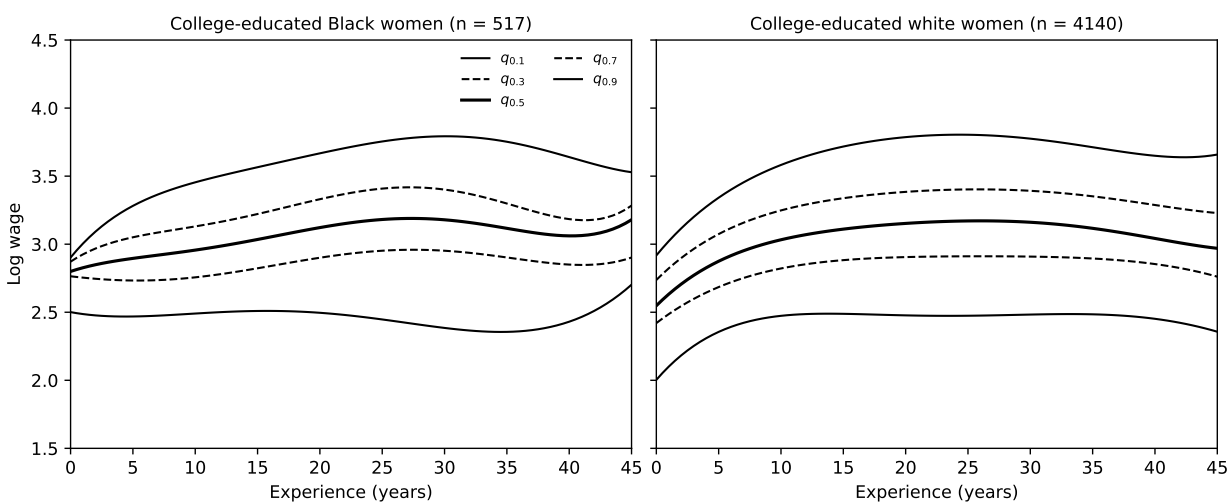
$n = 2980$ students in 60 schools; OLS slope 0.156. Clustered bootstrap (1000 replications, resampling schools):

τ	0.1	0.3	0.5	0.7	0.9
percentile	0.092 (0.008)	0.135 (0.008)	0.167 (0.013)	0.208 (0.013)	0.199 (0.013)
95% BC interval	[0.073, 0.107]	[0.119, 0.151]	[0.142, 0.192]	[0.181, 0.231]	[0.168, 0.221]
intercept	-0.50 (0.17)	1.46 (0.42)	3.80 (0.49)	6.64 (0.80)	14.57 (1.24)

Yes, meaningfully. The slope more than doubles from the 10th to the 70th quantile and stays high at the 90th. The difference $q_{0.9} - q_{0.1} = 0.107$ has paired bootstrap s.e. 0.016 (95% interval [0.073, 0.134]). Initial achievement therefore predicts end-of-year scores much more strongly for students who do well conditional on their start (upper quantiles) than for those who do poorly. Equivalently, the conditional distribution of final scores spreads out as initial percentile rises. Low-percentile students are uniformly low, while high-percentile students are more heterogeneous. This fans out the regression lines, and is consistent with the convexity of the mean found in Sections 19.21 and 20.16. As always, these are descriptive quantile relationships, not causal effects of initial scores.

Exercise 24.16

Following the book's code, experience is scaled by 40 and the polynomial is evaluated for 0–45 years. The samples are college-educated (education = 16) women: Black $n = 517$, white $n = 4140$.



Fitted log wages:

experience	Black women						white women					
	0	5	10	20	30	40	0	5	10	20	30	40
$q_{0.1}$	2.50	2.47	2.49	2.50	2.38	2.43	2.00	2.36	2.47	2.48	2.48	2.45
$q_{0.5}$	2.80	2.90	2.96	3.12	3.18	3.06	2.55	2.87	3.03	3.15	3.16	3.04
$q_{0.9}$	2.90	3.28	3.45	3.67	3.79	3.64	2.92	3.34	3.58	3.79	3.78	3.65

Findings. For both groups the median profile rises by about 30–60 log points over the first 20–30 years and then declines slightly. The quantile functions fan out with experience, as in Figure 24.6:

the 90–10 gap grows from 0.4 (Black) and 0.9 (white) at entry to about 1.2–1.4 log points after 20–30 years. So wage heterogeneity increases more than proportionately with experience.

Differences. At the top of the distribution the two profiles are very similar ($q_{0.9} \approx 3.8$ at 30 years for both). At the bottom they differ. The 10th percentile for Black women is essentially flat at about 2.5 (no wage growth with experience), whereas for white women it rises by almost 50 log points over the first ten years from a lower starting value. The higher entry values for Black women at low quantiles should be interpreted cautiously: with only 517 observations the 5th-order polynomial is imprecise, especially at the boundaries. The flat bottom quantile nonetheless suggests that low-wage Black college-educated women experience little wage progression, consistent with the book's observation that the lowest quantile peaks early.

Chapter 25

Binary Choice

Solutions to all 19 exercises of Chapter 25 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). The empirical exercises use `cps09mar` (full-time workers, March 2009). Probit models are estimated by maximum likelihood (Newton's method), with robust (sandwich) standard errors. Average partial effects (APE) are reported with delta-method standard errors; for dummy regressors they are average discrete changes.

Notation: $P[Y = 1 | X] = G(X'\beta)$ with G symmetric ($G(-u) = 1 - G(u)$), density $g = G'$, $Z = X$ if $Y = 1$ and $Z = -X$ if $Y = 0$, $h(u) = \frac{d}{du} \log G(u)$ and $H(u) = -\frac{d^2}{du^2} \log G(u)$.

Exercise 25.1

Jacob's variable is $1 - Y$, and $P[1 - Y = 1 | X] = 1 - \Phi(X'\beta) = \Phi(-X'\beta)$. His log-likelihood at b equals Emily's at $-b$, so his MLE is exactly $-\hat{\beta}$: every coefficient, including the intercept, flips sign, and the standard errors are identical. Marginal effects also flip sign. (This uses the symmetry of the link, so it holds for logit too but not for asymmetric links such as the complementary log-log.)

Exercise 25.2

With $X^* = X/1000$ the index is $X\beta = X^*(1000\beta)$, so the likelihood is the same function after reparameterization. By invariance of the MLE, Julie's coefficient on that regressor is exactly 1000 times Jackson's (standard error also $\times 1000$, t -ratio unchanged). All other coefficients, fitted probabilities and marginal effects per dollar are identical.

Exercise 25.3

With $P(X) = P[Y = 1 | X]$ and $e = Y - P(X)$: if $Y = 1$ (probability $P(X)$) then $e = 1 - P(X)$; if $Y = 0$ (probability $1 - P(X)$) then $e = -P(X)$, which is (25.1). Then $\mathbb{E}[e | X] = P(1 - P) + (1 - P)(-P) = 0$ and

$$\text{var}[e | X] = \mathbb{E}[e^2 | X] = P(1 - P)^2 + (1 - P)P^2 = P(1 - P)[(1 - P) + P] = P(X)(1 - P(X)),$$

which is (25.2).

Exercise 25.4

$\pi(Y | X) = G(X'\beta)^Y (1 - G(X'\beta))^{1-Y} = G(X'\beta)^Y G(-X'\beta)^{1-Y}$ by symmetry. If $Y = 1$ this equals $G(X'\beta) = G(Z'\beta)$; if $Y = 0$ it equals $G(-X'\beta) = G(Z'\beta)$.

Exercise 25.5

(a) $\Lambda'(x) = e^{-x}/(1 + e^{-x})^2 = \Lambda(x) \cdot \frac{e^{-x}}{1 + e^{-x}} = \Lambda(x)(1 - \Lambda(x))$. (b) $\frac{d}{dx} \log \Lambda = \Lambda'/\Lambda = 1 - \Lambda$. (c) $-\frac{d}{dx}(1 - \Lambda) = \Lambda' = \Lambda(1 - \Lambda)$. (d) $0 < \Lambda(1 - \Lambda) \leq \frac{1}{4}$ since $p(1 - p) \leq \frac{1}{4}$ on $[0, 1]$; in particular $|H_{\text{logit}}| \leq 1$.

Exercise 25.6

(a) $\frac{d}{dx} \log \Phi(x) = \phi(x)/\Phi(x) = \lambda(x)$. (b) Using $\phi'(x) = -x\phi(x)$,

$$-\frac{d}{dx} \lambda(x) = -\frac{\phi'(x)\Phi(x) - \phi(x)^2}{\Phi(x)^2} = \frac{x\phi(x)}{\Phi(x)} + \frac{\phi(x)^2}{\Phi(x)^2} = \lambda(x)(x + \lambda(x)).$$

Exercise 25.7

(a) $\ell_n(\beta) = \sum_i \log G(Z'_i\beta)$, so by the chain rule $S_n(\beta) = \sum_i Z_i h(Z'_i\beta)$. Differentiating again, $\mathbf{H}_n(\beta) = -\sum_i Z_i Z'_i h'(Z'_i\beta) = \sum_i Z_i Z'_i H(Z'_i\beta)$, and $Z_i Z'_i = X_i X'_i$ because $Z_i = \pm X_i$.

(b) For any $a \neq 0$, $a' \mathbf{H}_n(\beta) a = \sum_i (a' X_i)^2 H(Z'_i\beta) \geq 0$, with equality only if $a' X_i = 0$ for all i . Given full rank $\sum X_i X'_i > 0$ this is excluded, so $\mathbf{H}_n(\beta) > 0$ for every β .

(c) The Hessian of ℓ_n is $-\mathbf{H}_n(\beta)$, negative definite at every β , so ℓ_n is strictly concave on $\mathbb{R}^k$. Any stationary point is the unique global maximizer. (A maximizer exists unless the data are perfectly separated.)

Exercise 25.8

$0 = \mathbb{E}[Zh(Z'\beta_0)]$. Writing out the two cases, $Zh(Z'\beta) = X[Yg/G - (1-Y)g/(1-G)]$ with g and G evaluated at $X'\beta$. This simplifies to

$$\mathbb{E} \left[X \frac{g(X'\beta_0)}{G(X'\beta_0)(1-G(X'\beta_0))} (Y - G(X'\beta_0)) \right] = 0,$$

a weighted orthogonality condition between the regressors and the “residual” $Y - G(X'\beta_0)$.

Exercise 25.9

For the logit $g/(\Lambda(1-\Lambda)) = 1$, so $\sum_{i=1}^n X_i (Y_i - \Lambda(X'_i \hat{\beta}_{\text{logit}})) = 0$. In particular, with an intercept, the average fitted probability equals the sample frequency $\bar{Y}$.

Exercise 25.10

$\sum_{i=1}^n X_i \frac{\phi(X'_i \hat{\beta})}{\Phi(X'_i \hat{\beta})(1-\Phi(X'_i \hat{\beta}))} (Y_i - \Phi(X'_i \hat{\beta})) = 0$, or equivalently $\sum_i X_i \lambda(Z'_i \hat{\beta}) q_i = 0$ with $q_i = 2Y_i - 1$.

Exercise 25.11

By symmetry $h(-u) = g(u)/(1-G(u))$, so $h(Z'\beta) = Yg/G + (1-Y)g/(1-G)$ (all at $X'\beta$). Squaring, the cross term vanishes because $Y(1-Y) = 0$:

$$h(Z'\beta)^2 = \frac{g^2}{G^2} Y + \frac{g^2}{(1-G)^2} (1-Y) = \frac{g^2(Y-G)^2}{G^2(1-G)^2}.$$

The last equality checks both cases: $Y = 1$ gives $g^2(1-G)^2/(G^2(1-G)^2) = g^2/G^2$ and $Y = 0$ gives $g^2 G^2/(G^2(1-G)^2) = g^2/(1-G)^2$. For the logit $g = \Lambda(1-\Lambda)$, so the ratio $g^2/(G^2(1-G)^2)$ is one and $h(Z'\beta)^2 = (Y - \Lambda(X'\beta))^2$.

Exercise 25.12

By (25.1), $Y = \Phi(X'\beta) + e$ with $\mathbb{E}[e | X] = 0$ is a nonlinear regression, so

$$\hat{\beta}_{\text{nlls}} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \Phi(X_i'\beta))^2, \quad \text{first-order condition} \quad \sum_i X_i \phi(X_i'\hat{\beta}) (Y_i - \Phi(X_i'\hat{\beta})) = 0.$$

It is consistent under correct specification. Because $\text{var}[e | X] = \Phi(1 - \Phi)$ is heteroskedastic, standard errors must use the robust formula (23.6) with $\hat{m}_{\theta i} = X_i \phi(X_i'\hat{\beta})$. NLLS is less efficient than the MLE. Weighted NLLS with weights $1/\{\tilde{\Phi}_i(1 - \tilde{\Phi}_i)\}$ from a first-step estimate has the MLE's first-order condition (Exercise 25.10), and iterating it reproduces the MLE (iteratively reweighted least squares).

Exercise 25.13

(a) With $e_1 = \rho e_2 + \varepsilon$ and $e_2 = Y_2 - X'\gamma_1 - Z'\gamma_2$, substitute into the structural equation: $Y_1^* = X'\beta_1 + Y_2\beta_2 + \rho(Y_2 - X'\gamma_1 - Z'\gamma_2) + \varepsilon = \mu(\theta) + \varepsilon$.

(b) Conditional on (X, Z) , (e_1, e_2) is bivariate normal and $\varepsilon = e_1 - \rho e_2$ with $\rho = \sigma_{12}/\sigma_2^2$ has $\text{cov}(\varepsilon, e_2) = \sigma_{12} - \rho\sigma_2^2 = 0$. Jointly normal and uncorrelated implies independent. As $Y_2 = X'\gamma_1 + Z'\gamma_2 + e_2$ is a function of (X, Z, e_2) , ε is also independent of Y_2 given (X, Z) .

(c) Conditional on (Y_2, X, Z) , $\mu(\theta)$ is fixed and $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ with $\sigma_\varepsilon^2 = \text{var}(e_1) - \rho^2\sigma_2^2 = 1 - \sigma_{12}^2/\sigma_2^2$ (by (b)). Hence $Y_1^* | (Y_2, X, Z) \sim N(\mu(\theta), \sigma_\varepsilon^2)$ and $P[Y_1 = 1 | Y_2, X, Z] = \Phi(\mu(\theta)/\sigma_\varepsilon)$, which gives the likelihood in the text.

Exercise 25.14

(For a binary choice model the observation rule should read $Y = \mathbf{1}\{\{\} Y^* > 0\}$. As printed, $Y = Y^* \mathbf{1}\{\{\} Y^* > 0\}$ is a Tobit model, in which m and σ^2 are both identified.)

(a) $P(x) = P[m(x) + e > 0 | X = x] = \Phi(m(x)/\sigma(x))$.

(b) No. Only the ratio $r(x) = m(x)/\sigma(x) = \Phi^{-1}(P(x))$ is identified: $(cm, c\sigma)$ for any function $c(x) > 0$ gives the same response probability.

(c) Normalize $\sigma(x) \equiv 1$. Then $m(x) = \Phi^{-1}(P(x))$ is identified (equivalently, identify the ratio m/σ).

(d) Not nonparametrically. Any heteroskedastic model is observationally equivalent to a homoskedastic one with $m^* = m/\sigma$, so “allowing for heteroskedasticity” adds nothing unless m is restricted. In a parametric model, e.g. $m(x) = x'\beta$ with $\sigma(x) = \exp(z'\gamma)$, heteroskedasticity is identified only through the functional-form restrictions, and it is better viewed as a more flexible model for the response probability.

Exercise 25.15

$n = 29,140$. Only 2.25% of men in this extract report union membership, far below the published 2009 U.S. rate of about 12%. The *union* indicator evidently captures only part of membership in these data, so it is best to focus on relative magnitudes.

	coefficient	s.e.	APE (pp)	s.e.
intercept	−1.954	(0.100)		
age	0.0079	(0.0014)	0.04	(0.01)
education	−0.0255	(0.0057)	−0.14	(0.03)
Black	−0.054	(0.060)	−0.27	(0.29)
Hispanic	−0.298	(0.059)	−1.28	(0.20)

Membership increases with age (seniority, and declining unionization of younger cohorts) and decreases with education: unionized jobs are concentrated in blue-collar occupations. Hispanic men are about 1.3 percentage points less likely to be members, over half the base rate, even controlling for education and age. There is no significant difference for Black men.

Exercise 25.16

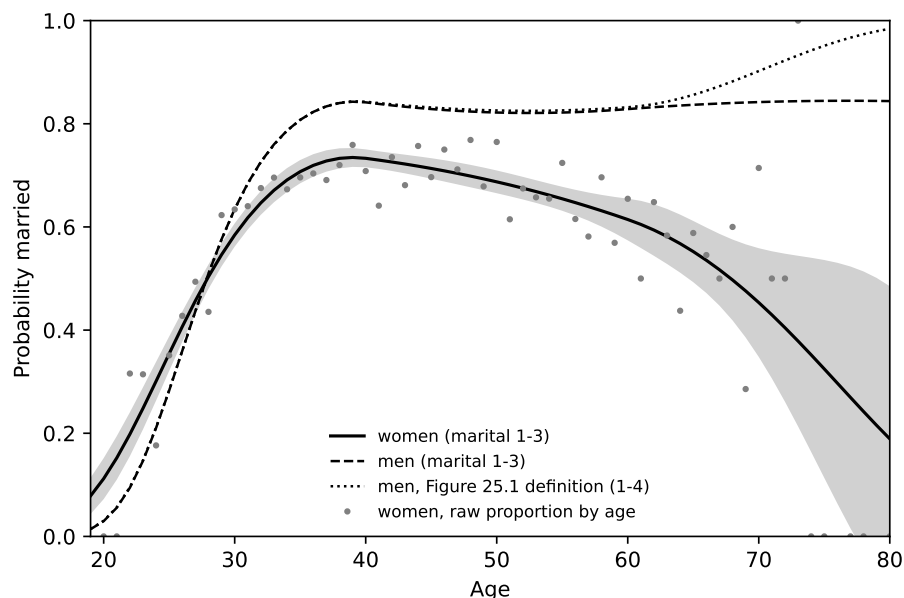
$n = 21,602$, membership rate 2.0%.

	coefficient	s.e.	APE (pp)	s.e.
intercept	−3.095	(0.141)		
age	0.0101	(0.0017)	0.05	(0.01)
education	0.0435	(0.0080)	0.21	(0.04)
Black	−0.047	(0.062)	−0.22	(0.28)
Hispanic	−0.148	(0.070)	−0.63	(0.27)

The age effect is similar to men's, but the education effect has the *opposite* sign: more educated women are more likely to be union members. This reflects the composition of female union membership, concentrated in the public sector and in professions such as teaching and nursing, whereas male membership is concentrated in manual occupations. The Hispanic gap is smaller than for men but still significant (−0.6 pp); Black women do not differ significantly.

Exercise 25.17

$n = 5199$, 63% married. Following Figure 25.1 we use a probit in a quadratic spline of age/100 with knots at 40 and 60. This gives a smooth, flexible age profile that can rise, plateau and fall, while the probit link keeps probabilities in $[0, 1]$. The spline improves substantially on a quadratic probit (AIC 6525 versus 6560).



Estimated probabilities: 0.20 at age 22, 0.35 at 25, 0.58 at 30, 0.71 at 35, 0.73 at 40, 0.69 at 50, 0.61 at 60, 0.45 at 70.

Comparison with men. College-educated men (same definition, $n = 6441$): 0.09 at 22, 0.28 at 25, 0.64 at 30, 0.84 at 40, and about 0.83 thereafter. Women marry earlier (higher probabilities before about 27), but their probability plateaus lower (0.73 versus 0.84) and declines steadily after 40, whereas men's stays flat. The decline for women reflects divorce and widowhood (widows are not counted as married here) with less remarriage, and women's longer life expectancy.

Remark. Coding note. Figure 25.1's code defines "married" as $\text{marital} \leq 4$, which includes widowed (code 4). With that definition the men's curve rises toward one at old ages (dotted line, 0.98 at 80), an artefact of counting widowers as married. With the exercise's definition (codes 1–3) it is flat after 40.

Exercise 25.18

Probit with the age spline of Exercise 25.17 (not reported; coefficients on $a = \text{age}/100$: 44.9, -57.0 , 60.5, -4.7), $n = 29,140$, 72% married:

	coefficient	s.e.	APE (pp)	s.e.
education	0.0263	(0.0031)	0.78	(0.09)
Black	-0.498	(0.029)	-16.4	(1.0)
Hispanic	0.005	(0.025)	0.1	(0.7)

For a white non-Hispanic man with mean education the probability of being married is 0.39 at 25, 0.65 at 30, 0.82 at 40, and stays around 0.81–0.84 through 70. Each additional year of education raises it by about 0.8 points. Black men are 16 points less likely to be married, a very large gap given the controls. Hispanic men do not differ from white men.

Exercise 25.19

$n = 21,602$, 59% married.

	coefficient	s.e.	APE (pp)	s.e.
education	0.0256	(0.0035)	0.92	(0.13)
Black	-0.645	(0.027)	-24.1	(1.0)
Hispanic	-0.132	(0.027)	-4.8	(1.0)

Profile for white non-Hispanic women with mean education: 0.39 at 25, 0.58 at 30, 0.72 at 40, 0.69 at 50, 0.62 at 60 and 0.44 at 70. It peaks around 40 and then declines, unlike men's. The education effect is similar to men's (about 0.9 points per year). The Black gap is even larger for women (-24 points), and Hispanic women are about 5 points less likely to be married, whereas Hispanic men are not. All results describe full-time workers, the population of the `cps09mar` extract, so they combine marriage patterns with selection into full-time work, which differs by sex and marital status.

Chapter 26

Multiple Choice

Solutions to all 18 exercises of Chapter 26 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). The empirical exercises use `cps09mar` and `Koppelman` (2779 business travellers choosing among train, air, bus and car). All models were programmed directly and estimated by (simulated) maximum likelihood, with standard errors from the inverse numerical Hessian.

- Simple multinomial probit (iid normal errors): exact one-dimensional Gauss–Hermite quadrature.
- Mixed logit: 400 (or 200) Halton draws per traveller.
- General multinomial probit: the GHK simulator with 300 Halton draws.

The conditional logit and simple probit columns of Table 26.1 are reproduced exactly. The nested and mixed logit columns are reproduced up to three apparent typos, discussed in Exercises 26.16 and 26.17, and the general probit column up to simulation error (Exercise 26.18).

Notation: P_j denotes response probabilities, $\mu_j = w'\beta_j + x'_j\gamma$ the systematic utility of alternative j , and J the base alternative.

Exercise 26.1

Each term $\exp(x'\beta_j)$ is strictly positive, so $P_j(x) > 0$. The numerator is one of the positive terms of the denominator, so $P_j(x) \leq 1$ (strictly less if $J \geq 2$). Summing over j , the numerators add up to the common denominator, so $\sum_j P_j(x) = 1$.

Exercise 26.2

Divide numerator and denominator by $\exp(x'\beta_J)$:

$$P_j(x) = \frac{\exp(x'(\beta_j - \beta_J))}{\sum_{\ell=1}^J \exp(x'(\beta_\ell - \beta_J))}.$$

Adding a common vector to every β_j leaves all probabilities unchanged, which is why one alternative must be normalized, e.g. $\beta_J = 0$.

Exercise 26.3

Write $D(x) = \sum_{\ell} \exp(x'\beta_{\ell})$, so that $\partial D / \partial x = \sum_{\ell} \beta_{\ell} \exp(x'\beta_{\ell})$. By the quotient rule,

$$\frac{\partial P_j}{\partial x} = \frac{\beta_j \exp(x'\beta_j)}{D} - \frac{\exp(x'\beta_j)}{D} \cdot \frac{\sum_{\ell} \beta_{\ell} \exp(x'\beta_{\ell})}{D} = P_j(x) \left(\beta_j - \sum_{\ell=1}^J \beta_{\ell} P_{\ell}(x) \right).$$

Exercise 26.4

Let $U_j = \mu_j + \varepsilon_j$ with ε_j i.i.d. Type I extreme value, $F(\varepsilon) = \exp(-e^{-\varepsilon})$, $f(\varepsilon) = e^{-\varepsilon} \exp(-e^{-\varepsilon})$. Conditioning on ε_j ,

$$P_j = \int f(\varepsilon) \prod_{\ell \neq j} F(\varepsilon + \mu_j - \mu_\ell) d\varepsilon = \int e^{-\varepsilon} \exp\left(-e^{-\varepsilon} \sum_{\ell=1}^J e^{\mu_\ell - \mu_j}\right) d\varepsilon.$$

With $t = e^{-\varepsilon}$ this is $\int_0^\infty \exp(-tS) dt = 1/S$, where $S = \sum_{\ell} e^{\mu_\ell - \mu_j}$. Hence $P_j = \exp(\mu_j) / \sum_{\ell} \exp(\mu_\ell)$, which is (26.8) with $\mu_j = w'\beta_j + x'_j\gamma$. (This is Theorem 26.1 with $\tau = 1$ applied to general indices μ_j .)

Exercise 26.5

x_j enters only μ_j , with $\partial\mu_j/\partial x_j = \gamma$. Hence

$$\frac{\partial P_j}{\partial x_j} = \gamma \frac{e^{\mu_j}}{D} - \frac{e^{\mu_j}}{D} \cdot \frac{\gamma e^{\mu_j}}{D} = \gamma P_j(1 - P_j), \quad \frac{\partial P_j}{\partial x_\ell} = -\frac{e^{\mu_j}}{D} \cdot \frac{\gamma e^{\mu_\ell}}{D} = -\gamma P_j P_\ell \quad (\ell \neq j).$$

Exercise 26.6

Divide numerator and denominator by $\exp(w'\beta_J + x'_J\gamma)$:

$$P_j(w, x) = \frac{\exp(w'(\beta_j - \beta_J) + (x_j - x_J)'\gamma)}{\sum_{\ell} \exp(w'(\beta_\ell - \beta_J) + (x_\ell - x_J)'\gamma)}.$$

Only utility differences matter. Adding a constant to all x_j (e.g. a common price change) has no effect, and the β_j are identified only relative to the base.

Exercise 26.7

By (26.9), $\text{AME}_{jj} = \mathbb{E}[\gamma P_j(W, X)(1 - P_j(W, X))]$. The plug-in estimator is

$$\widehat{\text{AME}}_{jj} = \hat{\gamma} \frac{1}{n} \sum_{i=1}^n \hat{P}_j(W_i, X_i)(1 - \hat{P}_j(W_i, X_i)), \quad \hat{P}_j(W_i, X_i) = P_j(W_i, X_i | \hat{\theta}),$$

with standard errors from the delta method, i.e. the gradient of this smooth function of $\hat{\theta}$ times the estimated covariance of $\hat{\theta}$.

Exercise 26.8

The two probabilities in (26.8) share the denominator $\sum_{\ell} \exp(w'\beta_\ell + x'_\ell\gamma)$, which cancels: $P_j/P_\ell = \exp(w'\beta_j + x'_j\gamma) / \exp(w'\beta_\ell + x'_\ell\gamma)$.

Exercise 26.9

Set $w'\beta_j = w'\beta_\ell = 0$ in (26.11): $P_j/P_\ell = \exp(x'_j\gamma) / \exp(x'_\ell\gamma) = \exp((x_j - x_\ell)'\gamma)$. The odds between two alternatives depend only on the difference in their attributes, not on any other alternative: independence of irrelevant alternatives.

Exercise 26.10

By Theorem 26.2, $P_{jk} = P_{k|j}P_j$, and the group probability cancels:

$$\frac{P_{jk}}{P_{j\ell}} = \frac{P_{k|j}}{P_{\ell|j}} = \exp\left(\frac{\mu_{jk} - \mu_{j\ell}}{\tau_j}\right).$$

This depends only on the utilities of k and ℓ . IIA therefore holds within a nest: a change in an alternative in another nest (e.g. a cheaper air fare) changes the probabilities of train and bus in the same proportion, so their relative odds are unchanged. The nested logit relaxes IIA only across nests.

Exercise 26.11

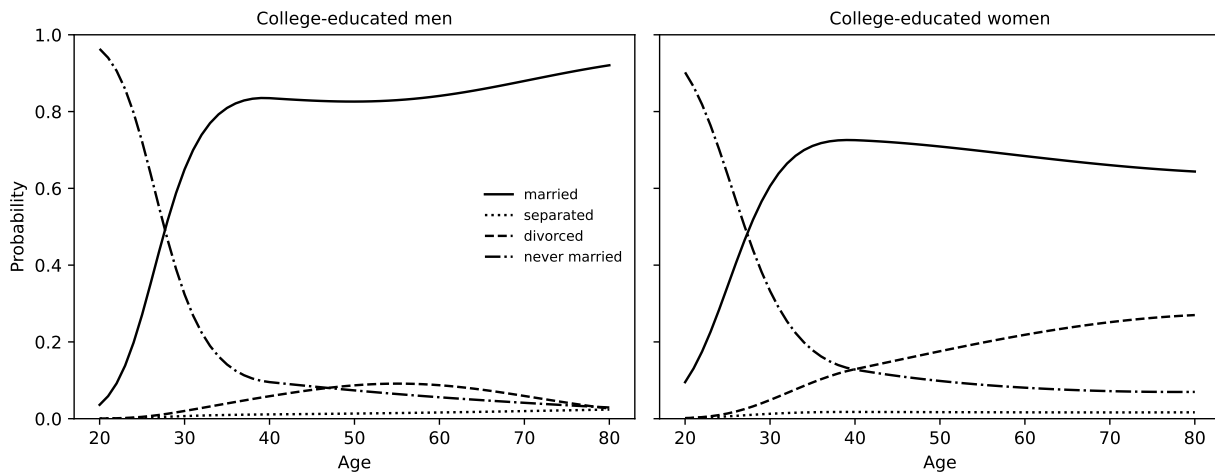
The group probabilities share a denominator, so

$$\frac{P_j}{P_\ell} = \frac{(\sum_m \exp(\mu_{jm}/\tau_j))^{\tau_j}}{(\sum_m \exp(\mu_{\ell m}/\tau_\ell))^{\tau_\ell}} = \frac{\exp(\tau_j I_j)}{\exp(\tau_\ell I_\ell)}, \quad I_j = \log \sum_m \exp(\mu_{jm}/\tau_j),$$

which involves only the alternatives in groups j and ℓ (through their inclusive values I_j, I_ℓ). IIA holds at the group level: the relative odds of two nests are unaffected by changes in a third nest. Within the model, substitution patterns are flexible only in the sense that alternatives in the same nest are closer substitutes.

Exercise 26.12

Following Figure 26.1 we use college graduates, four categories (married, which includes widowed as in the book's coding; separated; divorced; never married), and a quadratic spline in age/100 with a knot at 40. Samples: men $n = 6441$, women $n = 5199$.



age	men				women			
	married	separated	divorced	never	married	separated	divorced	never
25	0.27	0.00	0.01	0.72	0.35	0.01	0.01	0.64
30	0.65	0.01	0.02	0.32	0.61	0.01	0.05	0.33
40	0.84	0.01	0.06	0.10	0.73	0.02	0.13	0.13
50	0.83	0.01	0.09	0.07	0.71	0.02	0.18	0.10
60	0.84	0.02	0.09	0.06	0.68	0.02	0.22	0.08
70	0.88	0.02	0.06	0.04	0.66	0.02	0.25	0.07

For both sexes the probability of never having married falls steeply between 25 and 40. Men marry somewhat later (72% never married at 25 versus 64% for women) but then reach a much higher married share (0.84 versus 0.73 at 40). After 40 the profiles diverge. For men the married probability stays around 0.83–0.88 and the divorced share peaks below 10% and then falls (remarriage). For women the married probability declines slowly and the divorced share keeps rising, to 25% at 70. Divorced women are much less likely than men to remarry. Separation is rare for both.

Exercise 26.13

$n = 6645$ (married 48.8%, separated 2.7%, divorced 6.5%, never married 42.0%); base category: married.

	separated	divorced	never married
age	−0.041 (0.019)	0.054 (0.014)	−0.195 (0.007)
education	−0.247 (0.028)	−0.123 (0.020)	−0.023 (0.012)
constant	1.68 (0.66)	−1.91 (0.51)	5.74 (0.24)

Average marginal effects (probability points per year): age +3.3 (married), +0.1 (separated), +0.8 (divorced), −4.1 (never married); education +1.2 (married), −0.6 (separated), −0.6 (divorced), +0.05 (never married).

Interpretation. Among young women, each year of age moves about four points of probability from never married to married (and a little to divorced), the transition into marriage in the late twenties and early thirties. Education barely changes the probability of never having married, but among those who have married it strongly reduces separation and divorce. At age 35, for example, 14.4% of women with 12 years of schooling are divorced and 4.3% separated, versus 9.7% and 1.8% with 16 years, so college-educated women are more often currently married (0.72 versus 0.65).

Exercise 26.14

All women ($n = 21,602$), the same four categories and age spline, with alternative-specific coefficients and married as the base. The multinomial logit has log-likelihood −19823.24.

Choosing the nests. Alternatives that are close substitutes, i.e. share unobserved determinants, should be grouped together. Separated and divorced are both “formerly married” states (separation is often a step towards divorce), so the natural structure is {married}, {never married},

{separated, divorced}, with one dissimilarity parameter. We also tried {married, separated} versus {divorced, never married} (legally married or not) and {ever married} versus {never married}. Results:

nests	$\hat{\tau}$ (s.e.)	log-likelihood	LR vs. MNL
$\{M\}, \{N\}, \{S, D\}$	180 (109)	-19814.35	17.8
$\{M, S\}, \{D, N\}$	38.8 (34.0), 2.09 (0.72)	-19820.66	5.2
$\{M, S, D\}, \{N\}$	1.06 (7.8)	-19823.24	0.0

None of the groupings supports a nested structure. Consistency with random utility maximization requires $0 < \tau \leq 1$, and every unrestricted estimate exceeds one. Restricting $\tau \leq 1$, the profile likelihood for the preferred nesting increases monotonically in τ (-19823.86 at $\tau = 0.3$, -19823.59 at 0.6, -19823.33 at 0.9, -19823.24 at 1). So the constrained MLE is $\tau = 1$, i.e. the multinomial logit. With only individual-specific regressors (age) the dissimilarity parameter is also very weakly identified, as the huge standard errors show. The data give no reason to depart from IIA here. A nested model could still be motivated a priori, but it cannot be supported empirically with these regressors.

The transportation application (Exercises 26.15–26.18). The Koppelman data contain $n = 2779$ business travellers, each choosing among train, air, bus and car. Alternative-varying regressors are cost (dollars), in-vehicle time `intime` and out-of-vehicle time `outtime` (minutes). Individual-specific regressors are income and a dummy for urban origin/destination, which enter with alternative-specific coefficients (train is the base). All specifications below include the nine alternative-specific coefficients (income, urban and a constant for air, bus and car). The tables report only the cost and time coefficients, as the exercises request. LR denotes a likelihood ratio statistic.

Exercise 26.15

	(a) Table 26.1	(b) +outtime	(c) time = in+out	(d) logs
cost	-0.0218 (0.0035)	-0.0150 (0.0037)	-0.0153 (0.0037)	-4.67 (0.26)
intime	-0.0149 (0.0007)	-0.0178 (0.0008)		-0.44 (0.25)
outtime		-0.0309 (0.0027)		
time			-0.0176 (0.0008)	
car constant	1.86 (0.19)	-0.96 (0.30)	0.25 (0.19)	1.62 (0.20)
log-likelihood	-2100.6	-2026.8	-2040.8	-2082.3

In (d) the coefficients are on $\log(\text{cost})$ and $\log(\text{intime})$.

- Column (a) reproduces the conditional logit column of Table 26.1 exactly: cost -0.022, intime -0.015, and all nine alternative-specific coefficients agree to the printed digits. The implied value of in-vehicle time is $0.0149/0.0218 = \$0.68$ per minute, about \$41 per hour.
- Out-of-vehicle time is highly significant ($t = -11.4$; LR = 147.6 against (a)). Its coefficient is 1.7 times the in-vehicle coefficient, the classic transportation finding that waiting and access time is valued well above riding time. Controlling for it, the cost coefficient falls

by a third and the in-vehicle time coefficient rises slightly. The implied values of time are \$1.19/min (\$71/h) in-vehicle and \$2.06/min (\$124/h) out-of-vehicle. The car constant swings from +1.86 to −0.96. Car has no out-of-vehicle time, so in (a) its constant was absorbing car's access-time advantage.

- (c) Model (c) is model (b) with the restriction $\gamma_{\text{in}} = \gamma_{\text{out}}$. It fits much better than (a) but is rejected against (b): $\text{LR} = 2(2040.8 - 2026.8) = 28.0$, against χ_1^2 critical value 3.84. Adding the two times together is too restrictive, since travellers dislike out-of-vehicle minutes more.
- (d) The log specification makes utility depend on proportional changes. The cost coefficient −4.67 is now on the elasticity scale: the own-cost elasticity of P_j is $-4.67(1 - P_j)$, e.g. about −3.3 for a mode with a 30% share. The log-time coefficient becomes small and only marginally significant ($t = -1.7$). Once cost enters in logs, relative cost differences absorb most of what time was explaining. The fit improves on (a) by 18.3 log-likelihood points with the same number of parameters. So among the two-regressor specifications the data prefer logs, but both are dominated by specifications that include out-of-vehicle time.

Exercise 26.16

The model is the nested logit of Theorem 26.2. Within group g utilities are scaled by $1/\tau_g$, and the inclusive value is $I_g = \log \sum_{k \in g} e^{\mu_k/\tau_g}$. A singleton nest has no free τ . Consistency with random utility maximization requires $0 < \tau_g \leq 1$.

	(a) Table 26.1	(b) logs	(c) $\{C\}, \{T, B, A\}$	(d) $\{A\}, \{T, B, C\}$
cost	−0.0014 (0.0016)	−0.051 (0.009)	−0.0265 (0.0028)	−0.0333 (0.0055)
intime	−0.0054 (0.0010)	0.018 (0.005)	−0.0122 (0.0008)	−0.0190 (0.0009)
$\hat{\tau}$	0.236 (0.046)	0.0057 (0.0010)	0.664 (0.061)	3.81 (0.62)
log-likelihood	−2044.4	−2011.1	−2091.5	−2023.8

In (a) and (b) the groupings are {car, air} and {train, bus} with $\tau_{\text{train, bus}} = 1$, and $\hat{\tau}$ is $\tau_{\text{car, air}}$. In (c) and (d), $\hat{\tau}$ is the parameter of the three-alternative nest. In (b) the coefficients are on $\log(\text{cost})$ and $\log(\text{intime})$. The conditional logit log-likelihood is −2100.6.

- (a) With $\tau_{\text{train, bus}} = 1$ we reproduce the nested logit column of Table 26.1: intime −0.005, $\hat{\tau}_{\text{car, air}} = 0.24$ (0.05), air income 0.024, bus coefficients −0.049, −0.21, −1.55, and car coefficients 0.017, −0.58, 1.19. Two entries do not match:

- The cost coefficient is −0.0014 (0.0016), not −0.011 (0.002).
- The air–urban coefficient is −0.28 (0.08), not +0.28 (0.09).

Since every other coefficient agrees to the printed digits (which would be impossible if the cost coefficient really differed by a factor of eight), these are almost certainly typos in the table (−0.001 misprinted as −0.011, and a dropped minus sign).

Why the constraint is needed. Leaving $\tau_{\text{train, bus}}$ free, the profile log-likelihood is bimodal:

$\tau_{\text{train, bus}}$	0.1	0.26	0.5	1	1.5	3	10	30	100
log-lik.	−2045.8	−2043.5	−2043.9	−2044.4	−2044.5	−2043.3	−2038.5	−2037.0	−2037.9

There is a local maximum inside the admissible range, at $\tau_{\text{train,bus}} = 0.26$ (0.14), but it is only 1.0 log-likelihood points better than $\tau = 1$. The global maximum is near $\tau \approx 30$, far outside $(0, 1]$, where the model is not a random utility model. An unconstrained optimizer therefore drifts to meaningless values. The book imposes $\tau_{\text{train,bus}} = 1$, which the data do not reject (LR = 1.9 against the interior local maximum).

Interpretation. The car/air dissimilarity 0.24 implies a correlation of about $1 - 0.24^2 = 0.94$ between the car and air shocks. The likelihood gain over conditional logit is large: LR = 112. With nesting, the cost coefficient becomes small and insignificant, and the time coefficient falls by two-thirds. Much of the apparent price and time sensitivity in conditional logit reflects IIA-forced substitution patterns. Within the nest, utilities are scaled by $1/\hat{\tau} \approx 4.2$, so the within-nest cost and time sensitivities are -0.006 and -0.023 .

- (b) With logs, $\hat{\tau}_{\text{car,air}}$ collapses to 0.006. The within-nest choice between car and air becomes almost deterministic given the utilities. The log-likelihood rises to -2011.1 , but the log-intime coefficient is *positive* and significant (0.018, or $0.018/0.0057 = 3.1$ within the nest). A dissimilarity parameter at the edge of the parameter space together with a wrong-signed time effect indicates a model that fits by near-perfect prediction within the {car, air} nest, not an economically meaningful specification.
- (c) *Rationale:* car is private transport, while train, bus and air are common carriers with schedules, terminals and access and egress time. Their unobserved attributes (reliability, comfort, the inconvenience of fixed departure times) plausibly move together. The estimate $\hat{\tau} = 0.66$ lies in $(0, 1)$, so the model is RUM-consistent, and conditional logit is rejected (LR = 18.2). But the fit is much worse than the {car, air} nesting of (a), by 47 log-likelihood points with the same number of parameters. The data point to car and air as the closest substitutes, not the three public modes.
- (d) *Rationale:* air is the only fast mode, and the three ground modes might share unobservables such as tolerance for long journeys. But $\hat{\tau} = 3.81$ (0.62) is significantly greater than one, so the estimated model is not consistent with utility maximization and cannot be interpreted as a nesting. The high log-likelihood, and the implausibly large constants (car 6.15, air-urban -1.18), reflect the model using $\tau > 1$ to generate more substitution between air and the ground modes than IIA allows. This is the same car-air closeness captured by (a), achieved through an inadmissible parameter. The grouping does not make sense for these data.

Exercise 26.17

Simulated ML with Halton draws: 400 in (a), 200 in (b) and (c). With 100 draws, model (a) gives log-likelihood -2095.2 and coefficients identical to four digits, so simulation error is negligible.

	(a) Table 26.1	(b) time = in+out	(c) lognormal
cost	-0.0226 (0.0037)	0.0072 (0.0059)	-0.0221 (0.0036)
time coefficient: mean	-0.0168 (0.0010)	-0.0304 (0.0023)	-0.0159
time coefficient: s.d.	0.0048 (0.0011)	0.0168 (0.0021)	0.0034
lognormal (μ, σ)			-4.160 (0.054), 0.208 (0.052)
log-likelihood	-2095.5	-1994.7	-2098.0

- (a) These reproduce the mixed logit column of Table 26.1: cost -0.023 (0.004), $\sigma(\text{intime}) = 0.0048$ (0.0011), air 0.040, 0.35, -2.72, bus -0.050, -0.24, -1.82, and car 0.008, -1.01, 1.89, with matching standard errors. The exception is the mean intime coefficient, which we estimate at -0.0168 (0.0010), whereas the table and text print -0.014. All other entries, including the standard deviation, agree, so the printed value appears to be a typo (perhaps carried over from the conditional logit column, -0.015). The standard deviation is about 30% of the mean, so the implied probability of a wrong-signed time coefficient is $\Phi(-0.0168/0.0048) = \Phi(-3.5) \approx 0.0002$. Against conditional logit, LR = 10.2. Since $\sigma = 0$ lies on the boundary, the relevant null distribution is $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$, giving $p < 0.001$. Heterogeneity in time costs is statistically significant, but the other coefficients barely move.
- (b) Using total travel time, the random coefficient has mean -0.030 and a large standard deviation, 0.017. The fit improves greatly over the conditional logit with the same regressors (LR = $2(2040.8 - 1994.7) = 92$). But the cost coefficient turns positive and insignificant. Across modes, cost and travel time tend to move in opposite directions (air is expensive and fast). With a widely dispersed time coefficient, heterogeneous time valuation absorbs the variation that the cost coefficient captured before. The price effect is no longer identified in practice, and a value of time cannot be computed. This is a warning that mixed logit estimates can be fragile, not evidence that travellers ignore cost. A random cost coefficient or a sign restriction on it would be natural next steps.
- (c) The lognormal coefficient on nintime has median $e^{\hat{\mu}} = 0.0156$, mean $e^{\hat{\mu} + \hat{\sigma}^2/2} = 0.0159$ and s.d. 0.0034. So the intime coefficient has mean -0.016 and cannot take the wrong sign. The cost coefficient and the constants are essentially unchanged. *Comparison:* models (a) and (c) have the same number of parameters and are non-nested, so we compare log-likelihoods directly (equivalently AIC or BIC). The normal specification fits better by 2.5 points. The sign restriction buys nothing here, because the normal assigns only 0.02% probability to positive time coefficients. And the lognormal's right skew is not supported. Substantively, both models agree on the average disutility of time (-0.017 versus -0.016) and on modest heterogeneity (s.d. 0.005 versus 0.003).

Exercise 26.18

Identification and computation. Only utility differences matter. Take train as the base: the model depends on $\Omega = \text{var}(\varepsilon_{\text{air}} - \varepsilon_{\text{train}}, \varepsilon_{\text{bus}} - \varepsilon_{\text{train}}, \varepsilon_{\text{car}} - \varepsilon_{\text{train}})$, a 3×3 matrix with six elements. Fixing the scale by $\Omega_{11} = \sigma_{\text{air}}^2 = 2$ (the book's "scale alternative" air) leaves five free covariance parameters. We parameterize $\Omega = LL'$ with L lower triangular and $L_{11} = \sqrt{2}$, and evaluate

the three-dimensional choice probabilities with the GHK simulator (300 Halton draws). For each traveller the differences are taken relative to the *chosen* alternative.

A common shortcut assumes a train error independent of the others, with $\text{var} = 1$, and a free covariance among air, bus and car. It implies $\Omega - \mathbf{u}\mathbf{u}' \succeq 0$, a strict subset of the admissible Ω . That restricted model reaches only log-likelihood -2019.4 , and it does not reproduce the table.

	(a) ours	Table 26.1	(b) logs
cost	-0.0050 (0.0021)	-0.005 (0.002)	-4.12 (0.56)
intime	-0.0055 (0.0007)	-0.005 (0.001)	0.26 (0.24)
air: income, urban, const.	0.018, -0.38, 0.31	0.018, -0.38, 0.32	0.022, -0.18, 4.70
bus: income, urban, const.	-0.008, -0.13, -0.28	-0.008, -0.14, -0.23	-0.131, -0.26, -11.9
car: income, urban, const.	0.013, -0.79, 1.51	0.013, -0.79, 1.51	0.011, -1.11, 2.11
$\sigma_{\text{bus}}^2, \sigma_{\text{car}}^2$	0.41, 3.86	0.41, 3.8	40.2, 6.97
$\rho_{\text{air,bus}}, \rho_{\text{air,car}}, \rho_{\text{car,bus}}$	0.58, 0.99, 0.57	0.60, 0.99, 0.60	0.23, 0.93, 0.12
log-likelihood	-2017.5	—	-1979.0

In (b) the cost and intime coefficients are on the logarithms.

- (a) Column (a) reproduces the general probit column of Table 26.1, together with its standard errors (cost 0.0021, intime 0.0007, air constant 0.23, car constant 0.20, ...). Seven of the eleven coefficients agree to the printed digits, and all eleven agree to within a tenth of a standard error. The largest gap, the bus constant (-0.28 versus -0.23 , s.e. 0.58), is simulation error from a different simulator and different draws. The covariance estimates agree with the book's table of variances and correlations. Relative to the simple probit with i.i.d. errors, the five covariance parameters raise the log-likelihood by 91.8 (LR = 183.6, 5 degrees of freedom).

Interpretation. As the book notes, the cost and time coefficients are much smaller than in conditional logit, and so are the implied marginal effects. Their ratio, the value of in-vehicle time, is $0.0055/0.0050 = \$1.10$ per minute (\$66 per hour), versus \$41 per hour in conditional logit. The estimated correlation between the air and car shocks (relative to train) is 0.99. This is the probit counterpart of the nested logit's small $\hat{\tau}_{\text{car,air}} = 0.24$: car and air are close substitutes, so independent errors are misspecified.

- (b) With logs, the log-likelihood rises to -1979.0 , a gain of 38.5 with the same number of parameters. So the log specification fits much better, as it did for conditional logit (Exercise 26.15(d)). Both starting values converge to the same maximum (-1978.99). The restricted “independent train error” parameterization reaches -1979.03 , so the optimum is not a numerical artefact. The findings:

- Log cost is strongly significant (-4.12 , $t = -7.4$). Log in-vehicle time has the wrong sign and is insignificant (0.26 , $t = 1.1$). In every model with cost in logs, travel time loses most of its explanatory power (Exercises 26.15(d), 26.16(b)): proportional cost differences absorb what travel time explained in levels. The time effect is not robust to functional form.
- The air-car correlation remains high (0.93).

- The bus-difference variance explodes to 40, and the bus coefficients become meaningless (constant -11.9 , s.e. 14.3). Only 10 of the 2779 travellers (0.36%) chose the bus, so the bus utility, and in particular its scale, is barely identified. The likelihood is nearly flat in that direction, and nothing should be read into those parameters.

Chapter 27

Censoring and Selection

Solutions to all 11 exercises of Chapter 27 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 27.9–27.11 use CHJ2004, cps09mar and DDK2011.

- Tobit: maximum likelihood with robust (sandwich) standard errors.
- CLAD: Buchinsky's iterated LAD algorithm, with exact LAD by linear programming. Standard errors are nonparametric bootstrap (100 replications).
- OLS: heteroskedasticity-robust standard errors, clustered by school for DDK.

Notation: ϕ, Φ are the standard normal density and distribution function, $\lambda(x) = \phi(x)/\Phi(x)$ is the inverse Mills ratio, and $\mu = X'\beta$. We repeatedly use $\mathbb{E}[z\mathbf{1}\{\{z > a\}\}] = \phi(a)$ and $\mathbb{E}[z \mid z > a] = \phi(a)/(1 - \Phi(a))$ for $z \sim N(0, 1)$.

Exercise 27.1

Given $X, Y^* = \mu + \sigma z$ with $z \sim N(0, 1)$, and $Y^* > 0 \iff z > -\mu/\sigma$.

(27.2). $Y = Y^*\mathbf{1}\{\{Y^* > 0\}\}$, so

$$\begin{aligned}\mathbb{E}[Y \mid X] &= \mu\mathbb{P}[z > -\mu/\sigma] + \sigma\mathbb{E}[z\mathbf{1}\{\{z > -\mu/\sigma\}\}] = \mu\Phi(\mu/\sigma) + \sigma\phi(-\mu/\sigma) \\ &= X'\beta\Phi\left(\frac{X'\beta}{\sigma}\right) + \sigma\phi\left(\frac{X'\beta}{\sigma}\right).\end{aligned}$$

(27.3). $\mathbb{E}[Y^* \mid X, Y^* > 0] = \mu + \sigma\mathbb{E}[z \mid z > -\mu/\sigma] = \mu + \sigma\frac{\phi(\mu/\sigma)}{1 - \Phi(-\mu/\sigma)} = X'\beta + \sigma\lambda\left(\frac{X'\beta}{\sigma}\right)$, using the symmetry of ϕ .

Exercise 27.2

No. In the observed sample,

$$\mathbb{E}[Y \mid X, \text{observed}] = X'\beta + \mathbb{E}[e \mid e \leq \tau - X'\beta] = X'\beta - \sigma\lambda\left(\frac{\tau - X'\beta}{\sigma}\right),$$

since $\mathbb{E}[z \mid z \leq c] = -\phi(c)/\Phi(c)$. The correction term is always negative and grows in magnitude as $X'\beta$ approaches and exceeds τ . Units with high predicted Y are observed only when their error is sufficiently negative. The observed regression function therefore lies below $X'\beta$ and bends downward at the top, approaching τ . OLS estimates the best linear approximation to this concave function. It is inconsistent, with slopes typically attenuated toward zero (for normal regressors, Greene's result gives the factor $1 - \pi$, where π is the probability of truncation). The problem is that selection depends on Y , and hence on e .

Exercise 27.3

(a) Yes. Selection depends only on X , and e is independent of X , so $\mathbb{E}[e \mid X, X_1 > 0] = 0$. OLS on the selected sample converges to $\mathbb{E}[XX' \mid X_1 > 0]^{-1}\mathbb{E}[XY \mid X_1 > 0] = \beta$, provided $\mathbb{E}[XX' \mid X_1 > 0]$ is non-singular. Exogenous selection on regressors only changes the design distribution.

(b) No. Selection on Y is selection on e : $\mathbb{E}[Y \mid X, Y > 0] = X'\beta + \sigma\lambda(X'\beta/\sigma)$ by (27.3). The probability limit is

$$\begin{aligned}\text{plim } \hat{\beta} &= \mathbb{E}[XX' \mid Y > 0]^{-1} \mathbb{E}[X(X'\beta + \sigma\lambda(X'\beta/\sigma)) \mid Y > 0] \\ &= \beta + \sigma \mathbb{E}[XX' \mid Y > 0]^{-1} \mathbb{E}[X\lambda(X'\beta/\sigma) \mid Y > 0] \neq \beta.\end{aligned}$$

Since λ is decreasing, the omitted term is negatively correlated with $X'\beta$, and the slopes are attenuated. (The question's "consistent for $\hat{\beta}$ " should read "for β ".)

Exercise 27.4

Using all observations, including the zeros,

$$(\hat{\beta}, \hat{\sigma}) = \arg \min_{\beta, \sigma > 0} \sum_{i=1}^n \left(Y_i - X_i'\beta \Phi(X_i'\beta/\sigma) - \sigma \phi(X_i'\beta/\sigma) \right)^2.$$

It is consistent under the Tobit model (the regression function is correctly specified), but less efficient than the Tobit MLE. The errors are heteroskedastic, so robust standard errors (23.6) should be used. Starting values can come from OLS or Tobit.

Exercise 27.5

Using only the uncensored observations $Y_i > 0$,

$$(\hat{\beta}, \hat{\sigma}) = \arg \min_{\beta, \sigma > 0} \sum_{i: Y_i > 0} \left(Y_i - X_i'\beta - \sigma\lambda(X_i'\beta/\sigma) \right)^2,$$

with robust standard errors. Identification of σ separately from β relies on the nonlinearity of λ and hence on the normality assumption. The estimator can be poorly behaved when $X'\beta/\sigma$ is large for most observations, where λ is nearly linear.

Exercise 27.6

With $\mu_i = \beta_0 + X_i\beta_1$ and $\sigma_i = (\gamma_0 + X_i^2\gamma_1)^{1/2}$, a censored observation has probability $\mathbb{P}[Y^* \leq 0 \mid X_i] = \Phi(-\mu_i/\sigma_i)$ and an uncensored one has density $\sigma_i^{-1}\phi((Y_i - \mu_i)/\sigma_i)$. Hence

$$\ell_n(\beta_0, \beta_1, \gamma_0, \gamma_1) = \sum_{i=1}^n \left\{ \mathbf{1}\{Y_i = 0\} \log \Phi\left(-\frac{\mu_i}{\sigma_i}\right) + \mathbf{1}\{Y_i > 0\} \left[-\frac{1}{2} \log(2\pi\sigma_i^2) - \frac{(Y_i - \mu_i)^2}{2\sigma_i^2} \right] \right\}.$$

Exercise 27.7

$S = 1 \iff u > -X'\gamma$. Project $e = \sigma_{21}u + \varepsilon$ ($\text{var}(u) = 1$), where ε is normal, uncorrelated with u , hence independent of u (and of X). Then

$$\begin{aligned}\mathbb{E}[Y \mid X, S = 1] &= X'\beta + \sigma_{21}\mathbb{E}[u \mid u > -X'\gamma] + \mathbb{E}[\varepsilon \mid u > -X'\gamma] \\ &= X'\beta + \sigma_{21} \frac{\phi(-X'\gamma)}{1 - \Phi(-X'\gamma)} + 0 = X'\beta + \sigma_{21}\lambda(X'\gamma).\end{aligned}$$

Exercise 27.8

The same argument with selection $S = 1 \iff u > -Z'\gamma$: writing $e = \sigma_{21}u + \varepsilon$ with ε independent of (u, X, Z) , $\mathbb{E}[Y \mid X, Z, S = 1] = X'\beta + \sigma_{21}\mathbb{E}[u \mid u > -Z'\gamma] = X'\beta + \sigma_{21}\lambda(Z'\gamma)$.

Exercise 27.9

$n = 8684$ households. Mean in-kind transfers are 1.73 thousand pesos, the median 0.41. The slope below the kink is the income coefficient, and the slope above is the sum of the two coefficients:

	constant	income	Dincome	slope above 1
(a) OLS	2.70 (0.62)	−1.53 (0.63)	1.55 (0.63)	0.012
(c) OLS, tinkind > 0	3.56 (0.77)	−2.14 (0.79)	2.16 (0.79)	0.021
(d) Tobit	1.73 (0.80)	−1.56 (0.81)	1.57 (0.81)	0.009
(e) CLAD	0.96 (0.40)	−0.56 (0.40)	0.56 (0.40)	0.000

(b) 25.4% of households report zero in-kind transfers. Greene’s rule of thumb (27.4) suggests an attenuation of OLS slopes of roughly that order, so censoring bias is a potential concern.

(f) All estimators tell the same qualitative story: in-kind transfers decline with income at very low income, and the relationship is flat above 1000 pesos. Two features drive the differences.

- *The kink sits at the very bottom of the income distribution.* Only 82 households (0.9%) have income below 1000 pesos, so the “low-income” slope is estimated from a handful of observations, while Dincome is almost collinear with income.
- *The estimators target different objects.* OLS and Tobit are close. Censoring barely affects the slopes because income hardly changes the probability of a zero, so Greene’s normal-regressor bound is pessimistic here. Dropping the zeros (c) biases the intercept upward and exaggerates the low-income slope: selection on $Y > 0$. CLAD estimates the conditional median, which is much smaller than the mean because transfers are highly skewed (a few large transfers). Its slopes are about a third of Tobit’s, and the Tobit normality assumption is clearly violated by the long right tail.

Robustness: with the kink at 20 (20,000 pesos, as in Figure 20.2(b)) the below-kink slopes are OLS -0.032 (0.011), truncated -0.029 , Tobit -0.062 (0.014) and CLAD -0.015 (0.004), and all above-kink slopes are near zero. So the crowding-out of in-kind transfers is small in pesos and limited to the poorest households.

Exercise 27.10

23.5% of observations are capped. Estimates, with implied returns to schooling $\partial \mathbb{E} / \partial \text{edu} = \beta_1 + 2\beta_2 \text{edu}$:

	edu	edu ²	return at 12	at 16	at 20
(a) OLS, uncapped	0.118 (0.018)	−0.0000 (0.0006)	0.118	0.118	0.118
(b) OLS, capped	0.239 (0.013)	−0.0053 (0.0004)	0.113	0.070	0.028
(c) OLS, drop capped	0.211 (0.021)	−0.0052 (0.0007)	0.085	0.043	0.001
(d) Tobit	0.162 (0.020)	−0.0015 (0.0007)	0.125	0.113	0.101
(e) CLAD (cap 3.3)	0.112 (0.24)	0.0000 (0.008)	0.112	0.112	0.112

(a) The uncapped regression is essentially linear, with a constant return of about 12% per year of schooling.

- (b) With capped data one would conclude that returns fall sharply with education, from 11% at 12 years to under 3% at 20, a spurious concavity. The more educated are more likely to earn above the cap, so their mean is pulled down most.
- (c) Dropping capped observations makes it worse: returns fall from 8.5% to zero. Truncation removes exactly the high-wage workers among the educated.
- (d) Tobit largely undoes the censoring: returns of 12.5% declining mildly to 10%, close to (a). The remaining concavity reflects departures from homoskedastic normality.
- (e) CLAD reproduces the linear profile with a 11.2% return, since the median is unaffected by capping where the median is below the cap. The cap is lowered to 3.3 because at 3.4 the fitted median exceeds the cap for the highly educated, leaving too little uncensored information. Its bootstrap standard errors are large: few distinct education values lie in the uncensored region, so three coefficients are determined by essentially four conditional medians.
- (f) Censoring from above flattens the top of the regression (attenuation), and truncation flattens it further. Censored-regression methods recover the latent relationship: Tobit through the normality assumption (efficient but model dependent), CLAD through a median restriction (robust but imprecise).

Exercise 27.11

$n = 5304$; 57% of standardized scores are negative, so they are censored at zero in (b).

	tracking	percentile	percentile ²	constant
(a) OLS, testscore	0.163 (0.077)	0.0104 (0.0018)	0.00007 (0.00002)	-0.85 (0.06)
(b) OLS, ctest	0.089 (0.050)	-0.0020 (0.0011)	0.00011 (0.00001)	0.09 (0.03)
(c) OLS, ctest > 0	0.077 (0.062)	-0.0060 (0.0030)	0.00012 (0.00002)	0.75 (0.10)

- (a) Tracking raises end-of-year scores by 0.16 standard deviations (significant at 5%), and scores increase with the initial percentile, slightly convexly.
- (b) With censored data the tracking effect is roughly halved (0.09, significant only at 10%). The percentile profile changes shape: the linear term turns negative, suggesting a U-shaped relation. An unaware researcher would understate the effect of tracking and misread the role of initial achievement.
- (c) Dropping the censored observations shrinks the tracking effect further (0.08, insignificant) and distorts the percentile profile more, because the remaining sample is selected on the outcome.
- (d) With 57% censoring the attenuation is large. Greene's approximation predicts slopes multiplied by roughly $1 - 0.57 = 0.43$, i.e. $0.163 \times 0.43 \approx 0.07$, close to (b). Censoring removes all variation among low scorers, so effects that operate at the bottom of the distribution (and the curvature in percentile) are lost. Truncation additionally selects students with positive errors, whose conditional mean depends on the regressors. A censored regression method (Tobit gives a tracking effect of 0.21, or CLAD) would be needed to recover the latent effect.

Chapter 28

Model Selection, Stein Shrinkage, and Model Averaging

Solutions to all 12 exercises of Chapter 28 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercise 28.12 uses `cps09mar`. The code reproduces Table 28.1 (Asian women) exactly for the returns, standard errors, BIC, AIC and CV, and for FIC* in models 4–9; see the remark in Exercise 28.12.

Notation: $\hat{\theta} \sim (\theta, V)$ is $K \times 1$, $\text{wmse}[\tilde{\theta}] = \mathbb{E}[(\tilde{\theta} - \theta)'V^{-1}(\tilde{\theta} - \theta)]$, and $\lambda = \theta'V^{-1}\theta$.

Exercise 28.1

$\text{wmse}[\hat{\theta}] = \mathbb{E}[\text{tr}(V^{-1}(\hat{\theta} - \theta)(\hat{\theta} - \theta)')] = \text{tr}(V^{-1}V) = K$. For $\tilde{\theta} = 0$, $\text{wmse}[\tilde{\theta}] = (0 - \theta)'V^{-1}(0 - \theta) = \theta'V^{-1}\theta = \lambda$.

Exercise 28.2

WLS is linear: $\hat{m} = AY$ with $A = X(X'\Omega X)^{-1}X'\Omega$, $\Omega = \text{diag}(\omega_i)$. Its trace is $\text{tr} A = \text{tr}((X'\Omega X)^{-1}X'\Omega X) = k$. By (28.14),

$$C_p = \sum_{i=1}^n (Y_i - X_i'\hat{\beta}_{\text{wls}})^2 + 2\tilde{\sigma}^2 k,$$

the unweighted sum of squared WLS residuals plus the usual penalty. Under homoskedasticity the WLS weights are not efficient, but the criterion is still unbiased (after subtracting $e'e$) for $R = \mathbb{E} \sum (\hat{m}_i - m_i)^2$ by Theorem 28.5.

Exercise 28.3

For normal regression $\text{AIC} = n \log \hat{\sigma}^2 + 2K + \text{const}$. Removing regressor j changes K by -1 and the residual sum of squares from SSE to SSE_{-j} . The F statistic for $\beta_j = 0$ equals the squared classical t -ratio,

$$t_j^2 = F_j = \frac{\text{SSE}_{-j} - \text{SSE}}{\text{SSE}/(n-k)} \implies \text{SSE}_{-j} = \text{SSE} \left(1 + \frac{t_j^2}{n-k} \right).$$

Therefore $\text{AIC}_{-j} - \text{AIC} = n \log(1 + t_j^2/(n-k)) - 2$, a strictly increasing function of t_j^2 that is the same for every j . The omission with the smallest impact on AIC is the one with the smallest t_j^2 , i.e. the smallest $|t_j|$. (It lowers AIC iff $t_j^2 < (n-k)(e^{2/n} - 1) \approx 2$, the familiar $|t| < \sqrt{2}$, $p \approx 0.16$ rule; the book's $p = 0.32$ quote is the two-sided version.)

Exercise 28.4

Adding any one regressor raises the AIC penalty by the same amount, so the best addition is the one that most reduces $\text{SSE} = \|\hat{e}\|^2$, where $\hat{e} = MY$ and M is the annihilator of the active set. By

FWL, adding x gives the new residual $\hat{e} - \tilde{x}(\tilde{x}'\tilde{x})^{-1}\tilde{x}'\hat{e}$ with $\tilde{x} = Mx$, so

$$\text{SSE}_{\text{new}} = \|\hat{e}\|^2 - \frac{(\tilde{x}'\hat{e})^2}{\tilde{x}'\tilde{x}} = \|\hat{e}\|^2(1 - \cos^2 \angle(\tilde{x}, \hat{e})).$$

Geometrically, the reduction is the squared length of the projection of $\hat{e}$ on $\tilde{x}$: the regressor whose (partialled) direction is most parallel to $\hat{e}$ reduces SSE most. Since $\hat{e} \perp$ the active set, $\tilde{x}'\hat{e} = x'\hat{e}$, and $\cos \angle(\tilde{x}, \hat{e})$ is the correlation between $\hat{e}$ and $\tilde{x}$. So the optimal choice maximizes the absolute correlation between the residual and the regressor partialled on the active set. (With the intercept in the active set, centred correlations and centred residuals coincide with these cosines.) It coincides with the largest absolute correlation between $\hat{e}$ and x itself when the candidate regressors have equal norms after partialling out the active set, e.g. when they are standardized and orthogonal to the active set. This is the normalization used by LARS.

Exercise 28.5

With considerable caution. The reported model was chosen because its coefficients were significant, so the estimates are selected on their own realizations.

- They are biased away from zero (a winner's curse / pretest bias).
- The reported t -ratios overstate the evidence: conditional on selection their distribution is not $N(0, 1)$, so nominal p -values and confidence intervals are invalid. Section 28.17 shows that coverage can be far below the nominal level.
- The implicit multiple testing across specifications further inflates the false-positive rate.

A reader should discount the significance, ask to see all the estimated specifications (a specification curve or robustness table), and prefer estimates chosen by a pre-specified criterion (CV, AIC, FIC) or averaged across models.

Exercise 28.6

For $\tilde{\theta} = (1 - w)\hat{\theta}$: $\mathbb{E}[\tilde{\theta}] - \theta = -w\theta$ (28.21), $\text{var}[\tilde{\theta}] = (1 - w)^2 V$ (28.22), and

$$\text{wmse}[\tilde{\theta}] = \text{tr}(V^{-1}[(1 - w)^2 V + w^2 \theta \theta']) = K(1 - w)^2 + w^2 \lambda \quad (28.23).$$

- (1) $K(1 - w)^2 + w^2 \lambda < K \iff w[w(K + \lambda) - 2K] < 0 \iff 0 < w < 2K/(K + \lambda)$.
- (2) The derivative $-2K(1 - w) + 2w\lambda$ vanishes at $w_0 = K/(K + \lambda)$; the function is a convex quadratic, so this is the minimum.
- (3) $\text{wmse} = K\left(\frac{\lambda}{K + \lambda}\right)^2 + \lambda\left(\frac{K}{K + \lambda}\right)^2 = \frac{K\lambda(\lambda + K)}{(K + \lambda)^2} = \frac{K\lambda}{K + \lambda}$.

Exercise 28.7

Writing $\hat{\theta} = \theta + (\hat{\theta} - \theta)$, $\mathbb{E}[\hat{\theta}' V^{-1} \hat{\theta}] = \theta' V^{-1} \theta + 2\theta' V^{-1} \mathbb{E}[\hat{\theta} - \theta] + \mathbb{E}[(\hat{\theta} - \theta)' V^{-1} (\hat{\theta} - \theta)] = \lambda + 0 + \text{tr}(V^{-1} V) = \lambda + K$. Hence $\mathbb{E}[\hat{\lambda}] = \lambda$.

Exercise 28.8

With $V = I$ and $r = 0$, $\hat{\theta} - \hat{\theta}_R = P\hat{\theta}$ where $P = R(R'R)^{-1}R'$ is idempotent of rank q . The estimator is $\tilde{\theta} = \hat{\theta} - (q - 2)g(\hat{\theta})$ with $g(x) = Px/(x'Px)$. Then, with $Z = \hat{\theta} - \theta \sim N(0, I_K)$,

$$\text{wmse}[\tilde{\theta}] = \mathbb{E}\|Z\|^2 - 2(q - 2)\mathbb{E}[Z'g(\hat{\theta})] + (q - 2)^2\mathbb{E}\|g(\hat{\theta})\|^2.$$

Stein's lemma. For $X \sim N(\theta, I)$ and weakly differentiable g with $\mathbb{E}|\partial g_j / \partial x_j| < \infty$, $\mathbb{E}[(X - \theta)'g(X)] = \mathbb{E}[\text{div } g(X)]$ (integrate by parts coordinate by coordinate, using $\phi'(u) = -u\phi(u)$). Here

$$\text{div } g(x) = \frac{\text{tr } P}{x'Px} - \frac{2x'PPx}{(x'Px)^2} = \frac{q-2}{x'Px}, \quad \|g(x)\|^2 = \frac{x'PPx}{(x'Px)^2} = \frac{1}{x'Px}.$$

The integrability condition holds because $\mathbb{E}[1/Q] < \infty$ for $Q = \hat{\theta}'P\hat{\theta}$ when $q > 2$. Hence

$$\text{wmse}[\tilde{\theta}] = K - 2(q-2)^2\mathbb{E}[Q^{-1}] + (q-2)^2\mathbb{E}[Q^{-1}] = K - (q-2)^2\mathbb{E}[Q^{-1}].$$

Since P is idempotent of rank q , $Q \sim \chi_q^2(\lambda_R)$ with non-centrality $\theta'P\theta = \theta'R(R'R)^{-1}R'\theta = \lambda_R$. So $\mathbb{E}[Q^{-1}] = J_q(\lambda_R) > 0$ and $\text{wmse}[\tilde{\theta}] = K - (q-2)^2J_q(\lambda_R) < K = \text{wmse}[\hat{\theta}]$ for every θ : uniform improvement when $q > 2$.

Exercise 28.9

(a) $\tilde{\theta} = w\hat{\theta}_1 + (1-w)\hat{\theta}_2$ is unbiased with $\text{mse} = \text{tr var}[\tilde{\theta}] = w^2 \text{tr } V_1 + (1-w)^2 \text{tr } V_2$ (no cross term by uncorrelatedness). The first-order condition $w \text{tr } V_1 = (1-w) \text{tr } V_2$ gives $w = \text{tr } V_2 / (\text{tr } V_1 + \text{tr } V_2) = \frac{1/\text{tr } V_1}{1/\text{tr } V_1 + 1/\text{tr } V_2}$, which is the stated formula.

(b) Minimizing $\sum_m w_m^2 \text{tr } V_m$ subject to $\sum_m w_m = 1$ (Lagrangian) gives $w_m \text{tr } V_m = \text{const}$, so

$$\tilde{\theta} = \frac{\sum_{m=1}^M \hat{\theta}_m / \text{tr } V_m}{\sum_{m=1}^M 1 / \text{tr } V_m}, \quad \text{mse}[\tilde{\theta}] = \left(\sum_{m=1}^M \frac{1}{\text{tr } V_m} \right)^{-1}.$$

(c) Inverse-variance weighting: each estimator is weighted by its precision, as in GLS or meta-analysis. The combined MSE is smaller than that of the best single estimator. Nothing is gained from an imprecise estimator only if its variance is infinite.

Exercise 28.10

(a) Because the weights sum to one, $\hat{e}(w) = \sum_m w_m(Y - \hat{Y}_m) = Y - \sum_m w_m \hat{Y}_m$, and $\text{tr } P(w) = \sum_m w_m \text{tr } P_m = \sum_m w_m k_m$. Hence $C(w) = \hat{e}(w)' \hat{e}(w) + 2\sigma^2 \text{tr } P(w)$ is the stated expression.

(b) For nested models $\hat{e}_m - \hat{e}_M = (P_M - P_m)Y$ lies in the column space of X_M , orthogonal to $\hat{e}_M$. So $\|\hat{e}(w)\|^2 = \|\hat{e}_M\|^2 + \|\sum_m w_m(\hat{e}_m - \hat{e}_M)\|^2 \geq \|\hat{e}_M\|^2$, with equality at $w = (0, \dots, 0, 1)$. Without the penalty all weight goes to the largest model, the usual overfitting of in-sample fit.

Exercise 28.11

The JMA criterion is the sum of squared leave-one-out residuals of the averaged fit, $\sum_i (\sum_m w_m \tilde{e}_{mi})^2$ with $\tilde{e}_{mi} = Y_i - \tilde{Y}_{mi}$. Since $\sum_m w_m = 1$, $\sum_m w_m \tilde{e}_{mi} = Y_i - \sum_m w_m \tilde{Y}_{mi}$.

Exercise 28.12

Same nine regressions of log wage on married, three region dummies, an education specification (college dummy / linear spline with a knot at 9 / six education dummies), and experience polynomials of order 2, 4, 6. The focus is the return to experience $\mu = m(30) - m(0)$. AIC and BIC are in the book's units ($n \log(2\pi\hat{\sigma}^2)$ plus penalty with $K = k$). CV is the sum of squared leave-one-out errors, and $\text{FIC}^* = n(\tilde{\mu} - \hat{\mu})^2 + 2ns(\tilde{\mu})^2$ with $\hat{\mu}$ from model 9 and HC1 standard errors.

model	1	2	3	4	5	6	7	8	9
education	college	spline	dummy	college	spline	dummy	college	spline	dummy
experience	2	2	2	4	4	4	6	6	6
return (%)	25	35	35	37	47	47	38	47	47
s.e.	4	4	4	5	5	5	7	7	7
BIC	1811	1432	1465	1814	1434	1467	1830	1450	1483
AIC	1763	1378	1387	1754	1368	1377	1758	1372	1381
CV	860	756	759	857	753	756	857	754	757
FIC*	165	53	52	46	13	13	55	28	33

Selection. BIC selects model 2 (spline, quadratic experience; return 35%). AIC, CV and FIC all select model 5 (spline, quartic experience; return 47%), with model 6 essentially tied on FIC.

Interpretation. Two dimensions matter. First, education enters much better as a spline with a knot at 9 years than as a single college dummy: every criterion improves by hundreds of points, and the college-dummy models give much smaller returns to experience because education and experience are negatively correlated. The six dummies add parameters without improving fit. Second, the quadratic in experience is too restrictive: moving to a quartic raises the return from 35% to 47%, and AIC, CV and FIC agree. BIC's heavier penalty ($\log n \approx 8$) keeps the quadratic, but the BIC difference between models 2 and 5 is only 2 points. Given the agreement of AIC, CV and the focused criterion, the preferred model is model 5: the return to 30 years of experience for Hispanic women is about 47% (s.e. 5%), larger and more precisely estimated than the 37% found for Asian women. The flatness of CV across models 5, 6, 8 and 9 (all with the same 47% estimate) is reassuring. Once education is modelled flexibly and experience at least quartically, the estimate is robust.

Remark. Replication note. For Asian women our code gives exactly the book's returns, standard errors, BIC, AIC, CV and FIC* for models 4–9. For models 1–3 we obtain $\text{FIC}^* = 126, 77, 84$ rather than the printed 86, 48, 53; the printed values correspond to a reference estimate of about 0.385 instead of model 9's 0.45. The FIC choice (model 5) is the same. The book's AIC and BIC values omit the constant n and count $K = k$ rather than $k + 1$ as in (28.3)–(28.4), which shifts all models equally.

Chapter 29

Machine Learning

Solutions to all 10 exercises of Chapter 29 of Bruce E. Hansen, *Econometrics* (Princeton University Press, 2022). Exercises 29.9–29.10 use `cps09mar`. The Lasso is computed as in `glmnet`: standardized regressors, an unpenalized intercept, coordinate descent along a decreasing path of 100 penalties, and 10-fold cross-validation, reporting both the CV-minimizing penalty ($\lambda_{\min}$) and the “1se” penalty.

Notation: $A_\lambda = X'X + \Lambda$ (with $\Lambda = \lambda I_p$ unless stated otherwise), $\hat{\beta}_{\text{ridge}} = A_\lambda^{-1}X'Y$, $\hat{e}_i(\lambda) = Y_i - X_i'\hat{\beta}_{\text{ridge}}$, and $h_{ii} = X_i'A_\lambda^{-1}X_i$.

Exercise 29.1

The leave-one-out estimator is $\hat{\beta}_{-i} = (A_\lambda - X_iX_i')^{-1}(X'Y - X_iY_i)$. (The book’s display writes $\sum_{j \neq i} X_jY_j$; it should be X_jY_j .) By Sherman–Morrison,

$$(A_\lambda - X_iX_i')^{-1} = A_\lambda^{-1} + \frac{A_\lambda^{-1}X_iX_i'A_\lambda^{-1}}{1 - h_{ii}}.$$

Hence

$$\hat{\beta}_{-i} = \hat{\beta}_{\text{ridge}} - A_\lambda^{-1}X_iY_i + \frac{A_\lambda^{-1}X_i(X_i'\hat{\beta}_{\text{ridge}} - h_{ii}Y_i)}{1 - h_{ii}} = \hat{\beta}_{\text{ridge}} - \frac{A_\lambda^{-1}X_i\hat{e}_i(\lambda)}{1 - h_{ii}}.$$

Therefore $e_i(\lambda) = Y_i - X_i'\hat{\beta}_{-i} = \hat{e}_i(\lambda) + h_{ii}\hat{e}_i(\lambda)/(1 - h_{ii}) = \hat{e}_i(\lambda)/(1 - h_{ii})$, exactly as for least squares (Theorem 3.7), with the ridge “leverage” h_{ii} .

Exercise 29.2

The ridge fit is linear in Y : $\hat{m} = X\hat{\beta}_{\text{ridge}} = AY$ with $A = XA_\lambda^{-1}X'$, a function of X only. The Mallows criterion (28.14) is $\hat{e}'\hat{e} + 2\hat{\sigma}^2 \text{tr}(A)$, and $\text{tr}(A) = \text{tr}(XA_\lambda^{-1}X') = \text{tr}(A_\lambda^{-1}X'X)$. So $C(\lambda) = \sum_i \hat{e}_i(\lambda)^2 + 2\hat{\sigma}^2 \text{tr}((X'X + \Lambda)^{-1}X'X)$, which is (29.7). The trace is the “effective number of parameters”, decreasing from p at $\lambda = 0$ towards 0.

Exercise 29.3

With $Y = X\beta + e$ and $\mathbb{E}[e | X] = 0$, $\mathbb{E}[\hat{\beta}_{\text{ridge}} | X] = A_\lambda^{-1}X'X\beta = A_\lambda^{-1}(A_\lambda - \lambda I_p)\beta = \beta - \lambda A_\lambda^{-1}\beta$, so the bias is $-\lambda(X'X + \lambda I_p)^{-1}\beta$ (29.8). Since $\hat{\beta}_{\text{ridge}} - \mathbb{E}[\hat{\beta}_{\text{ridge}} | X] = A_\lambda^{-1}X'e$ and $\mathbb{E}[ee' | X] = D$ under independent sampling, $\text{var}[\hat{\beta}_{\text{ridge}} | X] = A_\lambda^{-1}X'DXA_\lambda^{-1}$ (29.9).

Exercise 29.4

$X^*X^* = X'X + \lambda I_p$ and $X^*Y^* = X'Y + \sqrt{\lambda}\mathbf{0} = X'Y$, so $(X^*X^*)^{-1}X^*Y^* = (X'X + \lambda I_p)^{-1}X'Y = \hat{\beta}_{\text{ridge}}$. Equivalently, the augmented sum of squared errors is $\|Y - Xb\|^2 + \lambda\|b\|^2$, the ridge criterion. The p pseudo-observations “pull” each coefficient toward zero.

Exercise 29.5

Least squares (weakly; strictly for $\lambda > 0$ unless the least squares coefficients are zero). It minimizes the sum of squared residuals over all b , so any other coefficient vector, including the ridge estimate, has a larger SSE and a lower R^2 (with the same intercept treatment). Ridge sacrifices in-sample fit for lower variance and better out-of-sample prediction, so a lower R^2 is not a reason to prefer least squares.

Exercise 29.6

The criterion is $\|Y - X(b_1 + b_2)\|^2 + \lambda(\|b_1\|^2 + \|b_2\|^2)$. For a given sum $s = b_1 + b_2$ the fit is fixed. The penalty is strictly convex, and $\|b_1\|^2 + \|b_2\|^2 \geq \frac{1}{2}\|s\|^2$ with equality iff $b_1 = b_2 = s/2$. So at the optimum $\hat{\beta}_1 = \hat{\beta}_2 = s/2$, where s minimizes $\|Y - Xs\|^2 + \frac{\lambda}{2}\|s\|^2$. This gives $s = (X'X + \frac{\lambda}{2}I_p)^{-1}X'Y$ and $\hat{\beta}_1 = \hat{\beta}_2 = \frac{1}{2}(X'X + \lambda I_p)^{-1}X'Y$. Ridge does not need X (here $\tilde{X}$) to have full column rank: $\tilde{X}'\tilde{X} + \lambda I_{2p}$ is invertible for any $\lambda > 0$, and the strictly convex penalty splits the weight equally between perfectly collinear regressors.

Exercise 29.7

The criterion is $\|Y - X(b_1 + b_2)\|^2 + \lambda(\|b_1\|_1 + \|b_2\|_1)$. By the triangle inequality $\|b_1\|_1 + \|b_2\|_1 \geq \|b_1 + b_2\|_1$, with equality iff $b_{1j}b_{2j} \geq 0$ for every j (same signs coordinatewise). Minimizing over $s = b_1 + b_2$ therefore gives the Lasso problem $\|Y - Xs\|^2 + \lambda\|s\|_1$, so $\hat{\beta}_1 + \hat{\beta}_2 = \hat{\beta}_{\text{Lasso}}$. But any split with $b_1 + b_2 = \hat{\beta}_{\text{Lasso}}$ and matching signs, e.g. $b_1 = \alpha\hat{\beta}_{\text{Lasso}}$, $b_2 = (1 - \alpha)\hat{\beta}_{\text{Lasso}}$ for any $\alpha \in [0, 1]$, attains the same minimum. The individual coefficients are indeterminate. The ℓ_1 penalty is not strictly convex, unlike ridge; the elastic net restores uniqueness by adding a small ridge component.

Exercise 29.8

No. A tree splits on thresholds of a single variable, $X \leq c$ versus $X > c$. Since $x \mapsto x^2$ is strictly increasing on $[0, \infty)$, $\{X^2 \leq c\} = \{X \leq \sqrt{c}\}$: every split available on X^2 is already available on X , and the fitted tree (partition and leaf means) is unchanged. Regression trees are invariant to monotone transformations of the regressors, so polynomial terms add no flexibility. They only change tie-breaking and the reported split values. Trees gain flexibility by adding genuinely different variables or more splits, not transformations.

Exercise 29.9

There are 27 candidate regressors, and some are redundant by construction. No worker in `cps09mar` has exactly 15 years of education (the coding jumps from 14 to 16), so that dummy is identically zero, leaving 26 usable regressors. The five marital dummies and the four region dummies each sum to one, so they are collinear with the unpenalized intercept. The Lasso still has a solution; as in Exercise 29.7 it simply picks one representation. With 10-fold cross-validation along a path of 100 penalties (λ in `glmnet` units, i.e. for the criterion $\frac{1}{2n}\text{SSE} + \lambda\|\beta\|_1$ on standardized regressors):

	$\lambda_{\min} = 0.00008$ (CV MSE 0.336)	$\lambda_{1se} = 0.062$ (CV MSE 0.353)
selected regressors	20 of 26	3 of 26
education	−0.0004	0.082
edu12, 13, 14	0.19, 0.34, 0.52	−0.083, −, −
edu16, 18, 20	0.85, 1.01, 1.24	0.016, −, −
experience powers	x : 2.64, x^2 : −5.78, x^3 : 4.13, x^6 : −0.71, x^9 : 0.09	none
marital	div. 0.06, sep. −0.15, wid. −0.05, never −0.03	none
region	NE 0.07, MW −0.14, S −0.07	none
union	−0.04	none

(Experience enters as $x = \text{experience}/40$. The $\lambda_{\min}$ fit implies a log-wage gain of 0.36 over the first 10 years and 0.35 at 30 years.)

Interpretation. The CV criterion is minimized at an almost negligible penalty. The $\lambda_{\min}$ model is essentially the least squares regression (residual variance 0.324 for OLS with all regressors), with only the redundant or weakest terms dropped. Education works through the dummy categories (the linear term is zeroed), and experience through a few low-order powers, a concave profile that peaks at about 15 years. The “1se” rule instead selects a very sparse model with education alone: a linear return of about 8% per year plus two tiny dummies, and no experience, marital, regional or union effects. With $n = 1149$ the CV curve is flat enough that the 1se rule shrinks away real but imprecisely estimated effects (the experience profile). For description $\lambda_{\min}$ is preferable; the 1se model is a parsimonious prediction model that sacrifices about 5% in CV MSE.

Exercise 29.10

	$\lambda_{\min} = 0.00003$ (CV MSE 0.311)	$\lambda_{1se} = 0.028$ (CV MSE 0.327)
selected regressors	21 of 26	11 of 26
education	0.028	0.071
edu12, 13, 14	0.16, 0.35, 0.44	−0.041, −, 0.009
edu16, 18, 20	0.55, 0.80, 1.06	0.046, 0.162, 0.230
experience powers	x : 2.15, x^2 : −2.35, x^4 : 0.88 x^5 : 0.26, x^7 : −0.44, x^9 : 0.09	x : 0.258
marital	married 0.07, sep. −0.07, wid. 0.15, never −0.09	married 0.056, never −0.115
region	NE 0.03, MW −0.03, S −0.10	S −0.047
union	0.31	0.098

The $\lambda_{\min}$ fit implies log-wage gains of 0.39 after 10 years and 0.58 after 30 years of experience; the 1se model implies 0.19 at 30 years.

Interpretation. Again $\lambda_{\min}$ is tiny and the fitted model is close to least squares (OLS residual variance 0.307). But with four times the sample size the 1se model is much richer than for Asian women. It keeps education, with large graduate-degree premia (MA +16%, PhD/professional +23% on top of a 7% per-year return), a linear experience term, the marriage premium (married +6%, never married −12%), a South discount, and a substantial union premium (+10% in the sparse model, +31% at $\lambda_{\min}$). The union premium is large for Hispanic men but was zeroed for Asian women, and marital status matters for Hispanic men but not for Asian women. The two samples

differ in both sex and ethnicity, so these contrasts are descriptive. They are in line with familiar patterns (a marriage premium for men, stronger union effects in blue-collar employment). Across both samples the Lasso's model choice depends strongly on the tuning rule and on the sample size, so the 1se model should not be read as evidence that the excluded variables have zero effect.